

SAFRAN CORTEX SERIES USER MANUAL



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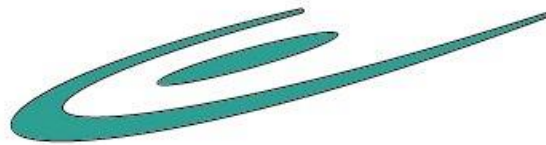
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CORTEX Series



HDR-4G+ USER'S MANUAL

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3.5	June 1, 2021	Safety instructions update

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ACRONYMS & ABBREVIATIONS

SYMBOLS:

%	Percent
±	Plus or minus
≥	Greater than or equal
≤	Less than or equal
&	And
≅	Approximately

PREFIXES:

n	Nano-
μ	Micro-
m	Milli-
k	Kilo-
M	Mega-
G	Giga-
T	Tera-

UNITS:

A	Ampera
b	Byte
bps	Bit per second
C	Celsius
dB	Decibel
dBc	dB with respect to the unmodulated carrier
F	Farad
Hz	Hertz
mm	Millimeter
rpm	revolutions per minute
Ω	Ohms
s	Second
sps	Symbol per second
s/s	Sample per second (ks/s, Ms/s,...)
V	Volt

ABBREVIATIONS:

AQPSK	Asynchronous QPSK
BCD	Binary Coded Decimal
BER	Bit Error Rate
BPL	BiPhase-L
BPSK	Binary Phase Shift Keying
BR	Bit Rate
CTL	Check To Lock
DMU	Demodulator Unit

DNRZ	Differential Non Return to Zero
DPU	Data Processing Unit
DRG	Data Recording Global
DRU	Data Recording Unit
DTR	Digital Tracking Receiver
DSP	Digital Signal Processor
FFT	Fast Fourier Transform
GMSK	Gaussian Minimum Shift Keying
HDR	High Data rate Receiver
H/W	Hardware
ID	Identifier
IF	Intermediate Frequency
LAN	Local Area Network
LDPC	Low Density Parity Check
LTS	Lock To Search
LSB	Least Significant Byte (Bit)
MCS	Monitoring & Control Sub-system
MSB	Most Significant Byte (Bit)
M&C	Monitoring & Control
NA	Not Applicable
NRZ-L	Non Return to Zero Level
NRZ-M	Non Return to Zero Mark
OQPSK	Offset QPSK
PBU	Playback Unit
PCM	Pulse Coded Modulation
PLL	Phase Locked Loop
PM	Phase Modulation
PSK	Phase Shift Keying
QPSK	Quadrature Phase Shift Keying
RAM	Random Access Memory
SDSY	Safran Data Systems
SW	Synchronization Word
TBD	To Be Defined
TM	TeleMetry
UQPSK	Unbalanced QPSK
UTC	Universal Time, Coordinated
VCXO	Voltage Controlled Crystal Oscillator
XDR	External Data Representation

SYMBOLS DESCRIPTION

The following symbols appear in this document and on the chassis:



ISO 7000-0434A
(2004-01)

Caution: The exclamation point within an equilateral triangle is intended to alert the user to the presence of important operating and maintenance (servicing) instructions in the literature accompanying the product.



ISO 7000-0434B
(2004-01)



IEC 60417-6042
(2010-11)

Caution, risk of electric shock: The lightning flash with an arrowhead symbol within an equilateral triangle, is intended to alert the user to the presence of uninsulated 'dangerous voltage' within the product's enclosure that may be of sufficient magnitude to constitute a risk of electric shock to persons.



IEC 60417-5019
(2006-08)

Protective earth; protective ground: terminal is intended for connection to an external conductor for protection against electric shock in case of a fault.



IEC 60417-5007
(2009-02)

"ON" (power): indicate connection to the mains.



IEC 60417-5008
(2009-02)

"OFF" (power): indicate disconnection from the mains.



IEC 60417-5032
(2002-10)

Alternating current: the equipment is suitable for alternating current only.



IEC 60417-6172
(2012-09)

Disconnection, all power plugs: The equipment has multiple power source. All power sources shall be disconnected before servicing to avoid shock hazard.



Altitude: Do not use at an altitude exceeding 2000m.



Tropical conditions: This product is not suitable for tropical conditions.

Documentation must be consulted in all cases where this symbol is marked.

CE MARKING

Markings are placed at the rear of the unit, regarding the power ratings and "CE" compliance:



The "CE" symbol on the PC chassis indicates that it is in compliance with the Electromagnetic Compatibility directive and the low Voltage directive.

This equipment is a: **Group 1 Class A product**

The following technical standards must be met:

EN 55022	Limits and methods of measurement of radio interference characteristics of information technology equipment
EN 61000-3-2	Limits for harmonic currents emissions"
EN 61000-4-2	Limits and methods of measurement - Section 2: electrostatic discharge immunity test
EN 61000-4-3	Limits and methods of measurement – Section 3: Radiated radio-frequency, electromagnetic field immunity test
EN 61000-4-4	Limits and methods of measurement - Section 4: Electrical fast transient / burst immunity test
EN 61000-4-5	Limits and methods of measurement - Section 5: Surge immunity test
EN 61000-4-6	Limits and methods of measurement - Section 6: immunity to conducted disturbances, induced by radio-frequency fields
EN 61000-4-8	Limits and methods of measurement - Section 8: Power frequency magnetic field immunity test
EN 61000-4-11	Limits and methods of measurement - Section 11: Voltage dips, short interruptions as voltage variations immunity tests

The unit also complies with all staff safety standards and decrees, in particular:

- Decree of 14.11.1989 and additions: Regulation of worker protection,
- Directive 2006/95/EC.

In accordance with European Community directives, a "Declaration of Conformity" has been made and is available by contacting technical support.

Recommendation concerning the cabling:

"CE" certification has been carried out with 3-meter interconnecting cables with remote equipment.

Safran Data Systems recommends the use of high quality interconnecting cables type. The high quality recommended user interconnecting cables are "shield twisted pair" or "coaxial cable" industrial type with cable shield. The cable shield will provide the necessary mechanical protection.

Not using this high quality designed cable could result in unexpected intersystem/extrasystem electromagnetic emission and susceptibility.

WEEE DIRECTIVE

Markings are placed at the rear of the unit and on all electronic subassemblies, regarding the WEEE directive:



DO NOT place the used equipment in regular waste bins.

All equipment marked with this logo must be recycled by an approved Waste Electrical and Electronic Equipment processing facility. If desired, the equipment can be returned at the customers expenses to Safran Data Systems for proper recycling, after asking for a RMA by sending a request to Safran Data Systems. Please consider the environmental cost of transportation of the used equipment and preferably use a local WEEE processing facility.

CAUTION

	CAUTION RISK OF ELECTRICAL SHOCK DO NOT OPEN	
CAUTION : TO AVOID SHOCK HAZARD, DO NOT REMOVE COVER (OR BACK). REFER SERVICING TO QUALIFIED PERSONNEL.		
	CAUTION RISK OF ELECTRICAL SHOCK MULTIPLE POWER SOURCES	
CAUTION : TO AVOID SHOCK HAZARD, ALL POWER SOURCES SHALL BE DISCONNECTED BEFORE SERVICING, FIRST DISCONNECT THE PLUG FROM THE OUTLET SOCKET, REFER SERVICING TO QUALIFIED PERSONNEL.		

IMPORTANT SAFETY INSTRUCTIONS

The safety precautions listed below represent warnings of certain dangers of which Safran Data Systems is aware. They must be observed during all phases of operation, service, and repair of this equipment. Failure to comply with these precautions or with specific warnings elsewhere in this manual violates safety standards of design, manufacture, and intended use of the equipment.

Safran Data Systems assumes no liability for the customer's failure to comply with these requirements. You, as the user of the product, should follow these warnings and all other safety precautions necessary for the safe operation of the equipment in your operating environment.

GENERAL



Take heed of warning and instructions

You should read all the safety and operating instructions before operating this apparatus. Retain this handbook for future reference and adhere to all warnings in the handbook or on the apparatus



This equipment shall not be operated in a place that may welcome children

This equipment shall not be used by children. Children should be supervised to ensure that they do not play with the equipment. This equipment is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge.

GROUNDING & ELECTRICAL INSTALLATION STANDARDS



Grounding

To minimize shock hazard, the equipment chassis and enclosure must be connected to an electrical ground.

Always start with plugging the mechanical ground (on the rear panel of the equipment). The product must be connected to a protective earth ground by a green/yellow cable or by a grounding strap with a minimum section of 1.0 mm² / 16AWG.

The mechanical ground cord must always be the last to be disconnected.

The external ground terminal must be equipped with a locking system (Glomer nut, locking washer, etc.). If you are using a ground lug, make sure to secure the grounding connection against accidental loosening of the conductor.

WARNING: this apparatus shall be connected to a mains socket outlet with a protective earthing connection.

The appliance must be connected to an electrical ground and the power cord is designed for that.

**Powers sources**

Only connect the apparatus to a power supply of the described in the operating instructions or as marked on the apparatus.



Turning off the power switch does not completely isolate this product from the power line so remove the plug from the socket if not using it for extended period of time.

Unplug this apparatus during lightning storms.

WARNING: The system must be connected to an electrical installation complying with the standards applicable in your country. It must be equipped with overcurrent, overvoltage and earth fault protections.

WARNING: The system must be plugged to electrical equipment in accordance with EN60950-1: 2006 2nd edition.

WARNING: This equipment has multiple power sources; all power sources shall be unplugged to isolate the equipment.

**Power supply cable and socket**

This apparatus must be used with a power supply cable of 1 mm² section minimum

WARNING: An isolator according to EN60947 standard serves as an isolating device. It must be easily accessible and installed near the power supply.

The power supply plugs are used as safety device.

The power supply socket must be located at a distance of less than 1.5 m from the equipment and easily accessible

RISK OF ELECTRIC SHOCKS**Opening of the product chassis**

To reduce the risk of electric shock, do not remove cover (or back).

No user serviceable parts inside; the product chassis should only be opened by qualified personnel for maintenance purpose (except for opposite instructions in the user's manual).

**Power cord protection**

Power supply cords should be routed so that they are not likely to be walked on or pinched by items placed upon or against them, paying particular attention to cords and plugs, and the point where they exit from the appliance

**Water and moisture**

To reduce the risk of fire or electric shock, do not expose this apparatus to rain or moisture. The presence of electricity near water can be dangerous. Do not use the apparatus near water.

INSTALLATION, ENVIRONMENT



Installation

Install in accordance with the manufacturer's instructions. Only use attachments and accessories specified by the manufacturer. This unit must be used in a 19" rack mount cabinet only



Lifting the unit

This apparatus weighs over 25kgs, so take extreme care when lifting or moving this unit. We recommend that two people are available to lift this unit.



Uncontrolled temperature:

Do not leave this equipment in an uncontrolled environment where the temperature is below -20°C (-4°F) or above 40°C (104°F). It may damage the equipment.



Heat & Moisture sources:

Do not install the equipment near a heat source or near a source of moisture. It may damage the equipment



Ventilation:

Do not block any ventilation openings. Ensure that the product ventilation is not impeded.



Object or liquid entry

Take care that objects do not fall and liquids are not spilled into the product chassis through any openings.



Explosive atmosphere

WARNING: Do not operate the equipment **in the presence of flammable gases or fumes**. Operation of any electrical equipment in such an environment constitutes a definite safety hazard.

ESD PROTECTIVE MEASURES



Electrostatic discharge can damage the electronic components within the system. Only personnel with knowledge of ESD protective measures may modify the system:

- Touch a grounded object to discharge any static in your body.
- When performing the work, make sure that the workspace is free of static, and wear an ESD wrist strap.
- If not used, ensure that the device is protected from electrostatic discharge, e.g., by storing it in its packaging.

CLEANING, MAINTENANCE



Cleaning

Unplug the unit before cleaning. Clean only with dry cloth.

This product should normally only require a wipe with a soft, damp, lint-free cloth.

Do not use paint thinner or other chemical solvents for cleaning.



Technical Maintenance

Refer all servicing to qualified maintenance personnel. Servicing is required when apparatus has been damaged in any way, such as power supply cord is damaged, liquid has been spilled or objects have fallen into the apparatus, the apparatus has been exposed to rain or moisture, does not operate normally, or has been dropped.

The qualified maintenance personnel must take reasonable precautions to face the obvious DANGERS.



Battery replacement and disposal

CAUTION: Risk of explosion if battery is replaced by an incorrect type. Dispose of used batteries according to the instructions.

INTERNATIONAL SECURITY STATEMENTS

France, Canada: L'appareil doit être connecté à une prise de terre

Suomi: Laite on liitettävä suojamaadoituskoskettimilla varustettuun pistorasiaan

Norge: Apparatet må tilkoples jordet stikkontakt

Sverige: Apparaten skall anslutas till jordat uttag

1 CORTEX HDR OVERVIEW

1.1 Scope

This manual contains the information necessary to install, configure, operate and maintain the SDSY PC-based **High Data Rate Receiver 4G+ (CORTEX HDR 4G+)**.

The document is divided into six main sections:

- ◆ **Section 1:** CORTEX HDR overview and performances,
- ◆ **Section 2:** Hardware description and I/O connections,
- ◆ **Section 3:** Functional description,
- ◆ **Section 4:** Operating procedures and description of the Graphical User Interface,
- ◆ **Section 5:** Hardware and software maintenance procedures,
- ◆ **Section 6:** Configuration examples for different satellites.

Project-specific information (if any) and additional data on the CORTEX HDR are annexed to the document:

- ☐ **Appendix 1:** Description of the TCP-IP Ethernet interface.
- ☐ **Appendices 2 & 3:** Layout of the Test Modulator board and Demodulator board.
- ☐ **Appendix 4:** Description of the PC compatible workstation.
- ☐ **Appendix 5:** Hardware and software configuration of the CORTEX HDR at delivery.
Project-specific data if any.

1.2 CORTEX HDR Main Features

High Data Rate Telemetry Processing

Fully compatible with HDR XL, XXL, 4G

Dual-demodulator capability with fully tunable IF @ 720 MHz, 1.2 GHz, 2,4GHz...

Supported demodulations: BPSK, QPSK, OQPSK (SQPSK), UQPSK (AQPSK, AUQPSK, AUSQPSK), 8PSK, GMSK, SOQPSK, 16QAM, 16APSK, 32APSK, 64 APSK 128APSK 256APSK

Digital filtering: RRC, GMSK, SOQPSK, User defined,

Equalizer, auto-adaptive mode (DEAF), SDSY advance algo fully compatible with all actual imperfection of transmission

Cross Polarization cancelation auto-adaptive mode (XDEAF), SDSY advance algo fully compatible with actual imperfection of dual polarization transmission

Antenna combining capability

Viterbi decoding: CCSDS, single or dual mode, $\frac{1}{2}$ (+ puncture) or $\frac{1}{4}$ rate, single or parallel processing

4D-TCM (Multi-Trellis Coded Modulation) decoding: CCSDS BEM

Data processing: frame synchronization, time-tagging, De-randomization, R-S decoding, CCSDS, DVB, ECSS, CRC, CCSDS Packet reconstruction and CFDP

Low Density Parity Check Code decoding (CCSDS LDPC 7/8, AR4JA LDPC $\frac{1}{2}$ $\frac{2}{3}$ $\frac{4}{5}$ etc...)

DVBS2 and DVBS2X capability: all ACM up to 32APSK for DVBS2 up to 256APSK in DVBS2X

SCCC and SCCC-X: all ACM up to 64APSK for SCCC and up to 256APSK for SCCC-X

Test Modulator IF: capable of generation of all demodulator capacities: example dual modulator, noise, XPOL effect,...

Data ingestion: data storage with programmable filtering and sorting for further transmission to remote clients or/and real-time data transmission to remote Clients (Data replay on LAN)

Data ingestion output: TCP/IP or UDT or file on 10G copper or fiber

Data replay on IF: capability to replay the recorded data (Playback mode) on the test modulator (IF)

Wide Band IF recorder and reproducer capacity see WBR user manual for more detail

High Integration with Drastically Reduced Hardware for Increased Availability

Extensive Use of Digital Signal Processing Techniques for Enhanced Performance, Upgradability and Flexibility

PC-based Architecture

Built-in Test & Simulation Capability

User-friendly and Intuitive Graphic User Interface

Wide Range of Configuration Parameters and Status

No Tuning, no Preventive Maintenance

1.3 Mission of the CORTEX HDR

1.3.1 Introduction

The Cortex High Data rate Receiver is dedicated to the reception of the high rate telemetry of the remote sensor satellites.

The Cortex HDR is part of the Cortex family products, all using the same architecture: a board with FPGA chips plugged into the 4U-high Cortex chassis (equipped with PC Server Board).

The Cortex HDR fulfills:

- ◆ Telemetry demodulation,
- ◆ Telemetry processing,
- ◆ Telemetry simulation for test purposes (optional),

Following, the different possible operating modes.

1.3.2 Operating Modes

Depending on the delivered hardware configuration, the delivered operational configuration (licenses dependent) and the selected start-up menu, the HDR can be operated in one of the following operating modes:

- ◆ 720 MHz +/-200MHz IF input
- ◆ 1,2GHz +/-300MHz IF input
- ◆ 2,4GHz +/-750MHz IF input
- ◆ Wideband: 500MHz to 2,4GHz IF input could be used for IF 1,5GHz+-750MHz
- ◆ All input could be dual IF: 2 input are available for dual polarization reception
- ◆ Number of demodulator from 2 up to 8 depending on menu

Performances and particularly the supported maximum bit rates are dependent on the operating mode.

Depending on the delivered configuration (licenses dependent) and the selected start-up menu, one or more data processing units can be connected at the output of demodulator units.

1.3.3 Telemetry Demodulation

- ☐ Input frequency:
 - **720 MHz nominal**, tunable between 530 and 910 MHz
 - **1.2 GHz nominal**, tunable between 910 MHz and 1490 MHz
 - **2.4 GHz nominal**, tunable between 1,65 GHz and 3.15 GHz
 - **1.25 GHz nominal**, tunable between 500 MHz and 2400 MHz
- ☐ Automatic carrier acquisition
- ☐ BPSK, QPSK, OQPSK (SQPSK), UQPSK (AQPSK, AUQPSK, AUSQPSK), 8PSK, GMSK, SOQPSK, 16QAM, 16APSK, 32APSK, 64APSK 128 APSK 256APSK demodulation
- ☐ Equalization filter: static or auto-adaptive (DEAF)
- ☐ Automatic compensation for cross-polarization perturbations(XDEAF)
- ☐ Adjacent channel rejection filters
- ☐ Bit synchronization
- ☐ Fully digital matched and shaped filters: Raised Cosine, Root Raised Cosine, GMSK, SOQPSK shaping filter (CCSDS and IRIG standard), User defined
- ☐ Viterbi decoding: single or dual mode, $\frac{1}{2}$ (+ puncture) or $\frac{1}{4}$ rate, single or parallel processing (stacked Viterbi)
- ☐ De-scrambler: CCITT V.35 and Intelsat V.35
- ☐ 4D-TCM (Trellis Coded Modulation) decoding
- ☐ DVB-S2, DVBS2X, SCCC and SCCC-X standards supported
- ☐ Multiple Demodulation capability up to 8

1.3.4 Telemetry Processing

- ☐ Multi-channel architecture (up to 8 parallel channels – Multi DPU processing)
- ☐ Transport Layer for RS DVB or LDPC decoders (Asynchronous decoding layer)
- ☐ CADU Frame synchronization (CCSDS Transfer layer)
- ☐ Data time-tagging (internal, external IRIG B signal or external NTP server)
- ☐ De-randomization, De-interleaving, Virtual filling, Reed-Solomon decoding (supported standard: CCSDS, DVB-S, ECSS)
- ☐ CRC frame checking
- ☐ CCSDS Low Density Parity Check decoding (LDPC 7/8, AR4JA $\frac{1}{2}$ 2/3 4/5)

- ☐ Programmable data filtering and sorting with dynamical creation of directories and files for recovery by remote clients using FTP
- ☐ Data output:
 - Raw data or frames (CADU / VCDU)
 - Real-time data or stored data
 - Data files
- ☐ Data interface:
 - Standard PCM outputs (differential Data plus Clock outputs): multi format provided:
 - output of the demodulator unit (DMU) or output of the data processing unit (DPU)
 - merged or split at the output of the DMU (2, 4 or 8 streams)
 - possibility of Soft Decision at the output of the demodulator(separate I and Q 3 bits or merged I/Q 6 bits)
 - Ethernet Network: TCP-IP. FTP server
- ☐ Data storage: internal or on external DAS (hardware dependent – consult SDSY)
 - Raw data, CADU, VCDU or Packet
 - CFDP protocol
- ☐ Replay of recorded data

1.3.5 Polarization and Antenna Combining

- ☐ Modes: Selection of the best channel or combining of two channels.
- ☐ Maximum combining gain: 3 dB (identical Eb/No on both channels)

1.3.6 Simulation & Self-testing

The CORTEX HDR incorporates powerful simulation & testing capabilities for functional and performance test purposes:

- ☐ Telemetry simulation
- ☐ Noise generation
- ☐ IF Modulation
- ☐ Doppler simulation
- ☐ XPO simulation
- ☐ Dynamic effect
- ☐ VCM simulation
- ☐ Automatic BER measurement
- ☐ Real time IF spectrum analysis
- ☐ Real time vector analysis
- ☐ Unit self-testing at boot

1.4 CORTEX HDR Architecture

1.4.1 Functional Block Diagram of a Demodulation Channel

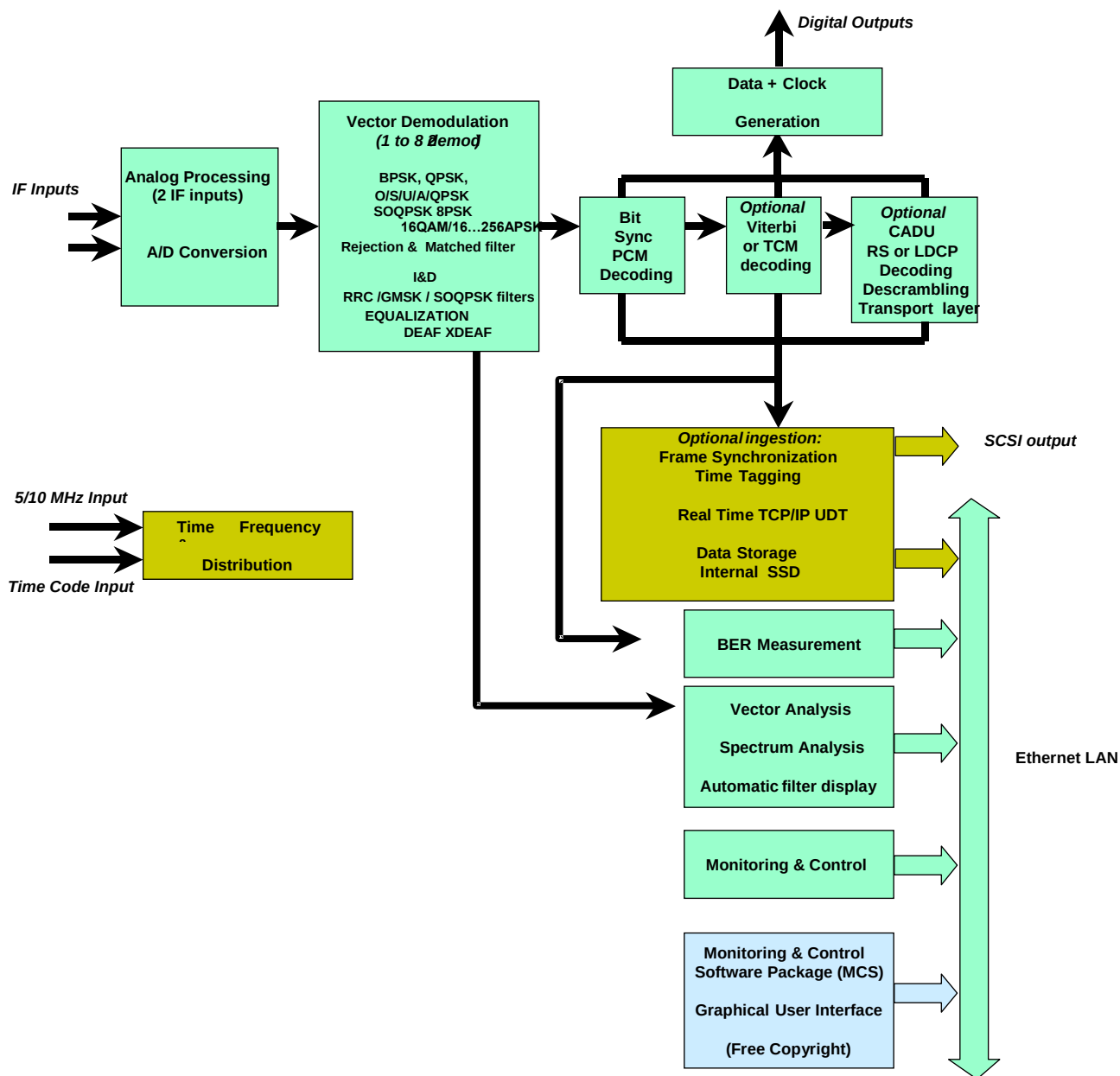


Figure 1: CORTEX HDR Demodulation and Ingestion Functional Block Diagram

1.4.2 Functional Block Diagram of a Modulation Channel

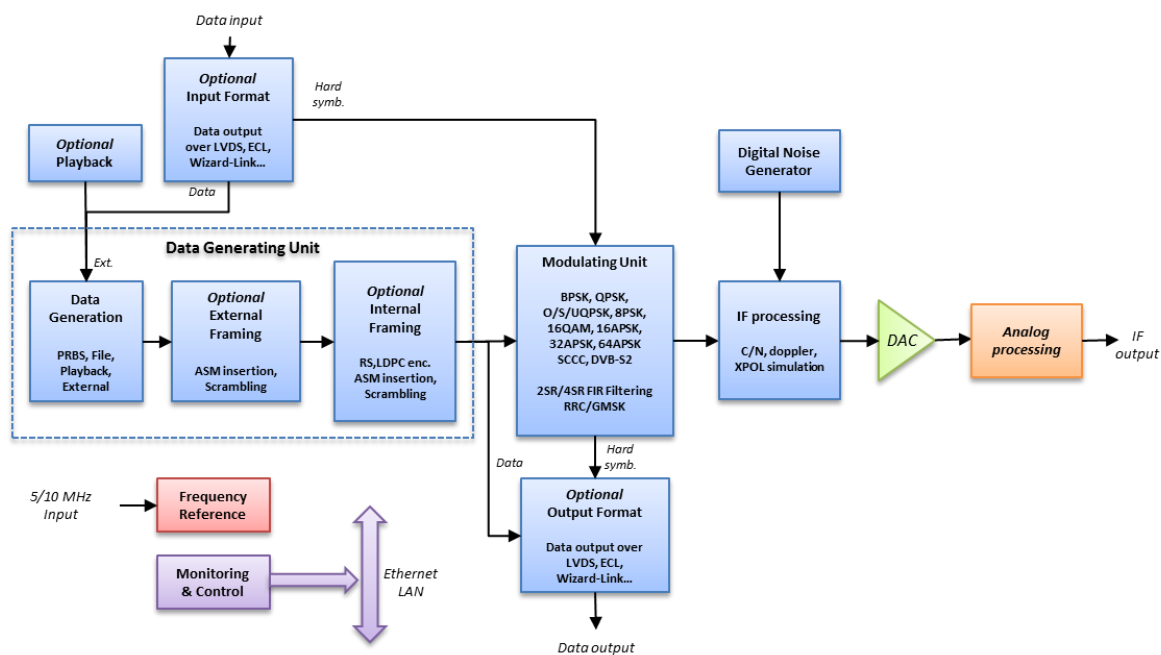


Figure 2: CORTEX HDR Test Modulator Functional Block Diagram (single channel, single IF)

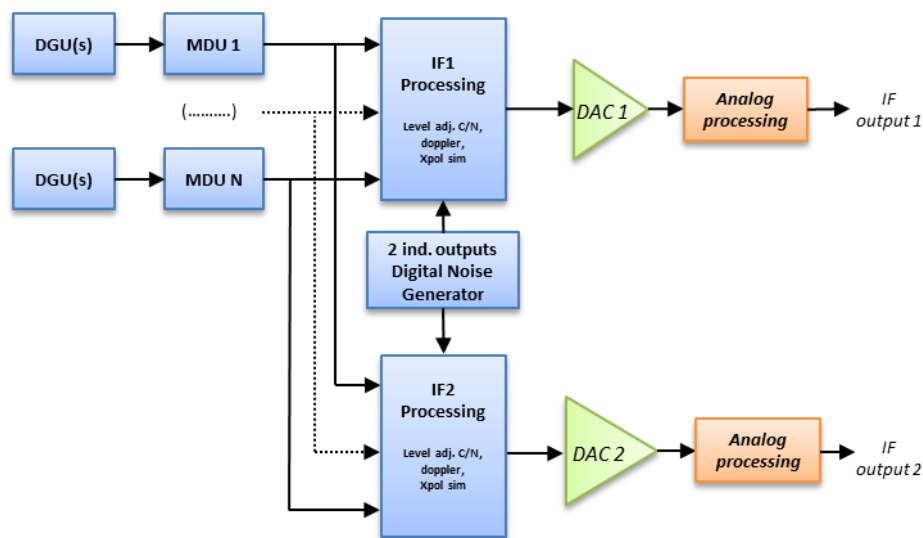


Figure 3: CORTEX HDR Test Modulator - Multi channel, multi IF operation

1.4.3 CORTEX HDR Hardware Architecture

The CORTEX HDR comprises the following elements:

- ❖ A PC-compatible workstation,
- ❖ A Demodulator board (SDSY),
- ❖ An optional Test Modulator board, including a noise generator and a telemetry simulator (SDSY).

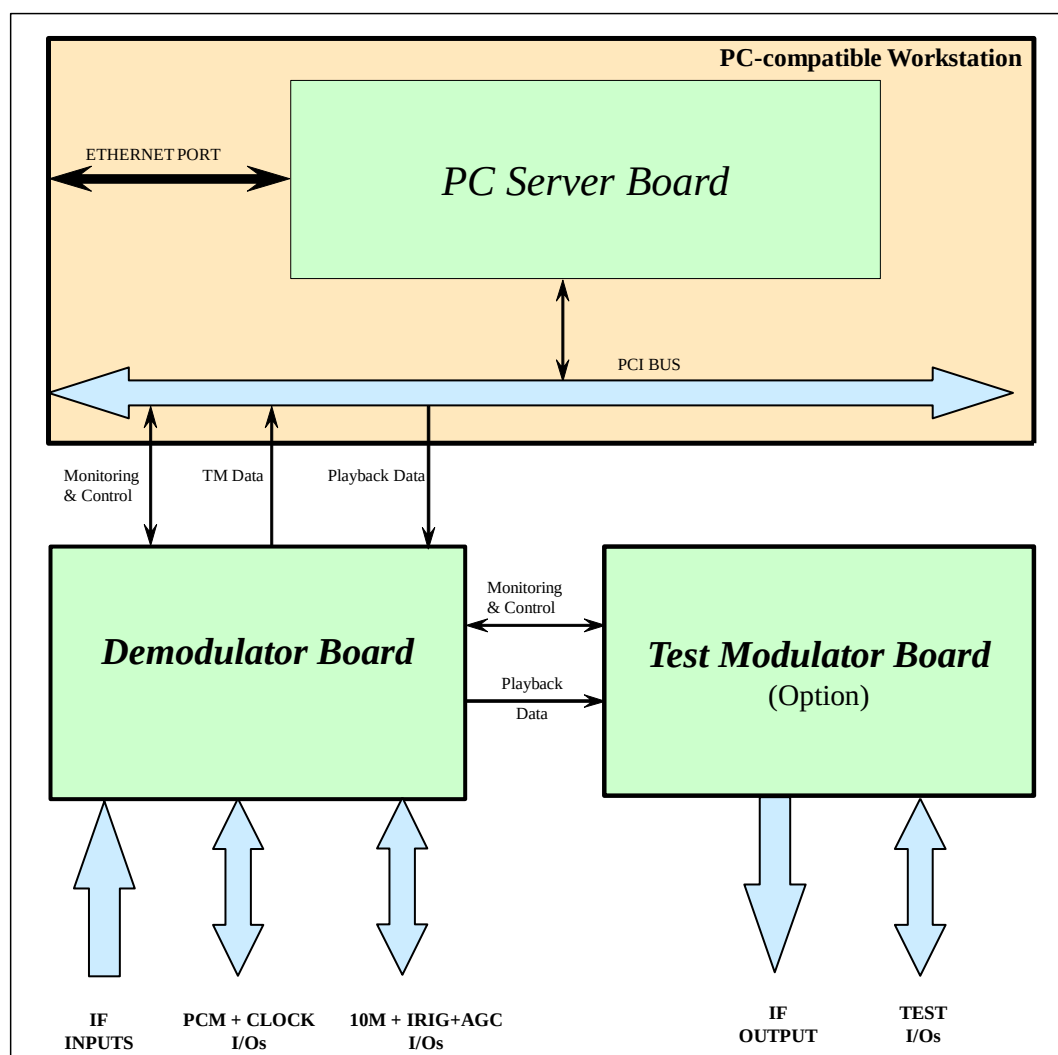


Figure 4: CORTEX HDR Hardware Block Diagram

1.4.3.1 PC-compatible Workstation

The PC-compatible workstation includes:

- ❖ A rack-mountable industrial PC chassis (19-inch width, 4 rack-space);
- ❖ A dual-Pentium CPU board with Dual port RJ45 Gigabit Base-T Ethernet ports;
- ❖ An integrated 8-inch color display;
- ❖ An extractable keyboard / pointing device for local operation;
- ❖ A hard disk for the operating system and application software;
- ❖ A set of hard disks for data recording (configuration dependent);
- ❖ Dual port 10 Gigabit RJ45 copper or LC fiber-optic connectors (configuration dependent);
- ❖ A DVD drive.

Refer to Annex 4 for more detail on the PC-compatible workstation.



Figure 5: Example of CORTEX HDR Workstation

1.4.3.2 Test Modulator Board

Test Modulator is an optional board for test purposes:

- PCM generation: internal data (pseudo-random or pattern from files) or external data (+ associated clock);
- PCM encoding;
- Frame encoding (CADU size) at two levels: External framing, internal framing. This enables transport layering;
- Randomization;
- Error code encoding (exclusive functions):
RS DVB transport layer,
CCSDS RS,
LDPC;
- Convolutional encoding;
- Carrier generation & modulation (programmable Bit rate and IF frequency):
BPSK, QPSK, OQPSK, UQPSK, (TCM) 8PSK, 16QAM, 16APSK, 32 APSK, 64 APSK;
- Baseband pulse shape filtering (RRC, GMSK);
- Modulated signal output with programmable power level, up to two IF inputs;
- Noise generation with programmable power level and perfect C/N generation;
- Doppler simulation;
- Cross polarization interference simulation;
- DVB-S2(-X) and SCCC(-X) modulation sequences simulation.

1.4.3.3 Demodulator Board

Main features of the **Demodulator board** are:

- ☐ Amplification, filtering and analog-to-digital conversion of the IF signal;
- ☐ Digital vector demodulation;
- ☐ Bit synchronization;
- ☐ Decoding: Viterbi/TCM/LDPC/....;
- ☐ PCM decoding;
- ☐ Frame synchronization;
- ☐ Time tagging;
- ☐ De-randomization, RS or LDPC decoding;
- ☐ Data and Clock output;
- ☐ BER measurement;
- ☐ Vector analysis.

The Demodulator board is monitored and controlled by the Server CPU board via the PCI-express bus.

1.4.3.4 I/O Connectors

SMA-type I/O connectors are located at the rear panel of the chassis (Demodulator I/Os and Test Modulator I/Os). The HDR can be delivered with LVDS outputs (option) or other.

See Section 2 for a detailed description of the I/O ports.

1.4.4 CORTEX HDR Software

The CORTEX HDR is supplied with two separate software packages:

- ◆ Signal Processing Software package (SPS)
- ◆ Monitoring & Control Software package (MCS)

These software are executed under the control of the Operating System. Software configuration at delivery is described in Annex 5.

1.4.4.1 Local or Remote M&C

The Signal Processing Software and Monitoring & Control Software communicate by means of connected sockets under TCP-IP protocol. As a result, the notion of local or remote M&C no longer applies to the CORTEX HDR: since the MCS software package can be installed (free copyright) into any PC-based computer(s), there is no difference between operating the equipment locally (front panel keyboard and screen) or remotely, provided that the CORTEX HDR and the PC-based computer(s) are connected to the same Ethernet network.

1.4.4.2 Password Protection

The following functions are password-protected:

- ☐ Cortex user login ;
- ☐ CORTEX HDR configuration;
- ☐ Reset of TCP-IP connections.

Default setting at delivery: **cortex** (login function). Other functions: no password.

1.4.4.3 Software Version

The software version at delivery is described in Annex 5 and, in more details, in the CORTEX HDR Acceptance Test Report (document reference: CRE101551).

- ❑ **PROMed software** (Demodulator and Test Modulator boards): see the factory acceptance test results (CRE101551).
- ❑ **Disk resident software (SPS & MCS):**
 - Signal Processing Software (FPGA software downloaded to the Demodulator board and Modulator board at boot-up): open the *CONFIG* window (see Section 4.5.3.4 for more details).
 - Signal Processing Software (Windows software): click on  at the bottom of the screen. Then select *About Cortex...* to display the software version.
 - Monitoring & Control Software: click on  or on *Help* in the top-level window, then select *About CrtxMcs...*:

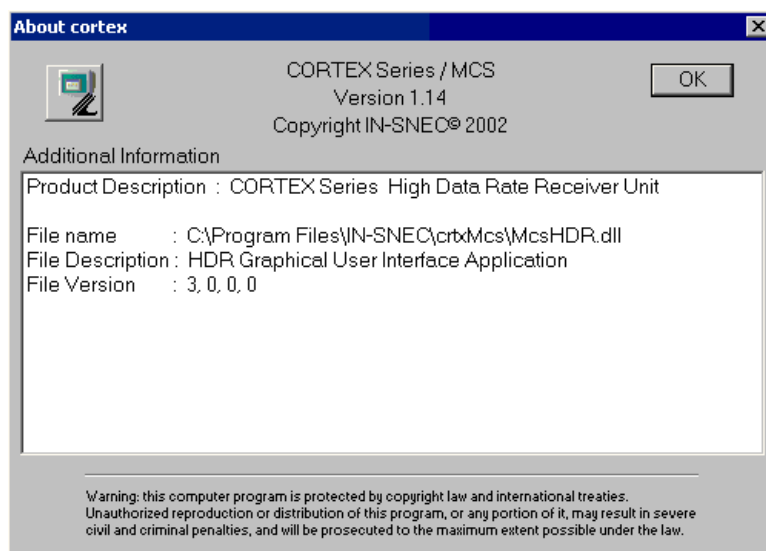


Figure 6: MCS Software Version

1.5 CORTEX HDR Interfaces

1.5.1 Mechanical & Electrical Interfaces

Refer to Section 1.6 and 2.2 for signal characteristics (bit rate, carrier frequency, modulation, PCM code, level, impedance, test points, etc...) and I/O connections.

1.5.2 Ethernet TCP-IP Interface

Monitoring & Control data & Received Data are exchanged between the Signal Processing Software (SPS) and remote M&C clients over any Ethernet port. The CORTEX HDR acts as a data server. Data are exchanged using the **connected-sockets** communication protocol.

Some data flows accept multiple clients at a time (monitoring, logging, spectrum analysis, vector analysis data), while other flows accept only one client at a time (equipment configuration and reset). Each type of transfer flow is allocated a specific port number which the clients use for connecting to the equipment. Port numbers are listed below:

DATA FLOW	PORT NUMBER	MAXIMUM NUMBER OF CONNECTIONS
CORTEX HDR Monitoring Data (MON)	3000	24
CORTEX HDR Control Data (CTRL)	3001	1
CORTEX HDR Remote Access Control (RAC)	3003	1
TCP-IP Reset (RST)	3002	1
Logging Data (LOG)	3040	5
Telemetry Data (TM)	3075 (Option)	10
Playback Data #1 (PBK-1)	3080	1
Playback Data #2 (PBK-2)	3081	1
Spectrum Analysis Data (SPA)	3050	3
Vector Analysis Data (VEC)	3055	3
Doppler or Analysis Data (DOP)	3060	3

Table 1: CORTEX HDR TCP-IP Port Addresses

A detailed description of this interface is available in Annex 1 (Documents reference: STI 100013 and STI 100013_HDR).

For Data transmission, the TCP/IP interface is recommended. As an option a UDT protocol could be use please consult the factory.

Next figure shows data flows between CORTEX HDR software components and between the CORTEX HDR and the outside world:

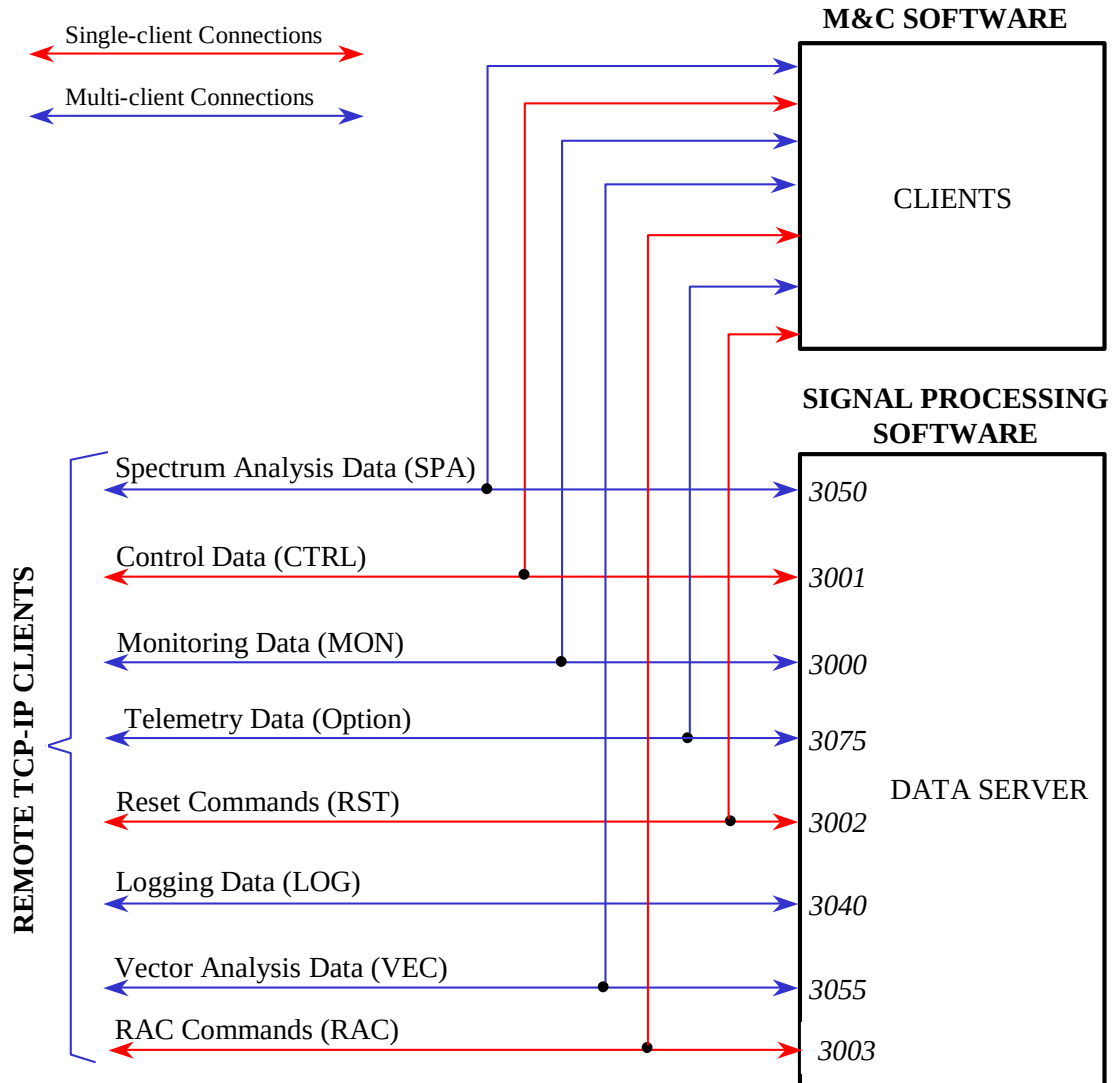


Figure 7: CORTEX HDR TCP-IP Interfaces

1.5.3 FTP Interface

Received data frames can be filtered and sorted, depending on the selected TM flow, the result of the Reed-Solomon decoding, the virtual channels, or User defined. The resulted data frames are embedded inside specified-names files and can be store in dedicated directories on an archiving system.

In order to recover the constructed data files through the network by external Clients, the CORTEX HDR is fitted with a powerful FTP server. External Clients can connect this server at the standard dedicated FTP port number.

1.6 CORTEX HDR Performances

WARNING:

PERFORMANCE DATA IN THIS SECTION ARE TYPICAL VALUES MEASURED IN BEST CASE CONDITIONS. SOME FUNCTIONALITIES MAY STILL BE IN THE DEVELOPMENT PHASE.
PLEASE CONTACT SDSY FOR MORE INFORMATION.

PROGRAMMING RESOLUTION: REFER TO ANNEX 1. CONFIGURATION PARAMETERS ARE SPECIFIED OVER 32-BIT INTEGERS OR SINGLE-PRECISION FLOAT (8-BIT EXPONENT AND TWO'S COMPLEMENT 24-BIT MANTISSA). HOWEVER, RESOLUTION CAN BE LIMITED BY THE HARDWARE DESIGN. LIMITATIONS ARE DESCRIBED IN THE GRAPHICAL USER INTERFACE.

1.6.1 Demodulation Unit

1.6.1.1 IF Reception

- ☐ Dual-demodulator capability.
- ☐ Input frequency:
 - **720 MHz nominal**, tunable between 520 MHz and 920 MHz
 - **1.2 GHz nominal**, tunable between 880 MHz and 1.52 GHz
 - **1.25 GHz nominal**, tunable between 520 MHz and 2.42 GHz
 - **2.4 GHz nominal**, tunable between 1,68 GHz and 3.12 GHz
 - **2.7 GHz nominal**, tunable between 2,28 GHz and 3.2 GHz
- ☐ Frequency setting step Single precision float + Single precision float resolution
- ☐ Input ports 2 (selectable) per IF inputs (Normal and Alternate)
- ☐ Input impedance 50 Ω
- ☐ VSWR < 1.3
- ☐ AGC range -10 dBm to -50 dBm (+10dBm nondestructive)
- ☐ AGC time constant 1 ms
- ☐ AGC test 0 Volt (IF level $\leq$ -60 dBm) to 3 Volts (IF level $\geq$ -10 dBm)
Minimum termination: 1000 Ω
- ☐ Carrier acquisition range $\pm$ 10 kHz to $\pm$ 2 MHz
- ☐ IF bandwidth Automatically adjusted from the symbol rate
- ☐ IF level variation $\leq$ 10 dB/sec

1.6.1.2 Demodulation

Demodulation (**notes 1**) BSPK, QPSK, OQPSK (SQPSK)

UQPSK (AQPSK, AUQPSK, AUSQPSK), 8PSK, GMSK, 16QAM, 16APSK, 32APSK, 64APSK, 128APSK, 256APSK

SOQPSK (MIL standard filter, IRIG TG filter, CCSDS A-type or B-type filters)

Bit rates (**notes 1, 2 & 3**)

Operating @ 1.25 GHz IF input:

Maximum actual symbol rate	1.6 Gsps
Maximum Bit rate in BPSK mode	1.6 Gbps
Maximum Bit rate in QPSK modes	3.2 Gbps
Maximum bit rate in 8PSK mode	4.8 Gbps
Maximum bit rate in 16QAM / 16APSK, 32APSK, 64APSK mode	6.4 Gbps

Operating @ 720 MHz IF input:

Maximum actual symbol rate per channel	320 Msps
Maximum Bit rate in BPSK mode per channel	320 Mbps
Maximum Bit rate in QPSK modes per channel	640 Mbps
Maximum bit rate in 8PSK mode per channel	960 Mbps
Maximum bit rate in 16QAM/16APSK mode per channel	1280 Mbps

Operating @ 1.2 GHz +/-300MHz IF input):

Maximum actual symbol rate	500 Msps
Maximum Bit rate in BPSK mode	500 Mbps
Maximum Bit rate in QPSK modes	1000 Mbps
Maximum bit rate in 8PSK mode	1500 Mbps
Maximum bit rate in 16QAM / 16APSK, 32APSK, 64APSK mode	2000 Mbps

Operating @ 2.4 GHz IF input:

Maximum actual symbol rate	1.2 Gsps
Maximum Bit rate in BPSK mode	1.2 Gbps
Maximum Bit rate in QPSK modes	2.4 Gbps
Maximum bit rate in 8PSK mode	3.6 Gbps
Maximum bit rate in 16QAM / 16APSK, 32APSK, 64APSK mode	4.8 Gbps

Synchronization threshold	$\leq 1\text{dB (Eb/No)}$ in QPSK, $\leq 3,5\text{dB (Eb/No)}$ in 8PSK
Acquisition time	≤ 0.25 second
Doppler measurement	Yes
Doppler rate	≤ 10 kHz/s
IF level measurement accuracy	± 2 dB
Cable Equalizer (note 1)	Yes. Amplitude equalization: 0 to 12 dB between 570 & 870 MHz

1.6.1.3 Implementation loss

Implementation losses:

- are given in dB to theoretical value;
- take into account all part HDR 4G part the unit filter/demodulator/decoding...;
- are given with actual BER measurement using a SDSY Test modulator assuming no implementation loss @ SDSY test modulator.

<1dB at @ 90% of max value of Bit Rate value
<0,5dB @ 50% of max value of Bit Rate value
<0,3dB @ 25% of max value of Bit Rate value

Please note that for link budget with actual satellite that implementation loss doesn't take into account imperfections of:

- on board modulator RF and antenna Multipath non-linearity ...
- non Gaussian transmission imperfection weather...
- ground imperfection: pointing accuracy RF & feed..

And for transmission above 300Mbps all those imperfections become critical.

Cortex DEAF and/or XDEAF will correct all those imperfections with residual loss from 0.1 to 1dB.

1.6.1.4 Bit Synchronization

Acquisition range	± 0.1 % of the symbol rate
PCM code	NRZ-L/M/S, BP-L/M/S
Differential decoder	DNRZ for QPSK and OQPSK
Eb/No measurement	Resolution: 0.1 dB Accuracy: 1 dB
Symbol clock display	Yes
Output ports	I Channel : Data & Clock Q Channel : Data & Clock Merged I+Q : Data & Clock
Output (electrical)	ECL Differential or LVDS
Output clock polarity	Normal or inverted
Output data polarity	Normal or inverted (+ swap I/Q)

1.6.1.5 *Filtering***(notes 1)**

Filter type	Finite Impulse Response Symmetrical (even or odd): 32 taps (even) or 31 taps (odd) Asymmetrical: 16 taps on direct path I and Q, plus 16 complex taps (Q to I and I to Q)
Sampling rate	2 or 4 time the symbol rate
FIR response	Characteristics defined in ASCII file Raised Cosine Root Raised Cosine (roll-off factor programmable) GMSK (BT factor programmable) Shaped filter for SOQPSK
Additional filters	Rejection filter HBF (sampling rate at 4 times the symbol rate with decimation by 2 at the output) Rejection low pass filter LPF +/- 128 MHz @ 720 MHz IF +/- 220 MHz @ 1.2 GHz IF +/- 440 MHz @ 2.4 GHz IF +/- 470 MHz @ 1.25 GHz IF
Auto-adaptability	Static or Auto-adaptive: "Digital Equalization Automatic Filter", with learn and save mode.

About DEAF please see the additional document....

1.6.1.6 Viterbi Convolutional Decoding

(notes 1)

1.6.1.6.1 Viterbi $\frac{1}{2}$

Convolutional decoding and error-correction comply to CCSDS recommendations 101.0-B-3 (Blue Book, 05/1992).

Code rate	0.5 information bit per encoded bit
Constraint length	7 bits
Code generators	G1 = 1111001; G2 = 1011011
Phase relationship	Programmable association of G1 or G2 with first symbol: Programmable inversion on output path of G2: G1 - G2 Inverted (CCSDS) (G1 is associated with first symbol) G1 - G2 G2 Inverted - G1 (G2 is associated with first symbol) G2 - G1
Decoding mode	Single decoder or separate decoders on I & Q (dual mode)
Puncturing	$\frac{2}{3}$ $\frac{3}{4}$ $\frac{4}{5}$ $\frac{5}{6}$ $\frac{6}{7}$ $\frac{7}{8}$ or programmable in the registry.
Parallel processing	1 to 16 parallel rate $\frac{1}{2}$ Viterbi decoders on each I & Q with symbol sequence compliant with NASA Space Network User's Guide (450-SNUG)
Maximum (encoded) bit rate	In single Viterbi mode: up to 240 Mbps or 1160 Mbps with the high rate menus In dual Viterbi mode: up to 480 Mbps or 1160 Mbps with the high rate menus (for Viterbi $\frac{1}{2}$. else puncturing rate dependent). In parallel processing mode (> 4xI / 4xQ): > 600 Mbps
Decoder status	Locked/unlocked. Symbol error rate (obtained by comparing the input data to the re-encoded decoded data).

1.6.1.6.2 Viterbi $\frac{1}{4}$

Application example: ERS-2 low bit rate satellite.

Code rate	0.25 information bit per encoded bit
Constraint length	4 bits
Code generators	G1 = 1000; G2 = 1100; G3 = 1010; G4 = 1001
Maximum (encoded) bit rate	240 Mbps

1.6.1.7 De-scrambler

The Cortex HDR supports several types of auto-synchronized de-scrambler (before frame synchronization):

- Specific satellite de-scrambler (consult SDSY for details)
- CCITT V.35 de-scrambler (standard and Intelsat modes selectable)

Note 1: License dependent (delivered licenses are listed in Annex 5)

Note 2: Software version dependent

Note 3: Bit rates are given at the input of the demodulator. Bit rates can be limited by the used decoder, refer to the Maximum bit rates tables given in the chapter “Licenses and Start-up Menus” or consult SDSY

1.6.1.8 Transport Layer (RS-DVB or asynchronous LDPC layer)

The Cortex HDR supports the “Transport layer” processing.

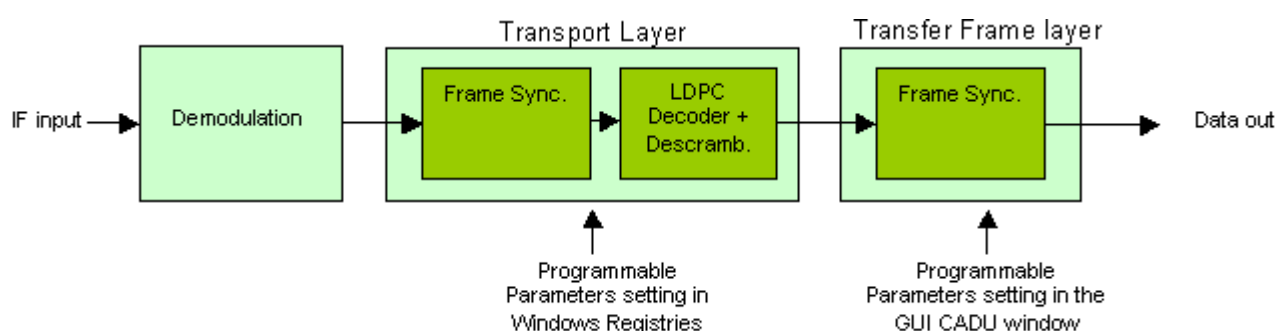
The Transport layer is implemented between the demodulator output and the input of the CADU frame synchronizer (CCSDS Transfer Frame layer).

The Transport layer can be used (enable or disable by configuration) with:

- the RS-DVB decoding process
- the LDPC decoding (7/8, 1/2, 2/3) process

The Transport layer acts as a first specific frame synchronizer. The parameters of this synchronizer can be set in Windows Registries (refer to the paragraph “Cortex HDR System Parameters in Registry”).

The following figure gives an example of the transport layer used with LDPC.



1.6.1.9 BEM 4D 8PSK TCM DVB-S R-S

This mode is compatible with the CCSDS report on BANDWIDTH-EFFICIENT MODULATIONS (**CCSDS 413.0-G-1, GREEN BOOK**, March 31, 2003) with completion for updating ECSS RF & Modulation standard ECSS E-50-05A and ECSS Telemetry channel coding ECSS E-50-01 draft 0.5.

This modulation is compatible with Pleiades, SMOS, Sentinel satellites.

- 8PSK - 4D TCM trellis: output rate 2.0, 2.5 or 2.75 bits/symbol (rate 2.25 is available on request)
- Decoder status: lock/unlock, BER estimation on corrected errors
- DVB-S Transport layer: synchronisation, randomization, **DVB-S RS** or **RS(239,255)** decoding , Interleaving
- Transport layer control parameter: ON/OFF, discard idle frames ON/OFF
- Transport layer status: lock/unlock, number of R-S correction, idle frames counter, corrected R-S CADU counter (good/bad CADU), dropout events.

1.6.1.10 VHR-DVB-S2 and CCSDS-SCCC standard

The Cortex HDR supports the future variable modulation and coding standards for high rate telemetry:

- DVB-S2 standard: **ETSI EN 302 307 V1.1.2**, Digital Video Broadcasting, second generation framing structure, channel coding and modulation systems for Broadcasting, Interactive Services, New Gathering and other Broad band satellite applications.
- S2-Extensions (DVB-S2X): **BlueBook A83-2 / EN302307-2**
- CCSDS SCCC standard: **CCSDS 131.2-R-1**, Flexible advanced coding and modulation scheme for high rate telemetry applications.
- CCSDS SCCC Extension (SCCC-X): **CCSDS 131.20-O**

1.6.1.11 OQPSK/8PSK variable modulation

The Cortex HDR supports variable QPSK/8PSK modulation scheme used by specific satellites.

The on line switching between the following modulation modes is supported:

- OQPSK with differential NRZ-M coding on each I and Q branches
- 8PSK with specific differential coding on the symbols.

It has to be noted that the input symbol rate is constant. The demodulator of the HDR recognizes the current modulation mode and switch automatically in the corresponding configuration (demodulation mode and differential decoding).

Whatever the current modulation can be at a given time, the demodulated data are output on the same two parallel ECL ports (Output mode = Split 2). The output clock rate is adjusted consequently.

This variable modulation mode is supported with the demodulator menu number 17.

1.6.2 Diversity Combining Unit and Antenna Combining

The Cortex HDR allows to combine two IF signals from different polarizations and from different delay.

Two combining modes are available:

- Automatic selection of the best channel
- Combining of the two channels, through optimal weighting. In this case up to 3 dB gain can be achieved when both signals have the same Eb/No (and/or same G/T for 2 antennas)

The combining system, also named Diversity Combining Unit (DCU), receives the data from two demodulators. It resolves phase ambiguity and delay between both channels, and deals with the different locking states possible. Its output feeds a FEC device, followed by a Format Output.

1.6.3 Data Processing Unit

1.6.3.1 Frame Synchronization

Synchronization word size	8, 16, 24, 32, 40, 48, 56, 64 or 128 bits
Frame size	16 to 1048576 bytes
Synchronization strategy	Error threshold on the synchronization word: SYN = 0 to 15 errors Check to Lock threshold: CTL = 0 to 15 frames Lock to Search threshold: LTS = 0 to 15 frames Bit slippage (0, 1 or 2 bits)

Automatic ambiguity resolution with feed-back on I & Q digital outputs

1.6.3.2 Data Time-tagging

Time code input	IRIG-B or External NTP server
Default time-tag format	64 bits (second of the year + μ s of the current second)
Time tagging accuracy	$\pm 50 \mu$ s (IRIG B)

1.6.3.3 CRC Frame checking

Polynomial	Programmable on 16 bits (CCITT compliant)
Preset value	Programmable on 16 bits

1.6.3.4 CCSDS and ECSS Decoding

Reed-Solomon decoding	CCSDS RS(255, 223) or RS(255, 239) Interleave factor automatically programmed (1 to 16) Virtual filling R-S counters (good and bad frames, BER)
Frame Header R-S decoding	CCSDS RS(10,6)
De-randomization	Polynomial, initialization and offset programmable
CRC verification	Frame by frame
CCSDS decommutation	Yes (CADU, VCDU and Packet)
CFDP protocol	Yes

1.6.3.5 LDPC Decoding

The LDPC (Low Density Parity Check code) decoders implemented inside the HDR are compatible with the LDPC codes recommended by CCSDS (**CCSDS 131.1-O-2**, September 2007). Are available:

- LDPC 7/8 (mono mode after I/Q merged or dual mode on separate I and Q outputs)
- LDPC 1/2 1K, 4K and 16K
- LDPC 2/3 4K and 16K
- LDPC 4/5 4K and 16K

The LDPC decoding is an iterative process with parity decoding. In this implementation, after the last iteration, the parity control is processed one more time. This control is able to detect bit error **before and after decoding**, and elaborates a quality "Good/Bad" status of the decoding CADU.

LDPC7/8 coding can be used in synchronous or asynchronous (alias slicing, alias transport layer... mode).

1.6.3.6 Data Output

Type of data	Raw data (frame synchronizer OFF) or frames (CADU/VCDU VC extraction capability) Real-time data or stored data
Data interface	dual 1-Gbit Ethernet (standard), dual 10-Gbit Ethernet (option) TCP-IP FTP (Integrated FTP server)
Ethernet data rate	in TCP/IP or UDT on LAN >700Mbps per 1Gigabit port (dual port) >6Gbps per 10Gbit port (dual port) Please note that such high throughput assume no other significant traffic on the LAN For WAN with large round trip time, UDT solution can use, please consult factory

1.6.4 Data Storage

Associated Hardware:

The Data storage system can be according the following configuration:

- Raid 0 with 4 Disk
- Raid 6 with 12 Disk
- Disk configuration are SSD or HDD
- Disk capacity are 1 Terrabytes or 2 Terrabytes

Data storage total capacity can be from: 4 Terrabytes to 20 Terrabytes

Read/Write Performance:

Raid 6 with 12 disk >10Gbit aggregate R/W

Raid 0 with 4 disk >4 Gbit aggregate R/W

File Associated Process for FTP access:

Files architecture and construction	Destination directories programmable Destination files names programmable Decommuration, filtering and sorting up to 5 fields in CADU Up to 64 destination files File size defined by the number of frames from 20 Mbytes to 4 Gbytes
Data recovery	FTP Read and Delete commands supported
Disk space management	Warning level programmable Alarm level for auto-erase programmable Delete of the oldest files (date programmable)
Auto tests	Auto test of directories and files

1.6.5 Playback

Operating Modes:	Test Modulator mode: replay at IF level using the Test Modulator board Data + Clock mode: Replay at the demodulator(s) PCM data outputs (Data + Clock outputs)
Data sources:	Recorded files from a previous satellite pass User files
Replay data rate :	Programmable up to 800 Mbps

1.6.6 Test Modulator

1.6.6.1 IF Modulation

IF carrier frequency	Programmable over the range 520 MHz to 2.2 GHz and 1.8 GHz to 3.2 GHz
IF output port	2
Output level	-5 dBm to -50 dBm (optimal -10 dBm to -40 dBm) (0.05 dB steps)
Output level setting accuracy	± 2 dBm
Output impedance / VSWR	50 Ω / 1.3
IF stability	As per reference clock (internal or external)
Unwanted emissions	Better than -40 dBc @ -10 dBm output level

1.6.6.2 Noise Source

Noise source	ON/OFF
Noise density	-93 dBm/Hz to -130 dBm/Hz (1 dB steps)
Noise bandwidth	2 GHz

1.6.6.3 Modulation

Modulation	BSPK , QPSK , OQPSK (SQPSK) UQPSK (AQPSK, AUQPSK, AUSQPSK) 8PSK, 16QAM, 16APSK, 32APSK, 64APSK
Symbol rate	Up to 2.0 Gbauds (may vary depending on selected encoding scheme)
Convolutional coding	Compliant to CCSDS recommendation (See characteristics in 0) Single or Dual encoding mode, rate $\frac{1}{2}$, parallel mode (stacked Viterbi) TCM

1.6.6.4 PCM Simulation

Operating mode	Fixed pattern: ASCII-coded file on hard disk (32768 bytes) Separate or merged I and Q data flows Programmable pseudo-random pattern: 2^{10} , 2^{11} , 2^{15} , 2^{23} , 2^{31} External Data + CLK (serial mode)
PCM coding	NRZ-L/M/S, DNRZ, BP-L/M/S Convolutional code (Viterbi) TCM (2/3, 2.25/3, 2.5/3, 2.75/3, 5/6 6D DSNG) CCSDS Reed-Solomon DVB Reed-Solomon LDPC 7/8
PCM modulation	ON or OFF
Polarity of transmitted NRZ-L	Normal or inverted
BER test capability	Yes (by the Demodulator) BER measurement from recorded data on external Data+Clock inputs
TM test ports	Data and Clock (I & Q) Level: ECL single ended or differential (hard-wired at delivery) Data: NRZ-L Clock phase: 0°
External Data & Clock	Level: ECL differential (hard-wired at delivery) Data: NRZ-L (serial mode) Clock phase: 180° Clock jitter: as per input clock jitter Modulator board termination must be either voltage terminated (unbalanced) or not (balanced). Factory setting: as per Customer

1.6.6.5 Doppler Simulation

Frequency range	0 to +/- 2MHz
Rate	0 to 100 KHz/s

1.6.7 Start-up Menus

The CORTEX HDR supports multiple start-up menus. See section 4.4.9 for more details on the functionalities supported by each menu.

1.6.8 Frequency Reference

1.6.8.1 External Reference

Input port	1 port for the demodulator and 1 port for the test modulator
Frequency	10 MHz sine or square
Impedance	50 Ω
Level	0.5 to 10 Vpp (if DC = 0) DC max. = ± 4 V

1.6.8.2 Internal Reference

Frequency	10 MHz
Frequency stability	$\leq \pm 5$ ppm ageing first year $\leq \pm 2$ ppm ageing following years $\leq \pm 15$ ppm vs temperature variations (10° to 40°)

1.6.9 Time Code Reference

Time code	IRIG-B or NASA 36
Input port	1
Amplitude	0.8 to 8 Vpp
Impedance	100 k Ω
Time tagging accuracy	± 50 μ s
Flywheel mode	Yes

1.6.10 Mechanical - Environment

Industrial PC chassis	19-inch rack-mountable (4 rack space) Depth: 550 mm
Test Modulator board	PCI, 1 slot (+ 1 I/O slot)
Demodulator board	PCI, 2 slots
Weight	≈ 25 kg
Operating temperature	+ 10°C to + 40°C
Storage temperature	- 20°C to + 60°C
Relative humidity	5% to 95% noncondensing

1.6.11 *Supply*

Supply (last generation)	100 - 240 VAC (-10%, +6%) 50 / 60 Hz
Nominal consumption	3.3 A – 1.4 A
Maximum consumption	9.0 A – 4.5 A

2 HARDWARE DESCRIPTION

2.1 CORTEX HDR INTEGRATION

2.1.1 Hardware Configuration

In its minimum hardware configuration, the HDR 4G unit consists in:

- one 4U-high PC-compatible workstation
- one demodulator module with its I/O interfaces (LVDS-ECL interfaces)

Depending on the delivered configuration, optional hardware parts can be added to the unit and consist in:

- test modulator board
- custom I/O interface board
- Dual port 10 Gigabit RJ45 copper or LC fiber-optic connectors (configuration dependent)
- System storage: raid0 raid6 4 or 12 disk from 1Terra or 2Terra byte disk

The position of the boards inside the chassis must not be changed. A sticker at the rear panel of the equipment shows how the boards must be mounted in the chassis. Do not change this setting.

2.1.2 PC-compatible Workstation

The PC-compatible workstation cabling is described in Annex 4.

2.1.3 Demodulator board

2.1.3.1 Overview

The demodulator assembly consists in:

- the mother board: PCI-e board with FPGA
- the IF input board
- the I/O interface board
- the set of cables and I/O connectors

The I/O mezzanine in its standard version provides both ECL I/O and LVDS I/O.

2.1.3.2 Separate IF inputs

The following figure gives the diagram of the IF input stage of the demodulator board.

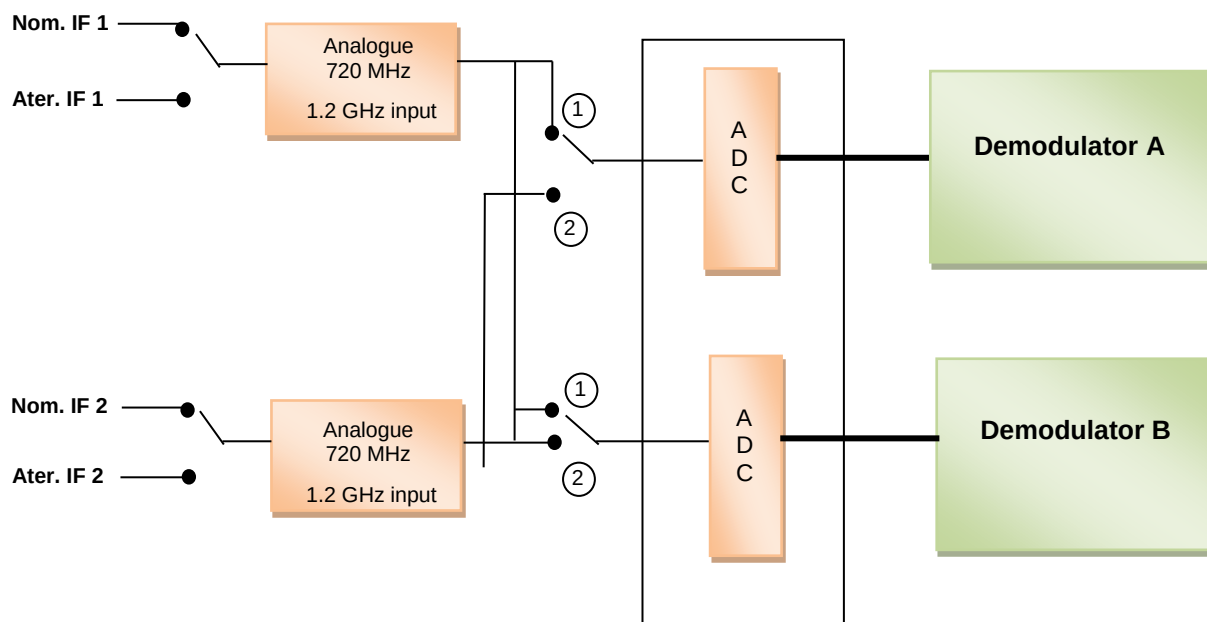


Figure 8: Demodulator input stage with separate IF inputs

The demodulator board provides double 720 MHz/1.2GHz/2,4GHz IF inputs. These configurations are exclusive. The selection between configurations is done by basic menu selection.

2.1.4 Test Modulator Board

The test modulator boards have the capacity to generate 2 outputs at any IF frequency.

The test modulator consists in one single board with FPGA, and two DACs.

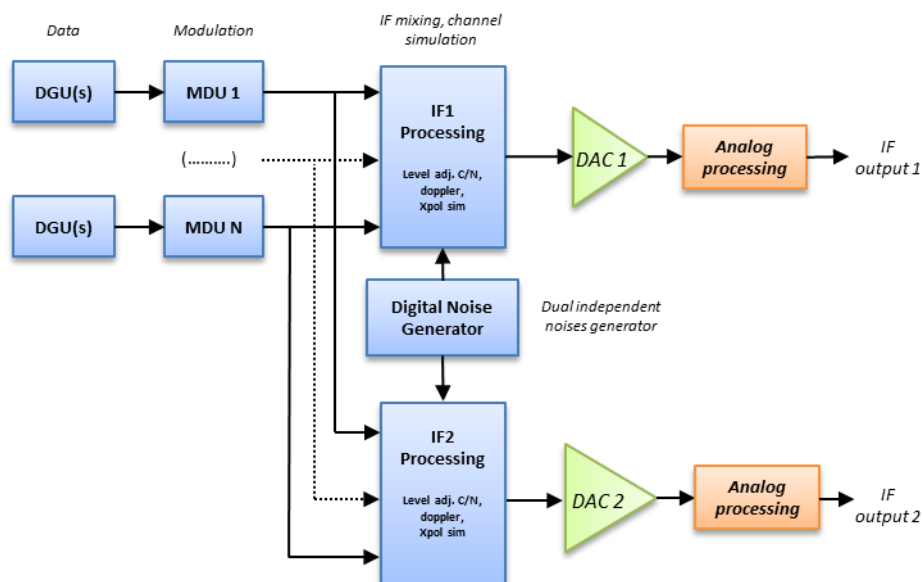


Figure 9: CORTEX HDR Test Modulator - Multi channel, multi IF operation

2.2 I/O CONNECTORS DEFINITION

2.2.1 CPU Board I/Os

The only operational I/O ports on the CPU board are:

- Dual port **RJ45 Gigabit Base-T Ethernet** ports, used for connection to the Ethernet network(s)
- Dual port **10 Gigabit RJ45 copper or LC fiber-optic connectors** (configuration dependent)

About detail see Intel® Ethernet Converged Network Adapter series X520 or consult factory.

See Annex 4 for more details on the CPU board.

2.2.2 Demodulator & Test Modulator I/Os

2.2.2.1 HDR 4G/4G+ I/Os

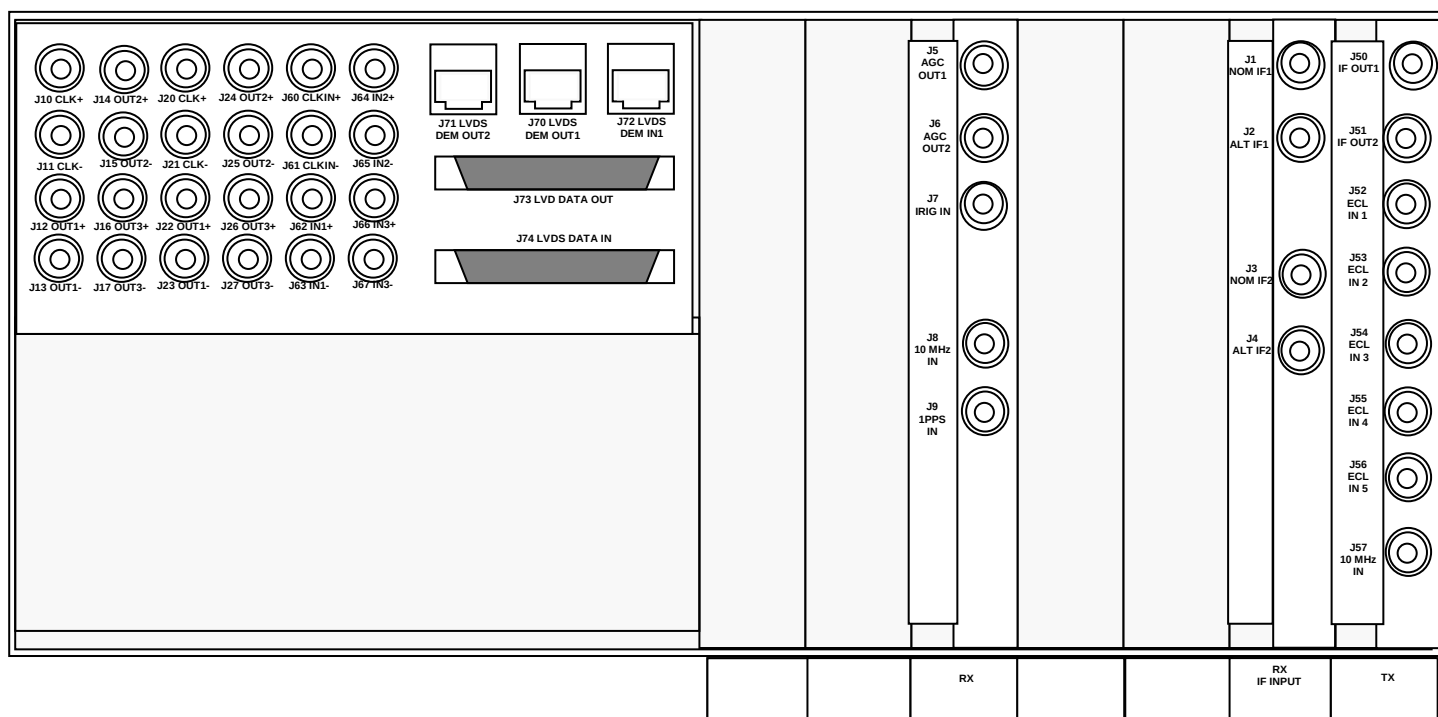


Figure 10: CORTEX HDR 4G/G+ I/Os

2.2.2.2 Demodulator I/Os

Unless otherwise specified, all I/Os are ECL with SMA connectors:

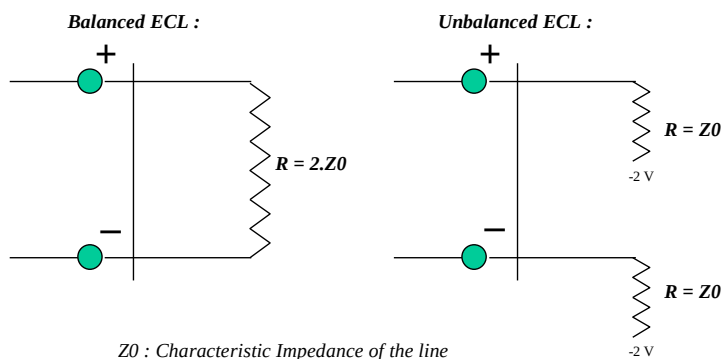
I/O SECTION	LABEL	ID	SIGNAL DEFINITION	ELECTRICAL
RX IF INPUT	NOM IF 1	J1	Nominal IF 1 input	-10 to -50 dBm
	ALT IF 1	J2	Alternate IF 1 input	
	NOM IF 1	J3	Nominal IF 2 input	
	ALT IF 1	J4	Alternate IF 2 input	
	X IN 1	J34	Optional	
	X IN 2	J35	Optional	
RX	AGC OUT1	J5	AGC voltage output	Between 0 and 3 V. -10 dBm => 3 V +/- 200 mV -50 dBm => 0.6 V +/- 200 mV
	AGC OUT2	J6		
	IRIG IN	J7	IRIG-B time code input	0.8 to 8 Vpp, 100 kΩ
	10 MHz IN	J8	Input Reference Clock	10 MHz, sine or square, 50 Ω 0.5 to 10 Vpp (if DC = 0) DC max = ± 4 V
	1 PPS IN	J9	1 PPS Reference Signal	50 Ω 0 V to 0.63 V for low level 1.17 to 1.8 V for high
DEMODULATOR CHANNEL A HW dependant	CLK + CLK -	J10 J11	Refer to paragraph (2.2.2.2.1)	ECL differential $V^+ > -1$ Volt, $V^- < -1.7$ Volt Adapted Termination required (note 1)
	OUT1 + OUT1 -	J12 J13	Refer to paragraph (2.2.2.2.1)	
	OUT2 + OUT2 -	J14 J15	Refer to paragraph (2.2.2.2.1)	
	OUT3 + OUT3 -	J16 J17	Refer to paragraph (2.2.2.2.1)	

DEMODULATOR CHANNEL B HW dependant	CLK + CLK -	J20 J21	Refer to paragraph (2.2.2.2.1)	ECL differential $V^+ > -1$ Volt, $V^- < -1.7$ Volt Adapted Termination required (note 1)
	OUT1 + OUT1 -	J22 J23	Refer to paragraph (2.2.2.2.1)	
	OUT2 + OUT2 -	J24 J25	Refer to paragraph (2.2.2.2.1)	
	OUT3 + OUT3 -	J26 J27	Refer to paragraph (2.2.2.2.1)	
DEMODULATOR INPUT	CLK IN + CLK IN -	J60 J61	Clock input	ECL differential $V^+ > -1$ Volt, $V^- < -1.7$ Volt 50 Ω termination required (note 2)
	IN1 + IN1 -	J62 J63	Data input <i>for BER measurement or data post-processing</i>	
	IN2 + IN2 -	J64 J65	Data input <i>for BER measurement or data post-processing</i>	
	IN3 + IN3 -	J66 J67	Data input <i>for BER measurement or data post-processing</i>	
LVDS I/Os HW dependant	LVDS DEMOD OUT A	J70	Data and clock outputs for demodulator A	LVDS standard characteristics (note 3)
	LVDS DEMOD OUT B	J71	Data and clock outputs for demodulator B	
	LVDS DEMOD IN	J72	Data and clock inputs	
	LVDS DATA OUT	J73	For customized interconnection panel	
	LVDS DATA IN	J74	For customized interconnection panel	

Table 2: Demodulator I/Os Definition

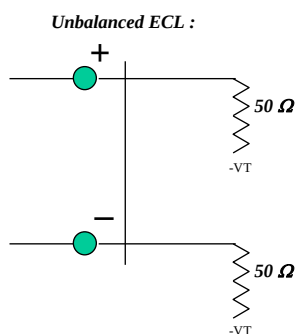
Note 1: ECL differential outputs must have this termination:

External termination :



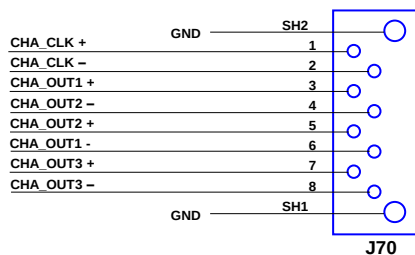
Note 2: ECL differential inputs have a $50\ \Omega$ termination:

External $50\ \Omega$ termination :



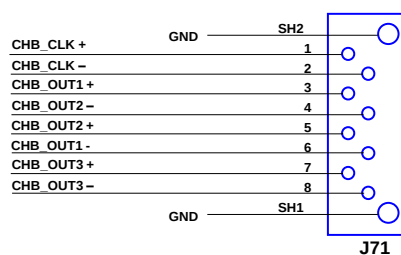
Note 3: LVDS Interface

DEMOM OUT 1



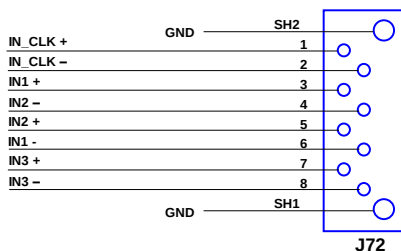
J70

DEMOM OUT 2



J71

DEMOM IN

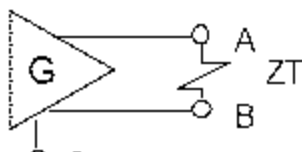


J72

LVDS Connector: EMBASE RJ45 CAT. 6

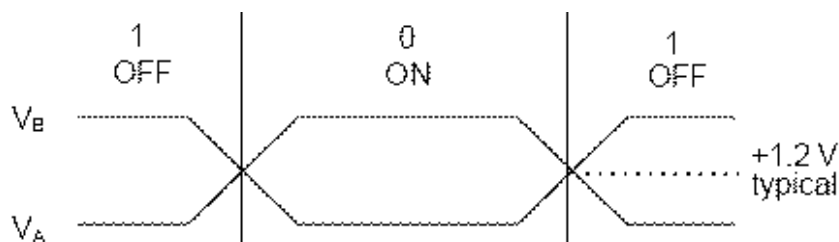


LVDS differential outputs must be terminated as follows:



ZT = Termination impedance = $100\ \Omega$

The output level is adjusted around the typical 1.2 V offset, with a peak-to-peak amplitude varying from 250 mV to 450 mV.



2.2.2.2.1 Channel A & Channel B I/Os description

The Outputs of the demodulators depend on:

- the selected option
- the demodulation mode
- the selected soft decision output
- the selected Data format output (CADU)
- and the setting of the **Output Mode** parameter

The following table shows the different configurations for both Demodulator A and Demodulator B (standard outputs).

OUTPUT MODE			BPSK	QPSK 8PSK 16QAM <i>SPLIT 2</i>	QPSK 8PSK 16QAM <i>I/QMERGE</i>	QPSK 8PSK 16QAM <i>SPLIT 4</i>	A/UQPSK
CHANNEL A ¹	CLK+ CLK-	J10 / J11	Clock @ BR Demod A	Clock @ BR/2 Demod A	Clock @ BR Demod A	Clock @ BR /4	Clock @ BR I Demod A
	OUT1+ OUT1-	J12 / J13	Data	Data I	Data	Data I -1	Data I
	OUT2+ OUT2-	J14 / J15	Reserved	Data Q	Data	Data Q -1	Data Q
	OUT3+ OUT3-	J16 / J17	Clock @ BR Demod A	Clock @ BR/2 Demod A	Clock @ BR Demod A	Clock @ BR/4	Clock @ BR Q Demod A
CHANNEL B ¹	CLK+ CLK-	J20 / J21	Clock @ BR Demod B	Clock @ BR/2 Demod B	Clock @ BR Demod B	reserved	Clock @ BR I Demod B
	OUT1+ OUT1-	J22 / J23	Data	Data I	Data	Data I - 2	Data I
	OUT2+ OUT2-	J24 / J25	Reserved	Data Q	Data	Data Q - 2	Data Q
	OUT3+ OUT3-	J26 / J27	Clock @ BR Demod B	Clock @ BR/2 Demod B	Clock @ BR Demod B	Reserved	Clock @ BR Q Demod B

Table 3: Data & Clock Outputs vs Demodulation

*** BPSK demodulation can be set to SPLIT2 mode with high rates hardware menus (single demodulator). In this case the data stream is split on the two outputs J12/ J13 and J22/J23.**

OUTPUT MODE			CADU-A + SYNC + USER	CADU-A + SYNC + GOOD	CADU-A SPLIT2 + SYNC	CADU-A SPLIT4
CHANNEL A ¹	CLK+ CLK-	J10 / J11	Clock @ BR Demod A	Clock @ BR Demod A	Clock @ BR / 2 Demod A	Clock @ BR/4 Demod A
	OUT1+ OUT1-	J12 / J13	Data	Data	Data I	Data I 1
	OUT2+ OUT2-	J14 / J15	Sync pulse	Sync pulse	Data Q	Data Q 1
	OUT3+ OUT3-	J16 / J17	Clock for User data Demod A	Status 1 = good 0 = bad	Sync pulse ²	Sync pulse
CHANNEL B ¹	CLK+ CLK-	J20 / J21	Clock @ BR Demod B	Clock @ BR Demod B	Clock @ BR / 2 Demod B	Clock for User data Demod A
	OUT1+ OUT1-	J22 / J23	Data	Data	Data I	Data I 2
	OUT2+ OUT2-	J24 / J25	Sync pulse	Sync pulse 2	Data Q	Data Q 2
	OUT3+ OUT3-	J26 / J27	Clock for User data Demod B	Status 1 = good 0 = bad	Sync pulse ²	Status 1 = good 0 = bad

Table 4: Data & Clock CADU Outputs

	QPSK NORMAL SOFT ³	QPSK MERGE SOFT
DEMODULATOR CHANNEL A		
CLK + CLK -	Clock @ BR/2 Demod A	Clock @ BR Demod A
OUT1 + OUT1 -	I bit 2 (MSB)	I/Q bit 5 (MSB)
OUT2 + OUT2 -	I bit 1	I/Q bit 4
OUT3 + OUT3 -	I bit 0	I/Q bit 3
DEMODULATOR CHANNEL B		
CLK + CLK -	—	0 = indicates I 1 = indicates Q
OUT1 + OUT1 -	Q bit 2 (MSB)	I/Q bit 2
OUT2 + OUT2 -	Q bit 1	I/Q bit 1
OUT3 + OUT3 -	Q bit 0	I/Q bit 0 (LSB)

Table 5: Data & Clock Outputs vs Demodulation (soft outputs) Hardware menu dependent

Note 1: Except for the QPSK mode with soft decision output, which involves only the demodulator A, the output configuration setting for each demodulator is done from the corresponding GUI windows (Demodulator Global Window: Output mode selection).

Note 2: Pulse corresponding to the first bit of the frame synchronizer.

Note 3: The coding of In and Qn used for the soft decision outputs is given in the following representation of the I/Q constellation.

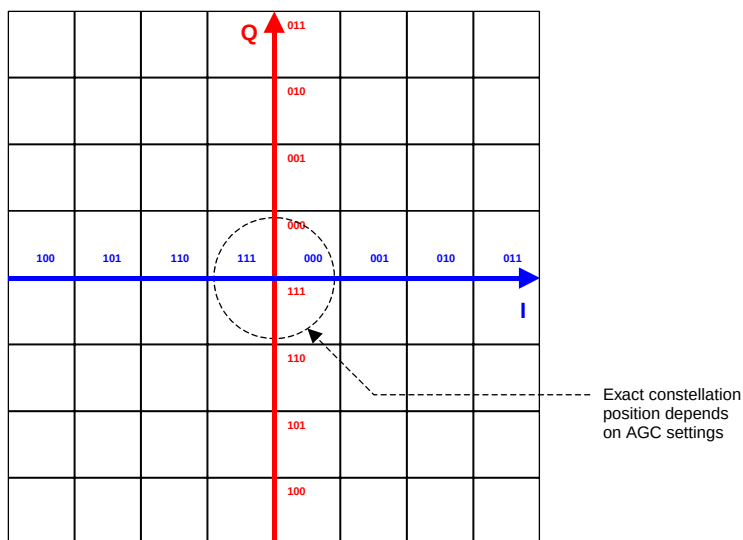


Figure 11: Soft Decision Coding vs I/Q Constellation (Normal Soft Output)

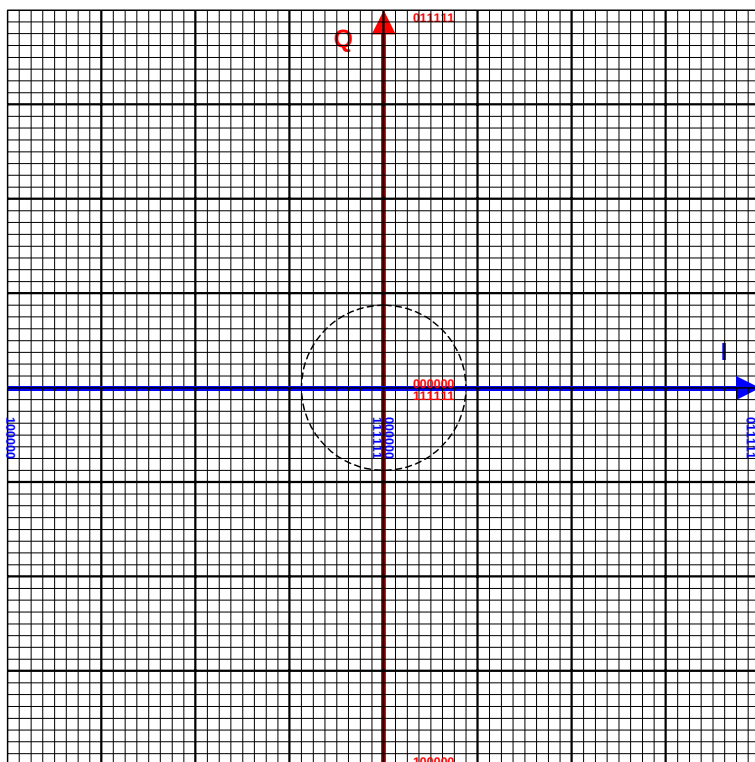


Figure 12: Soft Decision Coding vs I/Q Constellation (Merge Soft Output)

2.2.2.2.2 AIT Connector (LVDS Data Output - J73) description

This connector can be used to output data from demodulator A (single channel) in Split 8 and Split 16 modes.

The electrical format is standard LVDS.

PINS	SPLIT 8 MODE	SPLIT 16 MODE
6/8	Clock @ BR/8 +/-	Clock @ BR/16 +/-
2/4	Data I 1	Data I 1
14/16	Data Q 1	Data Q 1
10/12	Data I 2	Data I 2
22/24	Data Q 2	Data Q 2
18/20	Data I 3	Data I 3
30/32	Data Q 3	Data Q 3
26/28	Data I 4	Data I 4
38/40	Data Q 4	Data Q 4
34/36	Reserved	Data I 5
46/48	Reserved	Data Q 5
42/44	Reserved	Data I 6
54/56	Reserved	Data Q 6
50/52	Reserved	Data I 7
62/64	Reserved	Data Q 7
58/60	Reserved	Data I 8
70/72	Reserved	Data Q 8
66/68	Reserved	Reserved

Table 6: Data and Clock outputs on J73

2.2.2.2.3 Demodulator inputs description

Inputs into the frame synchronizer are available on SMA connectors in ECL electrical format or on RJ45 connector in LVDS electrical format.

The following tables show the different configurations for both single and dual channel input.

INPUT MODE		DATA SERIAL	DATA // 2
CLK IN+ CLK IN-	J18 / J19	Clock	Clock
IN1+ IN1-	J20 / J21	Data	Data I
IN2+ IN2-	J22 / J23	Reserved	Data Q
IN3+ IN3-	J24 / J25	Reserved	Reserved

Table 7: Single channel demodulator input modes

INPUT MODE		DATA SERIAL
CLK IN+ CLK IN-	J18 / J19	Clock DPU 1
IN1+ IN1-	J20 / J21	Data DPU 1
IN2+ IN2-	J22 / J23	Data DPU 2
IN3+ IN3-	J24 / J25	Clock DPU 2

Table 8: Dual channel demodulator input mode

2.2.2.3 Test Modulator I/Os

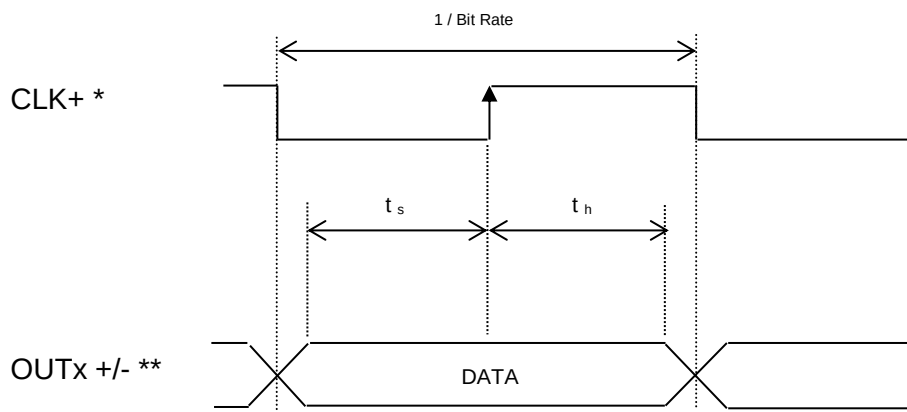
All I/O connectors are SMA:

I/O SECTION	LABEL	ID	SIGNAL DEFINITION	ELECTRICAL
	IF OUT 1	J50	IF output	-5 to -50 dBm
	IF OUT 2	J51		
	IN1 IN2 IN3 IN4 IN5	J52 J53 J54 J55 J56	Data input (note 1)	Single ended ECL > -0.81 V for high level < -1.95 V for low level ECL termination -2V, 56 Ω included (Not used in this version)
	10 MHz IN	J57	Input Reference Clock	10 MHz, sine or square, 50 Ω 0.5 to 10 V _{pp} (if DC = 0) DC max = ± 4 V
	MOD EXT IN 1	J58	IF Input of test mod Up-converter 1 Optional	0 dBm
	MOD EXT IN 2	J59	IF Input of test mod Up-converter 2 Optional	0 dBm
	MOD EXT OUT 1	J36	IF Output of test mod Up-converter 1 Optional	-10 to -40 dBm
	MOD EXT OUT 2	J37	IF Output of test mod Up-converter 2 Optional	-10 to -40 dBm

Table 9: Modulator I/Os Definition

2.2.3 Output Data Definition and Timing

2.2.3.1 Timing for all output types



For I/Q merged output, RS CADU output and LDPC CADU output:

$$t_s = 1/2BR - 500 \text{ ps}$$

$$t_h = 1/2BR - 500 \text{ ps}$$

For normal I and Q outputs:

$$t_s = 1/BR - 500 \text{ ps}$$

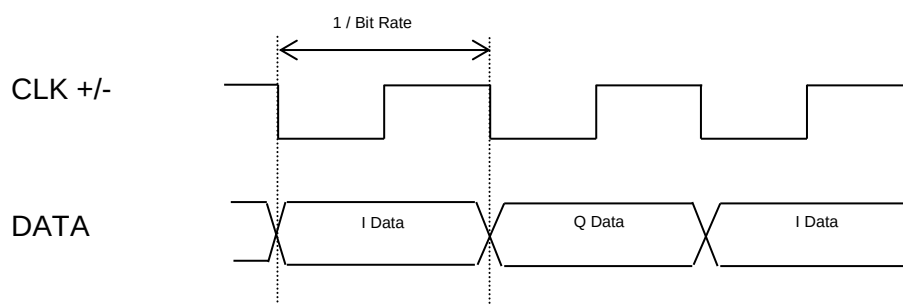
$$t_h = 1/BR - 500 \text{ ps}$$

* Clock polarity: Normal Output Clock

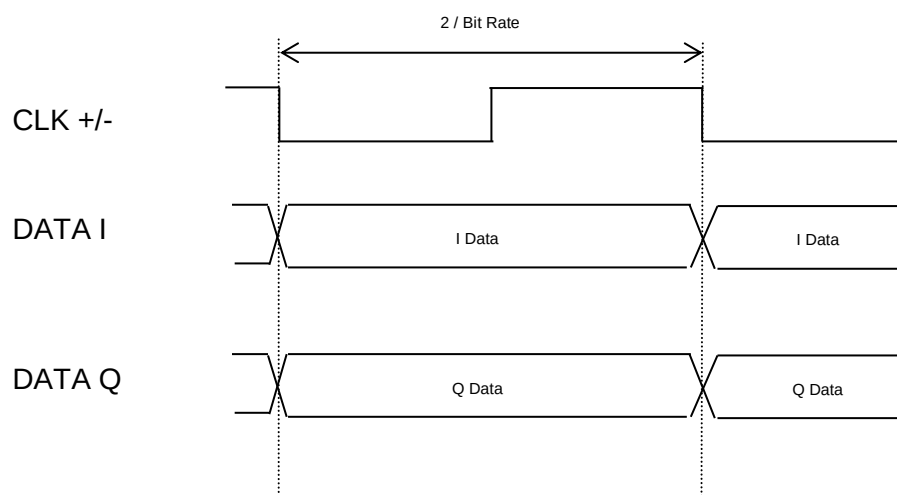
** Except when OUT3 = CLK

Remark: For data output after specific decoding process (example: TCM + RS-DVB, punctured Viterbi), “holes” inside the output data and clock signals can be present. These “holes” are due to the delayed check symbols and / or synchronization words. Nevertheless it has to be considered that for each data corresponds an edge of the clock.

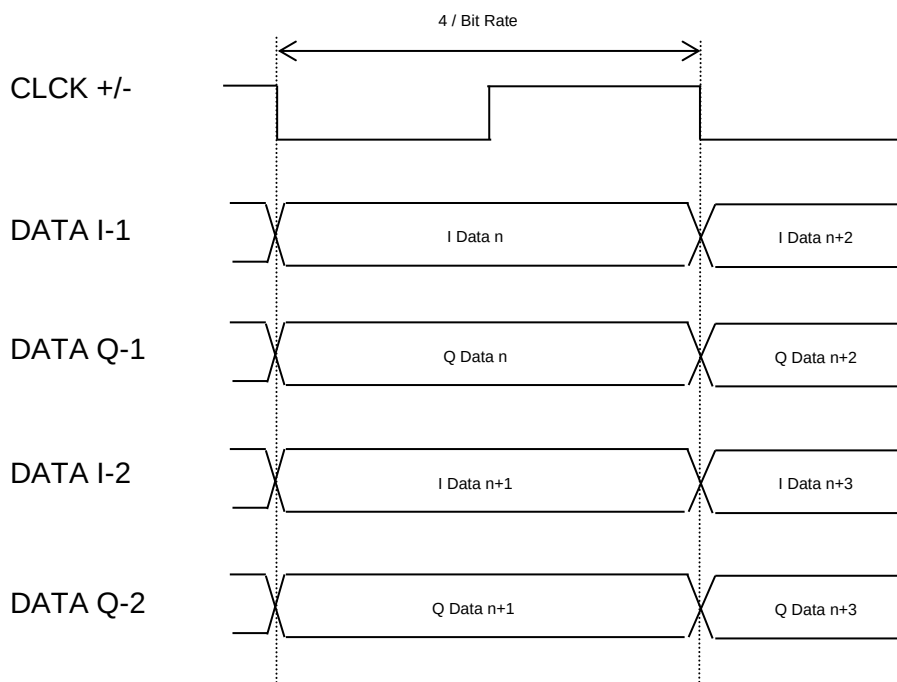
2.2.3.2 I/Q merged output



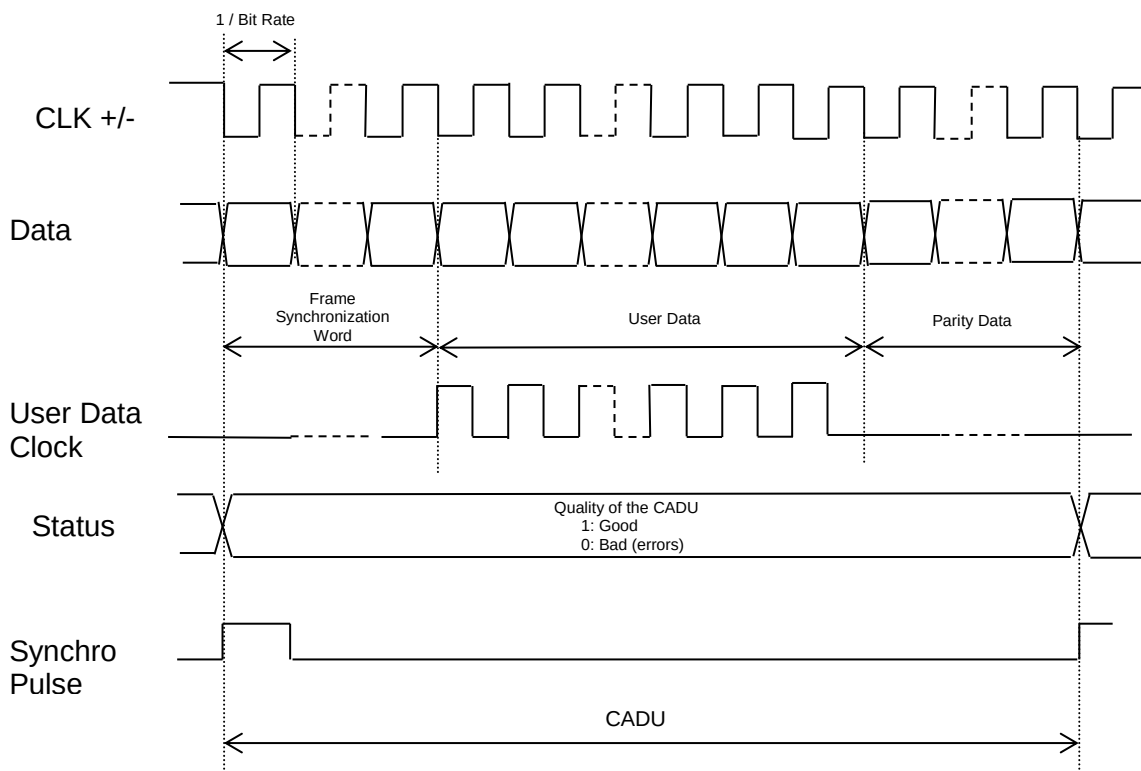
2.2.3.3 Split 2 output (I and Q normal output)



2.2.3.4 Split 4 output



2.2.3.5 CADU output formats



2.2.4 *Specific I/Os (Customers requirements)*

The Cortex HDR can be delivered with specific I/O for AIT or test purpose. These specific interfaces are described inside the specific DTU document. Please refer to the Annex 5 of the delivered documents folder.

3 FUNCTIONAL DESCRIPTION

3.1 Monitoring & Control

3.1.1 Monitoring

The CORTEX HDR supplies its status on request from a Monitoring Client. On reception of a monitoring request, the CORTEX HDR checks its validity. If the request is correct, it returns a monitoring message embedded in an Ethernet packet. In the case of an invalid request, the CORTEX HDR returns a negative acknowledgment message.

A monitoring message contains the current configuration of the unit (expected frequency, operating mode, etc.), dynamic status (IF level, Eb/No, frequency offset, etc.) and alarms (hardware alarm, etc.). The Monitoring Client can request:

- the high-level monitoring table of the CORTEX HDR,
- the Demodulator monitoring table,
- the Test Modulator monitoring table (if available),
- the Data Processing Unit table,
- the Data Recording Global table (if available),
- the Data Recording Unit table (if available),
- the project-specific monitoring table,
- the whole set of here above monitoring tables.

Up to eight Monitoring Clients can be simultaneously connected to the Monitoring port. For more detail on the monitoring messages, refer to Annex 1.

3.1.2 Control

Configuration commands received from a Control Client allow configuration of the equipment (example: change the uplink carrier frequency on the Test Modulator) or triggering and action (example: restart the carrier acquisition process on the Demodulator).

A detailed description of the Graphical User Interface is available in Section 4.

Simultaneous connection of several Control Clients to the Control port is not allowed for security reasons. For more detail on the configuration commands, refer to Annex 1.

3.2 Logging

The CORTEX HDR reports on the Logging port all changes in its configuration.

Up to five Logging Clients can be simultaneously connected to the Logging port. For more detail on the logging messages, refer to Annex 1.

3.3 Demodulation

3.3.1 IF Processing

Signal Processing:

- IF signal selection (Nominal or Alternate input).
- Automatic gain control: the signal level must be in the range -10 dBm to -50 dBm.
- Analog-to-Digital conversion of the received signal.
- IF acquisition: the programmed acquisition range must be compatible with the maximum frequency uncertainty (Doppler shift, oscillators drift,...).
- Cable equalizer filter (see Section 3.7): cable loss equalizer with programmable attenuation.
- IF filtering: matched filter, adjacent canal rejection or custom FIR filters (option).

Test & Monitoring:

- A number of status enable to monitor the demodulation process at IF: IF signal level, PLL status, carrier offset, etc...
- **Spectrum analysis:** allow to visualize the received signal spectrum.
- **Vector analysis:** check that the displayed constellation is correct and corresponds to the noise and signal characteristics.
- **Filter analysis:** display in real time the pulse and frequency responses of the digital filtering.

3.3.2 Bit Synchronization

3.3.2.1 Signal Processing

- Matched filters: basic supply: Integral & Dump. Optional: Root Raised, Raised filters with programmable roll-off factor, GMSK, user-programmable FIR filters (see next section).
- Symbol phase recovery

Test & Monitoring: a number of status enable to monitor the bit synchronization process: PLL status, signal to noise ratio (E_b/N_0), etc...

3.3.2.2 Customized FIR Filters

3.3.2.2.1 General

The filtering is based on one FIR for I channel and one FIR for Q channel.

The FIR sampling rate is either:

- twice the symbol rate (symbol rate < 480 Msps)
- or four time the symbol rate (only if symbol rate < 240 Msps)

Example: for a QPSK / 100 Mbps (50 Msps) data stream, the FIR filter sampling rate can be set to 100 MHz or 200 MHz.

Two types of filters are available:

- symmetrical filter: 32 taps (even symmetrical FIR) or 31 taps (odd symmetrical FIR)
- asymmetrical filter: 16 taps on direct path I and Q plus 16 complex taps (Q to I and I to Q)

The I & D filter design is usually the best answer to matched filtering needs. However, custom filters may improve the performances in some circumstances, like:

- reduction of adjacent channel interferences
- phase/amplitude equalization

3.3.2.2.2 Custom Filters Programming

Filter coefficients can be programmed from files on the hard disk. Up to 1000 files can be loaded to the disk. Files are ASCII-coded and can be edited using **Notepad** or any other text editor.

Each file contains 16-bit data (in hexadecimal format):

Line 1: FIR type selection:

0000 for asymmetrical filter
0001 for even symmetrical filter
0002 for odd symmetrical filter
Bit 2 set to 1 enables the complex filter
Bit 3 set to 1 enables the complex filter

Line 2: Sampling rate selection:

0000 for 2 time the symbol rate
0001 for 4 time the symbol rate

Line 3: Direct FIR length

Line 4: Cross FIR length

Line 5: Cross FIR delay

Line 6: Reserved line

Line 7 to line 32: Direct coefficients (16 bit) on I

Line 33: Reserved line

Line 34 to line 41: Crossed coefficients Q to I for complex filter

Line 42: Reserved line

Line 43 to line 68: Direct coefficients (16 bit) on Q

Line 69: Reserved line

Line 70 to line 77: Crossed coefficients I to Q for complex filter

Access path to the filter files is: **C:\Program Files\In-sneclrtxnt\FIR**. Files are named **fir_cust1** to **fir_cust1000**.

3.3.2.3 DEAF Filtering

3.3.2.3.1 General

The equivalent channel of a satellite communication system is constituted of:

1. the emitter filter: for frequency division multiplexing (FDM), the pass-band emitter filter is generally a cavity filter. Base-band filtering depends on the application.
2. the channel: the channel could be considered as an amplitude fading channel with additive White Gaussian Noise. Amplitude fading is resolved by automatic gain control. One can add the possibility of multi-paths created by the structure of the satellite that can be modeled by a Finite Impulse Response filter.
3. the receiver filter: there are many possibilities, commonly used filters are Integral & Dump and Root Raised Cosine (RRC) filters.

In order to get perfect recovery of transmitted pulses or symbols, the equivalent channel must verify the well-known Nyquist criterion (for a symbol rate $R = 2 \cdot f_c$ where f_c is the physical channel bandwidth, the impulse response of the equivalent channel should be equal to 0 for each multiple of $1/R$). This property is verified for a chain made of a RRC emitter filter and a RRC receiver filter, as RRC are well-known Nyquist filters. However, emitter filters at RF frequencies are very often cavity filter design to meet the regulatory rejection specifications, they are able to work with the signal coming from the high-power amplifier and they have a non-linear phase response. Moreover, changes of emitter filter characteristics (magnitude and phase) could not be avoided when there are large variations of temperature like those which take place in space. So, the equivalent channel seldom verify the Nyquist criterion and this induces inter-symbol interferences (ISI) that are harmful for performances.

A second aspect that is harmful for performances is noise. Two types of noises are present: the thermal noise and the interference noise, coming from close channels.

In order to reduce the ISI and the influence of noise, two types of receiver could be used:

- The Matched Filter
- The Wiener Filter

As they are, these filters are not useful because the equivalent channel is changing. So, they are implemented through conventional equalizer which are adaptive filters, but these filters need a training sequence which causes a payload loss.

A better solution for equalizing the incoming signal is to use an automatic adaptive equalizer.

3.3.2.3.2 Principle of the DEAF

The DEAF feature (DEAF stands for “**Digital Equalization Automatic Filtering**”) is a digital signal processing technique based on the adaptive filters theory. It performs a blind equalization for satellite communications of the incoming signal from the estimated statistics. It does not require additional knowledge of the desired response: whereas conventional equalization techniques requires a training period where a known data sequence is sent to the receiver in order to initialize the equalization device then switches to a standard transmission mode, a blind equalizer switches automatically and smoothly from the startup period to a standard transmission mode without training sequence.

The algorithm behind DEAF is a “Busgang-like” algorithm implemented on a fractionally-spaced equalizer for better performances:

Y is a vector of samples.

Z is the output vector of the equalizer.

C is the equalizer's taps vector.

First step: initialization / **C** is initialized at $C(0)$.

Second step: start-up / **C** is iteratively calculated / $C(n+1)=C(n)-\lambda \cdot f(Y,Z)$ where λ is the step-size and f a non-linear and zero-memory function of **Z**.

Third step: standard transmission mode / $C(\infty)$ is used to filter **Y**.

The algorithm used by the DEAF device has a convergence time lower than 100 ms.

3.3.2.3.3 Use of the DEAF

The automatic digital equalization function is particularly efficient in the following operations:

- satellite filter equalization (transmission channel)
- compensation of perturbations due to multi-path signal
- compensation of I/Q shift (OQPSK modulation) with LMS algorithm

The main result of the DEAF operation is an important improvement of the measured E_b/N_0 for high data rates telemetry.

The following screen shot of the constellation display of the HDR shows the advantage of the DEAF for filter equalization. The screen shot has been done during reception tests through an actual transmission filter. The demodulator #1 is set without the DEAF and demodulator #2 is set with the DEAF.

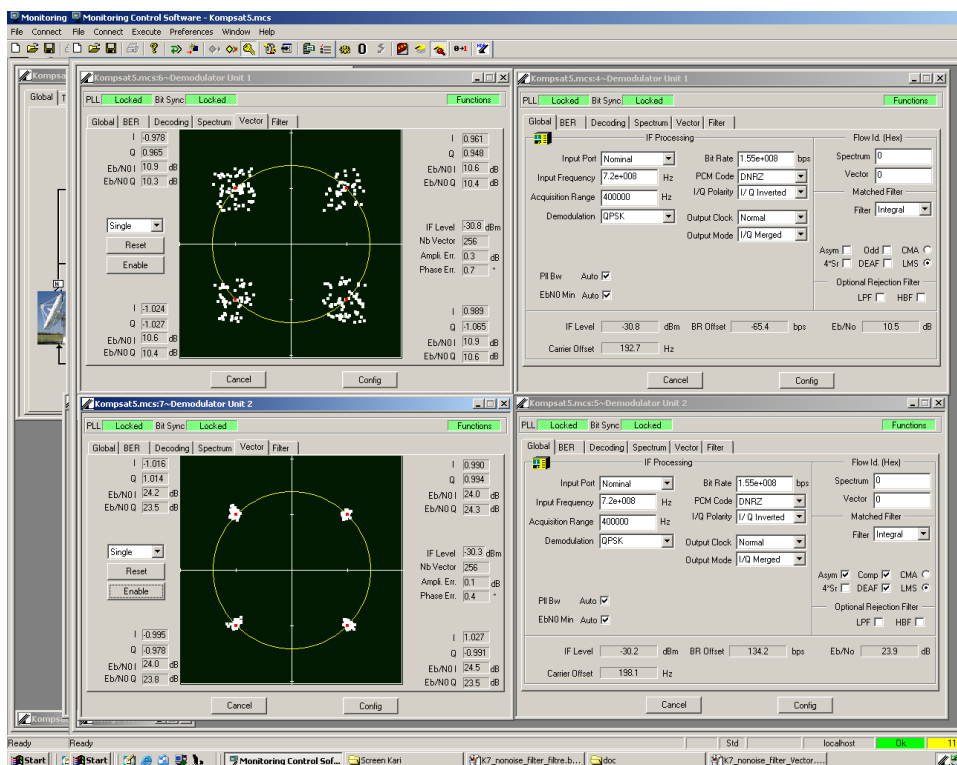


Figure 13: Effect of the DEAF.

3.3.2.4 XDEAF for reception of dual polarization data transmission

In order to increase the data transmission rate in a given bandwidth, some satellites use the polarization diversity. Two radio waves are then transmitted using orthogonal polarizations in the same frequency channel (typically RHCP and LHCP).

If we consider one channel (one polarization), we can see that this channel has a high probability to be perturbed by the other channel. And we can define a cross polarization discrimination coefficient as:

$$\text{XPD} = 10 \log (\text{Co-polar component} / \text{Cross-polar component})$$

In the transmission channel, several different sources of cross-polarization can occur:

- on-board antenna cross-polarization
- on-board antenna depointing
- propagation effect (rain, bad weather)
- ground antenna low elevation
- ground antenna depointing
- ground antenna cross-polarization

Typically and in case of bad conditions of transmission, XPD can be down to < 14 dB for X-band systems and < 8 dB for K-band systems.

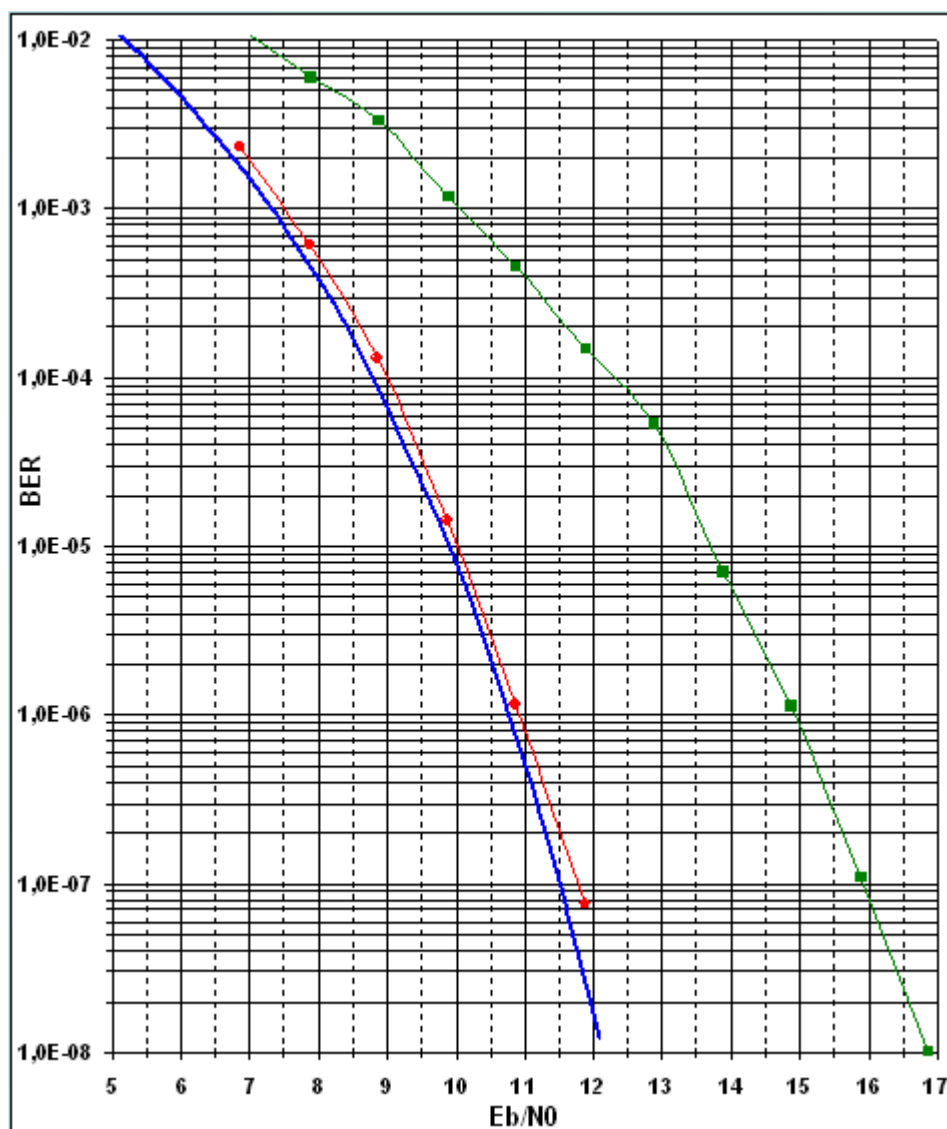
In order to compensate for the impact of the cross polarization effects on the data reception, the Cortex HDR provides the XDEAF function.

The XDEAF function is a real time and automatic digital filtering which takes into account the reception of both polarizations.

The XDEAF function can compensate for XPD down to 5 dB.

The following BER curves have been measured for a QPSK transmission at 200 Mbps using dual polarization. The XPD coefficient is 12 dB. The gain for a BER = 10^{-6} is 4 dB between the green curve (XDEAF disabled) and the red one (XDEAF enabled).

The following figure gives the effect of the XDEAF on the received constellation in the same previous conditions.



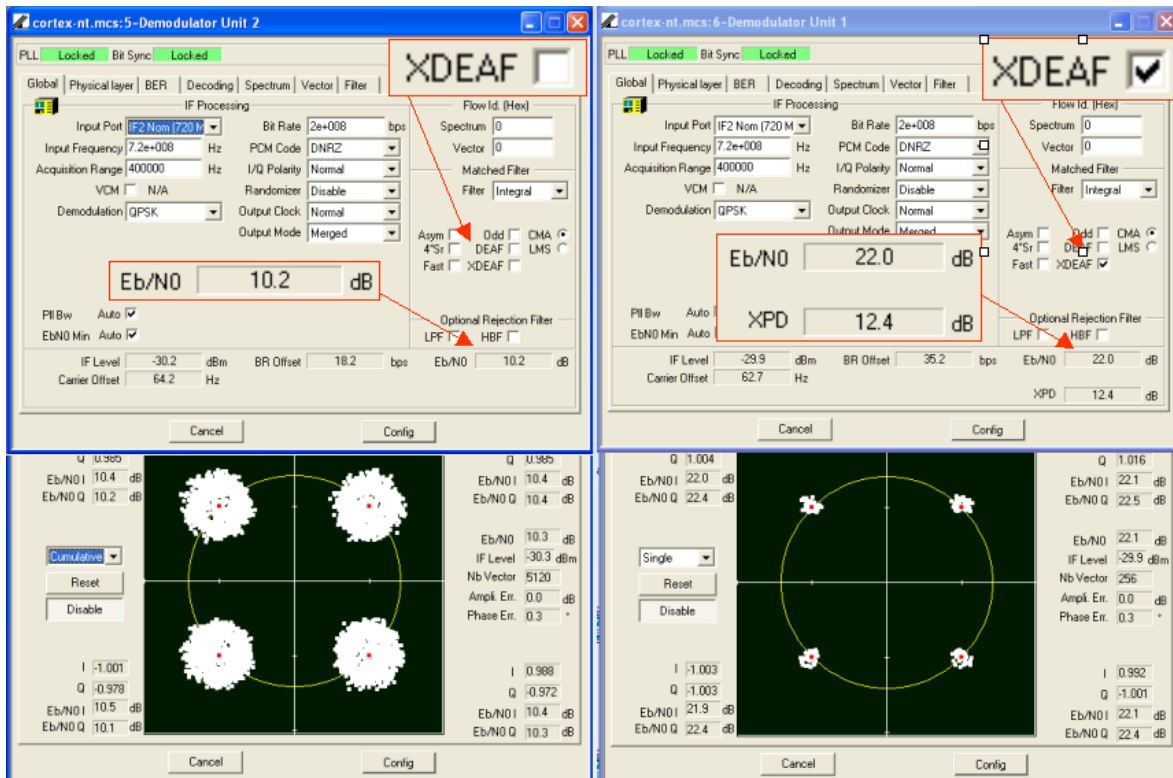


Figure 14: Effect of the XDEAF.

3.3.2.5 Adjacent channel rejection

In case of multi-channel transmission, the combination of successive filtering processes implemented inside the HDR:

- LPF rejection filter,
- HBF rejection filter,
- matched filter: RRC or customized equalizer filter for channel equalization,

Allows an efficient rejection of adjacent channels.

An interesting method offered by the HDR to customize the matched filter on the received signal is to use the combination of the rejection filters and the DEAF process. In a first time a standard filter is selected (example RRC filter) and the DEAF process is activated. When the frequency response of the filter is stabilized, it is possible to store the parameters of the filter inside a specific file. Then, the matched filter can be set up with the customized filter (file) constructed by the DEAF process. Refer to § 4.4.4.1 "Global" Window and § 4.4.4.6 "Filter" Window.

3.3.3 Viterbi Convolutional Decoding

3.3.3.1 General

Convolutional coding technique is well suited for telemetry channels with predominant Gaussian noise.

Convolutional decoding and error-correction by the CORTEX HDR complies to the CCSDS recommendation 101.0-B-3 (Blue Book - May 1992) and to the NASA Space Network User's Guide (450-SNUG, June 2002):

- Viterbi decoding of the data received from the bit synchronizer.
- Viterbi de-puncturing if required.
- Operating mode: single or dual Viterbi mode.
- Processing mode: single decoder or parallel decoders (up to 16 decoders on each I & Q).
- BER estimate by re-encoding the Viterbi decoded data and comparing the re-encoded data to the data fed to the Viterbi decoder. Note: if the Viterbi decoder(s) fails to lock (BER too high), check the Viterbi characteristics (Normal or inverted G2, single or dual Viterbi, G1→G2 or G2→G1 sequencing) and verify the setting of the I & Q output mode parameter (Demodulator output to the Viterbi decoder(s) input).

Test & Monitoring: a number of status enable to monitor the Viterbi decoding process: decoder lock status, BER (in %), etc...

3.3.3.2 Single or Dual Viterbi Decoder

The Viterbi decoding module can operate either in single mode or in dual mode:

- ❑ In the **single mode**, there is *only one* Viterbi decoder connected to the Demodulator module (I & Q output channels of the demodulator are connected to G1 & G2 inputs of the Viterbi decoder).
- ❑ In the **dual mode**, the Viterbi decoding module has two separate Viterbi decoders on I and Q Channels.

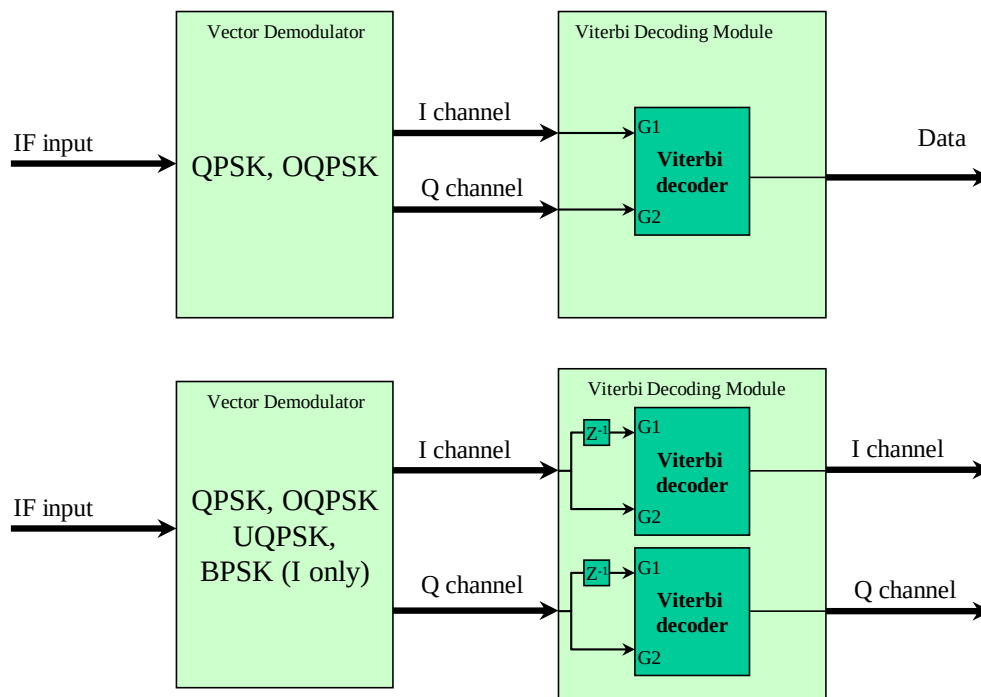


Figure 15: Viterbi Decoding Module. Single & Dual Operating Modes

3.3.3.3 Parallel Viterbi decoders

The HDR supports the rate $\frac{1}{2}$ Viterbi parallel processing as specified in the NASA SNUG (450-SNUG). The number of decoders can be set between $N = 1$ and 16 on I and Q channels. Dual Viterbi decoding as described in the previous paragraph can be seen as a particular case of the parallel processing with $N = 1$.

The supported symbols sequence is given in the following figure.

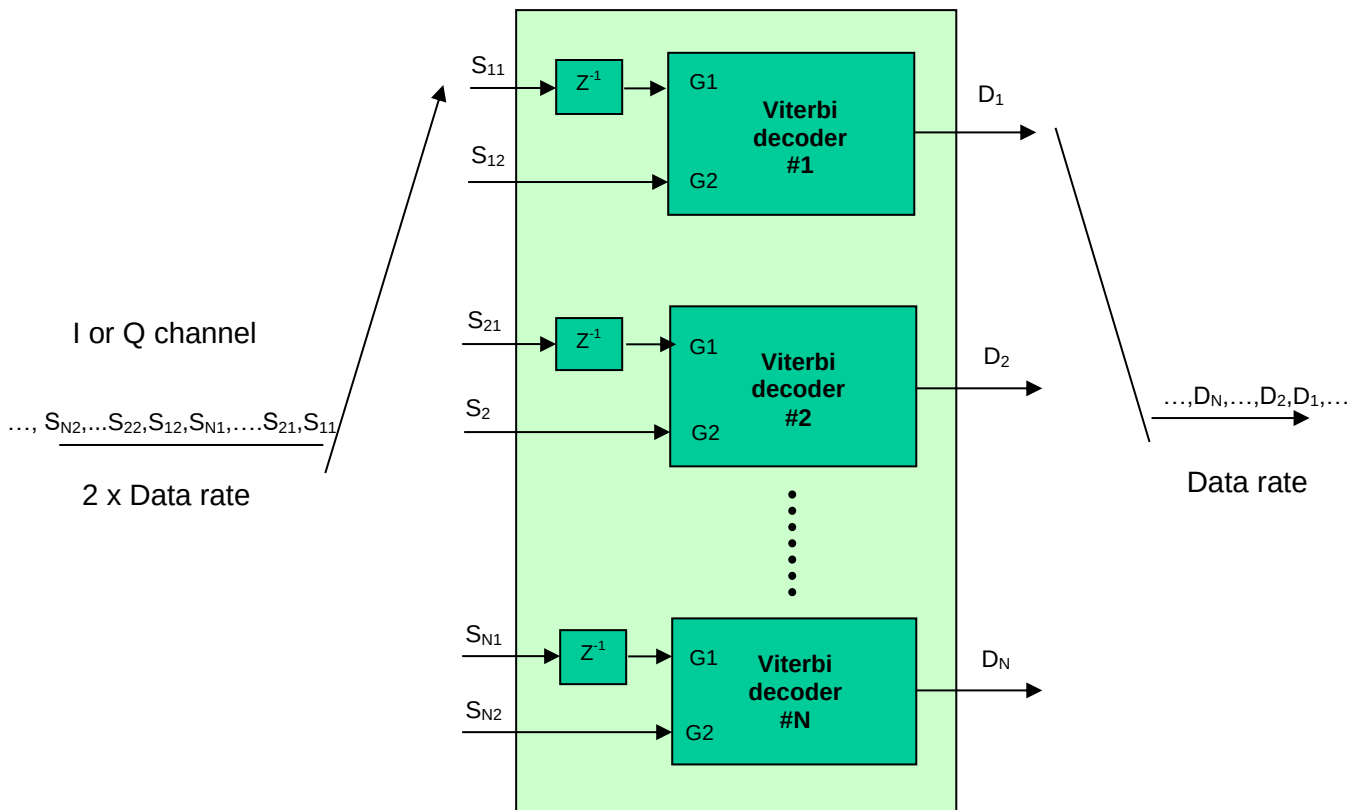
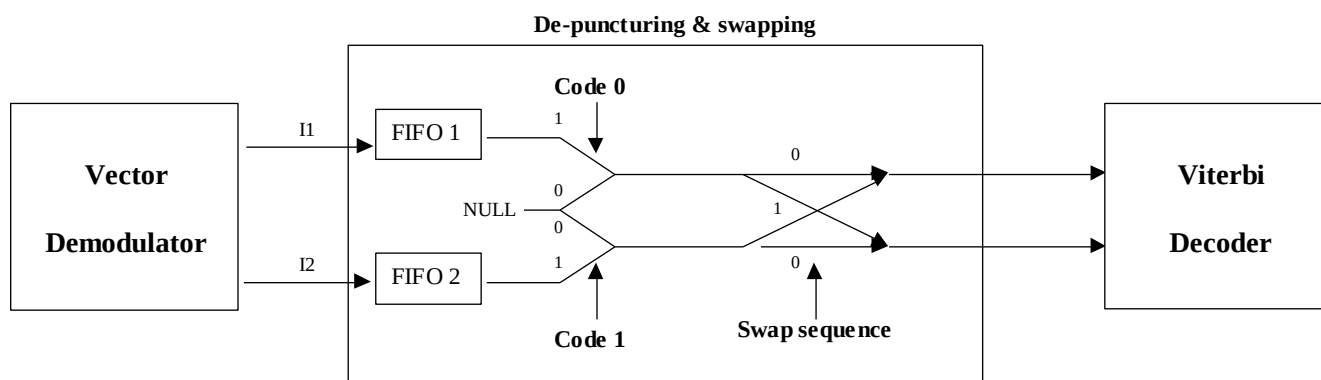


Figure 16: Parallel Viterbi Decoding Module.

3.3.3.4 Punctured Viterbi

Punctured Viterbi is supported for most puncturing rates: $\frac{2}{3}$ $\frac{3}{4}$ $\frac{4}{5}$ $\frac{5}{6}$ $\frac{6}{7}$ $\frac{7}{8}$

Puncturing codes (Code 0 and Code 1) and swap sequence must be programmed in the registry (see section 0).



Example: $\frac{3}{4}$ Punctured Dual Viterbi:

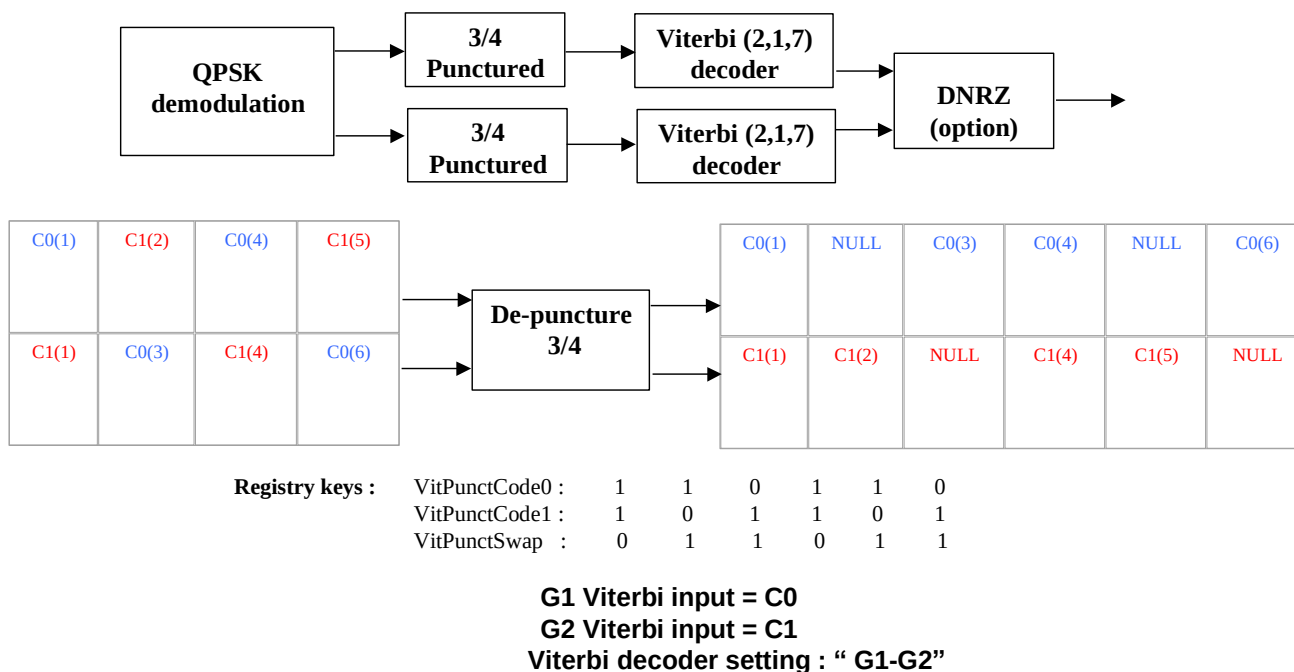


Figure 17: Viterbi De-puncturing Process.

There are two conventions for Viterbi puncturing codes. The first is the one which is defined by CCSDS recommendation and the second can be found on certain commercial encoders. The second convention only exists for puncturing ratios with an even denominator.

Example: 7/8 punctured Viterbi

Cod0: 1000101

Cod1: 1111010

Conv. 1 pattern: C0(1) C1(1) C1(2) C1(3) C1(4) C0(5) C1(6) C0(7) ...

Conv. 2 pattern: C0(1) C1(1) C1(3) C1(2) C1(5) C0(4) C1(7) C0(6) ...

Convention 2 pattern is convention 1 pattern with each pair of bit swapped apart from the first pair.

The convention can be defined with the registry key **VitPunctConv**.

The following table shows the effect of the different values of **VitPunctConv** registry key.

- 0:** depuncturing is defined by the VitPunctSwap registry key as described in the previous chapter
- 1:** VitPunctSwap registry key is ignored. VitPunctCode0 and VitPunctCode1 can be defined asymmetrically. Convention is 1.
- 2:** VitPunctSwap registry key is ignored. VitPunctCode0 and VitPunctCode1 can be defined asymmetrically. Convention is 2.

Next figure shows the BER performances of $\frac{3}{4}$ punctured Viterbi:

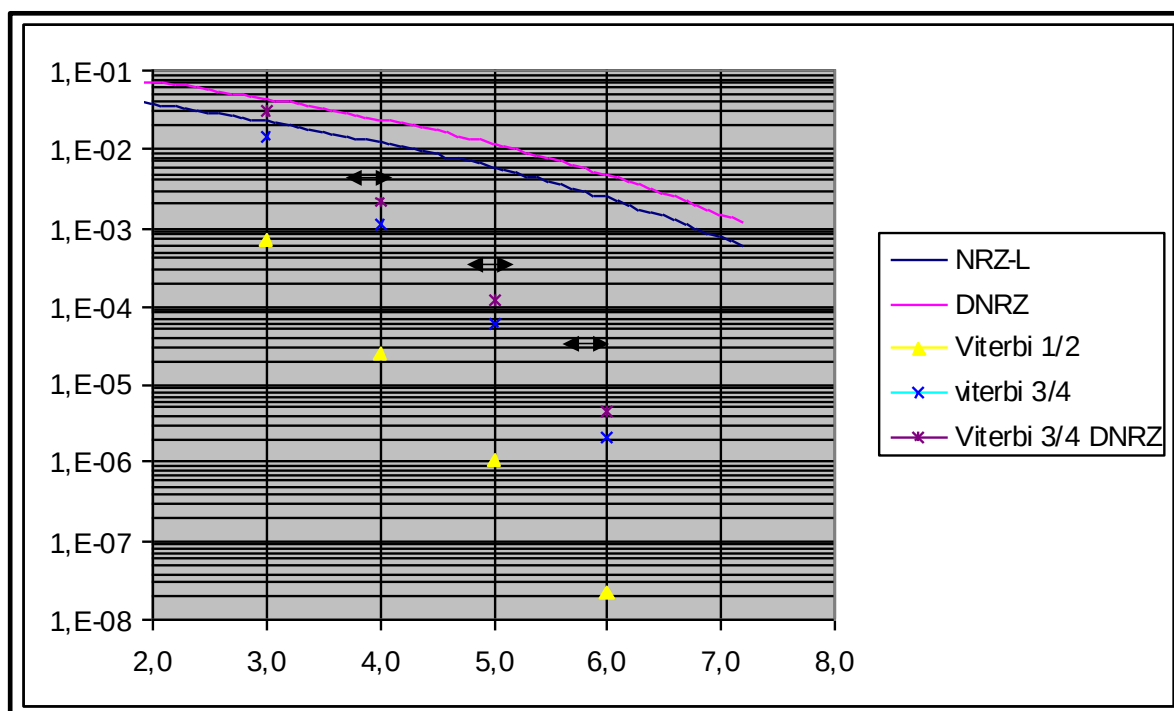


Figure 18: Punctured Viterbi BER Performances

3.3.4 PCM Decoding

The following codes are supported: NRZ-L/M/S, DNRZ, Bi-P-L/M/S.

When Viterbi decoding is not used, DNRZ is required for ambiguity resolution in QPSK and OQPSK mode. Several possible DNRZ schemes can be defined, according to the demodulation (QPSK, or OQPSK).

3.3.4.1 DNRZ, QPSK Decoding (DQPSK)

Several DNRZ algorithms are possible for ambiguity resolution in QPSK mode. The DNRZ code (algorithm) value can be programmed via Cortex HDR ICD parameter, using examples in next two sections (CCSDS recommended code and non-CCSDS codes).

3.3.4.1.1 CCSDS Recommended DNRZ Code in QPSK Mode

CCSDS recommendation for DQPSK is:

$$I_{out} = I_{in} \cdot Q_{in} \cdot I_{out-1} + \overline{I_{in}} \cdot \overline{Q_{in}} \cdot \overline{I_{out-1}} + \overline{I_{in}} \cdot Q_{in} \cdot I_{out-1} + I_{in} \cdot \overline{Q_{in}} \cdot \overline{I_{out-1}}$$

$$Q_{out} = I_{in} \cdot Q_{in} \cdot Q_{out-1} + \overline{I_{in}} \cdot \overline{Q_{in}} \cdot \overline{Q_{out-1}} + \overline{I_{in}} \cdot Q_{in} \cdot Q_{out-1} + I_{in} \cdot \overline{Q_{in}} \cdot \overline{Q_{out-1}}$$

This is equivalent to the following DQPSK decoding look-up table:

$I_{dl}[n-1]$	$Q_{dl}[n-1]$	$I_{dl}[n]$	$Q_{dl}[n]$	$\hat{I}_{in}[n]$	$\hat{Q}_{in}[n]$
0	0	0	0	1	1
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	1	0	0
0	1	0	0	0	1
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	1	1	0
1	0	0	0	1	0
1	0	0	1	0	0
1	0	1	0	1	1
1	0	1	1	0	1
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	1	0
1	1	1	1	1	1

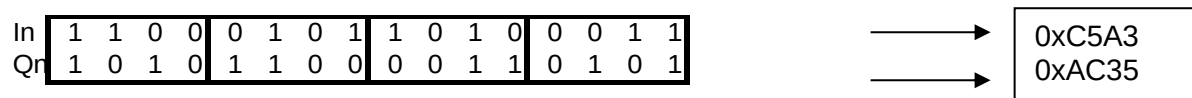
3.3.4.1.2 Other Possibilities for DNRZ Codes in QPSK Mode

Next table summarizes the 24 possibilities for DNRZ codes in QPSK Mode:

					In	Qn	Feature
code 1	11	10	01	00	C5A3	AC35	CCDS DS DNRZ
code 2	11	01	10	00	AC35	C5A3	⇔ CCDS DS DNRZ + Q;I
code 3	01	10	11	00	6996	AC35	
code 4	10	01	11	00	AC35	6996	
code 5	10	11	01	00	C5A3	6996	
code 6	01	11	10	00	6996	C5A3	
code 7	11	10	00	01	C5A3	9669	
code 8	11	00	10	01	AC35	9669	
code 9	00	10	11	01	6996	3A5C	
code 10	10	00	11	01	AC35	3A5C	⇔ CCDS DS DNRZ + Q;I
code 11	10	11	00	01	C5A3	53CA	⇔ CCDS DS DNRZ + I;Q
code 12	00	11	10	01	6996	53CA	
code 13	11	00	01	10	9669	AC35	
code 14	11	01	00	10	9669	C5A3	
code 15	01	00	11	10	3A5C	AC35	⇔ CCDS DS DNRZ + I;Q
code 16	00	01	11	10	3A5C	6996	
code 17	00	11	01	10	53CA	6996	
code 18	01	11	00	10	53CA	C5A3	⇔ CCDS DS DNRZ + /Q;I
code 19	00	10	01	11	53CA	3A5C	⇔ CCDS DS DNRZ + /Q;I
code 20	00	01	10	11	3A5C	53CA	⇔ CCDS DS DNRZ + /I;Q
code 21	01	10	00	11	53CA	9669	
code 22	10	01	00	11	9669	53CA	
code 23	10	00	01	11	9669	3A5C	
code 24	01	00	10	11	3A5C	9669	

3.3.4.1.3 DQPSK Configuration

Using the CCSDS DNRZ code and its decoding look-up table:



To select this code in the registry, set the Cortex HDR ICD parameter to **1** (code number in the left-hand column of the above table) or enter the DNRZ code value (0xC5A3AC35) as defined in the above decoding table).

Selection of the other codes follows the same rule.

3.3.4.2 DNRZ, OQPSK Decoding

Several DNRZ algorithms are possible for ambiguity resolution in OQPSK mode. The DNRZ code (algorithm) value can be programmed via a the Cortex HDR ICD parameter, using examples in next two sections (Most commonly used and Other possibilities).

3.3.4.2.1 Most Commonly Used DNRZ Code in OQPSK Mode

$I_{dl}[n-1]$	$Q_{dl}[n-1]$	$I_{dl}[n]$	$Q_{dl}[n]$	$\hat{I}_{in}[n]$	$\hat{Q}_{in}[n]$
0	0	0	0	0	1
0	0	0	1	0	0
0	0	1	0	1	0
0	0	1	1	1	1
0	1	0	0	1	1
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	0	1
1	0	0	1	0	0
1	0	1	0	1	0
1	0	1	1	1	1
1	1	0	0	1	1
1	1	0	1	1	0
1	1	1	0	0	0
1	1	1	1	0	1

3.3.4.2.2 Other Possibilities for DNRZ Codes in OQPSK Mode

Next table summarizes the 24 possibilities for DNRZ codes in OQPSK Mode:

					In	Qn	Feature
code 1	0	1	1	0	3C3C	9999	Most common
code 2	1	0	0	1	C3C3	6666	
code 3	0	0	1	1	0FF0	C3C3	
code 4	1	1	0	0	F00F	3C3C	
code 5	1	0	1	0	CC33	A5A5	⇔NRZ-S
code 6	0	1	0	1	33CC	5A5A	⇔NRZ-M
code 7	0	1	1	0	3C3C	6666	
code 8	1	0	0	1	C3C3	9999	
code 9	0	0	1	1	0FF0	3C3C	
code 10	1	1	0	0	F00F	C3C3	
code 11	1	0	1	0	CC33	5A5A	
code 12	0	1	0	1	33CC	A5A5	
code 13	0	1	1	0	9999	3C3C	
code 14	1	0	0	1	6666	C3C3	
code 15	0	0	1	1	C3C3	0FF0	
code 16	1	1	0	0	3C3C	F00F	
code 17	1	0	1	0	A5A5	CC33	
code 18	0	1	0	1	5A5A	33CC	
code 19	0	1	1	0	6666	3C3C	
code 20	1	0	0	1	9999	C3C3	
code 21	0	0	1	1	3C3C	0FF0	
code 22	1	1	0	0	C3C3	F00F	
code 23	1	0	1	0	5A5A	CC33	
code 24	0	1	0	1	A5A5	33CC	

3.3.4.2.3 DOQPSK configuration

Using the example of the most common DOQPSK code, and its decoding look-up table:

In	0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0
Qn	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1

3.3.5 I/Q Shift Compensation in OQPSK Mode

In OQPSK mode, I and Q symbols are ideally shifted by half symbol.

BER degradation is expected if the I-to-Q phase offset is not exactly half symbol.

OQPSK co-timing compensation can be automatically and precisely performed by the DEAF using the LMS algorithm. The impulse response of the digital filter is indeed able to adapt to the co-timing error.

3.3.6 CCITT V.35 De-scrambler

The Cortex HDR supports the V.35 descrambler.

In the demodulation channel, this descrambler is implemented after the Viterbi decoder and the differential decoder if used and before the frame synchronizer.

The V.35 descrambler is basically a 20-stages shift register with XORED specific outputs. In addition, a specific operation is used inside the process to guarantee an output transition once every 32 bits for long sequences of 0 or 1.

Two standards are implemented inside the Cortex HDR:

- the CCITT V.35 standard
- the INTELSAT V.35 standard

The descrambler can be enabled or disabled by set-up (from the Cortex HDR GUI or by TCP/IP remote control). The selection of the standard is done also by set-up.

3.3.7 BEM: 4D-8PSK-TCM-DVB-S Reed Solomon Decoding

This mode is compatible with the CCSDS REPORT ON BANDWIDTH-EFFICIENT MODULATIONS (**CCSDS 413.0-G-1, GREEN BOOK**, 31 March 2003) with completion for updating ECSS RF & Modulation standard ECSS E-50-05A and ECSS Telemetry channel coding ECSS E-50-01 draft 0.5

This modulation is compatible with the PLEIADES satellites.

3.3.7.1 Introduction

The L-dimensional MPSK-TCM (LD-MPSK-TCM) belongs to a family of modulations first introduced by G. Ungerboeck and improved by S. Pietrobon with the introduction of the multidimensional techniques. The MPSK-TCM are based on MPSK modulations with the use of convolutional coding to introduce "authorized sequences" between signal points linked by the trellis of the code. Note that single constellation MPSK modulations may also be referred to as "bi-dimensional" in reference to the representation of the MPSK constellation points in a signal space defined by orthogonal I and Q vectors.

In any case, the application of this procedure to several parallel constellations of the same size is referred to as L-dimensional TCM, denoted LD-MPSK-TCM (with $L > 1$ and $M = 8$). The trellis is constructed to maximize the minimum Euclidian distance between different paths originating and merging to the same state. The construction of the optimum trellis code and partitioning for the M points in the constellation is based on heuristic rules proposed in G. Ungerboeck. "Channel Coding with Multilevel/Phase Signals", IEEE Transactions on Information Theory, vol. IT-28, no. 1, pp.55-67, January 1982 and S. Pietrobon et al. "Trellis-Coded Multidimensional Phase Modulation", IEEE Transaction on Information Theory, vol. 36, no. 1, pp.63-89, January 1990.

With 4D-8PSK-TCM, the combination of convolutional coding, multiphase modulation and multidimensional techniques offers a substantial power gain together with bandwidth conservation or reduction, in comparison to their separate utilization as it is done frequently with binary or quaternary modulations (i.e., sequential implementation of convolutional coding). The result is an improvement of the performances in terms of BER versus signal to noise ratio for the same or better bandwidth efficiency, compared with the uncoded (or even convolutionally coded) OQPSK or QPSK modulations.

Example: Assuming the bit rate of input data equal to 100 Mbps, the 4D-8PSK-TCM channel symbol rate is 50 Msps for 2 bits/channel-symbol or 40 Msps for 2.5 bits/channel-symbol.

The performances of 4D-8PSK-TCM scheme can be substantially improved when concatenated with Reed Solomon (R-S) on bytes block coding with a proper interleaving so as to ensure a good word error splitting (at the output of the convolutional trellis decoder) and then good working conditions for the R-S decoder.

3.3.7.2 4D-8PSK-TCM Coder

The 4D-8PSK Trellis-Coded Modulator consists of a serial to parallel converter, a differential coder, a trellis encoder (convolutional coder), a constellation mapper and an 8PSK modulator (see Figure 19).

Note that in this figure, w_i (with index $i = 1, \dots, m$) represent the uncoded bits and x_j (with index $j = 0, \dots, m$) are the coded bits. The trellis encoder is based on a 64 state systematic convolutional coder and can be considered as the inner code if an outer block code is introduced. Carrier phase ambiguity is resolved by the use of a differential coder located prior to the trellis encoder. Spectral efficiencies of 2, 2.25, 2.5, and 2.75 bits/channel-symbol are achieved with four possible architectures of the constellation mapper. The output switch addresses successively one of the four symbols ($Z(0) - Z(3)$) from the constellation mapper to the 8PSK modulator (which is in fact a QPSK modulator for the 2 first efficiencies values).

The present standard is based on the use of a 4D-8PSK-TCM characterized by the following parameters:

- Size of the constellation: $M = 8$ phase states (8PSK)
- Number of signal set constituents: $L = 4$ (shown as $Z(0) \dots Z(3)$ in Figure 19)
- Number of states for the trellis encoder: 64
- Rate of the convolutional coder used for the construction of the trellis: $R = \frac{3}{4}$
- Rate of the modulation: $R_m = m/(m+1)$ selectable to 8/9, 9/10, 10/11, or 11/12
- Efficiency of the modulation:
 - ❖ $R_{eff} = 2$ bits per channel-symbol (for $R_m = 8/9$)
 - ❖ $R_{eff} = 2.25$ bits per channel-symbol (for $R_m = 9/10$)
 - ❖ $R_{eff} = 2.5$ bits per channel-symbol (for $R_m = 10/11$)
 - ❖ $R_{eff} = 2.75$ bits per channel-symbol (for $R_m = 11/12$)

In fact, the serial to parallel conversion is a byte (8 bits) to symbol (8 to 11 bits) conversion that will be described later.

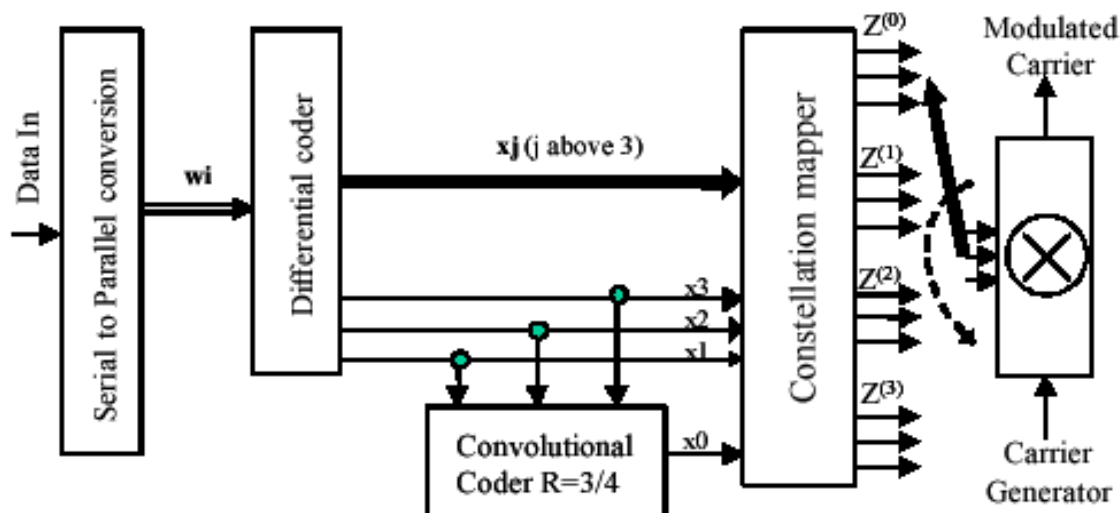


Figure 19: Structure of the 4D 8PSK-TCM Coder/Mapper

3.3.7.2.1 Differential Coder

The differential coder depicted in Figure 20 is used to eliminate phase ambiguity on carrier synchronization for each modulation efficiency. Table 10 gives the bit reference at input and output of the differential coder in each case.

Efficiencies in bits /channel-symbol							
2		2.25		2.5		2.75	
bit IN	bit OUT	bit IN	bit OUT	bit IN	bit OUT	bit IN	bit OUT
w1	x1	w2	x2	w3	x3	w4	x4
w5	x5	w6	x6	w7	x7	w8	x8
w8	x8	w9	x9	w10	x10	w11	x11

Table 10: Bit Mapping for Differential Coder

An example of differential encoder connections is given in Figure 20 for the 2 bits/channel-symbol case. The structure of the modulo 8 adder is also shown; it is applicable to both the coder mapper and differential coder.

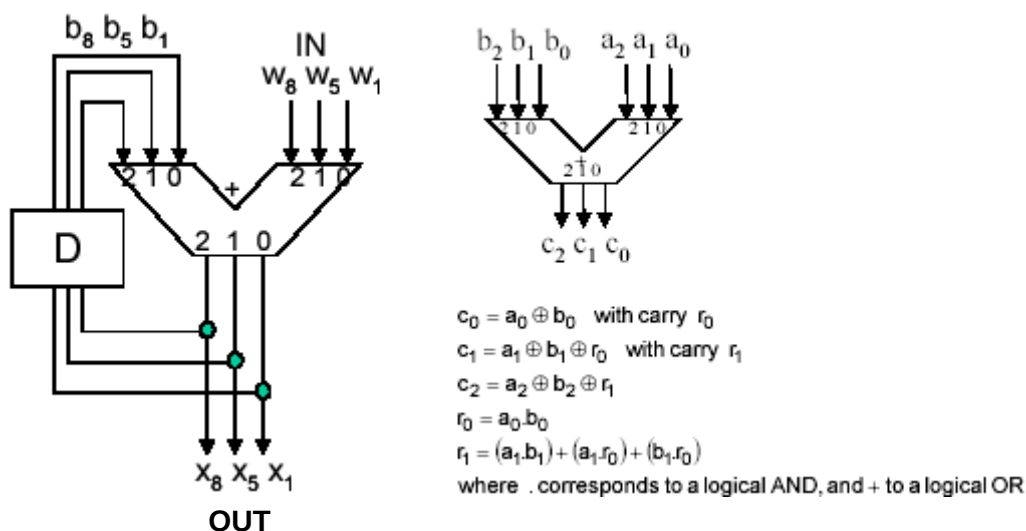


Figure 20: Differential Coder and Modulo-8 Adder Principle

For all the bits non concerned by the differential encoding, we have then: $x_k = w_k \quad k \leq m+1$

3.3.7.2.2 Convolutional Coder

The convolutional coder used to implement the trellis is depicted in Figure 21. This code corresponds to one of the “best” codes for phase transparency.

The systematic coder is implemented with the following characteristics:

- number of states: 64 states,
- constraint length: $K = 7$

The convolutional encoder is specified by the following polynomial in octal: $h_3 = 050$, $h_2 = 024$, $h_1 = 006$, $h_0 = 103$. Figure 21 shows the recommended convolutional encoder. The shift registers of the encoder are clocked at the rate of $R_{Chs}/4$.

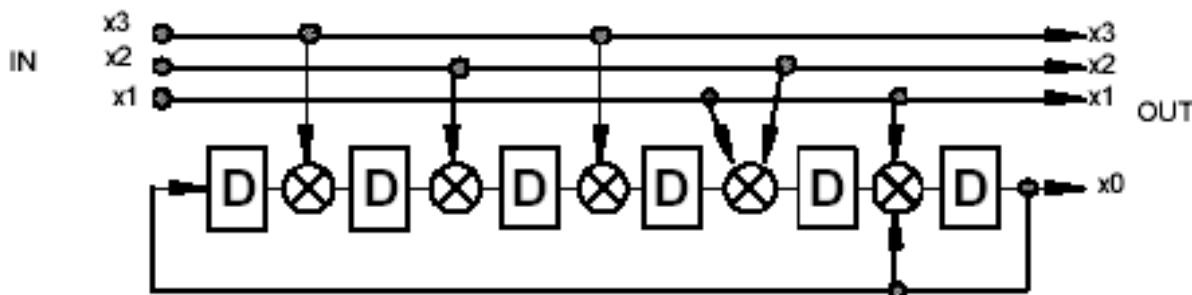


Figure 21: Convolutional Coder Recommended for High Data Rates

The number of coded bits is the same for the four modulation efficiencies (i.e., the same structure is used for 2, 2.25, 2.5, and 2.75 bits/channel-symbol), only the number of uncoded bits is changed. The advantage of this coder is its optimized performance and the reduced internal rate which is equal to 1/8, 1/9, 1/10, or 1/11 of the information rate.

3.3.7.2.3 Constellation Mapper for 4D-8PSK-TCM

The constellation mapper principles are given in next figures for the four possible efficiencies of this modulation (i.e., 2 bits/channel-symbol, 2.25 bits/channel-symbol, 2.5 bits/channel-symbol and 2.75 bits/channel-symbol). These mappers implement the straightforward logical mapping described in the equations below. The correspondence between the signals $Z^{(i)}$ at the input of the modulator and the 8PSK phase states of the constellations follows a natural mapping (i.e., 0, 1, 2 ..., 7).

If $Z^{(i)}$ represents the signals (three lines) at the input of the modulator with $Z^{(0)}$ being the signal set of the first constellation and $Z^{(3)}$ being the signal set of the fourth constellation, the signal set $Z^{(i)}$ is represented by the following equation. This representation shows that the bits which are common to each vector set (shown in the first part of right-hand side of each equation) are sensitive to a phase rotation of $\pi/4$ and will be differentially encoded.

(i) Equation for 2 bits/channel symbol efficiency:

$$\begin{bmatrix} Z^{(0)} \\ Z^{(1)} \\ Z^{(2)} \\ Z^{(3)} \end{bmatrix} = \left[\left(4x_8 + 2x_5 + x_1 \right) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ x_7 \\ x_6 \\ x_7 + x_6 + x_4 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ x_3 \\ x_2 \\ x_3 + x_2 + x_0 \end{pmatrix} \right] \bmod 8$$

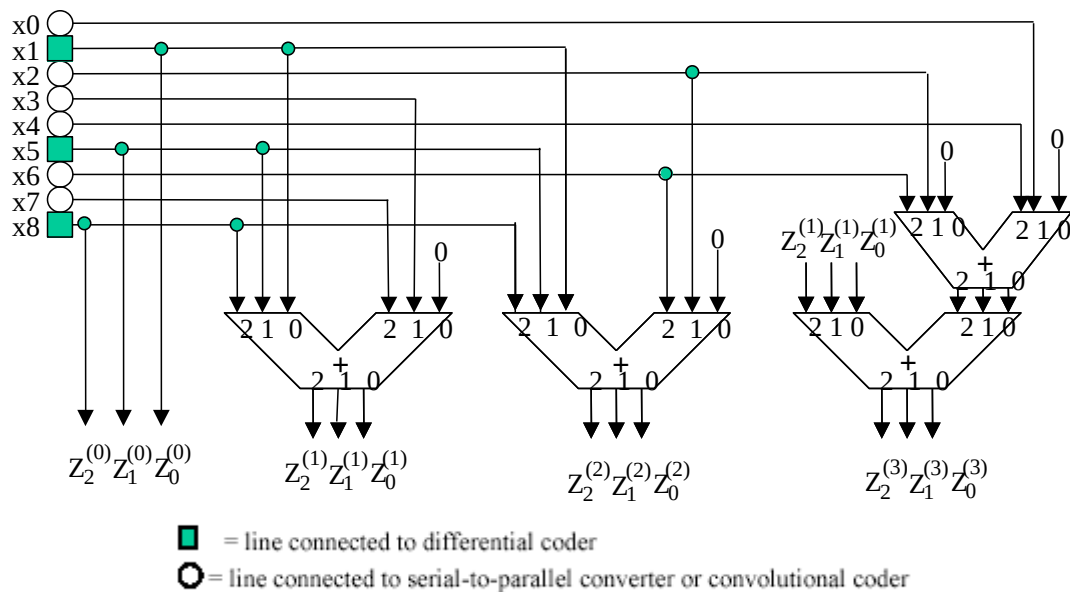
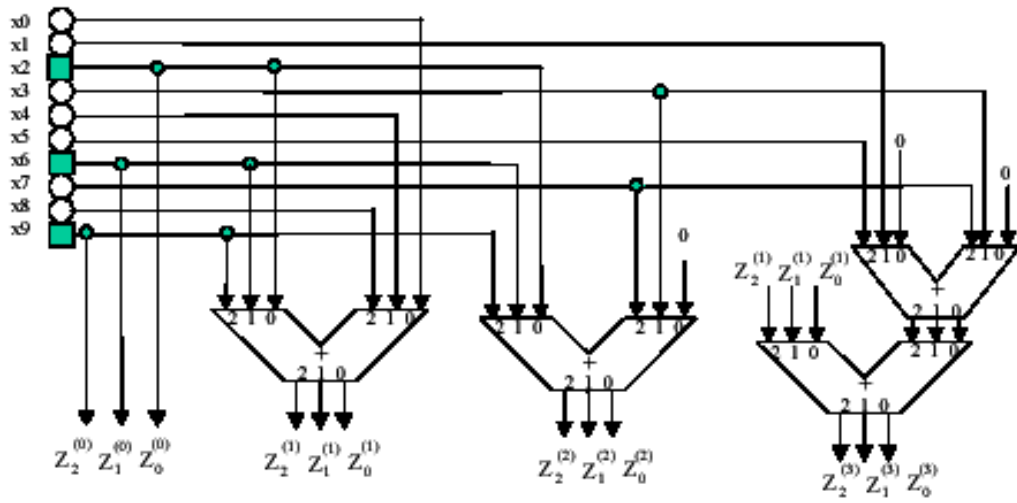


Figure 22: Constellation Mapper for 2 bits/channel-symbol

(ii) Equation for 2.25 bits/channel symbol efficiency:

$$\begin{bmatrix} Z^{(0)} \\ Z^{(1)} \\ Z^{(2)} \\ Z^{(3)} \end{bmatrix} = \left[\begin{matrix} (4x_9 + 2x_6 + x_2) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ x_8 \\ x_7 \\ x_8 + x_7 + x_5 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ x_4 \\ x_3 \\ x_4 + x_3 + x_1 \end{pmatrix} + \begin{pmatrix} 0 \\ x_0 \\ 0 \\ x_0 \end{pmatrix} \right] \bmod 8$$



■ = line connected to differential coder
○ = line connected to serial-to-parallel converter or convolutional coder

Figure 23: Constellation Mapper for 2.25 bits/channel-symbol

(iii) Equation for 2.5 bits/channel symbol efficiency

$$\begin{bmatrix} Z^{(0)} \\ Z^{(1)} \\ Z^{(2)} \\ Z^{(3)} \end{bmatrix} = \left[\begin{matrix} (4x_{10} + 2x_7 + x_3) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ x_9 \\ x_8 \\ x_9 + x_8 + x_6 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ x_5 \\ x_4 \\ x_5 + x_4 + x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ x_1 \\ x_0 \\ x_1 + x_0 \end{pmatrix} \right] \bmod 8$$

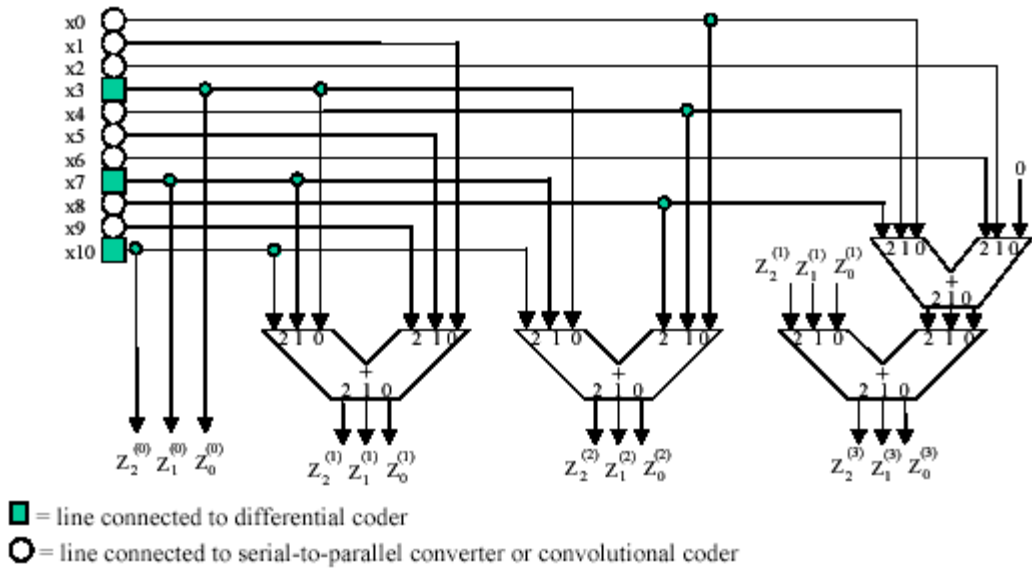


Figure 24: Constellation mapper for 2.5 bits/channel-symbol

(iv) Equation for 2.75 bits/channel symbol efficiency

$$\begin{bmatrix} Z^{(0)} \\ Z^{(1)} \\ Z^{(2)} \\ Z^{(3)} \end{bmatrix} = \begin{bmatrix} (4x_{11} + 2x_8 + x_4) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ x_{10} \\ x_9 \\ x_{10} + x_9 + x_7 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ x_6 \\ x_5 \\ x_6 + x_5 + x_3 \end{pmatrix} + \begin{pmatrix} 0 \\ x_2 \\ x_1 \\ x_2 + x_1 + x_0 \end{pmatrix} \end{bmatrix} \bmod 8$$

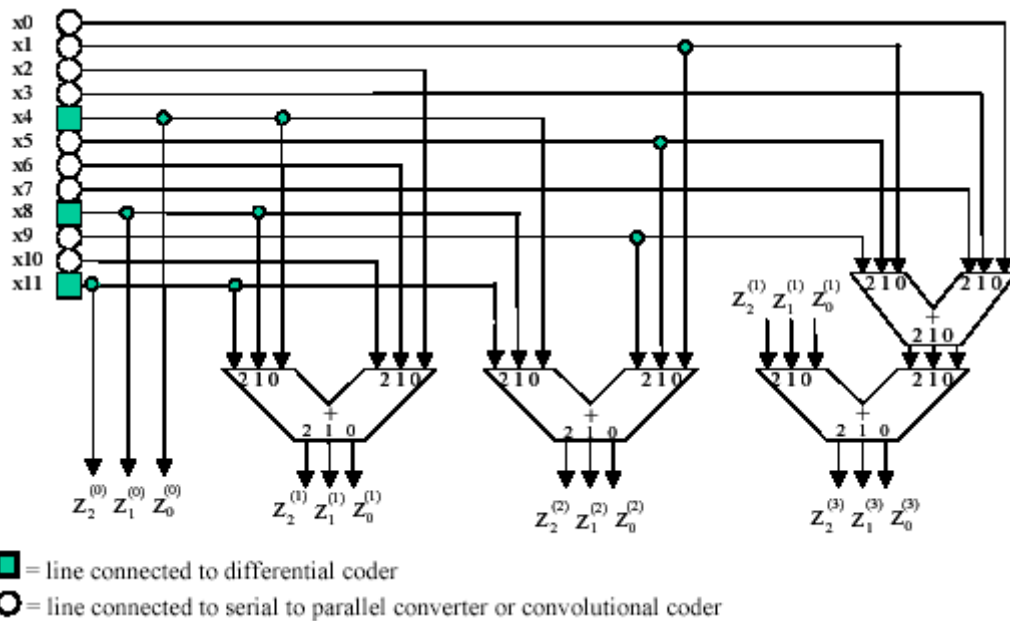


Figure 25: Constellation Mapper for 2.75 bits/channel-symbol

3.3.7.2.4 Coder/Mapper Implementation at 2, 2.25, 2.5 & 2.75 bits/channel-symbol Efficiency

The principle of the coder-mapper for 2, 2.25, 2.5, and 2.75 bits/channel-symbol efficiency is given in next figures:

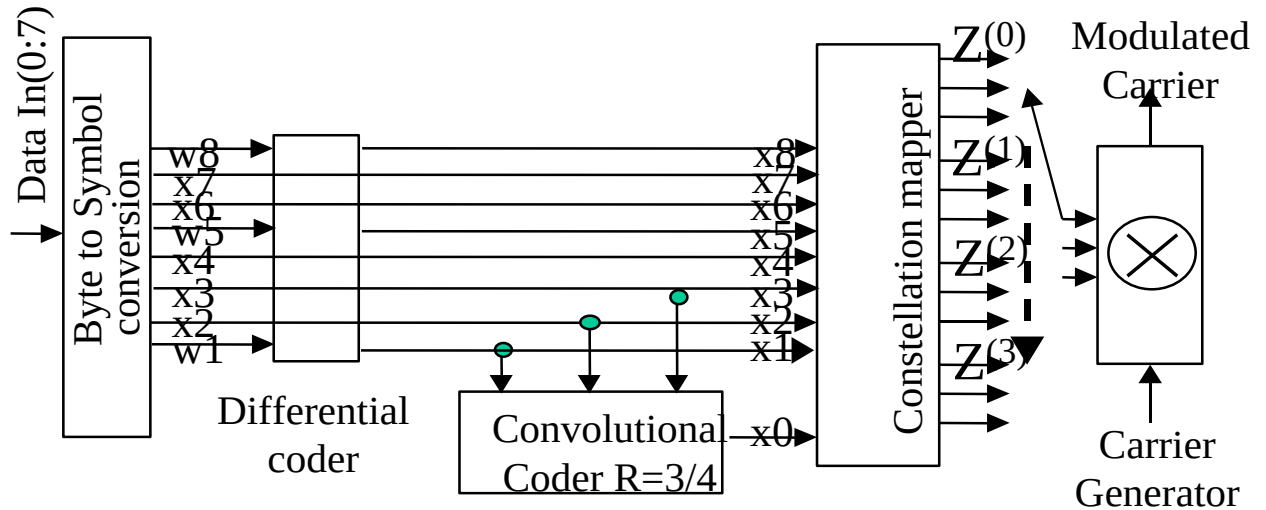


Figure 26: Coder and Mapper Implementation at 2 bits/channel-symbol Efficiency

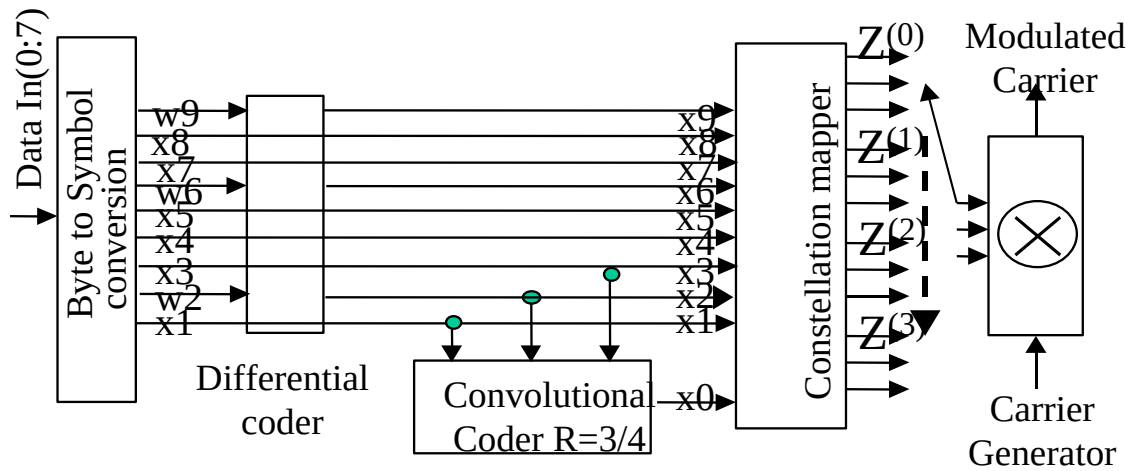


Figure 27: Coder and Mapper Implementation at 2.25 bits/channel-symbol Efficiency

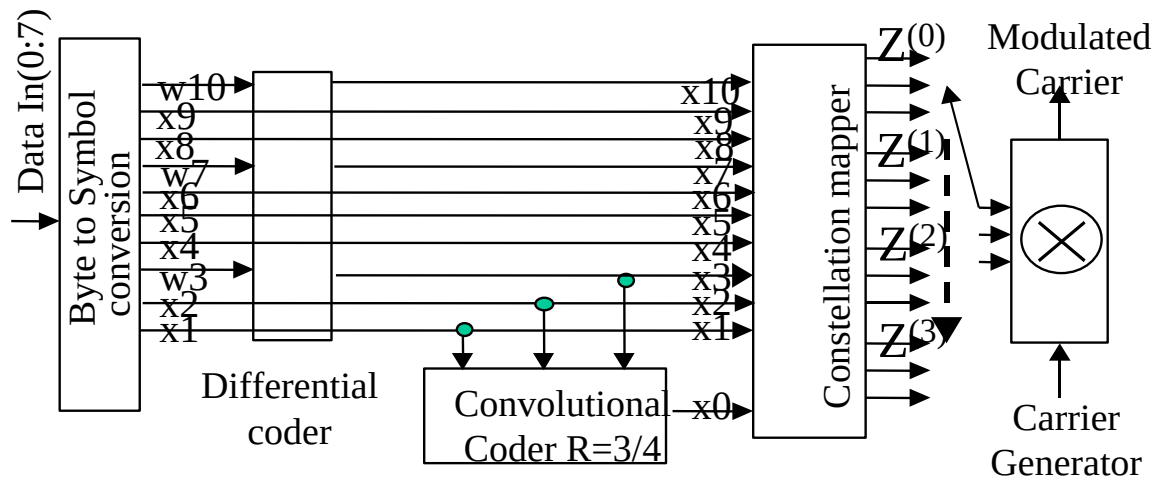


Figure 28: Coder and Mapper Implementation at 2.5 bits/channel-symbol Efficiency

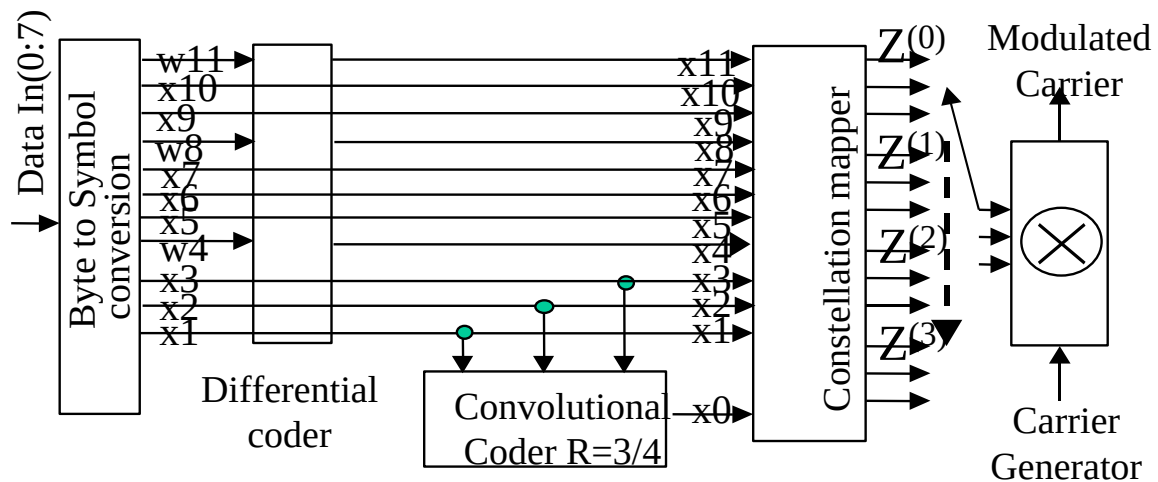


Figure 29: Coder and Mapper Implementation at 2.75 bits/channel-symbol Efficiency

3.3.7.3 DVB-S Transport Layer: Reed Solomon Code

As defined in § 4.7 of ECSS-E50-01 draft 0.5, the concatenation principle correspond to the following processes but with some interaction between them as depicted in next figure:

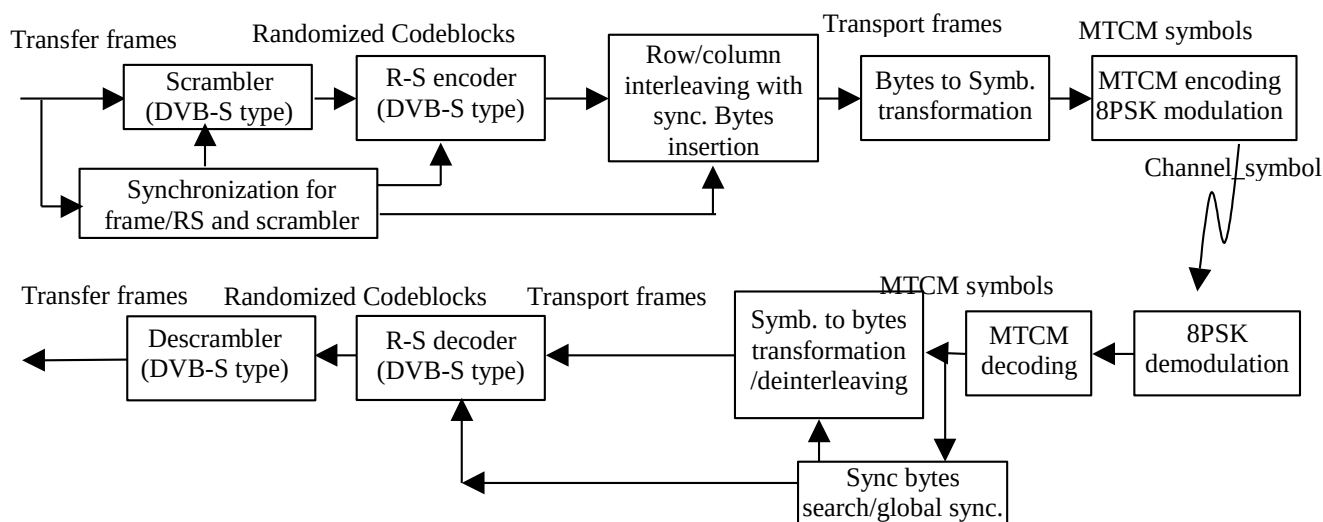


Figure 30: Concatenation Principle

3.3.7.3.1 Synchronized Randomization

Within the transmitter, the input digital data is randomized with a PRBS whose characteristics are:

- Generating polynomial: $1+x^{14}+x^{15}$
- Initialization sequence: 100101010000000

This sequence is loaded at the beginning of each transport frame. Due to the chosen polynomial, it is possible to output the data by bytes with the help of a shifted calculation, i.e. out bit1 or MSB of byte = reg15 EX-OR reg14, out bit2 = reg14 EX-OR reg13, out bit3 = reg13 EX-OR reg12, out bit4 = reg12 EX-OR reg11, out bit5 = reg11 EX-OR reg10, out bit6 = reg10 EX-OR reg9, out bit7 = reg9 EX-OR reg8 and out bit8 or LSB of byte = reg8 EX-OR reg7 and then shift by 8 bits the contents of the register before calculating the PRBS sequence for the next byte.

It should be noticed that in order to avoid tedious rate changes, the data have to be captured at the interface by group of 238 bytes with one blank byte in front and 16 blank bytes at the rear. During the blank bytes, the randomization is shifting but the enable bit must be downed to zero.

The randomization processing is described on next figure.

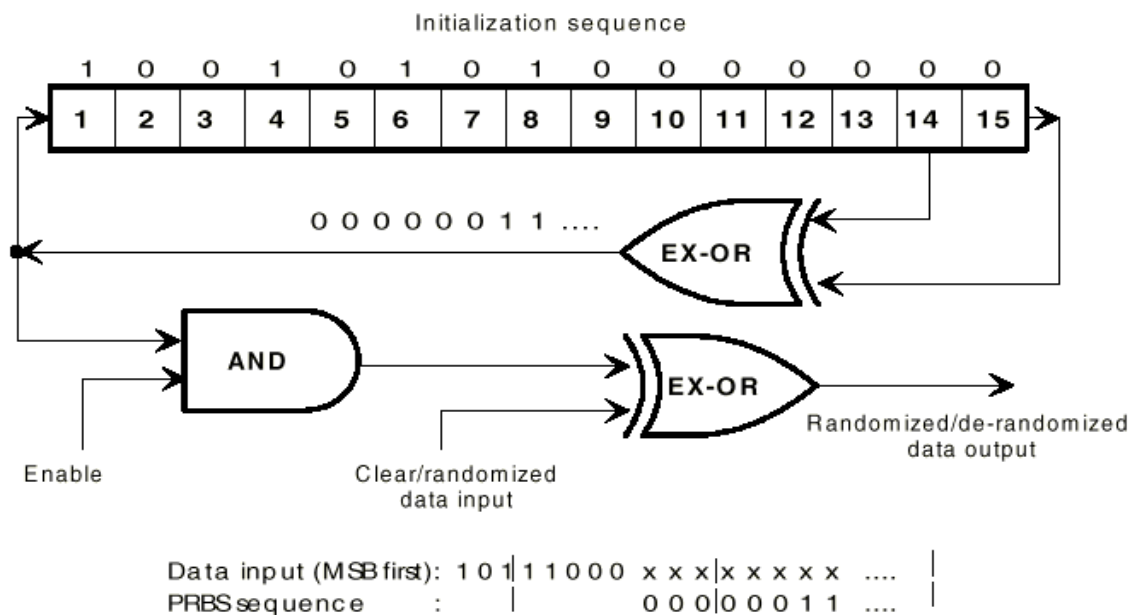


Figure 31: Randomization Principle

3.3.7.3.2 Reed-Solomon encoding

Then, the data shall be encoded by using a shortened Reed-Solomon encoder (238,254) type

The encoding process is the following:

- first, one byte (00 hexa.) is added in front of the 238 data bytes
- then a standard (239,255) R-S code (from DVB-S standard) is applied by computing 16 redundancy bytes.
- finally the 00 extra-byte is removed

The code generation polynomial is: $g(x) = (x+\lambda^0)(x+\lambda^1)(x+\lambda^2)\dots(x+\lambda^{15})$ with $\lambda=02$ (hexa)

The primitive polynomial is: $p(x) = 1+x^2+x^3+x^4+x^8$

3.3.7.3.3 Packets interleaving and synchronization byte insertion

After R-S encoding, the data shall be interleaved (depth 8 or 16) and formatted in transport frames.

The interleaving is performed by using a matrix of (8)16 rows and 254 columns. The 254-bytes packets are written row by row, the first byte being put in the first cell. When filled, the matrix is read column by column by packets of 254 bytes separated by synchronization markers. The process goes on with the following group of 16(8)*254 data bytes creating then a complete frame of 16(8)*255 bytes.

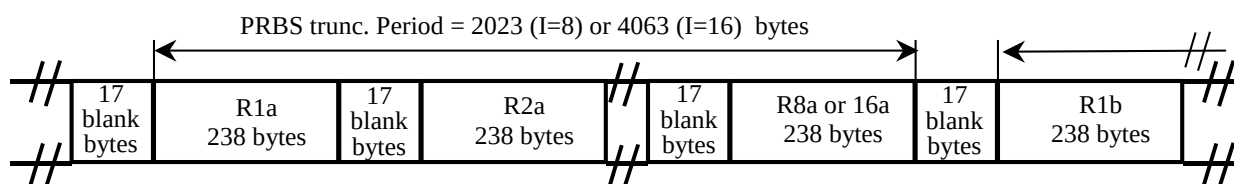
The 16 (8) blocks of 255 bytes (254 data bytes plus one synchro. byte) form the transport frame. This frame accounts for 4080 (2040) bytes or 32640 (16320) bits.

The synchronization marker of the frame (corresponding to the first block of 254 bytes) is B8 (in hexadecimal) and the separation between the other 15 (7) blocks of the frame is made by a marker whose value in hexadecimal is 47 when the medium data rate is used.

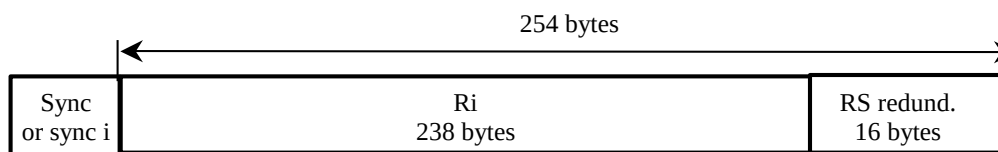
For high data rates interface, the equipment has the capacity for inserting filling frames. The synchronization marker for those filling frame are specific. 96 (in hexadecimal) for the first block and 69 for the 15 (7) following blocks.

It should be noticed that at the receiver side, those filling packets have to be removed.

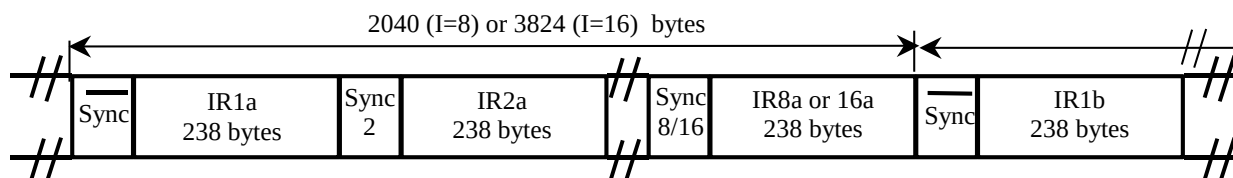
The succession of the three processes described here above give the way to:



Source information in packets : Blank bytes and randomized block Ri



Reed Solomon RS (254,238, t=8) error protected packet

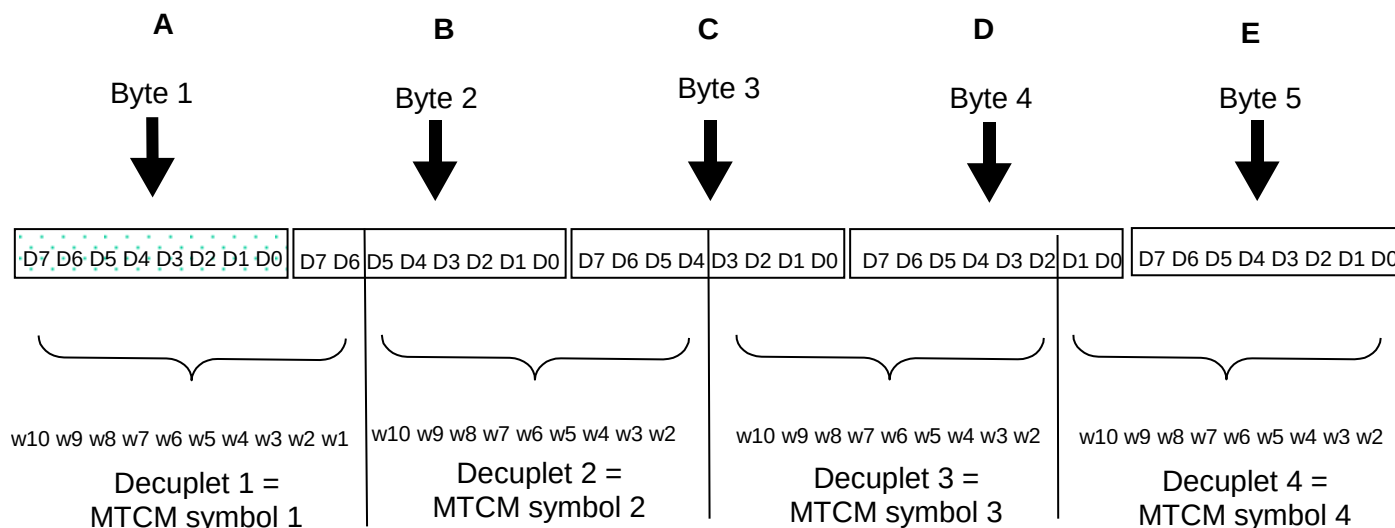


Randomized transport packets : Sync bytes and interleaved randomized block IRI

3.3.7.3.4 Interface between Trellis Coded Symbols and Bytes

As the size of trellis coded symbol can vary from 8 to 11 bits, there must be a definition of the arrangement that gives the bytes (8 bits fixed) for the R-S concatenated part.

The convention is the following, depicted for 4D-8PSK-TCM 2.5 with the arrangement of the 10 bits with regard to the 8 input bits is described on next figure.



Notes: * Byte 1 is the first byte transmitted on the Air interface,
* The UW will be every time positioned on the byte 1.

Figure 32: Data Arrangement in TCM 2.5 bits/channel_symbol

3.3.8 PRN-BER Computation

Depending on the mission, BER measurement in PRN mode can be done either on separate or merged I and Q channels (see 4.2.9.2.3).

3.3.8.1 BER on Separate I & Q Channels

This method is based on PRN codes property: The splitting of a PRN data flow into I and Q provides two PRN data flows (same PRN).

When restarting BER measurement (see 4.5.4.4), each ambiguity is tested (~1s), before selecting the ambiguity which provides with the lower BER. The BER is then measured applying this ambiguity. This method is particularly interesting when the ambiguity is not resolved by the demodulation (example: Metop satellite configuration).

Example 1: QPSK, NRZ-L demodulation. Demodulation remaining ambiguities:

If natural QPSK de-mapping is used: IQ, /QI, Q/I, /I/Q.

If inverse QPSK de-mapping is used: QI, I/Q, /IQ, /Q/I.

Even with these ambiguities, BER measure can be done in this mode.

Using this method, BER I and BER Q can be separately displayed (see 4.2.9.2.3 , value 1), or averaged (see 4.2.9.2.3 , value 0).

Example 2: 8PSK, NRZ-L demodulation, using a PRN code for each 3 8PSK data flows.

Demodulation remaining ambiguities:

If natural 8PSK de-mapping is used: 8 phase ambiguities.

If inverse 8PSK de-mapping is used: 8 other phase ambiguities.

Providing that the right 8PSK de-mapping (normal or inverse, natural or Gray) has been selected, BER measure can be done in this mode.

3.3.8.2 BER on Merged I & Q Channels

This method applies to merged I and Q data in QPSK or OQPSK mode, or single I data in BPSK mode, or single I + Single Q in UQPSK mode, or single1 + single2 + single3 in 8PSK mode.

In this mode, no ambiguity tests are done. Data which are used for BER measure are an image of output data (channel A for Demodulation Unit # 1, channel B for Demodulation Unit # 2, see 2.2.2). Ambiguities can then be manually tested, using the 8 first values of **OUTPUT MODE** parameter (see 4.5.4.1).

REGISTRY SELECTION 4.2.9.2.3	QPSK OQPSK	QPSK & OQPSK + SINGLE VITERBI	BPSK	UQPSK AQPSK	8PSK
0	<u>Method 1:</u> IQ/IQ,I/Q,I/Q, QI/QI, Q/I,Q/I GUI: 1 BER value (I,Q average)	<u>Method 1:</u> IQ,I/Q (IQ merged before Viterbi decoding) GUI: 1 BER value	<u>Method 1:</u> I/I GUI: 1 BER value	<u>Method 1:</u> I/I, on I Q/Q on Q GUI: 2 BER values (BER I & BER Q)	<u>Method 1:</u> 8 phase ambiguities GUI: 1 BER value (averaged BER)
1	<u>Method 1:</u> IQ/IQ,I/Q,I/Q, QI/QI, Q/I,Q/I GUI: 2 BER values (BER I & BER Q)				
2	<u>Method 2:</u> GUI: 1 BER value (BER IQ)	<u>Method 2:</u> GUI: 1 BER value	<u>Method 2:</u> GUI: 1 BER value (BER I)	<u>Method 2:</u> GUI: 2 BER values (BER I & BER Q)	<u>Method 2:</u> GUI: 1 BER value (averaged BER)

Figure 33: BER Measurement Methods: Tested ambiguities and GUI display

3.3.9 Digital Outputs

Digital outputs characteristics can be configured:

- **Clock polarity:** normal or inverted.
- **Data (I/Q) polarity:** normal or inverted.
- **Data swap:** Normal I & Q, I inverted & Q, I & Q inverted, I inverted & Q inverted
+ Swap: Q & I, Q inverted & I, Q & I inverted, Q inverted & I inverted
- **Output mode:** Normal or split (separate I & Q output) or I/Q merged or CADU output (refer to the Hardware Description chapter)

3.3.10 Doppler and Integrated Phase Measurement

3.3.10.1 Introduction

The HDR provides Doppler measurements (measurements of the offset between the set up IF value and the actual one) and integrated phase measurements. These measurements are available for the user through a specific TC/IP port (refer to the STI 100013 HDR document).

The elementary Doppler measurements (IF offset measurements) are generated every 13.1072 ms. They are sampled at the rate specified by the parameter « Doppler measurement sampling rate » (every 100 ms to every 10 seconds). As the sampling rate and the generation rate are asynchronous, a **linear interpolation** is applied for improved measurement accuracy. Samples are stored, for further transmission, in a measurement block which size is programmable between 1 and 1000 elementary measurement(s).

The Elementary integrated phase measurements are generated every 13.1072 ms. They are sampled at the rate specified by the parameter « Doppler measurement sampling rate » (every 100 ms to every 10 seconds). As the sampling rate and the generation rate are asynchronous, a **linear interpolation** is applied for improved measurement accuracy. Samples are stored, for further transmission, in a measurement block which size is programmable between 1 and 1000 elementary measurement(s). The integrated phase resolution is $2\pi/2^{32-N}$ radian, with N tunable (see user's manual for more information). The 64-bit phase counter capacity is $2^{32+N}/\text{IF}$ seconds. As an example, N = 15 gives a counter capacity of 1 day 8 hours and 46 minutes at 1.2GHz.

The following demodulator menus support advanced Doppler measurements, integrating the phase of the carrier oscillator: Menus # 5, 7, 8, 9, 10, 11, 12, 13, 15, 16, 23, 24, 39, 46, 57.

A single Doppler port is used for transmitting IF offset measurements or integrated phase measurements.(refer to the STI 100013 HDR document).

3.3.10.2 Required setup for optimal performance

Accurate phase and Doppler measurements need a very precise and stable sampling period. Moreover, the internal oscillators must also run at a very stable frequency. Even though it is possible to get rough Doppler and phase measurements without any external reference, it is strongly recommended to connect the demodulator to a 10MHz reference to obtain accurate results.

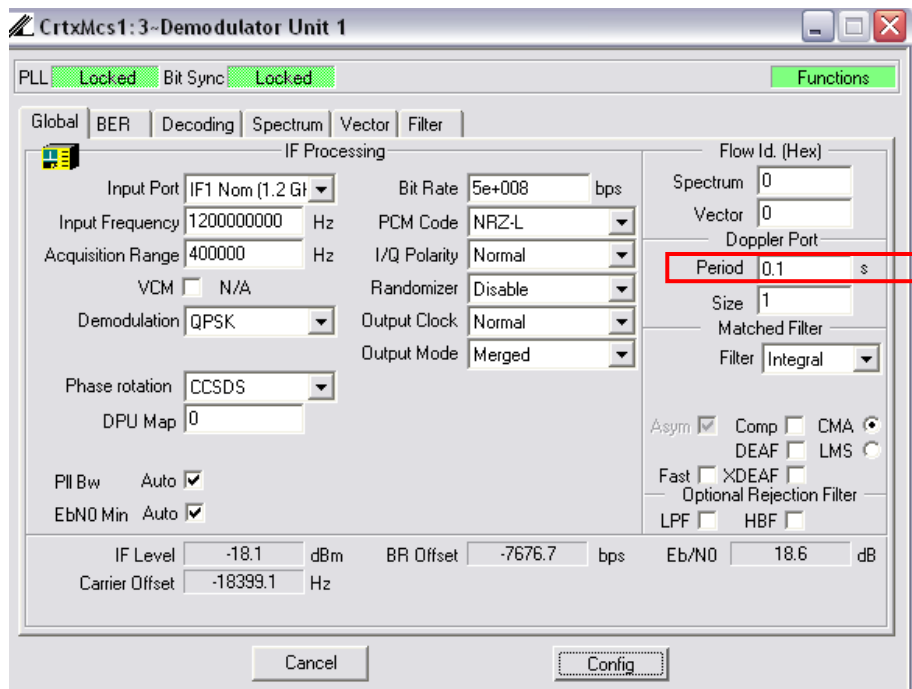
Doppler and phase measurements are time-tagged. For precise datation of Doppler and phase measurements, it is strongly recommended to connect the demodulator to an IRIG-B generator. In such case, the precision of the time tag is 1μs.

3.3.10.3 Sampling rate

Elementary accumulated phases are sampled at the rate specified by the parameter « Doppler measurement sampling rate » (every 100 ms to every 10 seconds, see STI). Phase generation period depends on the sampling rate value.

If the sampling rate is a multiple of 10 ms, the accumulated phases are generated each 10 ms in order to take advantage from raw data. Thus, Doppler measurement is based on exact values of phase.

If the sampling rate is not a multiple of 10 ms, the raw accumulated phases are sampled every 10 ms. To obtain the accumulated phase at the desired date, an interpolation process is used. The following figure shows how to set sampling rate for the Doppler port:



3.3.10.4 Accumulated Phase Format

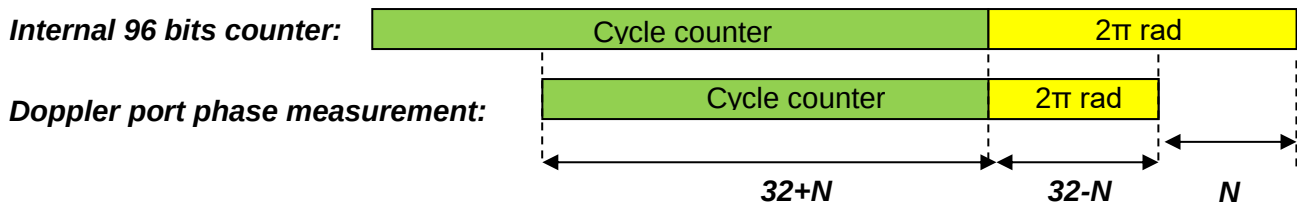
Elementary 32-bit phase values are accumulated in a 96-bit internal counter. The phase resolution of this counter is 2^{-32} cycle, or $2\pi/2^{32}$ radian = $8.4e-8^\circ$. The 32 LSB can thus be seen as a measurement of the phase, modulo 2π , while the 64 LSB count the number of cycles since the Doppler port started to emit:



A 64 bit counter, using 32 bits for the modulo part, and 32 bits for the cycle counting part, has a revolution period of 5s for an IF at 720 MHz. The overall 96 bits counter has a revolution period of more than 800 years. The Doppler port thus discards some of the MSBs and LSBs to output 64 bits phase measurements. The truncature operates as follows. The registry value:

HKEY_LOCAL_MACHINE\SOFTWARE\IN-SNEC\CrtxHdr\SetDefault\Signal Processing\DmuX\Integ_Phase_BinShift

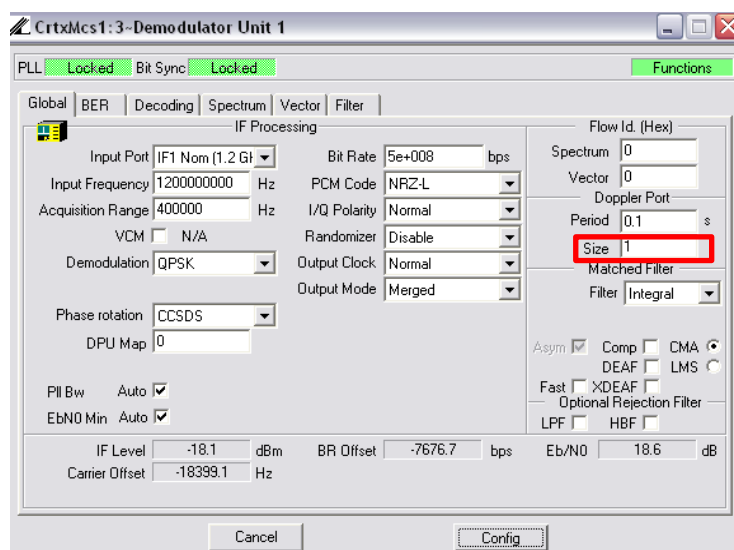
sets the number of LSBs that are discarded. Let N be the value of **Integ_Phase_BinShift**:



The phase resolution thus becomes $2\pi/2^{32-N}$ radian. The revolution period of this counter is $2^{32+N}/\text{IF}$ where IF is the input IF frequency. The default value for the registry key is 15 (decimal), 0xE (hex).

Table 11 and Table 12 give phase resolution and revolution period for different settings of N , with an IF of 720MHz, and 1.2 GHz.

Samples are stored for further transmission, in a measurement block which size is programmable between 1 and 1000 elementary measurement(s). The first measurement in the block is time-tagged. The figure below shows how to set the block size in the DMU window.



Default
registry value →

Truncature (N)	Phase resolution in deg	64-bits phase revolution time	Doppler resolution over 1s in Hz
26	5.6E+00	12 y 253 d 7 h 59 m 26 s	1.6E-02
25	2.8E+00	6 y 126 d 15 h 59 m 43 s	7.8E-03
24	1.4E+00	3 y 63 d 7 h 59 m 51 s	3.9E-03
23	7.0E-01	1 y 214 d 3 h 59 m 55 s	2.0E-03
22	3.5E-01	289 d 13 h 59 m 57 s	9.8E-04
21	1.8E-01	144 d 18 h 59 m 58 s	4.9E-04
20	8.8E-02	72 d 9 h 29 m 59 s	2.4E-04
19	4.4E-02	36 d 4 h 44 m 59 s	1.2E-04
18	2.2E-02	18 d 2 h 22 m 29 s	6.1E-05
17	1.1E-02	9 d 1 h 11 m 14 s	3.1E-05
16	5.5E-03	4 d 12 h 35 m 37 s	1.5E-05
15	2.7E-03	2 d 6 h 17 m 48 s	7.6E-06
14	1.4E-03	1 d 3 h 8 m 54 s	3.8E-06
13	6.9E-04	13 h 34 m 27 s	1.9E-06
12	3.4E-04	6 h 47 m 13 s	9.5E-07
11	1.7E-04	3 h 23 m 36 s	4.8E-07
10	8.6E-05	1 h 41 m 48 s	2.4E-07
9	4.3E-05	50 m 54 s	1.2E-07
8	2.1E-05	25 m 27 s	6.0E-08
7	1.1E-05	12 m 43 s	3.0E-08
6	5.4E-06	6 m 21 s	1.5E-08
5	2.7E-06	3 m 10 s	7.5E-09
4	1.3E-06	1 m 35 s	3.7E-09
3	6.7E-07	47.7 s	1.9E-09
2	3.4E-07	23.8 s	9.3E-10
1	1.7E-07	11.9 s	4.7E-10
0	8.4E-08	5.9 s	2.3E-10

Table 11: Counter capacity and resolution at 720MHz

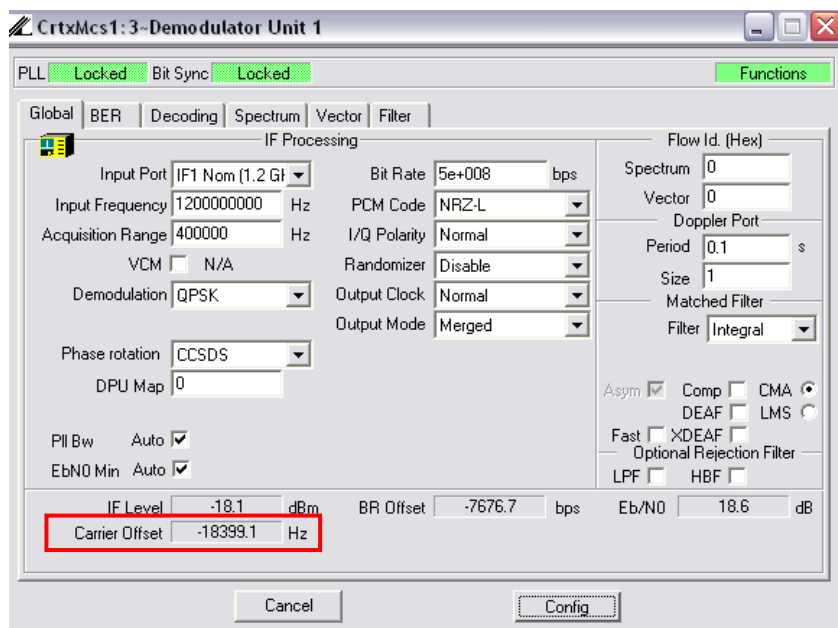
Default
registry value →

Truncature (N)	Phase resolution in deg	64-bits Phase revolution time	Doppler resolution over 1s (in Hz)
26	5.6E+00	7 y 241 d 4 h 51 m 50 s	1.6E-02
25	2.8E+00	3 y 303 d 2 h 25 m 55 s	7.8E-03
24	1.4E+00	1 y 334 d 1 h 12 m 57 s	3.9E-03
23	7.0E-01	349 d 12 h 36 m 28 s	2.0E-03
22	3.5E-01	174 d 18 h 18 m 14 s	9.8E-04
21	1.8E-01	87 d 9 h 9 m 7 s	4.9E-04
20	8.8E-02	43 d 16 h 34 m 33 s	2.4E-04
19	4.4E-02	21 d 20 h 17 m 16 s	1.2E-04
18	2.2E-02	10 d 22 h 8 m 38 s	6.1E-05
17	1.1E-02	5 d 11 h 4 m 19 s	3.1E-05
16	5.5E-03	2 d 17 h 32 m 9 s	1.5E-05
15	2.7E-03	1 d 8 h 46 m 4 s	7.6E-06
14	1.4E-03	16 h 23 m 2 s	3.8E-06
13	6.9E-04	8 h 11 m 31 s	1.9E-06
12	3.4E-04	4 h 5 m 45 s	9.5E-07
11	1.7E-04	2 h 2 m 52 s	4.8E-07
10	8.6E-05	1 h 1 m 26 s	2.4E-07
9	4.3E-05	30 m 43 s	1.2E-07
8	2.1E-05	15 m 21 s	6.0E-08
7	1.1E-05	7 m 40 s	3.0E-08
6	5.4E-06	3 m 50 s	1.5E-08
5	2.7E-06	1 m 55 s	7.5E-09
4	1.3E-06	57.6 s	3.7E-09
3	6.7E-07	28.8 s	1.9E-09
2	3.4E-07	14.4 s	9.3E-10
1	1.7E-07	7.2 s	4.7E-10
0	8.4E-08	3.6s	2.3E-10

Table 12: Counter capacity and resolution at 1.2GHz

3.3.10.5 Displaying the carrier offset of the Doppler port in the DMU window

When using menus that support integrated phase measurement (see above), the display of the carrier offset in the DMU window (see figure below) can be modified to take advantage of the accurate estimate of the Doppler port.



By default, in these menus the carrier offset is computed using the default estimator.

The integrated phase estimator can be activated and the sampling period can be modified through the Windows registry key:

HKEY_LOCAL_MACHINE\SOFTWARE\IN-SNEC\CrtxHdr\SetDefault\Signal Processing\DMUX\Integ_Phase_Freq_Offset

With the following possible values:

- 0: Default estimator
- 1: Integrated phase estimator over 10ms period
- 2: Integrated phase estimator over programmable period (same as the Doppler port)

3.3.10.6 Interpolation Process

In some cases, when the sampling rate and the generation rate are asynchronous, a linear interpolation is applied to provide the accumulated phases at the required sampling frequency.

Such an interpolation generates an error. Let's call φ_{inter} the interpolated phase, φ_{err} the phase error, T the 10-ms period and ε the time between 0 and T when the phase is interpolated.

$$\varphi_{\text{inter}}(t+\varepsilon) = \varphi(t) + \frac{\varepsilon}{T} [\varphi(t+T) - \varphi(t)]$$

The Taylor expression of the phase at $t+\varepsilon$ is:

$$\varphi(t+\varepsilon) = \varphi(t) + [F + D(t)]\varepsilon + D'(t) \times \frac{\varepsilon^2}{2} + R_2(t+\varepsilon), \text{ where } D(t) \text{ is the input Doppler and } R_2(t) \text{ the rest such that:}$$

$$\|R_2(t+\varepsilon)\| \leq \frac{\varepsilon^3}{3!} \sup_{x \in [t, t+\varepsilon]} \|D''(x)\| \text{ (Lagrange overestimation).}$$

$$\varphi_{\text{err}}(t+\varepsilon) = \varphi_{\text{inter}}(t+\varepsilon) - \varphi(t+\varepsilon) = \frac{\varepsilon}{T} [\varphi(t+T) - \varphi(t)] - [\varphi(t+\varepsilon) - \varphi(t)] = -\frac{D'(t)}{2} \varepsilon(T-\varepsilon) + \frac{\varepsilon}{T} \times R_2(t+T) - R_2(t+\varepsilon)$$

$$\text{so } \|\varphi_{\text{err}}(t+\varepsilon)\| \leq \frac{T^2}{8} \times \sup_{x \in [t, t+T]} \|D'(t)\| + \frac{T^3}{3} \times \sup_{x \in [t, t+T]} \|D''(x)\|$$

For instance, a Doppler rate of 1000 Hz/s with a second-order Doppler variation of 10 Hz.s⁻² give a **phase error lower than (0.0125 + 3.33×10⁻⁶) cycle**.

IN MOST CASES, USER WOULD NOT USE CRANKY VALUES FOR THE SAMPLING RATE. THUS, NO INTERPOLATION WOULD BE APPLIED, BECAUSE ACCUMULATED PHASE ARE GENERATED SYNCHRONOUSLY WITH THE SAMPLING RATE DATES.

3.3.10.7 Doppler Computing

The Doppler measurements are computed owing to the accumulated phase interpolated every T_s seconds according to the formula $\Delta f(t + \frac{T_s}{2}) = \frac{1}{2^{32}} \times \frac{\varphi(t+T_s) - \varphi(t)}{T_s} - f_{\text{IFR}}$, where Δf is the computed

Doppler, f_{IFR} is the IF Receiver center frequency and φ the accumulated phase with a resolution of 2⁻³² cycle.

$$\text{The Doppler error is } f_{\text{err}}(t + \frac{T_s}{2}) = \frac{\varphi(t+T_s) - \varphi(t)}{T_s} - \varphi'(t + \frac{T_s}{2})$$

$$\text{With } \varphi(t+T_s) = \varphi(t) + T_s \times \varphi'(t) + \frac{T_s^2}{2} \times \varphi''(t) + \frac{T_s^3}{6} \times \varphi^{(3)}(t) + R\varphi_3(t+T_s)$$

$$\text{where } \|R\varphi_3(t+T_s)\| \leq \frac{T_s^4}{4!} \times \sup_{x \in [t, t+T_s]} \|\varphi^{(4)}(x)\|,$$

$$\text{and } \varphi'(t + \frac{T_s}{2}) = \varphi'(t) + \frac{T_s}{2} \times \varphi''(t) + \frac{(T_s/2)^2}{2} \times \varphi^{(3)}(t) + R\varphi'_2(t + \frac{T_s}{2})$$

$$\text{where } \|R\varphi'_2(t + \frac{T_s}{2})\| \leq \frac{(T_s/2)^3}{3!} \times \sup_{x \in [t, t+T_s]} \|\varphi^{(4)}(x)\|.$$

$$\text{So } \|f_{\text{err}}(t + \frac{T_s}{2})\| \leq \frac{T_s^2}{24} \times \sup_{x \in [t, t+T_s]} \|D^{(2)}(x)\| + \frac{T_s^3}{16} \times \sup_{x \in [t, t+T_s]} \|D^{(3)}(x)\| \text{ where } D(t) \text{ is the input Doppler.}$$

A typical second-order Doppler variation of 10 Hz.s^{-2} and third-order Doppler variation of $10^{-4} \text{ Hz.s}^{-3}$ give a **phase error lower than $(4.16 \times 10^{-3} + 6.25 \times 10^{-9}) \text{ Hz}$** for a sampling period T_s of 0.1 s, and lower than $(41.6 + 6.25 \times 10^{-3}) \text{ Hz}$ for a sampling period T_s of 10 s.

3.3.10.8 Impact of Phase Errors on Doppler Computing

If φ_{err} is the error on the accumulated phase, the error on the computed Doppler is lower than $\frac{1}{2^{32}} \times \frac{2\varphi_{\text{err}}}{T_s}$, where T_s is the sampling period.

So, the maximum error on the Doppler computing due to the linear interpolation on the second-order phase variation is $\leq \frac{DR}{T_s} \times \left(\frac{T}{2}\right)^2$.

For example, with a Doppler rate of 1000 Hz/s, the maximum error will be 0.43 Hz for a sampling period of 0.1 s and 0.0043 Hz for a sampling period of 10 s.

The relation between range rate and Doppler is: $\frac{\Delta f}{f} = -\frac{2v_r}{c}$, where f is the satellite carrier frequency, v_r the radial velocity (range rate) and c the light celerity.

For a 2.2-GHz beacon, the corresponding **range rate error will be 2.93 cm/s for a sampling period of 0.1 s** and 0.293 mm/s for a sampling period of 10s.

3.3.10.9 Sampling Period Resolution

3.3.10.9.1 Error Due to Computing

The accumulated phase and Doppler sampling is processed owing to an internal 64 bits- NCO running at $T = 10 \text{ ms}$. The effective sampling period is: $T_{\text{eff}} = \frac{2^{64}}{\text{incr}} \times T$ where incr is the NCO phase increment.

So the resolution on the period is $2^{64} \times T \times \left(\frac{1}{\text{incr}} - \frac{1}{\text{incr}+e} \right) \approx e \times \frac{2^{64} \times T}{\text{incr}^2} = \frac{e \times T s^2}{2^{64} \times T}$ where e is the error made on the phase increment computing (in this computing T_{eff} is approximated by T_s , and e is considered as $\ll \text{incr}$).

Due to double-float format limitations, $e = 2^{14}$ is higher than the possible error (11 bits come from the difference between the 64 bits and the 53 bits mantissa of the double-float representation, on which up to 3 bits may be lost in the computing process).

When $T_s = 10 \text{ s}$ (worst case), the resolution is $6.8 \times 10^{-12} \text{ s}$.

This resolution is effective when the programmed sampling period is a value for which the floating representation does not generate any error (examples: 10 s, 1 s or 0.125 s).

An error ε made on the sampling period generates an error on the Doppler estimation:

$$\frac{\varphi(t) - \varphi(t - T_s)}{T_s} - \frac{\varphi(t) - \varphi(t - T_s)}{T_s + \varepsilon} = \varphi(t) - \varphi(t - T_s) \frac{\varepsilon}{T_s(T_s + \varepsilon)} \approx \frac{\varphi(t) - \varphi(t - T_s)}{T_s} \times \frac{\varepsilon}{T_s} = f \left(\frac{t - T_s}{2} \right) \times \frac{\varepsilon}{T_s},$$

where f is the current IF frequency.

With a typical value of 720 MHz, the **Doppler computing error will be 39 mHz** with $T_s = 0.125 \text{ s}$.

3.3.10.9.2 Error Due to Floating Representation

The floating representation may generate an error in $T_s \leq T_s \times 2^{-23}$ (for example, $0.1 + 1.49 \times 10^{-9}$ s in the case of 0.1 s programming), which could have generated an error on Doppler estimation of

$$f\left(t - \frac{T_s}{2}\right) \times 2^{-23}.$$

But this floating representation is converted into double, owing to rounding processing, so that this error is not added to the previous one (provided that the programmed sampling period does not have more than 5 digits in its fractional part).

3.3.11 Es/N0 estimators in DVB-S2 and SCCC environment

3.3.11.1 Warning

The HDR unit has been designed for optimal demodulation and decoding of IF signals. As such, it also provides some estimates of typical useful parameters such as Es/N0, IF Level and so on. However these estimates are not intended to compete with dedicated constellation analysers in terms of accuracy and precision.

3.3.11.2 Standard Es/N0 estimation

Let's consider a QPSK modulation.

The standard method for Es/N0 estimation is to compute on the constellation the following values:

- Mean Radius of the constellation. The square of this value leads to Es
- Variance of the I/Q vector. This leads to N0.

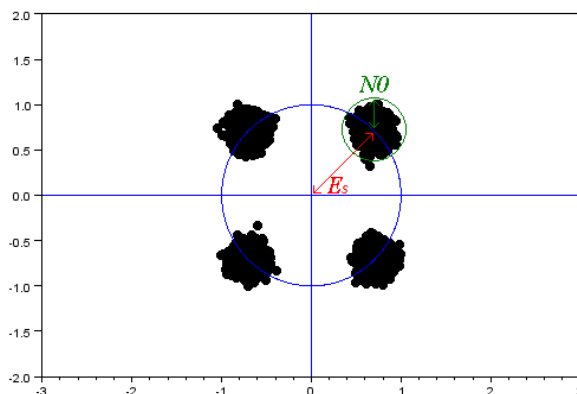


Figure 34: Typical Es/N0 estimation over the constellation

This simple method can be applied in a similar way to multilevel modulations such as 16APSK and 32APSK.

This method is simple, but is biased at low SNRs (or Es/N0), when I/Q points easily cross the decision regions, especially in high density modulation where those regions are very small. Below is a typical example of this effect in different modulation schemes.

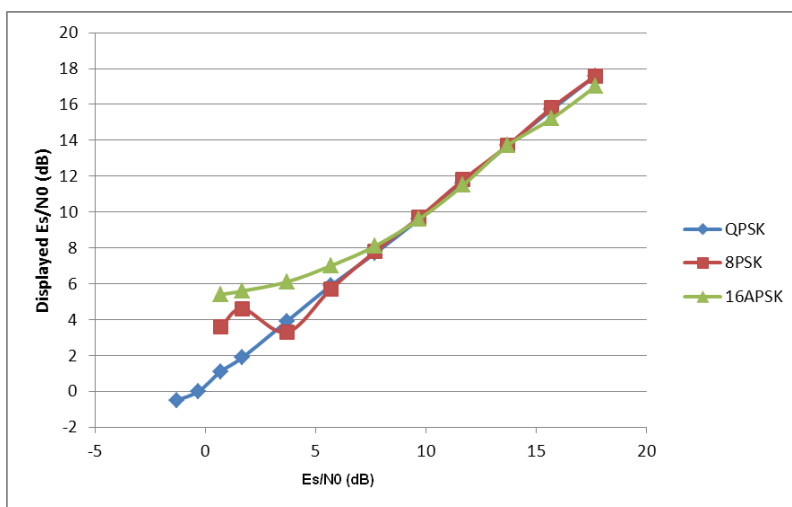


Figure 35: Estimator behavior VS Es/N0 and modulations

This estimate is available in the HDR through the following means:

- In the TCP/IP interface (See STI100013_hdr document), in Demodulator Unit Table (base component 0x3010, offset 43).
- In the GUI, in the main demodulator pane:

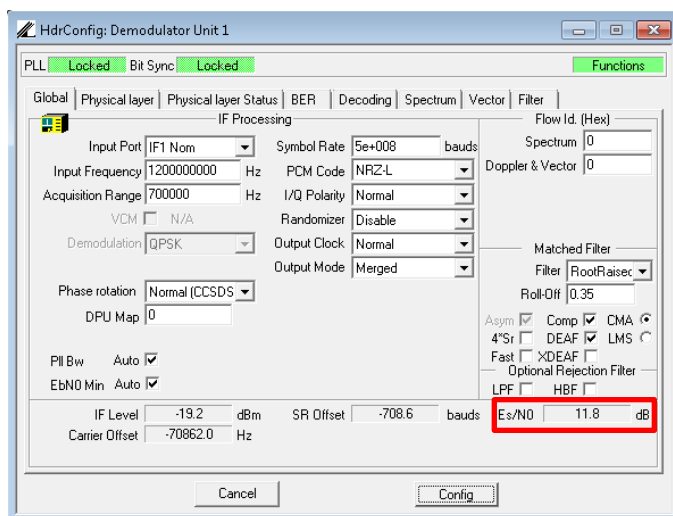


Figure 36: Standard Es/N0 estimate in the GUI

C. In the GUI, in the Vector pane:

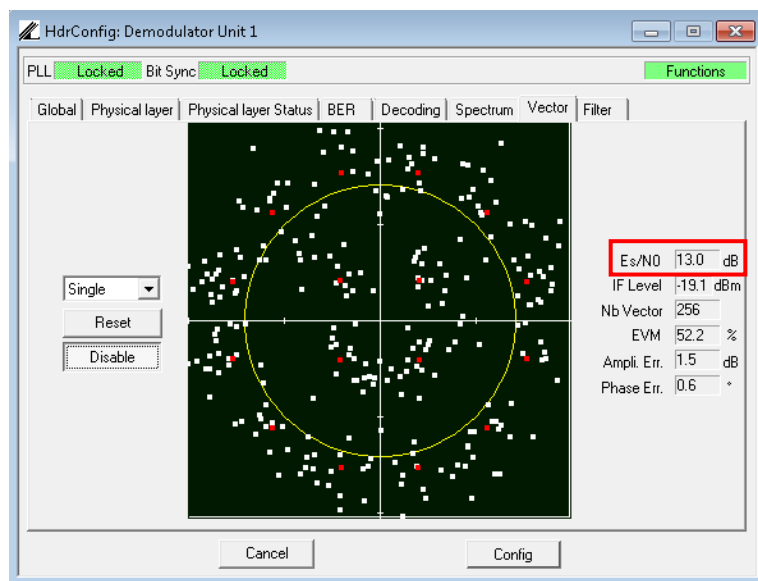


Figure 37: Standard Es/N0 estimate in the vector pane

3.3.11.3 Data-aided Es/N0 estimation in SCCC and DVB-S2

SDSY worked on an improved version of the estimator, specifically for the SCCC and DVB-S2 modulation schemes. These evolved modulation scheme are based on symbol-framing, with robust headers. In both standards, the headers are Pi/2 BPSK-modulated, and are guaranteed to be error-free well-below the threshold of the lowest ACM.

The developed algorithm recovers and decodes these robust headers to estimate Es/N0. This provided a better estimation of Es/N0 for low values, as seen on next figure.

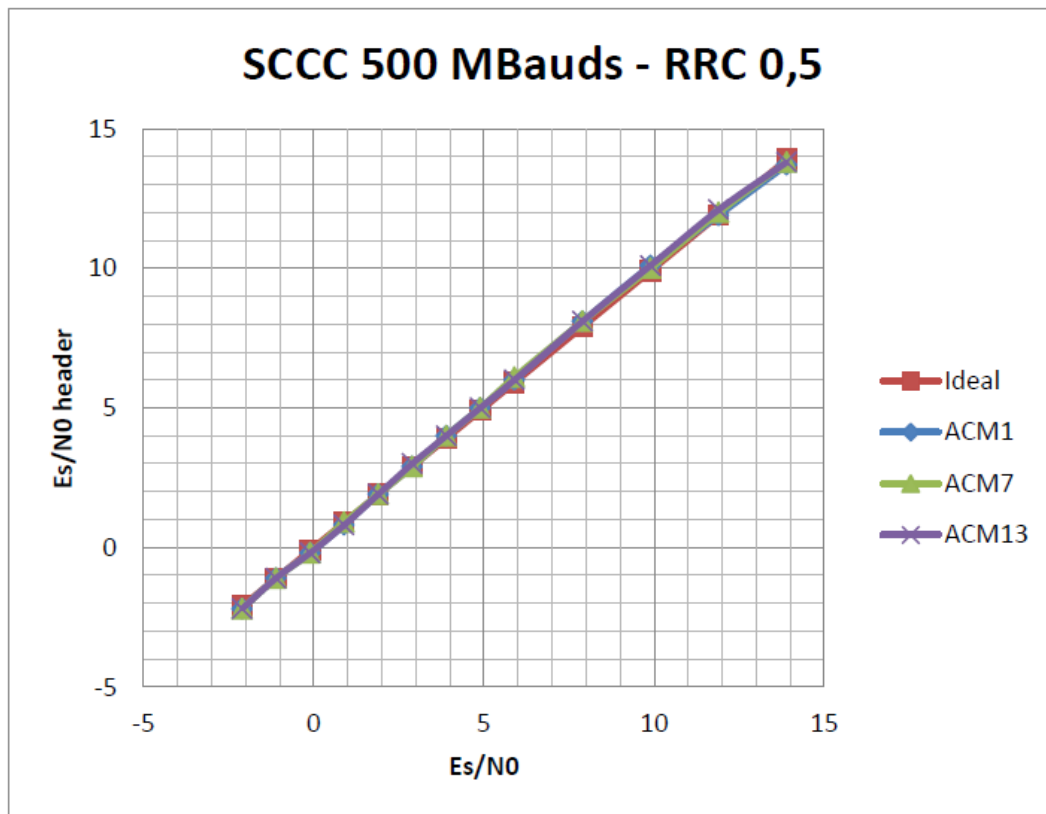


Figure 38: data-aided estimation of $Es/N0$

This estimator is also faster to converge than the standard estimator. It is advised for use in real operation.

This estimator can be accessed by the following means:

- In the TCP/IP interface (See STI100013_hdr document), in Demodulator Unit Extended Table (base component 0x30A0, offset 112).
- In the GUI, in the main Physical Layer pane of the demodulator window:

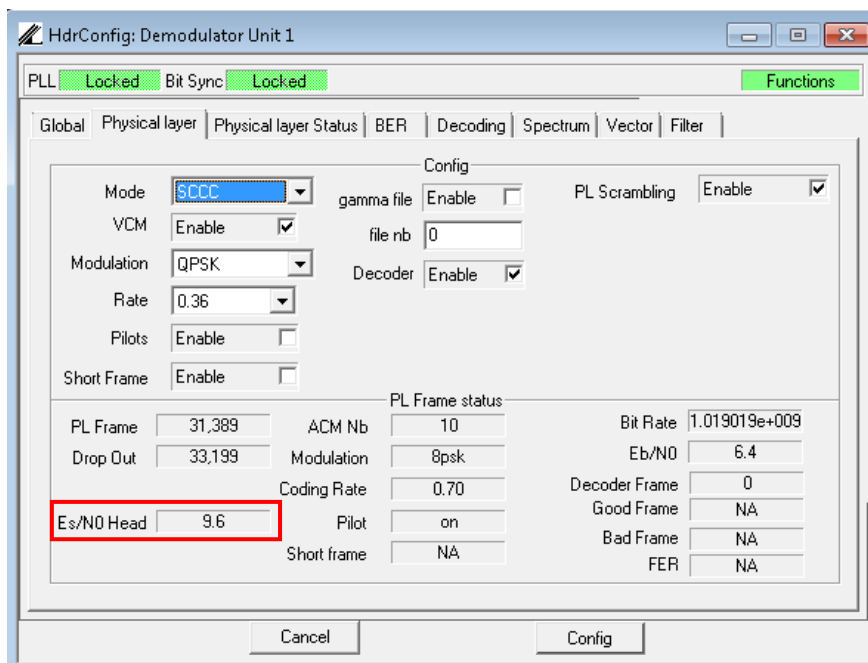


Figure 39: data-aided estimator in the GUI, Physical Layer pane of the demodulator

3.4 Data Processing

3.4.1 Functional Breakdown of the Data Processing Unit

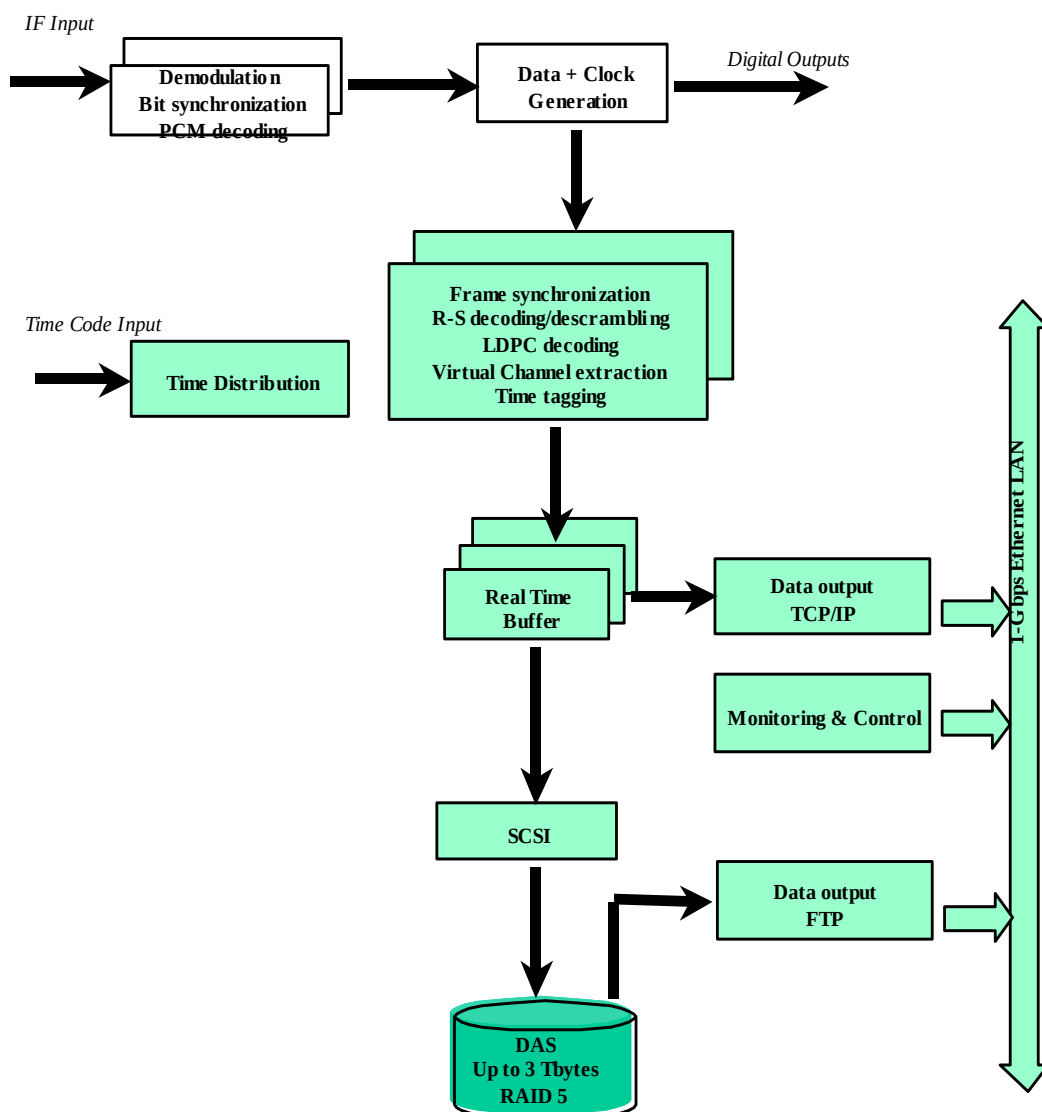


Figure 40: Data Processing Unit Block Diagram

3.4.2 Frame Synchronization & Ambiguity Resolution

Frame synchronization is performed only if parameter **Frame Synchronizer** is set to **on**. If **Frame Synchronizer** is set to **off**, raw data mode is activated (used with ingestion capability).

3.4.2.1 Frame Synchronization Strategy

The frame synchronizer strategy consists in finding the synchronization word which marks the beginning of the frame in the signals (I in BPSK, I or Q or I & Q in QPSK, 1 & 2 & 3 in 8PSK...) output by the bit synchronizer and in reacting correctly to a loss of lock. This process is illustrated by next figure:

❑ **SEARCH phase (S):**

In this mode, the frame synchronizer compares the signal bit by bit, until it matches the expected synchronization word to within n bits where « n » is a programmable error factor, referred to as the synchronization threshold (SYN). The synchronization threshold can vary from 0 to 15 (inclusive).

A programmable mask can also be applied to the synchronization word to ignore corrupted bits in the synchronization word resulting from a spacecraft failure.

❑ **CHECK phase (C):**

Before moving onto the LOCK mode, the frame synchronizer will check « i » consecutive occurrences of the synchronization word (to within n errors). The contents of the telemetry block between two synchronization words is not checked. During this phase, any occurrence of an incorrect synchronization word returns the frame synchronizer to the SEARCH mode. « i » is programmable between 0 and 15 (inclusive) and referred as the CTL (Check-To-Lock) threshold. To switch from SEARCH to LOCK, the frame synchronizer must receive $i+1$ correct synchronization words except when $i = 0$ (in that case, the frame synchronizer will transmit frames starting from the frame containing the first occurrence of a correct synchronization word).

❑ **LOCK phase (L):**

In the LOCK mode, the frame synchronizer merely checks the occurrence of a new synchronization word (to within n errors) every frame. The window in which the synchronization word should be found is enlarged by $\pm N$ bits ($N = \text{bit slip parameter} = 0, 1 \text{ or } 2 \text{ bits}$) to cope with potential bit slip in the telemetry demodulation process.

❑ **FLYWHEEL phase (F):**

Transmission errors (signal heavily affected by noise) or loss of PSK demodulator lock may occur: the frame synchronizer will then only return to SEARCH mode after « j » consecutive occurrences of incorrect synchronization words (number of errors greater than the synchronization threshold). « j » is programmable between 0 and 15 (inclusive) referred to as the LTS (Lock-To-Search) threshold.

The frame synchronizer synchronizes the frame in accordance with the SYN, CTL, LTS and Bit Slip parameters provided to it:

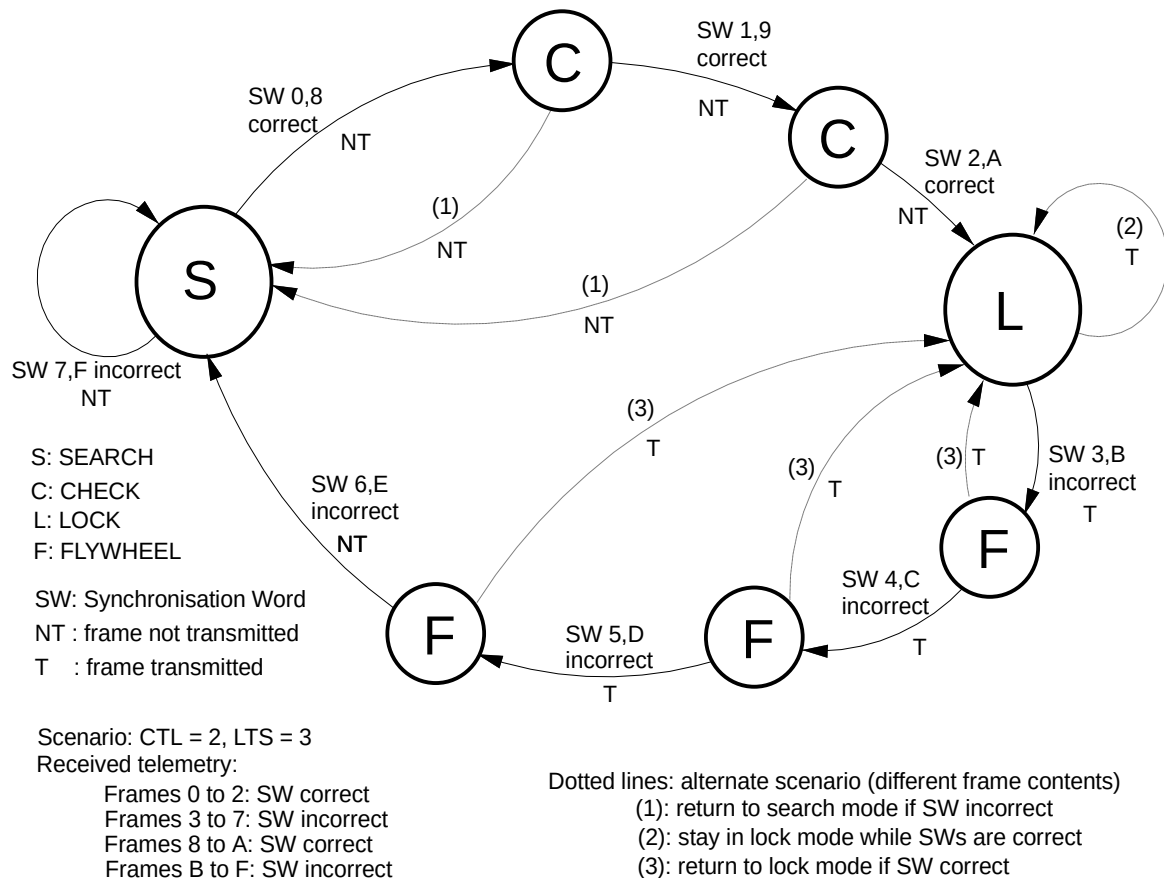


Figure 41: Frame Synchronization Process

3.4.2.2 Frame Synchronizer Input

Frame synchronizer input can be either I or Q or merged I & Q. Depending on the demodulation and decoding scheme, the following setting must be programmed (refer to parameter **Input Selection** in the Data Processing Unit. Section 4).

3.4.2.3 Automatic PSK Ambiguity Resolution

Automatic PSK Ambiguity Resolution is performed only if parameter **IQ output polarity** in the Demodulator Unit Monitoring Table is set to **Ambiguity solved by DPU** (see section 4.5.4.1.1.1). Should the data at the bit synchronizer output be ambiguous, good data can be obtained by using this Ambiguity Resolution Mode.

Phase or channel ambiguity resolution is then automatically performed by the frame synchronizer during the SEARCH phase (and **only during this phase**). If the comparison between the expected synchronization word and a given window on the received data stream fails, another phase & channel configuration is checked and the process is repeated until the frame synchronization word is identified. The frame synchronizer switches then to the CHECK phase.

When the synchronization input is set to I (respectively Q) (see section above)., the frame synchronizer searches the synchronization word on I and Q channels (at Frame synchronizer input) and automatically route the selected channel to the I (respectively Q) ECL output.

MODULATION & DECODING	NRZ-L, BP-L	DNRZ	NRZ-M/S BP-M/S
QPSK	Ambiguity resolved	Specification (note 2)	Ambiguity resolved
OQPSK		Not Applicable	Specification (note 2)
BPSK			Specification (note 2)
AQPSK (note 1)			Ambiguity resolved
UQPSK			Specification (note 2)
QPSK + Viterbi single			Specification (note 2)
BPSK + Viterbi			Specification (note 2)
QPSK + Dual Viterbi		Ambiguity resolved	Ambiguity resolved
OQPSK + Dual Viterbi		Not Applicable	Specification (note 2)
8PSK	Ambiguity resolved (note 3)	Not Applicable	
8PSK + TCM	Specification (note 2)		

Note 1: Same bit rate but asynchronous I and Q clocks.

Note 2: Enables to retrieve the original data stream even if the on-board data process is unknown to the operator.

Note 3: Providing that correct 8PSK de-mapping (normal or Gray) has been selected.

3.4.3 CCSDS Data De-randomizer

Randomization technique can be used:

- to maintain bit or symbol synchronization on the telemetry processing chain, by always having a minimum bit transition density on the received telemetry signal,
- to avoid a false carrier lock that can occur on a modulated signal with periodic data,
- to avoid ambiguous data patterns that allow decoder false lock.

De-randomization by the CORTEX HDR complies to the CCSDS recommendation 101.0-B-3 (Blue Book - May 1992). The pseudo-random sequence is generated using the following polynomial:

$$h(x) = x^8 + x^7 + x^5 + x^3 + 1$$

Customized de-randomization can also be used (refer to the following paragraph).

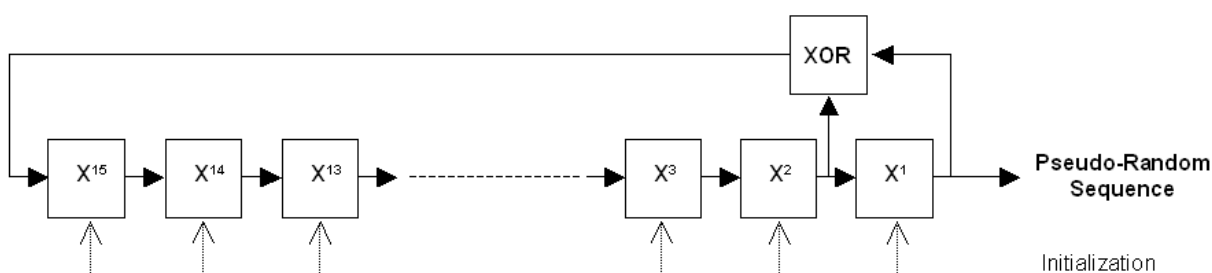
3.4.4 Customized Data De-randomization

The Cortex HDR provides a fully programmable de-randomization function.

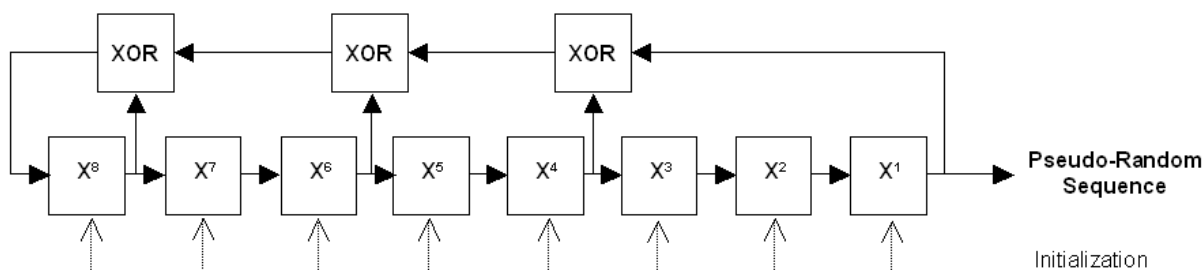
Are programmable:

- the polynomial of the generator of the pseudo-random sequence The supported maximum degree of the polynomial is 16 (refer to the examples below)
- the initial value of the generator (fully programmable between 0x0000 and 0xFFFF).

Example 1: Polynomial: $X^{15} + X + 1$ (polynomial set-up value: 0x8003)



Example 2: Polynomial: $X^8 + X^7 + X^5 + X^3 + 1$ (polynomial set-up value: 0x1A9)



For modulation schemes with separate I and Q, each de-randomization function is configurable independently of each other.

The customized de-randomization can be performed before or after the associated decoding (Reed Solomon or LDPC). That is programmable (refer to CADU window).

3.4.5 CRC Frame Checking

Reconstructed frames will be verified only if parameter **Frame Checking** is set to **CRC**.

The CRC is user-programmable via two parameters: **CRC Polynomial** (CRC16, CCITT, etc...) and **CRC Preset Value** (usually all bits set to 0 or to 1).

The CRC value is supposed to be the last two bytes of each frame. The **CRC Preset Value** is added to the computed checksum.

3.4.6 (255,223) Reed-Solomon Decoding

Reed-Solomon coding technique is well suited for burst error correction. It can be used in conjunction with convolutional encoding to allow data reconstruction with the lowest error probability. Reed-Solomon decoding and error-correction by the CORTEX HDR complies to the CCSDS recommendation 101.0-B-3 (Blue Book - May 1992). The following coding characteristics should be used:

- ❖ J = 8 bits per R-S symbol
- ❖ Conventional and dual basis supported
- ❖ E = 16 R-S symbol error correction capability within a R-S codeword (coded to decoded information ratio = 255/223),
- ❖ Interleaving depth: auto-adjusted from 1 to 8 (frame-length dependent),
- ❖ Virtual filling for shorten RS code
- ❖ Other characteristics: see CCSDS recommendation 101.0-B-3.

The Virtual fill length and Interleaving depth are automatically deduced by the HDR from the CADU length set by the operator with the following formula:

$$\begin{array}{llll} \text{For } l = 1: & 240 + 4 & \leq & CL = (255-V) + 4 \leq 255 + 4 \\ \text{For } 1 < l \leq 8: & 240 \times l + 4 & < & CL = (255-V) \times l + 4 \leq 255 \times l + 4 \end{array}$$

V max = 15
CL max = 2044
CL min = 244

l: Interleaving depth
CL: CADU length (in bytes)
V: Virtual fill length (in bytes)
Vmax: Virtual fill length max (in bytes)

The R-S decoder status includes two counters (good frames and bad frames), the number of corrected errors (per frame) and the BER.

3.4.7 (255,239) Reed-Solomon Decoding

This (255,239) Reed Solomon is recommended by the ECSS for links which have 4D-8PSK-TCM modulation:

- ❖ J = 8 bits per R-S symbol
- ❖ Conventional and dual basis supported
- ❖ E = 8 R-S symbol error correction capability within a R-S codeword (coded to decoded information ratio = 255/239),
- ❖ Interleaving depth: 1 to 8
- ❖ Virtual filling for shorten RS code
- ❖ Other characteristics: refer to: **ECSS - Space data links – Telemetry synchronization and channel coding ECCS-E-50-01.**

Interleaving depth and virtual filling length are automatically deduced from the CADU length set by the operator with the same formula as for RS (255,223).

3.4.8 (10,6) Reed-Solomon Frame Header Decoding

This CCSDS shorten (10,6) Reed Solomon can be optionally used for transfer frame header error control:

- ❖ J = 4 bits per R-S symbol
- ❖ E = 2 R-S symbol error correction capability within a R-S codeword,

Other characteristics: refer to: CCSDS – AOS Space Data Link Control- CCSDS 732.0-B-1 Blue Book.

3.4.9 Specific Decoding

The HDR provides specific decoding scheme, i.e. separate de-randomizer and Reed Solomon decoding on each I and Q outputs of the demodulator.

On each I and Q separate outputs, the supported format of a CADU is the following:

- CADU length: 9424 bytes
- 19 RS frames per CADU
- RS frame format: 2 interleaved (223,255)RS blocks with 7 bytes of virtual filling per RS block
- CADU synchronization word: RS encoded
- De-randomizer: CCSDS type.

3.4.10 LDPC Decoding

3.4.10.1 Introduction

Low Density Parity Check codes are powerful linear block codes. They are defined by their parity check matrix. The construction of these codes involves the juxtaposition of smaller cyclic sub-matrices. Nowadays, iterative algorithms, which offer near-optimum performance at a manageable complexity, are used for the decoding of large linear LDPC codes.

In conformity with the CCSDS experimental specification, the CORTEX HDR supports:

- the 7/8 LDPC code optimized for near-earth applications
- the 1/2 and 2/3 LDPC codes optimized for deep space applications

For more details on these codes, refer to the following CCSDS document:

Low Density Parity Check codes for use in near-earth and deep space applications
Experimental Specification - CCSDS 131.1-O-2
September 2007

3.4.10.2 LDPC 7/8 for near-earth applications

As specified by the CCSDS, the implemented (8160, 7136) LDPC code is a shortened version of the baseline (8176, 7156) LDPC code, i.e.

- information block length k: 7136 bits (892 bytes)
- code block length n: 8176 bits (1022 bytes)

3.4.10.3 AR4JA LDPC 1/2, 2/3, 4/5 codes for deep space applications

The Cortex HDR supports a subset of the LDPC codes family optimized for deep space applications and specified by CCSDS. These codes are also known as AR4JA LDPC codes.

The supported codes are the following:

- 1/2 rate LDPC 1K code
- 1/2 rate LDPC 4K code
- 1/2 rate LDPC 16K code
- 2/3 rate LDPC 4K code
- 2/3 rate LDPC 16K code
- 4/5 rate LDPC 4K code
- 4/5 rate LDPC 16K code

3.4.10.4 Implementation

The LDPC decoder is implemented after a frame synchronizer and after the de-randomizer if the randomization is used. The LDPC is a block decoder that can be synchronized to CADU processing and when it is, from this aspect it is identical to the Turbo decoding and very close to the Reed Solomon decoding. The LDPC code can also be asynchronous from the CADU (refer to § 1.6.1.7 Transport layer (RSDVB or asynchronous LDPC layer)).

- Any phase ambiguity at the output of the demodulation has to be resolved by the frame synchronizer before the decoding
- The decoder uses soft decision at the output of the bit synchronizer
- If a de-randomizer is used, this one has to work on soft decision

The LDPC decoding is implemented as an iterative process with parity decoding.

In the HDR implementation, after the last iteration the parity control is processed one more time. This control is able to detect bit errors before and after decoding. This enables to elaborate a BER estimation before and after the decoding and also a quality indication "Good/Bad" of the decoded CADU.

In the Cortex HDR, the LDP decoder can be enabled in the CADU layer or in the Transport Layer if this one is used. The selection can be done from the GUI.

3.4.10.5 *Performances*

<0.3dB from CCSDS reference curve.

3.4.11 *Recording and Data Transmission over Ethernet*

3.4.11.1 *General consideration*

Data transmission:

- ❖ Transmitted in real time through Ethernet, the internal buffer is 1 to 2 Gbyte
- ❖ Store in hard disk, buffer from few Gbyte to Tbyte
- ❖ Transmitted recorded data of disk in quasi real time (as soon as a file are closed delay <0.5sec)
- ❖ A mix of all those Store and transmission

Performance see performance of Ethernet transmission and storage performance.

Assuming an 12 disk and 10Gbit option, a typical 5Gbps on air application could achieved:

- A real time transmission: 5Gbps on LAN
- A record on Disk full speed: 5Gbps
- A simultaneous quasi real time with no loose of data in case of LAN/WAN issue

3.4.11.2 *Data Storage*

Data can be recorded on a hard disk.

The Data storage system can be according the following configuration:

- Raid 0 with 4 Disk
- Raid 6 with 12 Disk
- Disk configuration are SSD or HDD
- Disk capacity are 1 Terra byte or 2 Terra byte

3.4.11.3 TCP-IP Data Recovery

In TCP-IP mode, up to 6 clients can connect to the telemetry port (port # 3075) for receiving **real-time or recorded data**. Connected clients must send a data request specifying:

- ☐ The data source (Data Processing Unit number: A, B or C),
- ☐ The message size (number of TM Blocks per TCP-IP message)
- ☐ The requested Virtual Channel(s), for CCSDS applications
- ☐ The first TM Block to transmit: first block in the HDR real-time buffer or TM block with specified time-tag (real-time or recorded data).

Next figure illustrates the data formatting for real-time and recorded TM Messages. See Annex 1, STI 100013_HDR Section 2.6 for more details on the Ethernet interface.

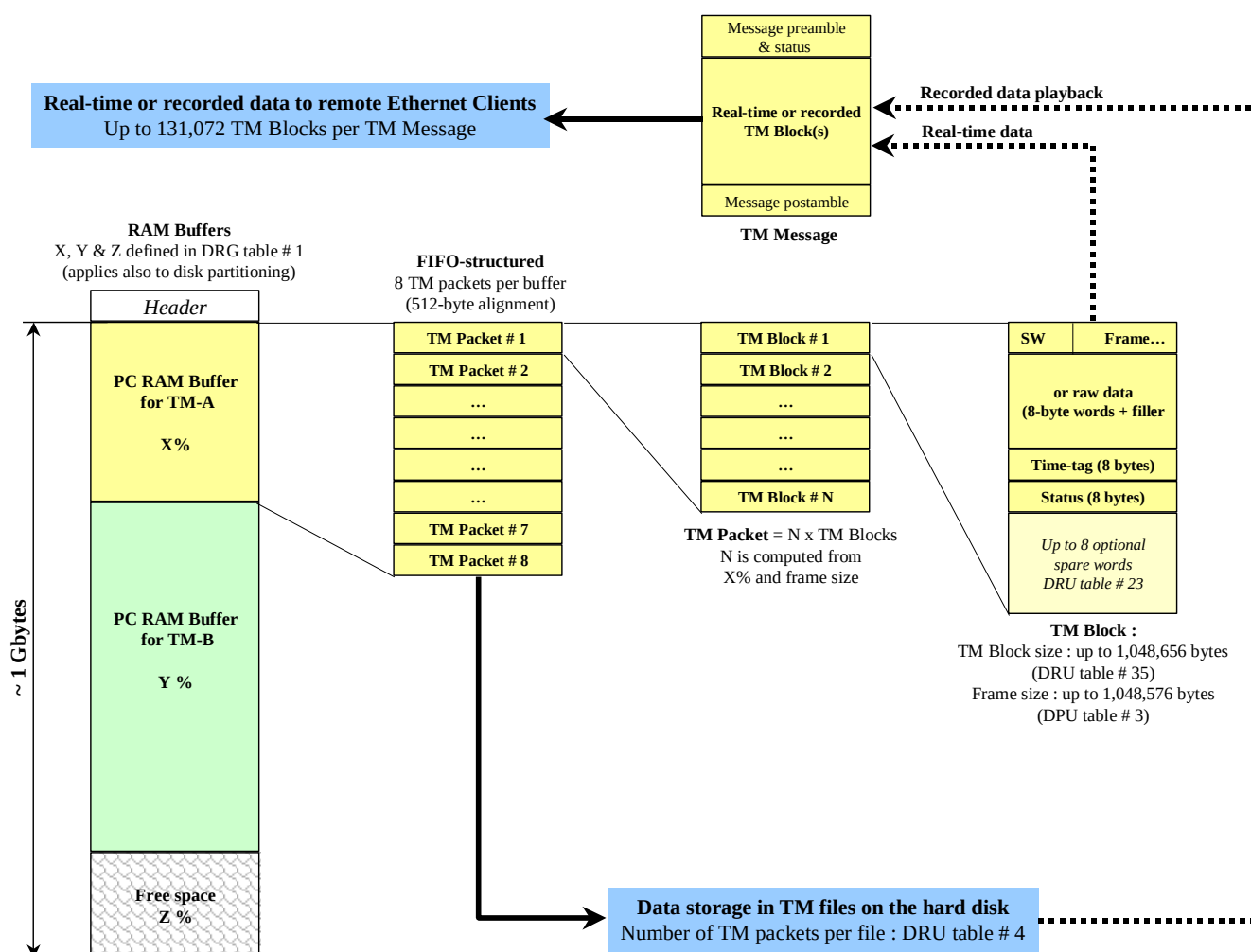


Figure 42: Data Transmission over TCP-IP. Message Formatting

3.4.11.4 Archiving and FTP Recovery

3.4.11.4.1 Generality

3.4.11.4.1.1 Overview

The main objective of the functionality is to record the down-link data coming from a satellite during its successive passes and to place them at the disposal of different Users.

The data are recorded at the frame level after the Reed-Solomon decoding if used (CADU). The data are recorded on an external DAS archiving system connected with the CORTEX HDR unit.

Before being recorded, successive filtering and sorting operations can be applied on specific words inside the incoming data frames in order to store into dedicated directories the frames which are useful for the Users. Nevertheless the other frames are temporary recorded in a specific “trash” directory. These operations are fully programmable and can be individually selected or not.

There are basically 6 successive filtering and sorting operations:

- ☐ RS filtering, allows to discard or not the Reed-Solomon code blocks declared as non-corrected,
- ☐ Filter A,
- ☐ Sort A,
- ☐ Sort B,
- ☐ Sort C.

The filtered and sorted frames are recorded inside **binary files**. To be easily recovered by the Users, the binary files are written inside specific **dedicated directories** of which tree construction and names are fully programmable and are in relation with the successive filtering and sorting operations.

For each recording session, a **pass report file** is created. At the end of a recording session, the corresponding text file provides information about the recorded data files. These report files are also placed at the disposal of the Users in a dedicated directory.

The names of the files are automatically defined at the starting of a recording session. The names are divided into several fields providing the start date of the recording session, specific markers and programmable characters strings associated with the different filtering and sorting operations.

The HDR unit is fitted with a standard FTP server “FileZillaServer” and the users can recover the useful files from the dedicated directories using a standard remote FTP client in a simple read operations.

The files storage durations are programmable and depend on the directory type: the directories containing useful data files have a nominal storage duration and the “trash” directories have a temporary storage duration.

For the setting and operation of the functionality basically there are two steps. The first step is the static set-up and it is done one time for the mission of the unit,. The static set-up consists in the definition of the partition of the memory space for data and report directories, the definition of the rules to construct the dedicated directories and the definition of the roots of the files names. A static set-up is defined inside a set of configuration files which are text files, there can be several different sets of configuration files. The second step is the dynamic set-up and it is done before a pass of a given satellite. It consists in selecting the filtering and sorting operations for the given pass, its set of configuration files and if necessary the generation of the associated directories tree, in programming fields to complete the construction of the files names, in giving a start and a stop dates for the recording. This dynamic set-up

is done through the TCP/IP Control TCP/IP port of the HDR unit. Detailed instructions to program the functionality are given in the **“Recording and FTP Recovery” User Manual DTU 100381**.

3.4.11.4.1.2 Data files structure

The recorded data files are binary files.

A data file contains a variable number of TM blocks.

The maximum number of TM blocks per file is defined by the current configuration of the CORTEX HDR. The effective number of TM blocks per file can be less than the maximum only when the recording session is stopped.

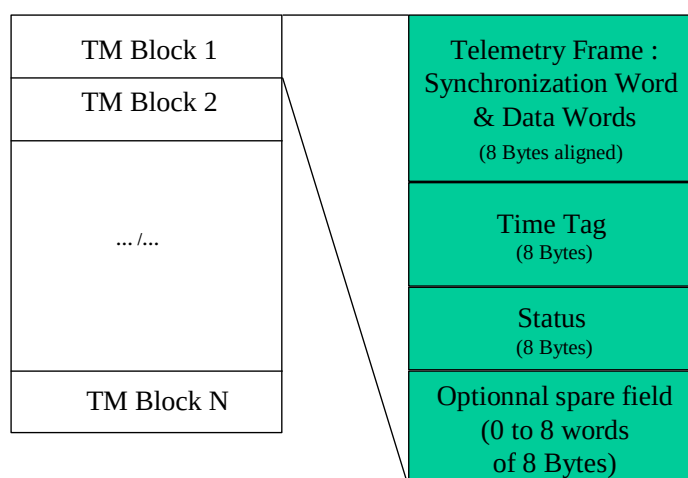


Figure 43: Recorded Data File Structure

When the frame synchronizer is disabled, the Status field is never included inside the TM blocks. On the other hand and independently, the insertion of the Time Tag field can be disabled or not (it is a set-up parameter, refer to the “Real Time” window in chapter 4).

For example, in order to record **raw data** without any added bytes inside the binary files, the following set-up conditions must be satisfied:

1. the frame synchronizer must be disabled
2. the selected TM data block size must be set to N x 8 bytes (example: 1024 bytes)
3. the insertion of the Time Tag must be disabled
4. the number of added spare words must be set to 0.

3.4.11.4.1.3 Report files structure

The pass report files are ASCII files.

The CORTEX HDR generates a pass report file in the result passes directory. The name of the report file is reconstructed as the TM file name, except for its extension.

There is one line per TM file in the ASCII report file. Each line is composed of 5 fields separated by ' ; ' .

- The number of TM blocks in the file,
- The number of TM blocks not corrected in the file,
- The date of the first TM block in the file,
- The date of the last TM block in the file,
- The full path and name of the file.

3.4.11.4.2 *Standard FTP CCSDS mode*

The standard mode delivered with the FTP functionality is the sorting and storage of received CCSDS VCDU.

In this mode, 64 directories for the storage of the data are created (VC_00 to VC_63) per defined reception channel. Each directory corresponds to a given VCID.

Only the Sort A operation is used (the other filtering and sorting operations are disabled by set-up in the FTP window). The Sort A operation consists in extracting from each received CADU its VCID and comparing it with the different possible VCID values (from 0 up to 63), its operation is defined inside the configuration file: SORT_A_0.txt. The VCDU are stored inside dedicated files in their corresponding directories and put at the disposal of remote Users using a FTP client software.

For each session a corresponding report file is generated including information on the recorded files. These files are stored in specific directories.

The created directories arborescence is given by the following figure:

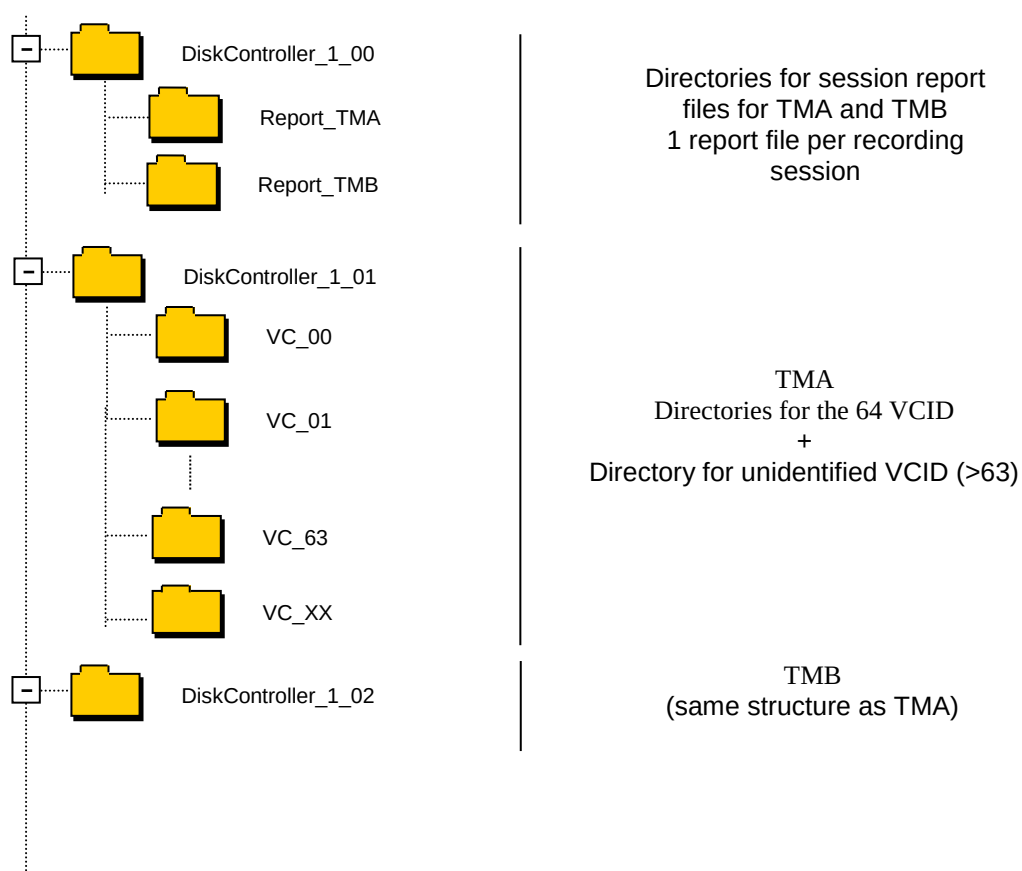
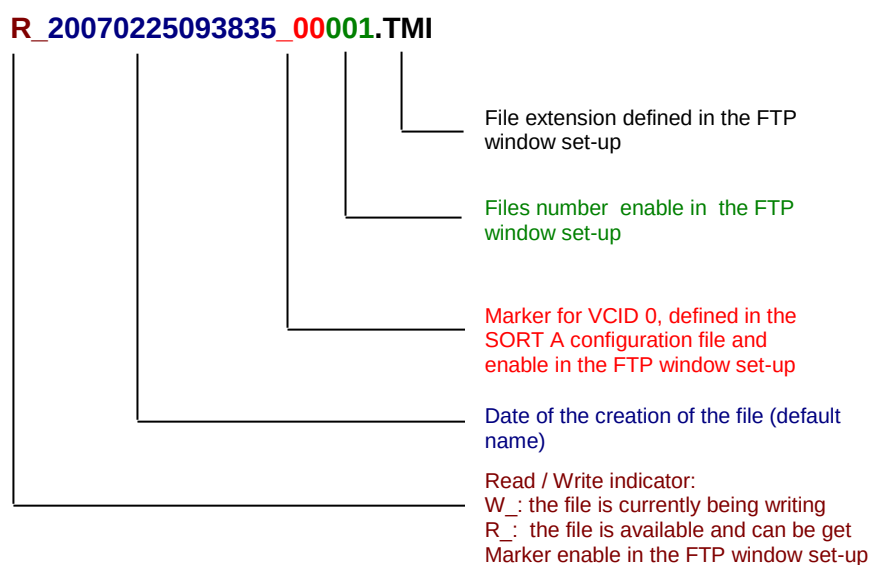
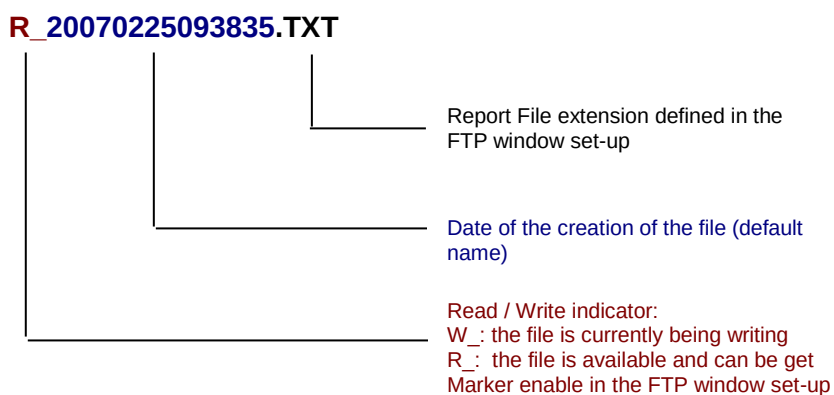


Figure 44: CCSDS FTP Mode Storage Directories

The names of the recorded data files are defined by the SORT_A_0.txt configuration file and by the set-up in the FTP window, an example of a file name is given below:



The contents of the data files are defined in the paragraph **3.4.6.4.1.2**.
The names of the session report files are defined by the set-up in the FTP window:



The contents of the session report files are defined in the paragraph **3.4.6.4.1.3**.

3.4.11.4.3 FTP user defined mode

The FTP functionality is fully programmable and open device: filtering and sorting operations, directories arborescence and names of the files. The previously described standard CCSDS mode is a simple example from which a number of more complex modes can be constructed.

For detailed explanation on how to program and operate the FTP functionality refer to the **“Recording and FTP Recovery” User Manual DTU 100381**.

3.4.12 Playback function

3.4.12.1 Operating Modes

The Cortex HDR provides two different operating modes for the Playback function:

- ❑ Replay of data using the Test Modulator (Test Modulator mode). In this case the data are sent to the Test Modulator for modulation and re-transmission at the IF level (720 MHz or 1.2 GHz). The data can be demodulated by the Demodulators just as in real time reception.

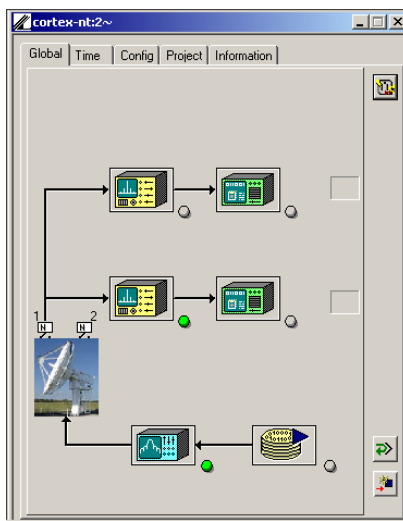


Figure 45: Playback function: Test Modulator operating mode

- ❑ Replay of data at the ECL outputs (Data + Clock mode). In this case the data are sent directly to the ECL outputs of the selected demodulator(s) in the selected output format. The Test Modulator is not used nor the demodulation function of the Demodulators.

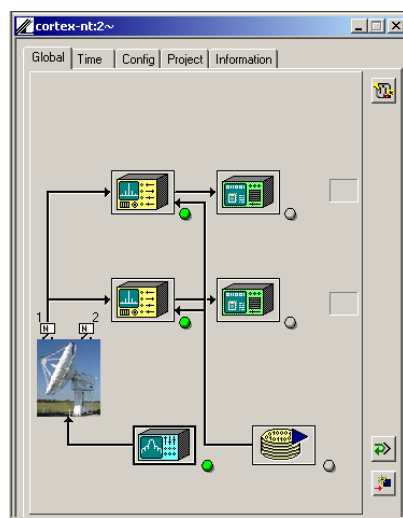


Figure 46: Playback function: Test Modulator Data + Clock mode

3.4.12.2 Sources of the data to replay

The data to be replayed can be provided from files coming from different sources:

- ☐ Recorded data by the Cortex HDR itself during a pass of a satellite (Report TM mode)
- ☐ Files containing User data
- ☐ PRN data or data from simulation files (BER analysis)

In the first case, all the data contained inside files recorded during a pass of a satellite and corresponding to a report are replayed in the chronological order. It is also possible to select only one specific file of a selected report.

In the second mode (User Data), in order to be easily selected from the Playback Window, the data files must be written in a sub-directory called "**PLAYBACK**" in a selected recording partition (example: TMA, TMB,... refer to the recording section). The files can be written directly in this directory, then it is necessary to change their names by 1, 2, 3 or inside sub-directories, in this second case the names of the subdirectories have to be named: 1, 2, 3... The format of the User files must be compatible with the standard HDR files, i.e. binary files.

During the playback, the recording function can be used simultaneously.

3.4.13 Multi-DPU processing

3.4.13.1 Generality

The Cortex HDR provides the multi-DPU (Data Processing Unit) capabilities.

The multi-DPU capabilities (4 DPU or 8 DPU) are **license dependent**.

The multi-DPU functionalities are implemented inside a specific demodulator menu (**menu 52**) and a specific test modulator menu (**menu 5**), which can be configured by parameters setting.

Basically this menu supports the following configurations described in the following figures:

- One channel QPSK demodulation with 2 DPU connected to the demodulator outputs (DPU A and DPU C)
- One channel QPSK or 16QAM demodulation with 4 DPU connected to the demodulator outputs (DPU A, DPU C, DPU E and DPU G)
- One channel 16QAM demodulation with 8 DPU connected to the demodulator outputs (DPU A, DPU B, DPU C, DPU D, DPU E, DPU F, DPU G and DPU H)
- One channel QPSK or 16QAM demodulation with 1 DPU (DPU A) connected to the demodulator merged output, essentially used for BER measurement purpose

For each configuration, the input is done at 720 MHz IF level, the demodulated and processed data at the output of each DPU can be recorded inside the internal hard disks of the unit. The data can be recovered using FTP.

For each possible configuration, the test modulator can provide:

- the symmetrical functions as described in the following figures
- or a single data channel test modulation as described in the last figure

The differential decoding is a NRZ-M to NRZ-L decoding

Each DPU (Data Processing Unit) consists in the following CCSDS compliant functions:

- a frame synchronizer
- a de-randomizer
- a Reed Solomon RS(223,255) decoder with supported interleaving and virtual filling
- a VCDU detector

In the different configurations, each function (differential coder/decoder, randomiser/de-randomiser, CCSDS encoder/decoder can be enabled or disabled.

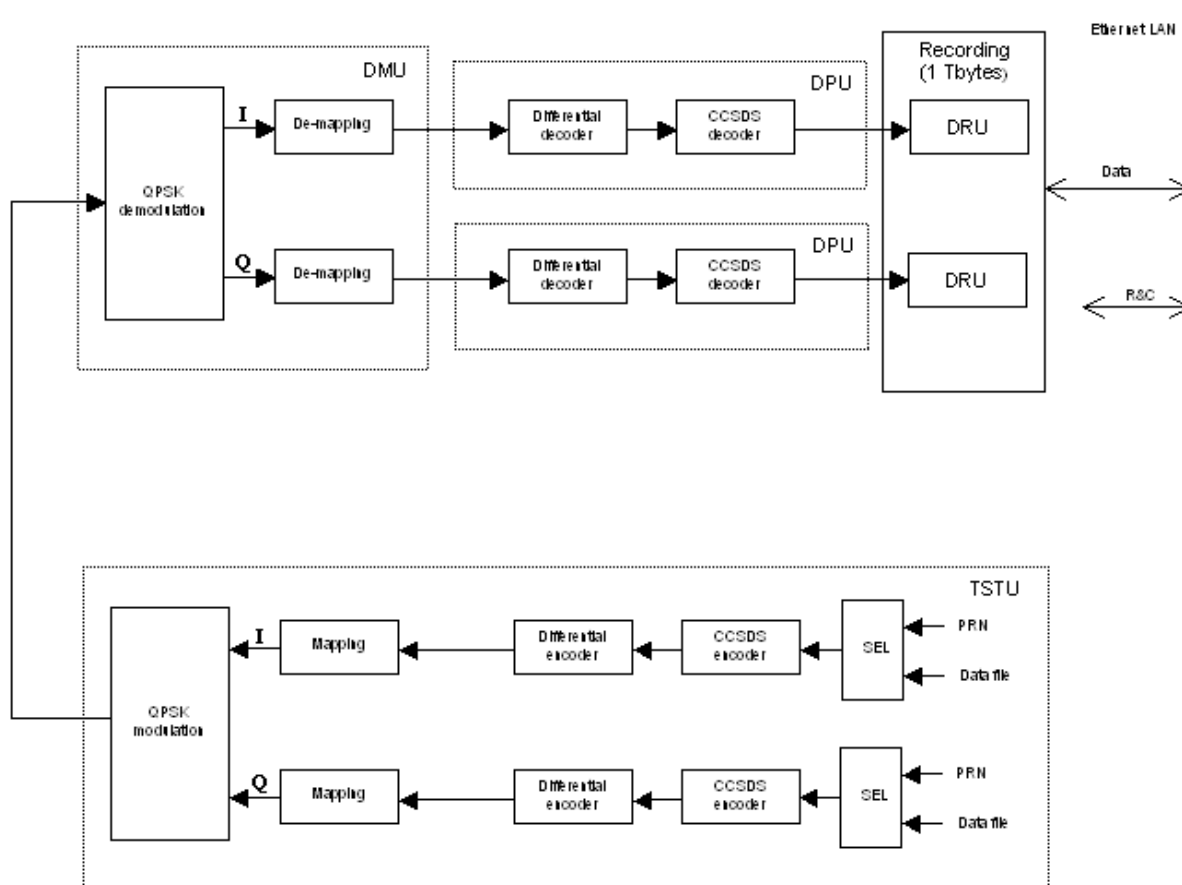


Figure 47: QPSK demodulation with 2 DPU

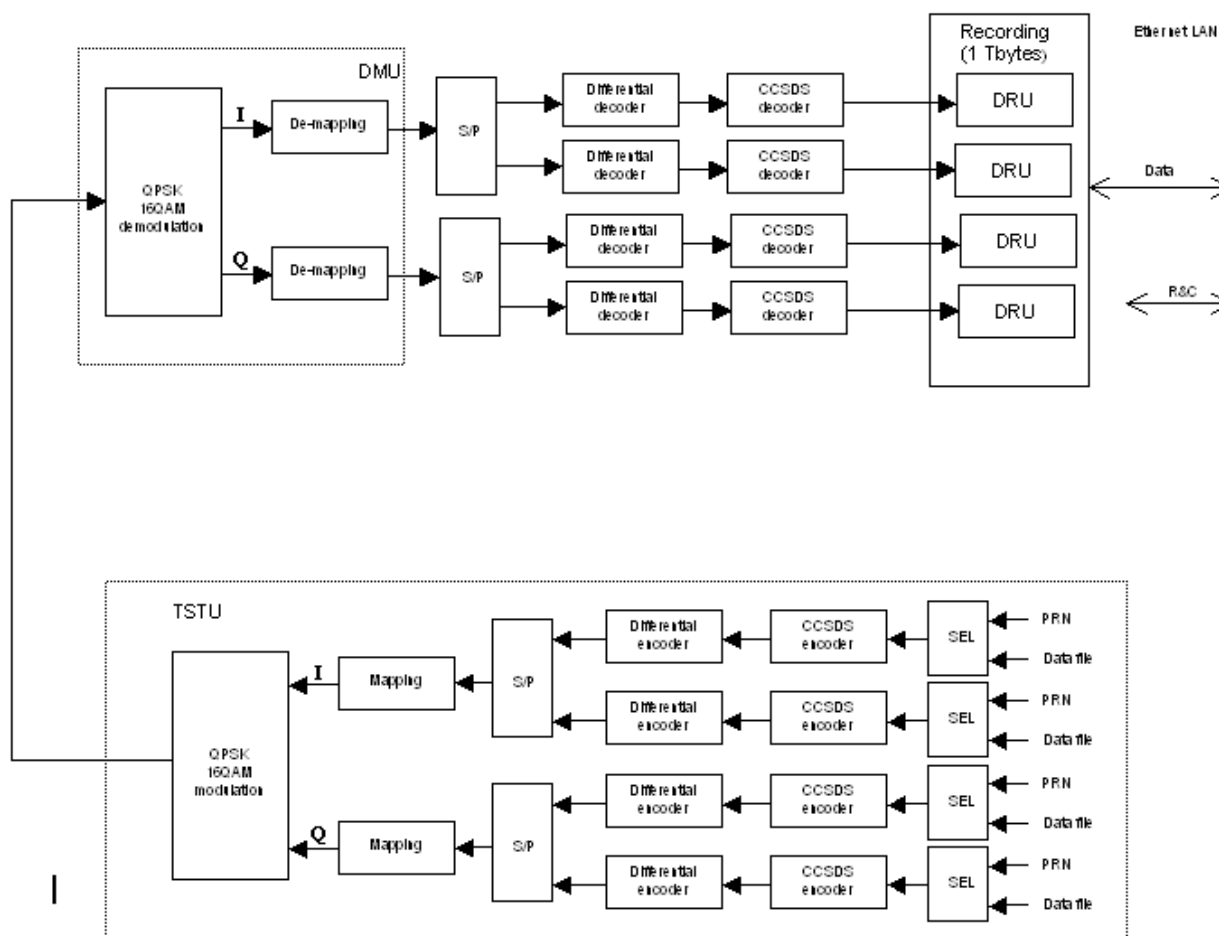


Figure 48: QPSK or 16QAM demodulation with 4 DPU

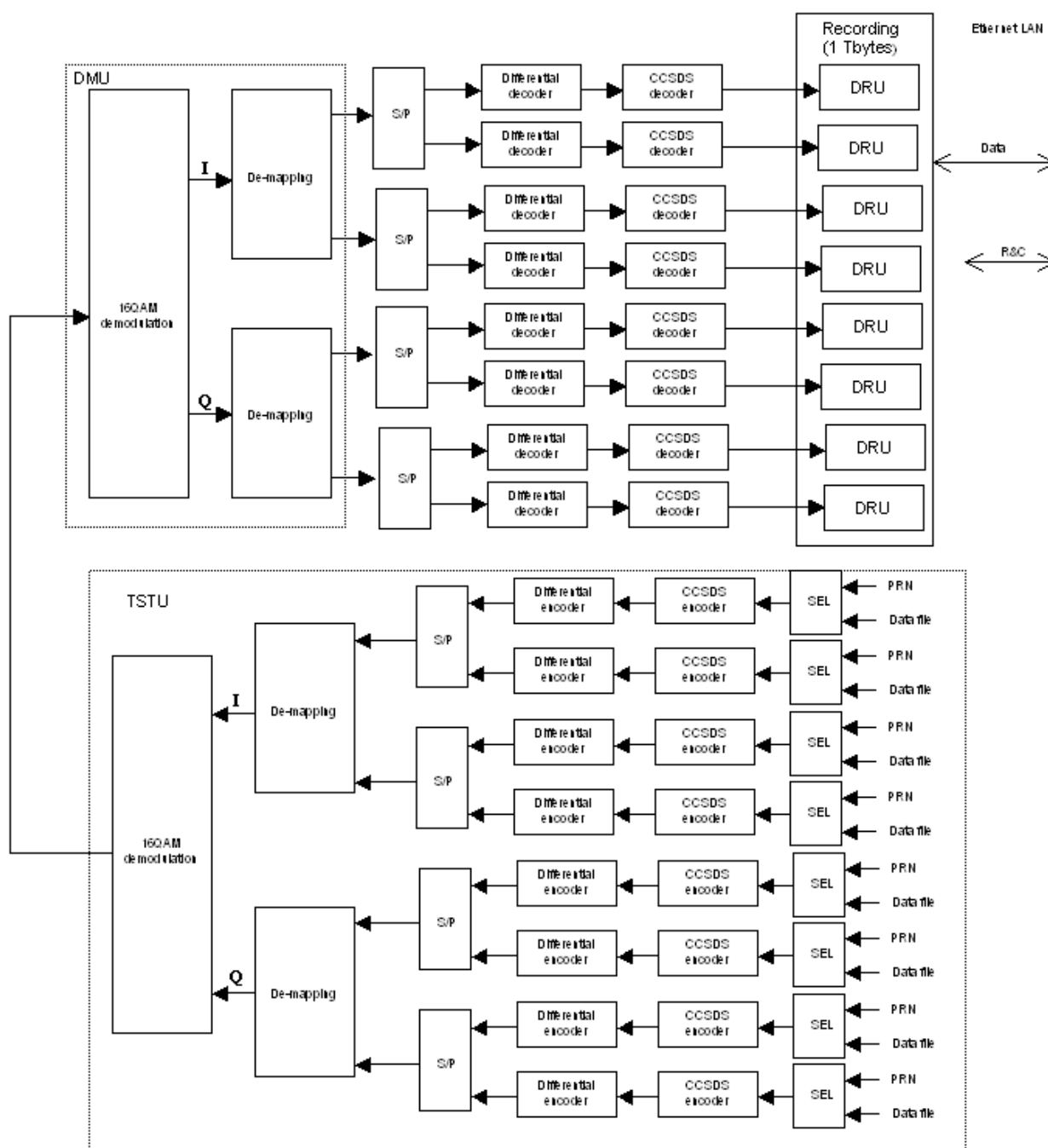


Figure 49: 16QAM demodulation with 8 DPU

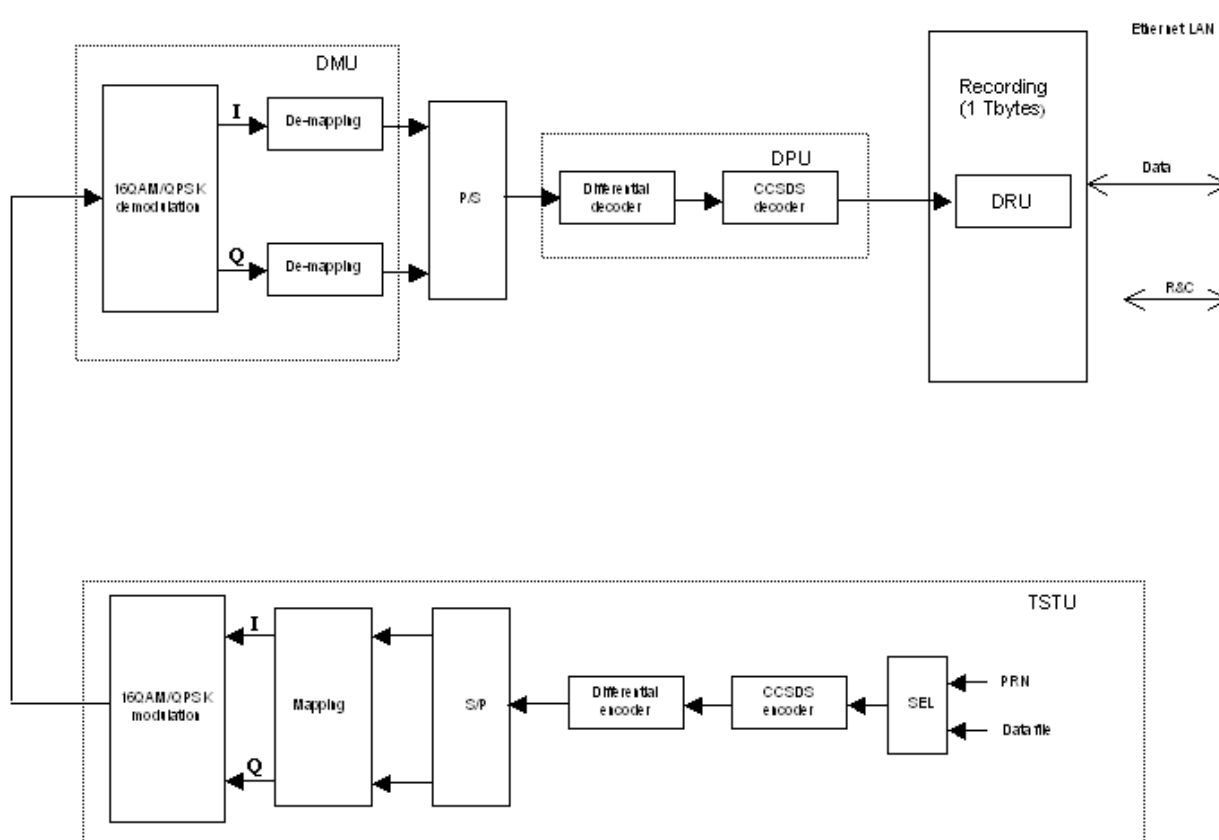


Figure 50: QPSK or 16QAM demodulation with 1 DPU

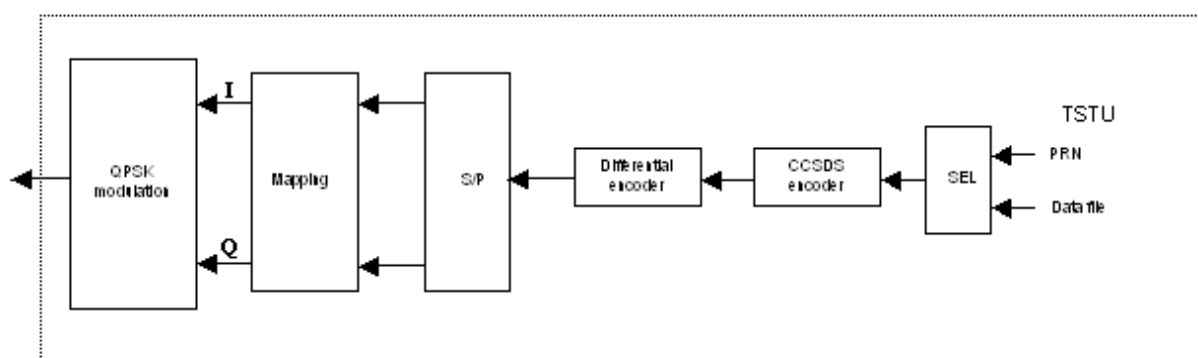


Figure 51: QPSK demodulation with 2 DPU

3.4.13.2 Ambiguity resolution by the DPUs

The ambiguity resolution is done by a specific process consisting in analysing the timing of the received CADU at the input of the DPUs and allowing a feed-back towards the DMU to route the data flows to the right DPUs. For that, The DPUs expect to receive the CADUs with a specific arrangement.

For example with the 16 QAM – 8 DPU mode, the received data flows at the inputs of the DPUs is given in the following figure. An offset of 1 byte is inserted between the successive CADUs.

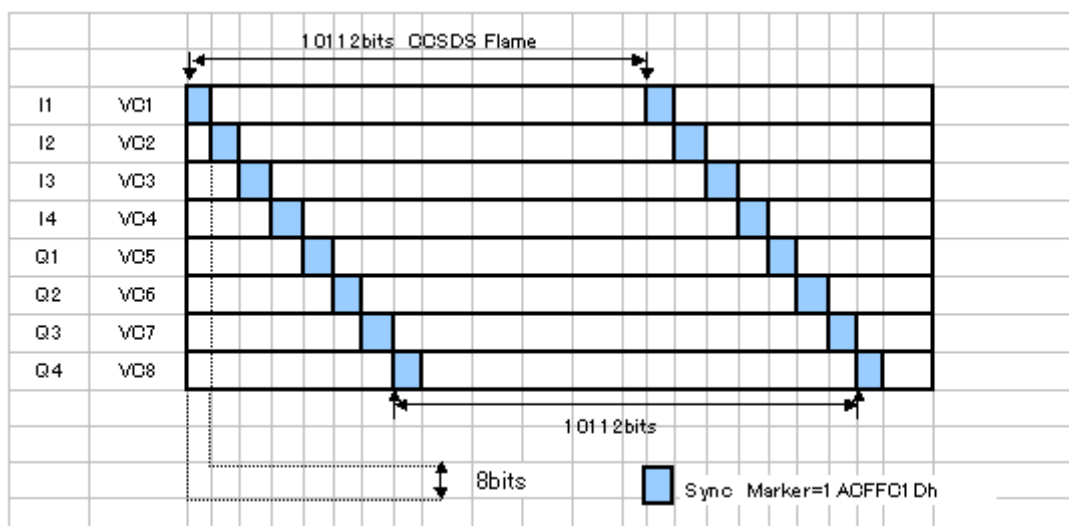


Figure 52: 8 DPU Data Flow for automatic ambiguity resolution

An identical mechanism is implemented for the QPSK – 4 DPU mode and the QPSK – 2 DPU mode.

The following tables give the correspondence between the offsets and the DPUs for the different modes. For each mode, two mapping are today available and selectable from the GUI (parameter: **DPU Map** in the DPU CADU window).

16QAM – 8 DPU, DPU Map = 0

Bit number	odd-b1	even-b1	odd-b2	even-b2	odd-b3	even-b3	odd-b4	even-b4
DPU-X	DPU-A	DPU-C	DPU-E	DPU-G	DPU-B	DPU-D	DPU-F	DPU-H
Offset (bytes)	0	1	2	3	4	5	6	7

16QAM – 8 DPU, DPU Map = 1

Bit number	odd-b1	odd-b3	even-b1	even-b3	odd-b2	odd-b4	even-b2	even-b4
DPU-X	DPU-A	DPU-C	DPU-E	DPU-G	DPU-B	DPU-D	DPU-F	DPU-H
Offset (bytes)	0	4	1	5	2	6	3	7

QPSK – 4 DPU, DPU Map = 0

Bit number	odd-b1	even-b1	odd-b2*	even-b2*
DPU-X	DPU-A	DPU-C	DPU-E	DPU-G
Offset (bytes)	0	1	2	3

QPSK – 4 DPU, DPU Map = 1

Bit number	odd-b1	odd-b2	even-b1	even-b2
DPU-X	DPU-A	DPU-C	DPU-E	DPU-G
Offset (bytes)	0	4	1	5

QPSK – 2 DPU, DPU Map = 0 (No ambiguity resolution process)

Bit number	b1	b2
DPU-X	DPU-A	DPU-C
Offset (bytes)	N/A	N/A

QPSK – 2 DPU, DPU Map = 1

Bit number	b1	b2
DPU-X	DPU-A	DPU-C
Offset (bytes)	0	4

3.4.13.3 BER measurements

The BER can be done at the DMU (Demodulator Unit) outputs and at each DPU (Data Processing) output. The BER can be done at different levels inside the DPU by enabling or not the different operations (Differential encoder/decoder, Frame synchronizer, CCSDS encoder/decoder).

3.4.13.4 Data output

Data at the output of **each** DPU can be recorded in specific directories of the internal hard disk named "T" or/and output at the ECL connectors as a single data stream with its clock.

The recording directories architecture is the following:

The **report files** are recorded in:

T:\DiskController_1_00\Report_TMA for DPU A
T:\DiskController_1_00\Report_TMB for DPU B
T:\DiskController_1_00\Report_TMC for DPU C
T:\DiskController_1_00\Report_TMD for DPU D
T:\DiskController_1_00\Report_TME for DPU E
T:\DiskController_1_00\Report_TMF for DPU F
T:\DiskController_1_00\Report_TMG for DPU G
T:\DiskController_1_00\Report_TMH for DPU H

The **data files** are recorded in:

T:\DiskController_1_01\.... for DPU A
T:\DiskController_1_03\.... for DPU C
T:\DiskController_1_05\.... for DPU E
T:\DiskController_1_07\.... for DPU G
T:\DiskController_1_02\.... for DPU B
T:\DiskController_1_04\.... for DPU D
T:\DiskController_1_06\.... for DPU F
T:\DiskController_1_08\.... for DPU H

If the CCSDS Virtual Channel sorting is disabled (TFP window), the data files are recorded inside the sub-directories CV_XX of each directory.

If the CCSDS Virtual Channel sorting is enabled, the data files are recorded inside the specific sub-directories: CV_NN with NN = 00 to 63 being the Virtual Channel ID number.

Multi-DPU outputs on ECL connectors are described in chapter 2.

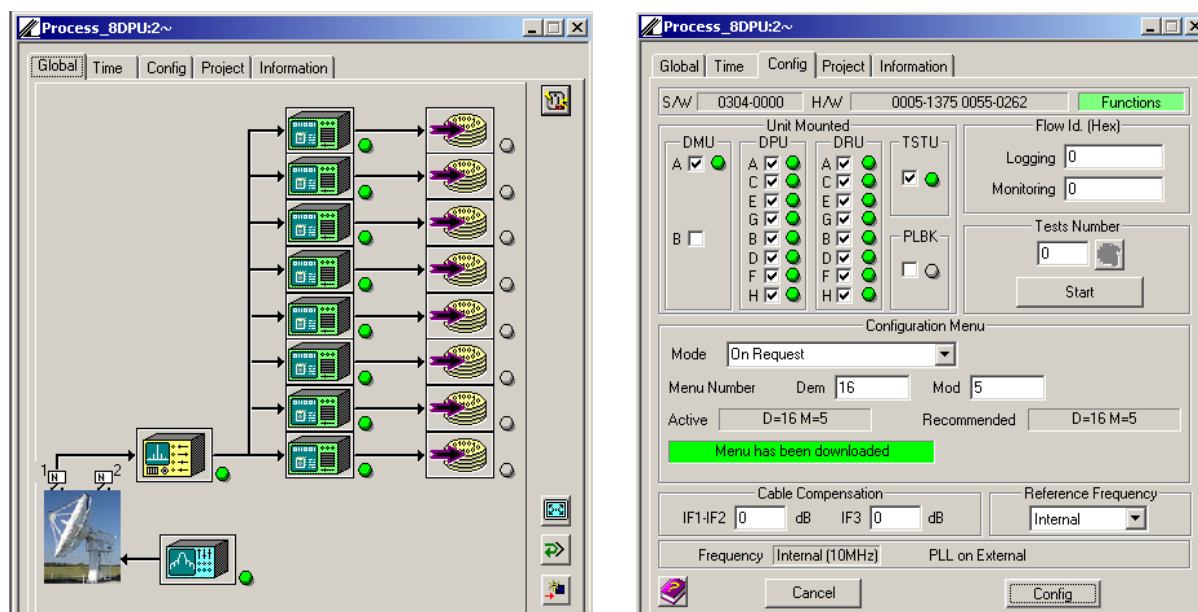
3.4.13.5 Guide for multi DPU Operation

CONFIGURATION

The Configuration window enables to mount and connect up to 8 DPU (Data Processing Unit) with the DMU (Demodulation Unit A).

From the following windows examples it can be seen that the hardware menu (FPGA basic menu) number is **5 for the test modulator and 16 for the demodulator**. These menus support the multi-DPU processing.

In the following first example, the 8 DPU and associated 8 DRU (Data Recording Unit) are enabled. This selection corresponds to the 16QAM – 8DPU configuration.



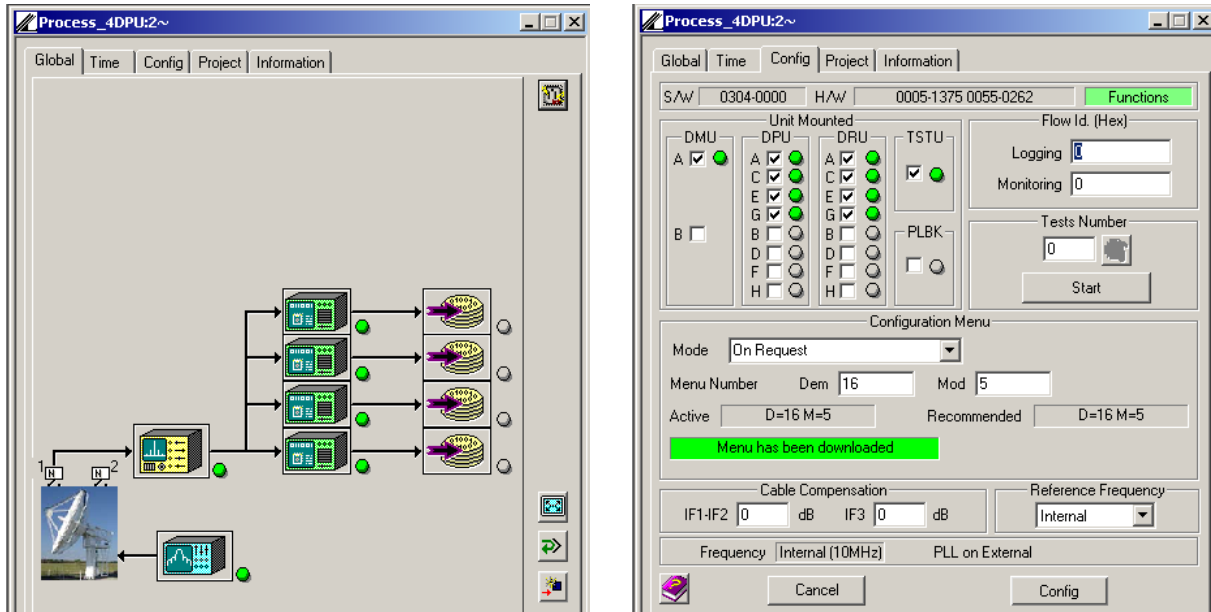
16QAM – 8DPU Configuration

In the following second example, 4 DPU and associated 4 DRU are enabled. This selection corresponds to the QPSK or 16QAM – 4DPU configuration.

Remark that with the 4 DPU configuration, must be selected: DPU and DRU A, C, E and G.

For the standard QPSK- 2 DPU configuration, must be selected only DPU and DRU A and C .

Only one DMU (DMU A) is available with this demodulator hardware menu number 16.



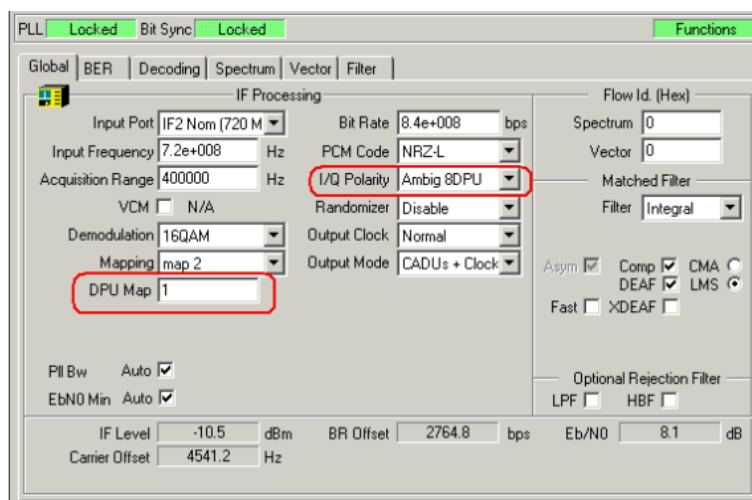
QPSK– 4DPU Configuration

DEMODULATION UNIT

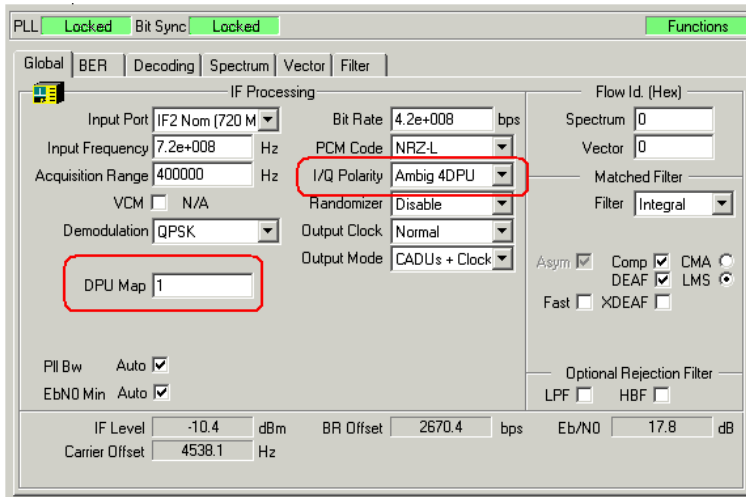
The demodulation function contains a specific feature, this is at this point of the data processing chain that the data flow is arranged in order to meet the 8 bits symbol shift special feature for ambiguity resolution purpose. To properly work it receives the frame synchronisation signals from the “Frame Sync”. This way it can reorder bits to route the right channel to the right DPU.

The configuration field in the MCS is the I/Q Polarity filed. If the used configuration used is 16QAM, 8 channels we must select the “**Ambig 8DPU**” item, if the used configuration used is QPSK 4 channels we must select the “**Ambig 4DPU**” item and if the used configuration used is QPSK 2channels we must select the “**Ambig DPUA+C**” item.

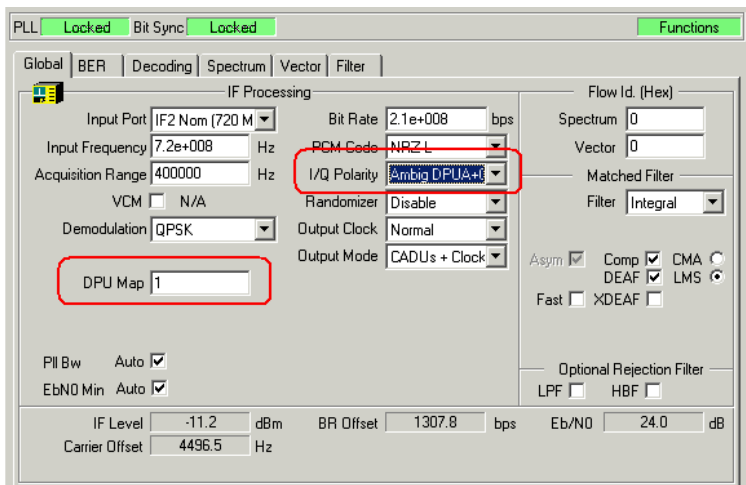
The setting of the demodulator for the 16QAM 8 DPU mode is the following:



The setting of the demodulator for the QPSK 4 DPU mode is the following:



The setting of the demodulator for the QPSK 2 DPU mode is the following:



Remark 1: The ambiguity resolution for multi DPU works only if all of the concerned channels are valid (e.g. the 2, 4 or 8 Frame sync are locked), if one of these channels becomes and stays unlocked during equivalent time of 15 frames, the ambiguity resolution state machine will consider itself in a bad ambiguity resolution state and then will search for a new one. As a result, if one of the channels does not work broken for any reason, it will impact all the channels.

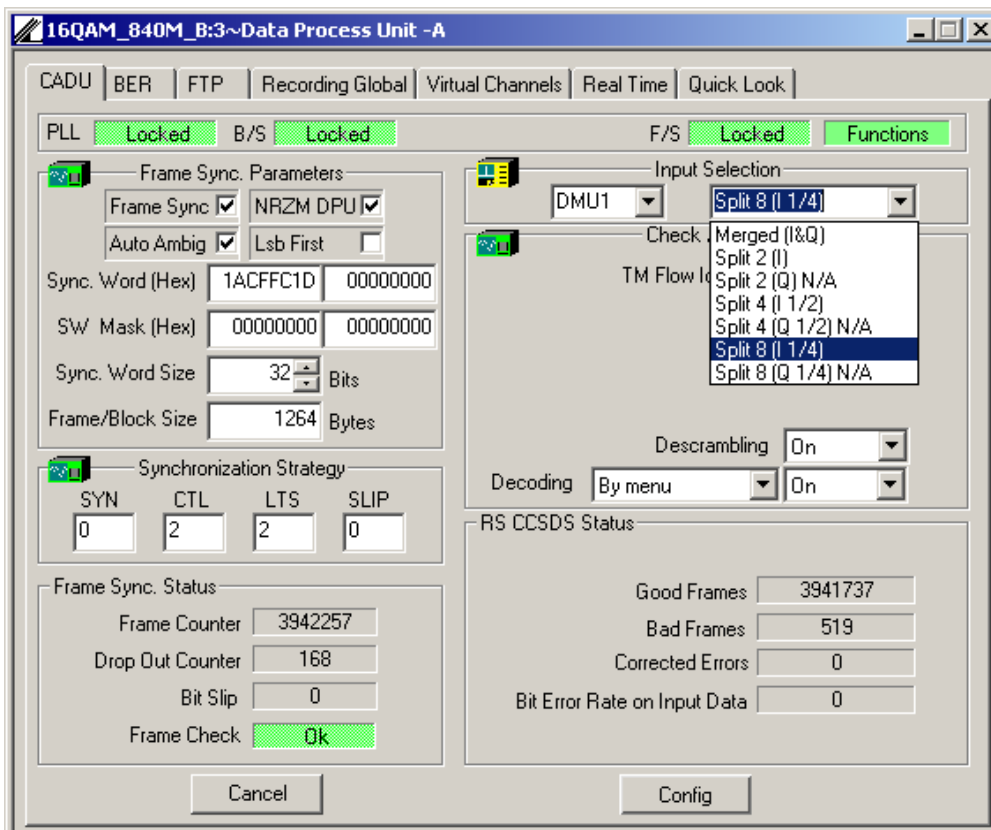
Remark 2: The mode “8DPU Ambig” does work only for 16 QAM's mapping: map 2.

DATA PROCESSING UNIT

For each DPU, a specific parameter (Input Selection) must be set. The role of this parameter is to split the data flow coming from the demodulator and depending on the DPU configuration. This is at this stage that the data flow is split in 2, 4 or 8 depending on the configuration needed.

The configuration fields in the MCS are those corresponding to “**Input selection**” section.

The NRZ-M to NRZ-L decoding function is activated in the MCS by ticking the “**NRZM DPU**” field.

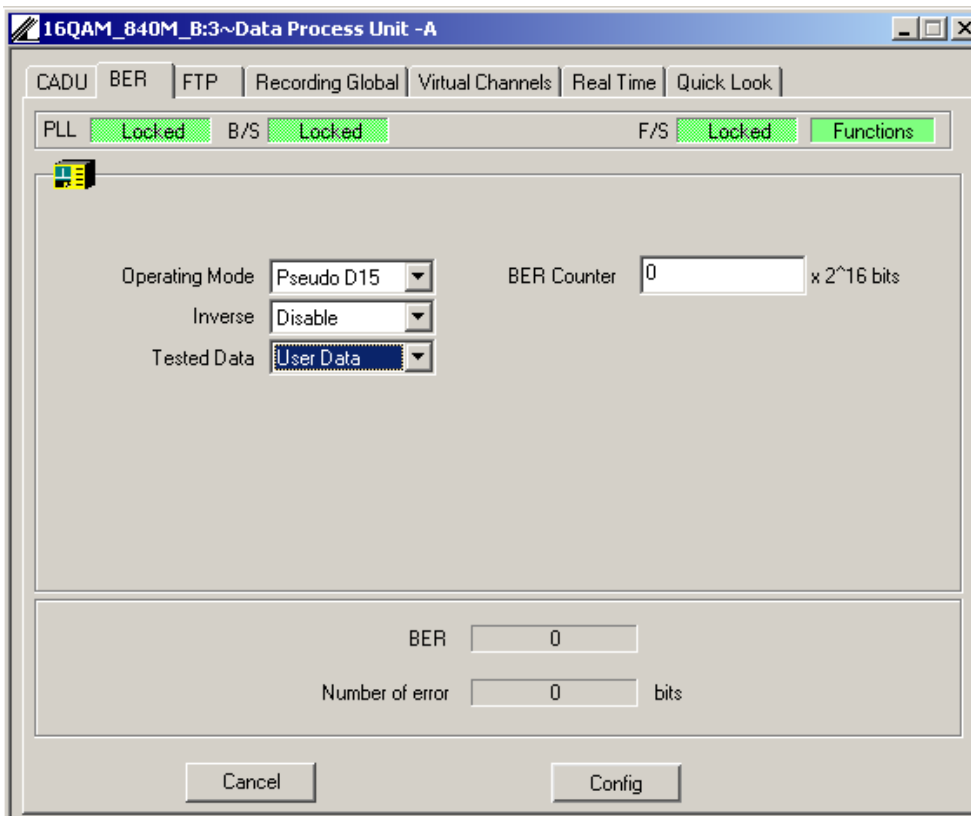


BER TESTER

It is possible to perform BER testing at the output of each DPU.

The PRN tester has been added at the DPU level. It can be configured via the MCS "Operating mode" field:

- Possibility to test an inverse PRN has been added too (it is done by inverse the binary flow before the PRN tester). It can be configured via the MCS "Inverse" field.
- Possibility to test whole data or only user data has been added. "Whole data" consists in the entire data flow with the Synchronization word and the RS parity bytes. "User data" excludes synchronization word and RS parity bytes. It can be configured by the field "Tested Data".
- Possibility to test only a portion of the CADU frame. The user can define a portion of the frame to be taken into account for BER calculation in a ASCII file stored on the hard drive.



The screenshot shows a software window titled "16QAM_840M_B:3~Data Process Unit -A". It has several tabs: CADU, BER (selected), FTP, Recording Global, Virtual Channels, Real Time, and Quick Look. Below the tabs, there are status indicators for PLL (Locked), B/S (Locked), F/S (Locked), and a Functions button. The main area contains configuration options for BER testing: Operating Mode (Pseudo D15), Inverse (Disable), and Tested Data (User Data). A BER Counter is shown with a value of 0 and a multiplier of $\times 2^{16}$ bits. At the bottom, there are fields for BER (0) and Number of error (0) bits. The window has Cancel and Config buttons at the bottom.

3.5 Polarization Combining

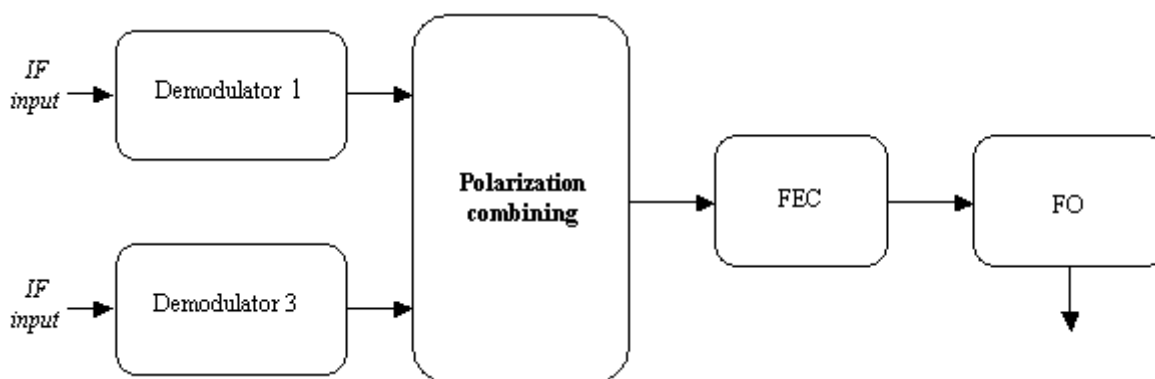
3.5.1 General

The CORTEX HDR allows to combine two IF signals from different polarizations in post-detection mode. Two combining modes are available:

- Automatic selection of the best channel
- Combining of the two channels, through optimal weighting. In this case up to 3 dB gain can be achieved when both signals have the same E_b/N_0 .

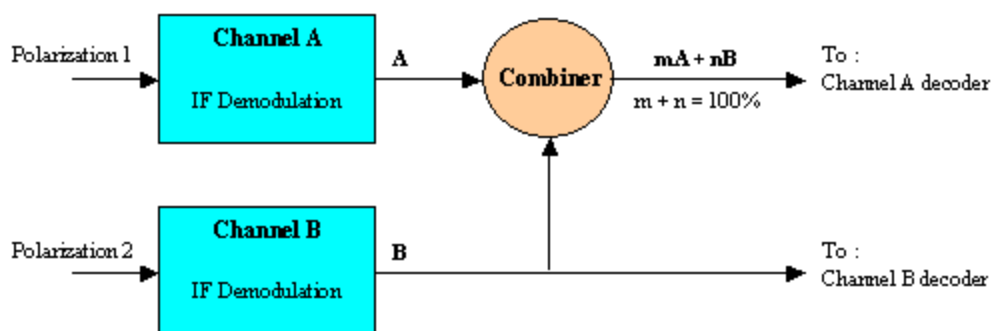
The combining system, also named Diversity Combining Unit (DCU), receives data from two demodulators. It resolves phase ambiguity between the two channels, and deals with the different locking states possible. Its output feeds a FEC device, followed by a Format Output.

Note: Only DMU1 and DMU3 can be combined in the HDR.



3.5.2 Data Path

When active, the combining system output is routed to the FEC input of the first demodulator while the data path of second demodulator remains unchanged.



Three choices are possible concerning the output routing:

- m = 100 %, n = 0 %: channel A is directly routed to the output. No combining is performed.
- m = 0 %, n = 100 %: channel B is directly routed to the output. No combining is performed.
- m and n are dynamically adjusted by the combining system. This is described in the "Signal processing" section.

3.5.3 Signal Processing

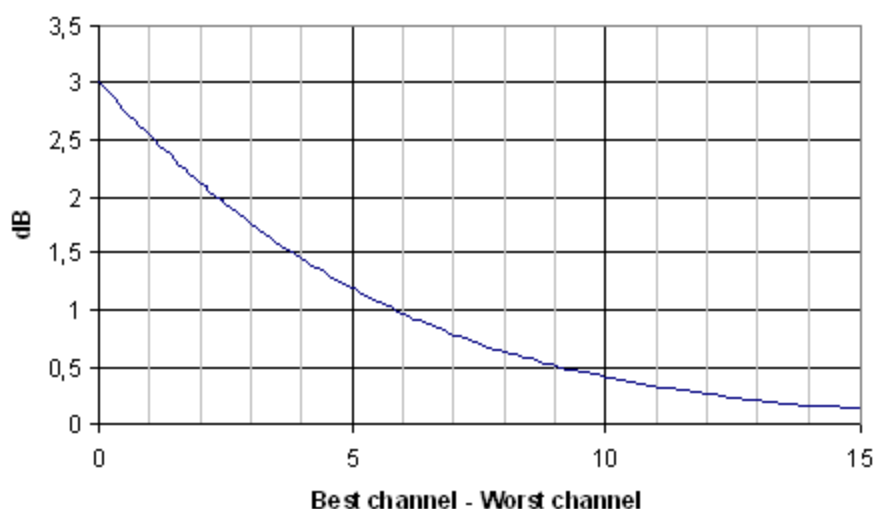
When both channels are locked, the combining system identifies and compensates the relative phase ambiguity before combining. The supported modulations are QPSK, OQPSK and 8PSK.

Two combining modes exist:

- **Best channel:** only the signal of the channel with the best E_b/N_0 is fed through the combining system.
- **Combining:** signals of both channels are combined using optimal weighting. In this case, the combining gain versus the best channel is:

$$\left(\frac{E_b}{N_0}\right)_{(dB)_{combining}} - \left(\frac{E_b}{N_0}\right)_{(dB)_{best_channel}} = 10 \log_{10} \left(1 + 10^{\frac{Level(dBm)_{worst_channel} - Level(dBm)_{best_channel}}{10}} \right)$$

Combining gain



The relative amplitude of a channel (A or B) is given by:

$$\frac{Amplitude_Channel}{Amplitude_Channel_A + Amplitude_Channel_B} \times 100$$

The sum of the relative signal amplitudes is 100. The best case (3-dB gain) is achieved when both IF Receivers are locked onto equal-strength signals.

Moreover, the system can be in different states:

- **N/A** if Channel A or Channel B has been chosen as output of the DCU.
- **Unlocked** if the output is set to combining, and both demodulators are unlocked.
- **Single channel** if only one demodulator is locked.
- **Acquisition** if both demodulators are locked, and the system searches for a fit between both signals. The system identifies the differential ambiguity between both signals.
- **Locked** if combining is performed normally.
- **Check** if the system verifies the differential ambiguity found during the acquisition phase. This happens when the observed combining gain is lower than the theory by more than 0.3dB. This is also the normal working state when "Best channel" mode is selected.

Specific attention should be paid to the following points otherwise warnings will be issued:

- modulations must be identical, and must be one of the following: QPSK, OQPSK, 8PSK.
- bitrates must be identical.
- PCM codes must be identical.
- IF inputs must be different (no gain can be achieved with the same signal feeding both inputs).
- Configuration of matched filters must be identical (data paths before the DCU must be equivalent).

3.6 Test modulation

The test modulator allows to test the carrier acquisition, demodulation and decoding processes:

- **Functional tests:** Telemetry reception and processing without noise (IF demodulation, bit synchronization, BER measurement)
- **BER tests:** BER measurement in the presence of noise. Bit-by-bit comparison of the simulated PCM to the regenerated PCM from the bit synchronizer (File mode or pseudo random mode).

3.6.1 Data Generation

The Test Modulator allows generation of test data with the following scheme:

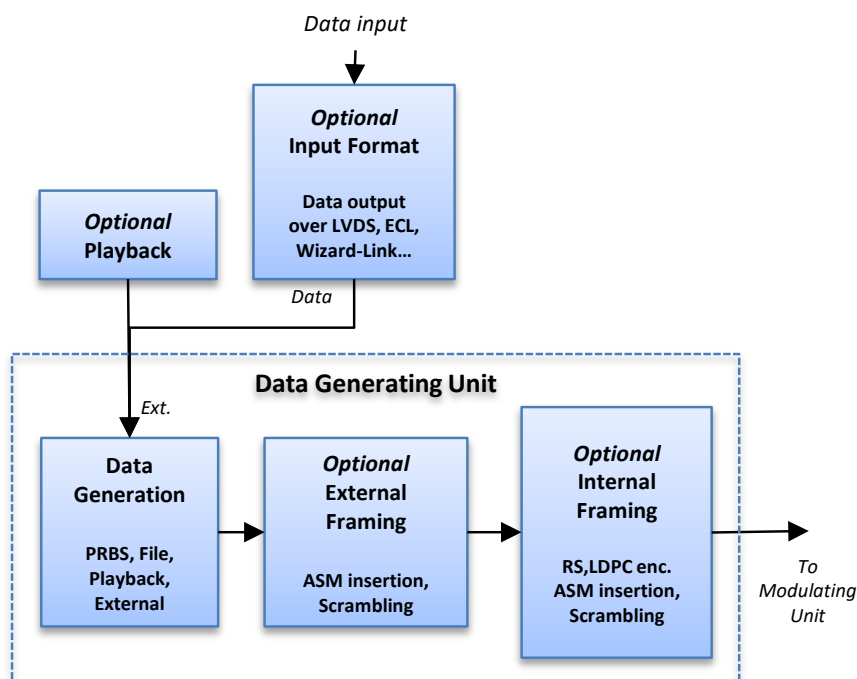


Figure 53: Data Generation Scheme

3.6.1.1 Data Sources

The Test Modulator allows to generate (or encapsulate) test data from the following sources.

Four data sources are available:

- Pseudo-random binary sequences (PRBS for BER testing), of order 7, 10, 11, 15, 23 and 31,
- Data from hexadecimal-coded files on the hard disk (up to 32768 bytes)
- External data (typically PCM + clock).
- Playback of high size binary files from attached storage (either internal or external)

3.6.1.1.1 Pseudo Random Data

Six PRN generators (Pseudo Random Noise) providing pseudo-random bit sequences of programmable length and period can be selected for BER analysis. Data sequences are sequences of maximum length which are generated by means of feedback shift registers.

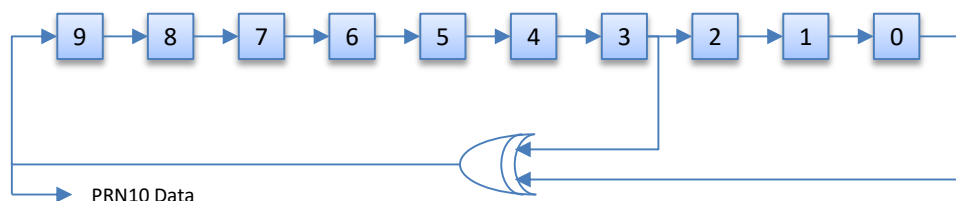


Figure 54: 10-bit generator with feedbacks after register 3 and 0

Next table describes all PRBS generators available:

PRBS Generators	Length in bits	Feedback after	Comments
PRBS 7	$2^7 - 1 = 127\ 127$	Register 1, 0	
PRBS 10	$2^7 - 1 = 127\ 1,023$	Register 3, 0	
PRBS 11	$2^7 - 1 = 127\ 2,047$	Register 2, 0	
PRBS 15	$2^7 - 1 = 127\ 32,767$	Register 1, 0	PRBS 15 is configured according to CCITT Rec. 0.151/152/153
PRBS 23	$2^7 - 1 = 127\ 8,388,607$	Register 5, 0	PRBS 23 is configured according to CCITT Rec. 0.151/152/153
PRBS 31	$2^7 - 1 = 127\ 2,147,483,647$	Register 3, 0	

Table 13: PRBS Generators for BER Analysis

3.6.1.1.2 Data from File

3.6.1.1.2.1 File Format

Data can be retrieved from files on the disk.

Files are hexadecimal-coded and can be edited using *Notepad* or any other text editor.

Access path to the simulated telemetry files is:

D:\ZDS\Data\HDR\SPS\BER\

3.6.1.1.2.2 Cycle Length

Parameter **Cycle Length** specifies which part of the file must be modulated. It can be set to any value from 32 to $2^{22} - 1 = 4,194,303$ bytes.

3.6.1.1.3 External Data & Clock (Input Format)

One should ensure the data & clock timing comply to the CORTEX HDR requirements (see Section 1.6.6).

- Data-to-clock timing requirements:
- Square 180° clock
- 50% duty cycle (worst case: 45/55)
- Tset_time < 1 ns
- Thold_time < 0.6 ns

The phase jitter should be as low as possible ($< \pm 1$ ns). One recommends connecting the input clock before selecting the External Data mode. In this mode, the external clock must be permanently provided.

3.6.1.1.4 Playback

Please refer to the dedicated Playback section 3.4.12 .

3.6.1.2 Internal and External framing

The test modulator supports up to two framing layers (see Figure 53: Data Generation Scheme):

- *External framing* which is the first framing encountered by the data. It can be ON or OFF.
- *Internal framing* which is the second framing encountered by the data, as it immediately follows the external framing. It can also be ON or OFF.

Both layers are asynchronous.

3.6.1.2.1 Framing features

Each framing supports:

- ASM insertion (CCSDS 131.0-B-2, up to 128 bits SW)
- Scrambling (CCSDS 131.0-B-2 or custom)
- FECF insertion (CCSDS 100.0-G-1, or custom)
- NRZ-M encoding

3.6.1.2.2 Framing and Cycle Length

When data are generated from a file, if the Cycle Length is compatible with the frame length i.e. if the actual number of user data needed for each frame (thus excluding the Synchronization Word, the Parity bits and the FECF) is equal to the Cycle Length, the first bit of the file is the first user bit of the frame.

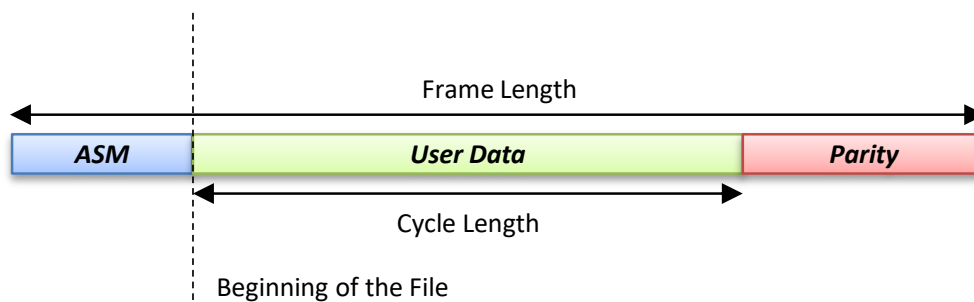


Figure 55: Frame aligned with File

3.6.1.2.3 Encoding

Data encoding is possible at the internal framing level. The following encoders are supported (according to CCSDS 131.0-B-2):

- RS (223,255), and RS (239,255)
- LDPC (8160,7136) or “NASA-shortened”. This code is known as LDPC 7/8 in the GUI.
- LDPC 1/2, 2/3, 4/5 with frame sizes of 1024, 4096 and 16384 bytes
- DVB and DVB DSNG

The test unit supports all the features of the demodulator. Please refer to section 3.4 for more detailed presentation of the supported features.

Due to their “modulation scheme” nature, DVB-S2 and SCCC are described in the Modulation chapter.

3.6.1.2.4 Scrambling

In addition to the CCSDS Scrambling, the Scrambling can be entirely customized in terms of:

- Initialization
- Generator polynomial
- Offset (in bits)

When the offset is set to a non-zero value n , it means that the scrambling process is reinitialized n bits after the beginning of the frame.

3.6.1.2.5 FECF

The FECF insertion can be entirely customized (Cf. CCSDS 100.0-G-1) in terms of:

- Initialization
- Generator polynomial

3.6.2 Modulation Unit

3.6.2.1 Main features

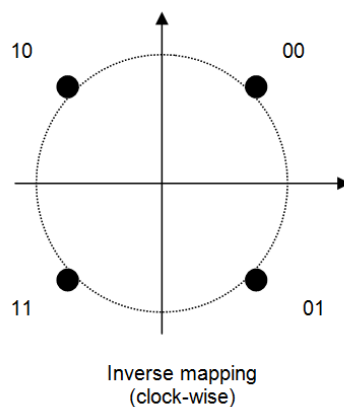
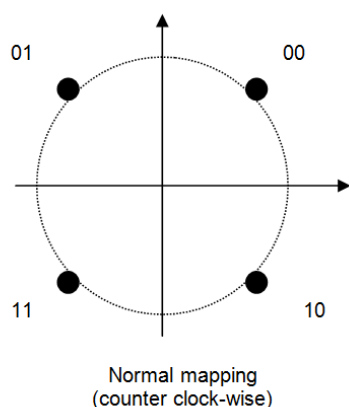
The data streams can be modulated with the following characteristics:

- Various basic modulation schemes: BPSK, (U/O)QPSK, 16APSK, 16QAM, 32APSK, 64APSK, GMSK. The modulation can be ON or OFF
- Carrier Frequency is configurable
- Advanced modulation schemes: DVB-S2 and SCCC(-X) (according to CCSDS 131.2-B-1 and CCSDS 131.20-O)
- 2SR and 4SR FIR pulse-shaping filter (typ. RRC and GMSK filter with adjustable roll-off/BT)
- NRZ-L/M/S, Bi-Phase and programmable DNRZ support
- Convolutional encoding schemes for (O)QPSK (single, dual, parallel up to 16, with many puncturing available) and 8PSK (TCM with rates 2/3, 2.25/3, 2.5/3, 2.75/3, 4D-DSNG)
- Symbol rate up to 2 Gsps depending on the configuration

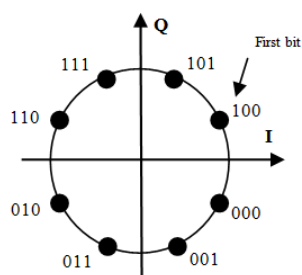
The test unit supports all the features of the demodulator. Please refer to section 3.3 for a complete overview of the Viterbi and puncturing schemes, TCM encoding and so on.

3.6.2.2 Constellation Mapping

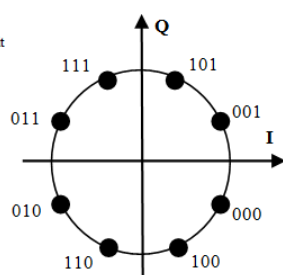
In QPSK mode, two mapping are supported:



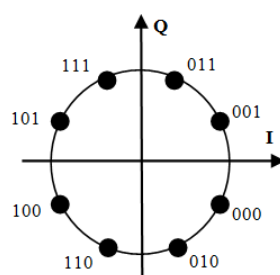
In 8PSK, the following mappings are supported:



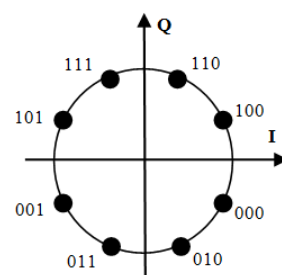
Inverted Gray 1



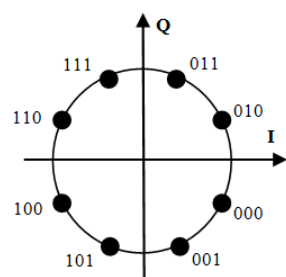
Inverted Gray 2



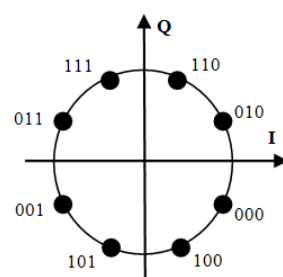
Inverted Gray 3



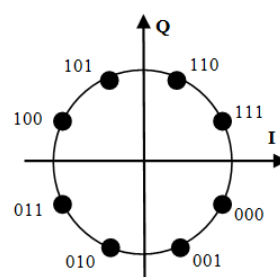
Inverted Gray 4



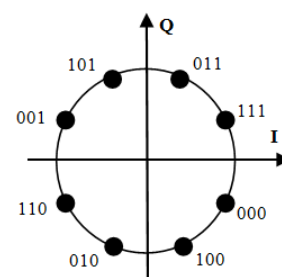
Inverted Gray 5



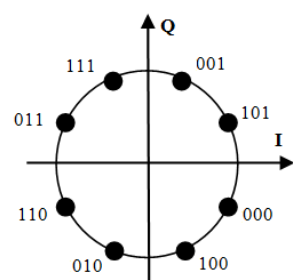
Inverted Gray 6



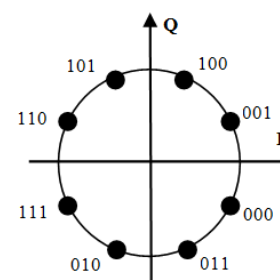
Inverted Natural 1



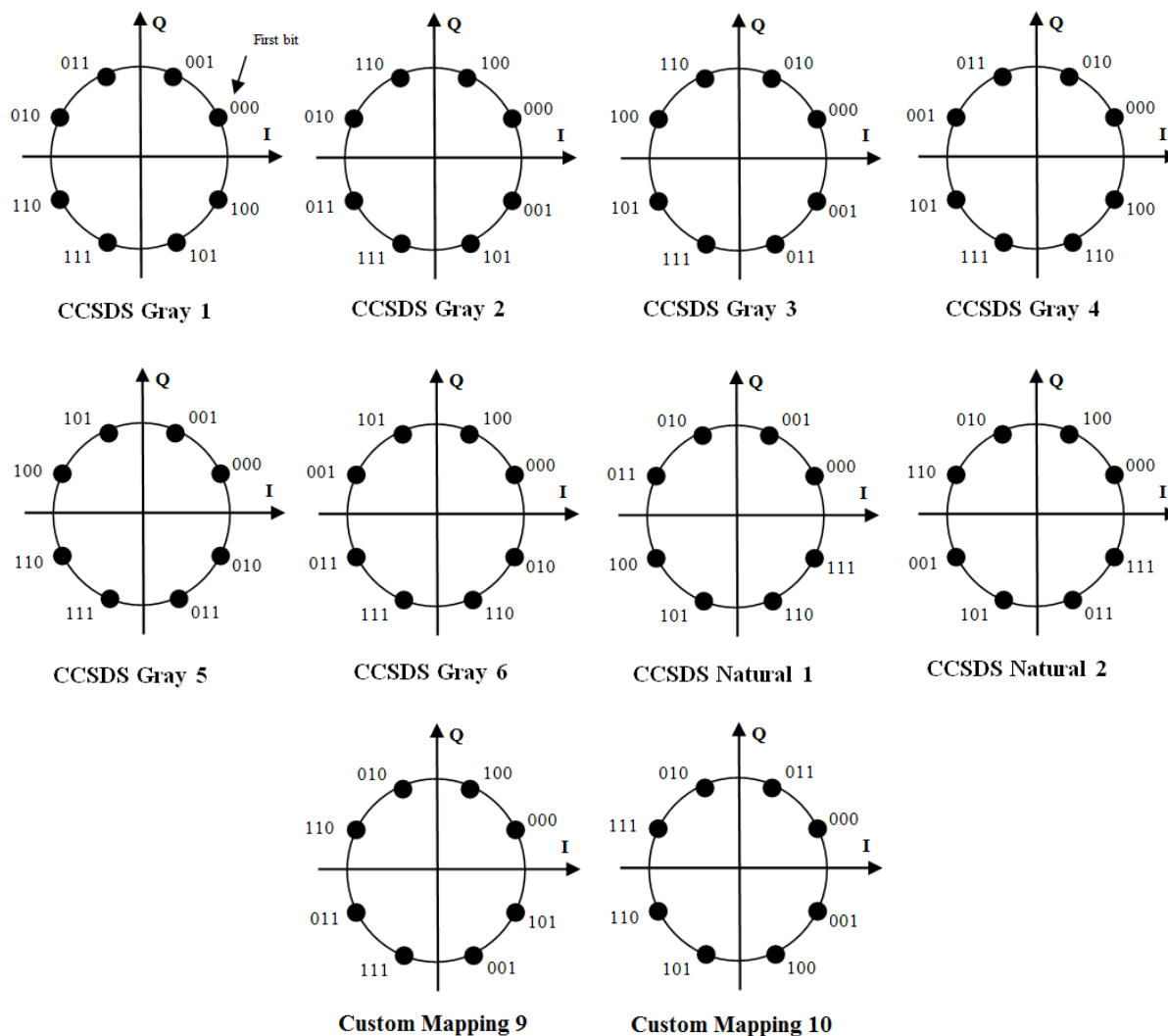
Inverted Natural 2



Inverted Mapping 9

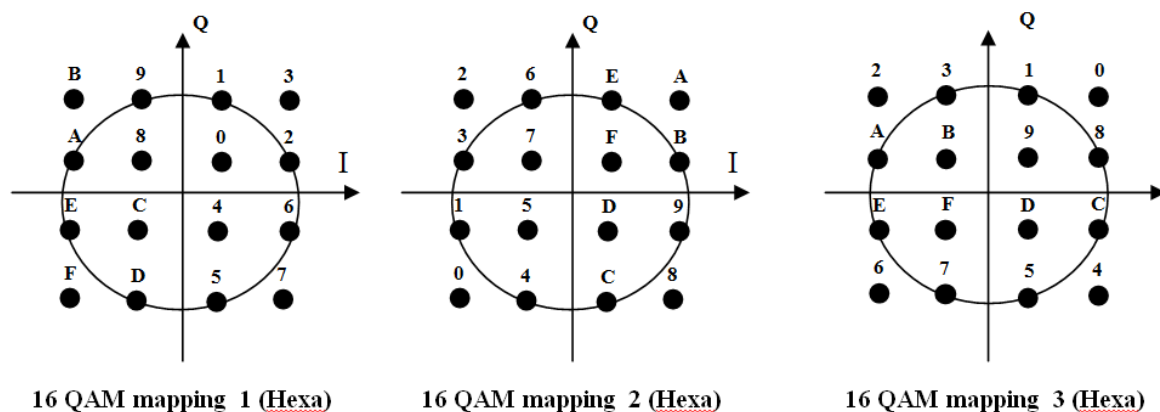


Inverted Mapping 10



Remark: The 8PSK are inverted in the revisions of the User Manual preceding the revision E9R5.

In 16QAM the following mappings are supported:



In 16APSK the following mappings are supported:

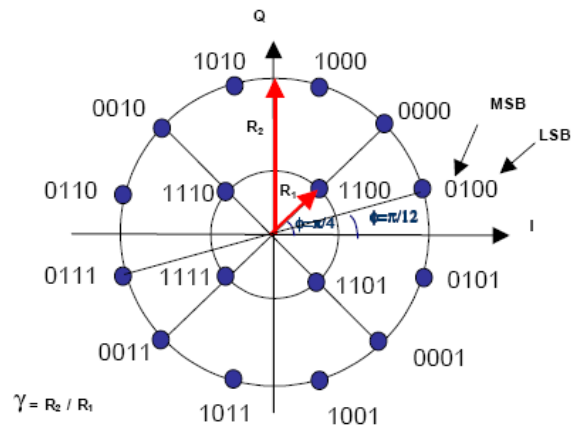


Figure 56: 16APSK Mapping 1

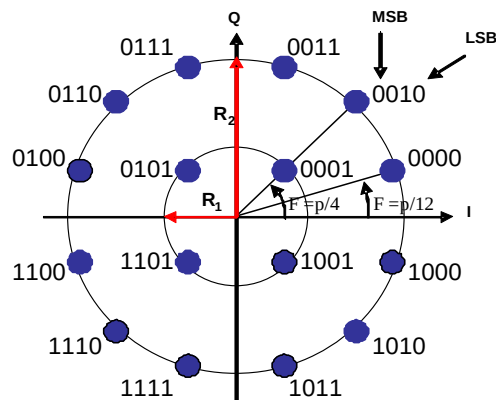
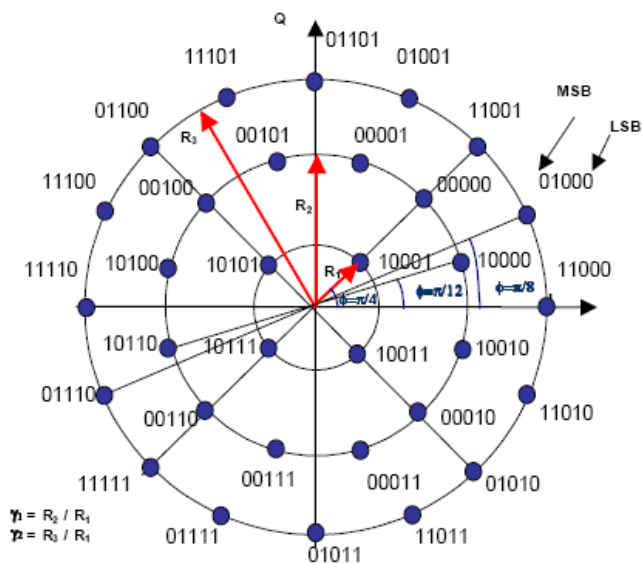
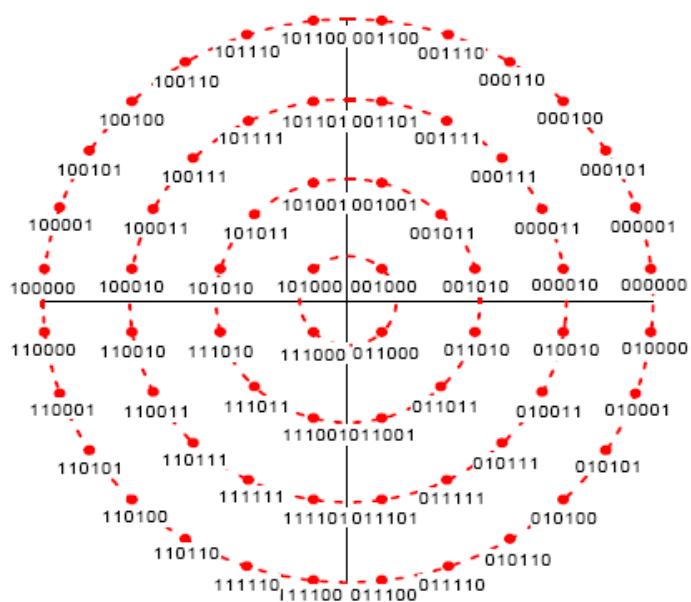


Figure 57: 16APSK Mapping 2

In 32APSK the following mapping is supported:



In 64APSK the following mapping is supported:



3.6.2.3 DNRZ Coding

Several DNRZ algorithms are possible for ambiguity resolution in QPSK and OQPSK modes. The DNRZ code (algorithm) value can be programmed via the Cortex HDR ICD parameter "DNRZ Code".

To select a particular DNRZ code, set the parameter "DNRZ Code" to a value between 0x0 and 0x18 (24_D) which refers to a pre-programmed type of DNRZ Code. These pre-programmed values are compatible with the pre-programmed values of DNRZ Code in the demodulator unit.

	DNRZ Code in QPSK	DNRZ Code in OQPSK
Code 1	CA53A3C5	3C3C6969
Code 2	AC35C5A3	C3C36969
Code 3	6A59A965	0FF0C3C3
Code 4	6C39C963	F00F3C3C
Code 5	C69363C9	CC33A5A5
Code 6	A69565A9	33CC5A5A
Code 7	C96393C6	3C3C9696
Code 8	9C36C693	C3C39696
Code 9	596A9A56	0FF03C3C
Code 10	5C3ACA53	F00FC3C3
Code 11	C5A353CA	CC335A5A
Code 12	95A6569A	33CCA5A5
Code 13	9A56A695	5A5A6969
Code 14	A96595A6	A5A56969
Code 15	3A5CAC35	C3C30FF0
Code 16	396C9C36	3C3CF00F
Code 17	93C6369C	AA55C3C3
Code 18	A3C535AC	55AA3C3C
Code 19	53CA3A5C	5A5A9696
Code 20	35AC5C3A	A5A59696
Code 21	63C9396C	3C3C0FF0
Code 22	65A9596A	C3C3F00F
Code 23	569A6A59	AA553C3C
Code 24	369C6C39	55AAC3C3

It is also possible to set directly the explicit DNRZ Code. When the parameter "DNRZ Code" to a value above 0x18, it is automatically assumed that the value is an explicit DNRZ Code.

Example: Code 6 QPSK

I*[n-1]	Q*[n-1]	I[n]	Q[n]	I*[n]	Q*[n]
0	0	0	0	1	1
0	0	0	1	0	0
0	0	1	0	1	0
0	0	1	1	0	1
0	1	0	0	1	0
0	1	0	1	0	1
0	1	1	0	0	0
0	1	1	1	1	1
1	0	0	0	0	1
1	0	0	1	1	0
1	0	1	0	1	1
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	1	1
1	1	1	0	0	1
1	1	1	1	1	0

The last two columns have to be read from bottom to top:

I_n : 1010 0110 1001 0101 = 0xA695

Q_n : 0110 0101 1010 1001 = 0x65A9

Then the explicit DNRZ Code is 0xA69565A9.

3.6.2.4 Multi-DGU support

The test modulator has firmware-dependent support of multiple data sources. This enables simulation of transmitting schemes with multiple interleaved logical data units. Here are a few examples of the available schemes:

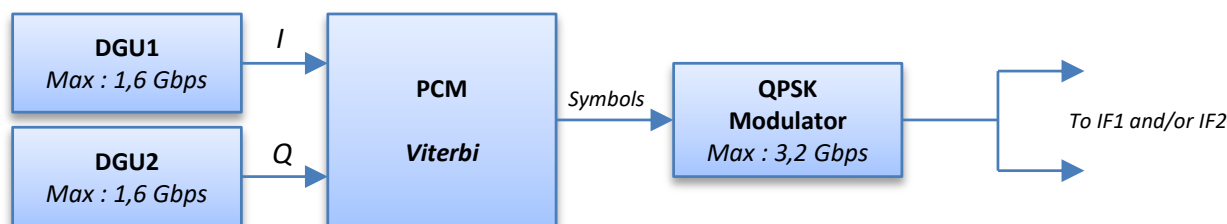


Figure 58: Example of 2 DGUs merging for simulation of independent I/Q transmission in QPSK

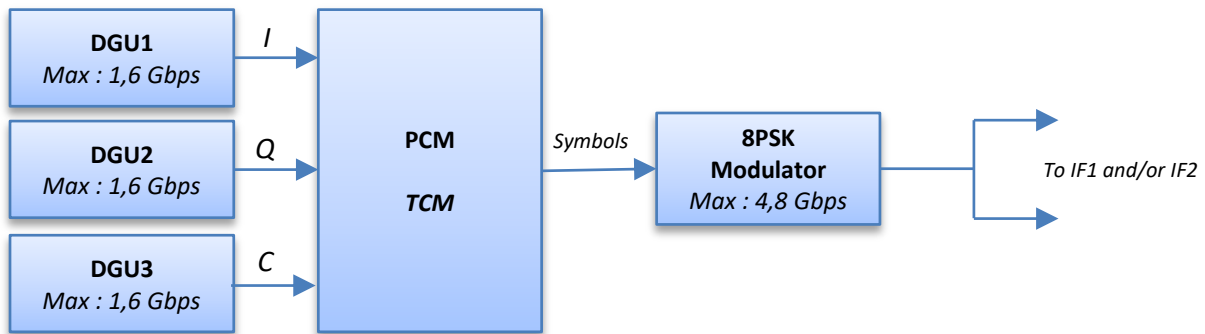


Figure 59: Example of 3 DGUs merging for simulation of independent I/Q/C transmission in 8PSK

The total number of data units that can be connected simultaneously to a modulating unit can be 1, 2, 3 or 8.

Given a modulation order n (ex: 3 in 8PSK):

- I/Q/C... synchronization is guaranteed only when the number of DGUs is equal to n
- the convention for bit to symbol mapping is: DGU1 $\rightarrow$ I, DGU2 $\rightarrow$ Q, DGU3 $\rightarrow$ C (or Z) and so on.

Multi-DGU support is enabled by adjusting the number of DGUs in the modulator's GUI or the TCP/IP interface.

Each data unit has an adjustable starting offset, in bits. This enables to offset multiple data streams with a predefined scheme in order to ease multi-channel ambiguity resolution at the receiving end.

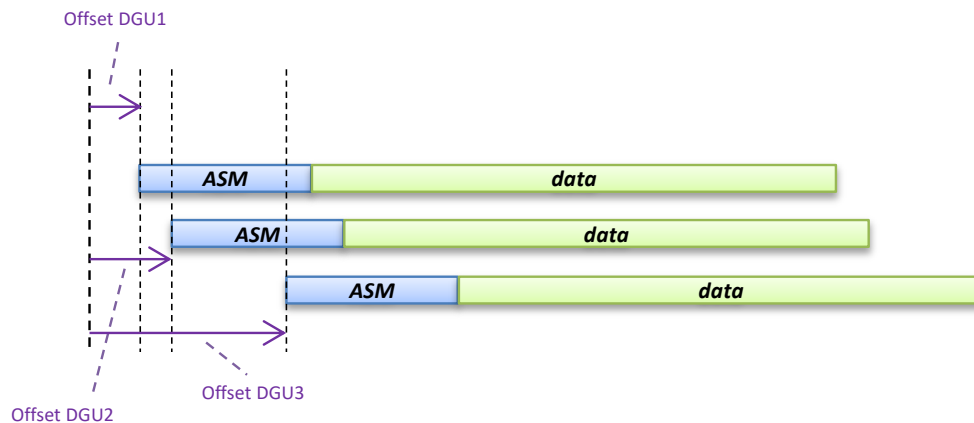


Figure 60: Example of 3 DGUs with different offsets

3.6.2.5 DVB-S2 & SCCC

The test modulator supports advanced modulation schemes such as SCCC and DVB-S2. These modes are entirely compliant to:

- The CCSDS specification 131.2-B-1 (SCCC)
- The CCSDS specification 131.20-O (SCCC-X)
- The functioning of the Demodulator Unit

These two modes support a VCM functioning.

VCM can be based on:

- **Frame:** The configuration (ACM, modulation, coding rate, pilots ...) is changed after a determined number of frames
- **Time:** The configuration is changed after a certain amount of times. Even in this mode, it is guaranteed that the configuration is not changed in the middle of a frame.

The sequences of the different configurations are described inside ASCII Files available in:

D:\ZDS\Data\HDR\SPS\VCM

Each file contains 1 line per ACM.

It begins with #frame# when the VCM is frame-based. If not, it is time-based.

Each line must have the following syntax:

Nb Frames ; modulation OFF ; stop ; ACM nb ; pilot ; short frame (Frame-based mode)
time (seconds); noise level (dBm/Hz) ; ACM nb ; pilot ; short frame (Time-based mode)

Lines beginning with '#' character and blank are ignored.

See 4.5.8.2.2.3 for more details.

Note: Even in a CCM, a change of configuration (ACM) does not generate any errors due to the reconfiguration.

3.6.3 IF Signal Generation

3.6.3.1 IF1 and IF2

For each Modulation Unit, the following characteristics of the simulated telemetry signal can be configured:

- Carrier: Adjustable Level and ON/OFF switch
- Noise: Adjustable Level and ON/OFF Switch

Note:

The "raw" signal-to-noise ratio (SNR or E_b/N_0) can be computed from the following formula:

$$\frac{E_b}{N_0} = C_{carrier} - C_{noise} - 10 \log_{10}(BitRate)$$

Where:

$C_{carrier}$: Carrier Level in dBm

C_{noise} : Noise Level in a 1-Hz bandwidth in dBm/Hz

Note: For DVB-S2 and SCCC, E_s/N_0 makes more sense due to VCM support.

$$\frac{E_s}{N_0} = C_{carrier} - C_{noise} - 10 \log_{10}(SymbolRate)$$

Each Modulation Unit and the Noise Generator can contribute independently to both IF1 and IF2 as it is shown on Figure 61: IF Signal Generation

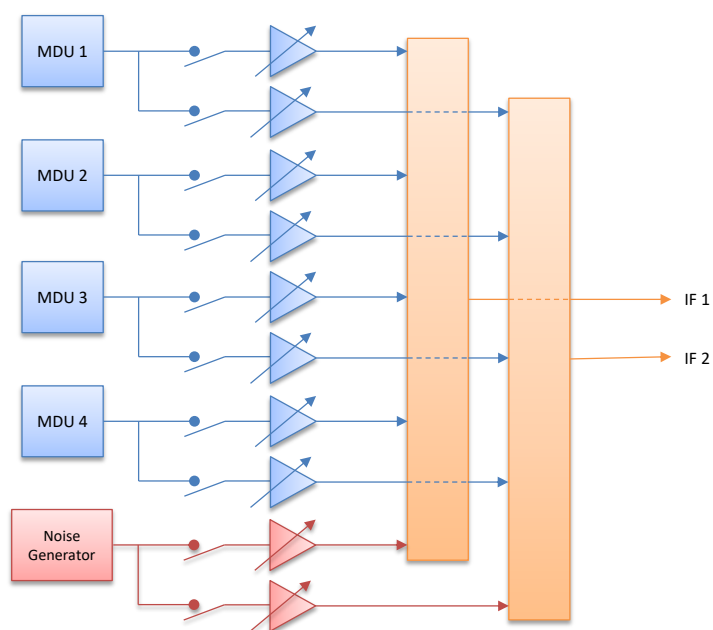


Figure 61: IF Signal Generation

This makes the test modulator very flexible in terms of routing of logical modulating units.

Here is a snapshot of the routing GUI:

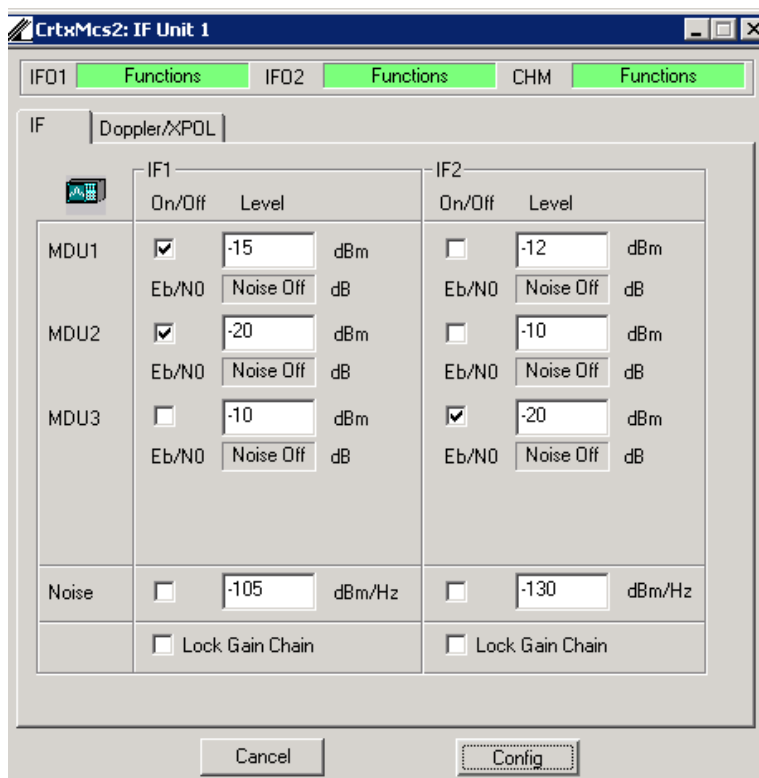


Figure 62: GUI: IF Unit

See 4.5.8.3.1 for more details on the GUI.

Here are some examples of possible configurations.

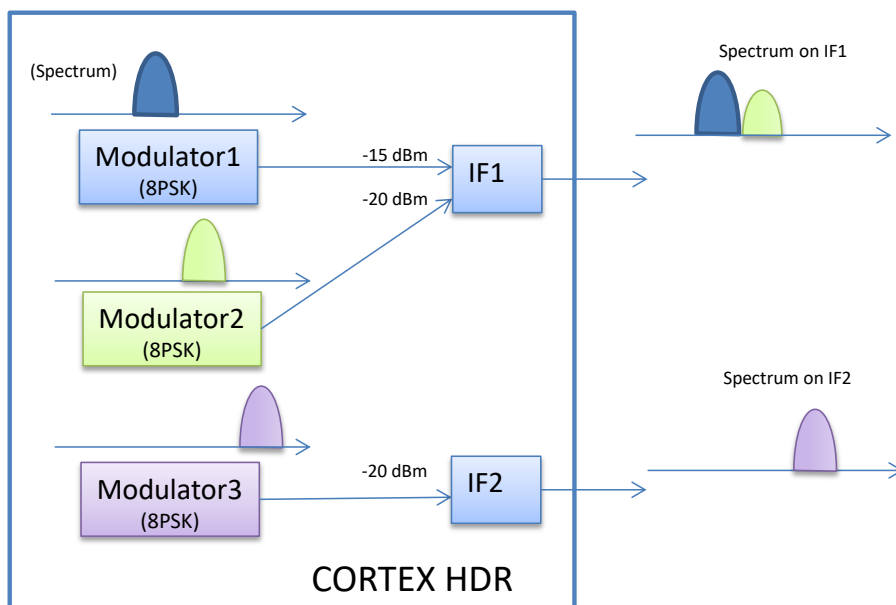


Figure 63: Example 1: 2 adjacent channels on IF1 – 1 single channel on IF2

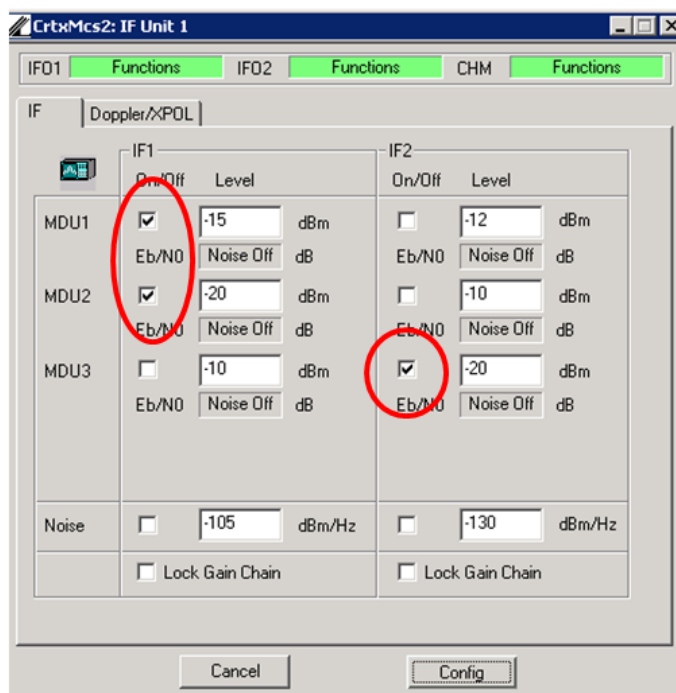


Figure 64: Example 1: 2 adjacent channels on IF1 – 1 single channel on IF2 (GUI)

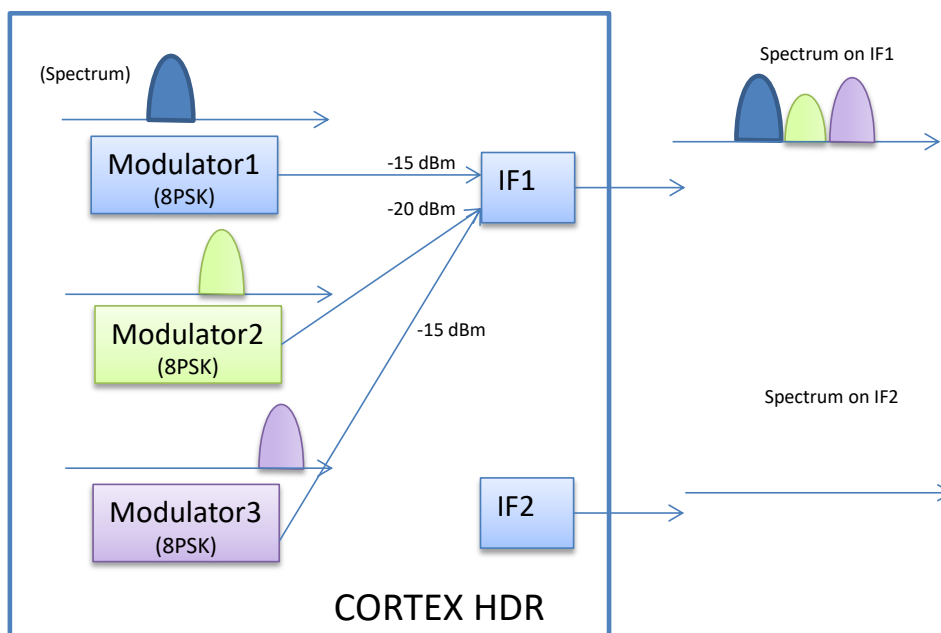


Figure 65: Example 2: 3 adjacent channels on IF1 – Nothing on IF2

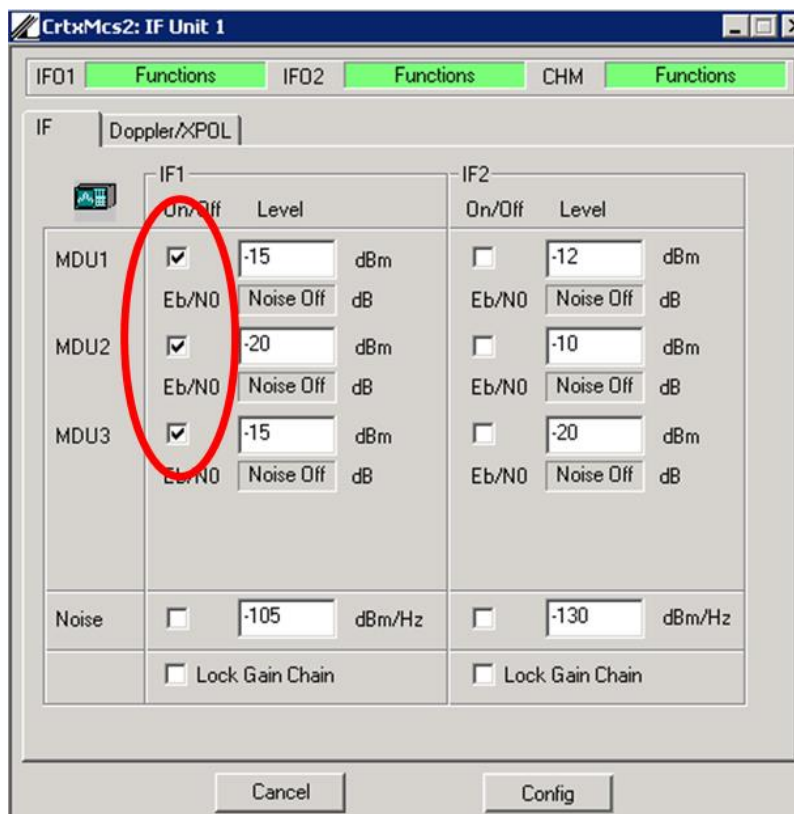


Figure 66: Example 2: 3 adjacent channels on IF1 – Nothing on IF2 (GUI)

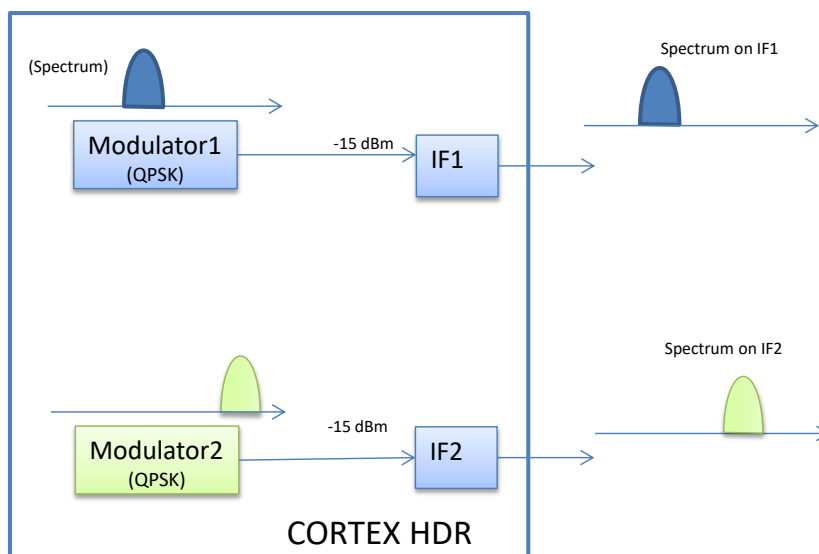


Figure 67: Example 3: 1 channel on IF1 – 1 channel on IF2

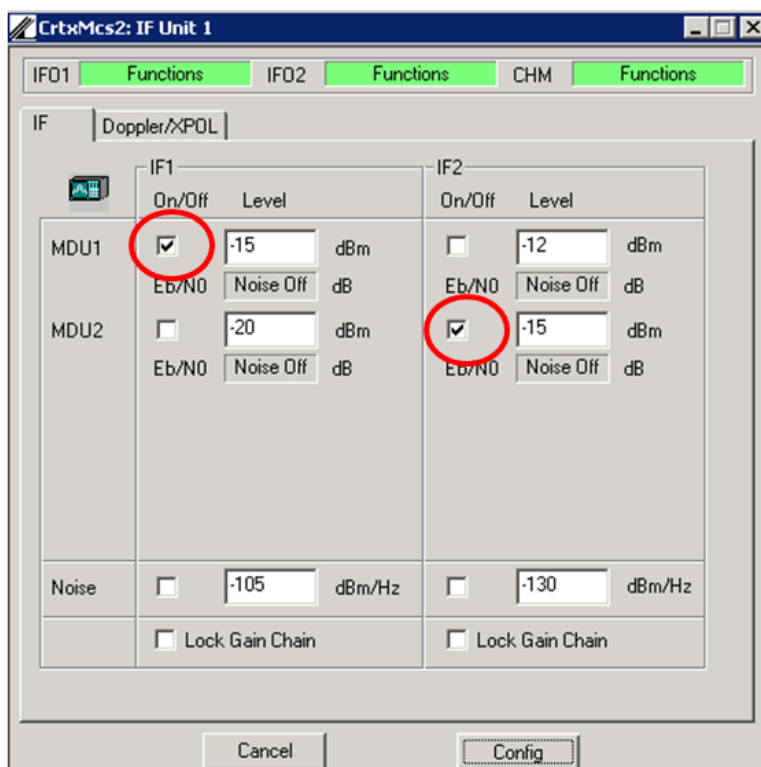


Figure 68: Example 3: 1 channel on IF1 – 1 channel on IF2 (GUI)

3.6.3.2 Doppler

A Doppler-effect generator is integrated to the test modulator.

- Rate is configurable between 0 and 100 kHz/s
- Range is configurable between 0 and 2 MHz
- The “shape” of Doppler can be chosen between a Triangle, a sinusoid or an arctangent function. It also can be customized thanks to an editable File.

Depending on the firmware, it can also be configured to be identical or independent on both IFs (IF1 and IF2) of the Unit.

3.6.4 Modulation of external data

3.6.4.1 Direct input to the modulator

3.6.4.1.1 Principle

The test modulator enables direct modulation of incoming data over SMA connectors J52 to J56. The electrical standard is single ended ECL or differential ECL depending on the chosen option at purchase.

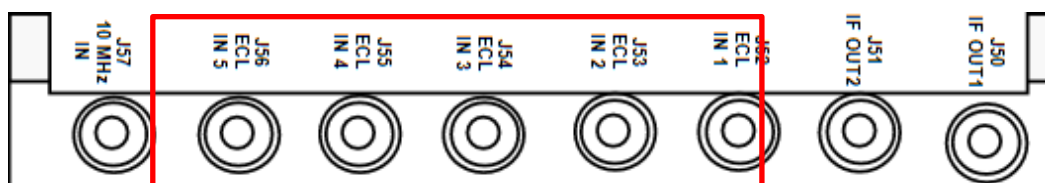


Figure 69: ECL inputs on test modulator board – Single Ended ECL

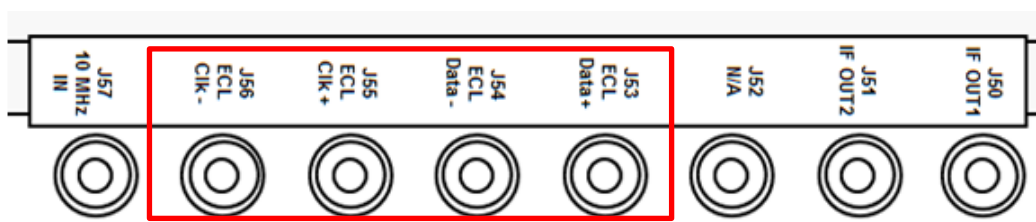


Figure 70: ECL inputs on test modulator board - Differential ECL

When the board is fitted **with single-ended ECL** two different input modes are supported:

- **Hard Symbol mode:** in this mode, each ECL connector is linked to a bit of the resulting symbol. Typical schemes are I/Q in QPSK, and IQZ in 8PSK. The symbol rate is the input clock rate, and the modulation is set in the modulating unit linked to the format input unit.
- **Data mode:** in this mode, the alignment of the symbol with respect to the data is not guaranteed. This enables to use various input parallelisms with all supported modulations (for example split 2 in 8PSK). Data can be input using parallelism 1 (Serial), 2, 3 and 4. The overall raw data rate is obtained by multiplying the input clock rate by the selected parallelism.

When the board is fitted with **differential ECL** the only mode available is **Serial data mode** as only one data input is available. In this case the serial data mode is forced, regardless of the setting that is set in the configuration table.

Here are the possible schemes depending on the modulation and the data type:

MODULATION	HARD SYMBOL (SINGLE ENDED ECL ONLY)	OTHERS
BPSK	Yes, I data	Yes
QPSK	Yes, IQ data	Yes
OQPSK	Yes, IQ data	Yes, single clock, offset introduced internally to the modulator.
8PSK	Yes, IQZ data	Yes
16QAM	Yes, IQZX	Yes
16APSK	Yes, IQZX	Yes
32APSK	No	Yes
64APSK	No	Yes

Table 14: modulations and input modes

Here is the table of connector allocation depending on the data scheme, **for single ended ECL**:

LABEL	ID	HARD SYMBOL	SERIAL	//2 DATA	//3 DATA	//4 DATA
ECL IN 1	J52	I data ⁽¹⁾	Data	D1 ⁽²⁾	D1 ⁽²⁾	D1 ⁽²⁾
ECL IN 2	J53	Q data ⁽¹⁾	-	D2 ⁽²⁾	D2 ⁽²⁾	D2 ⁽²⁾
ECL IN 3	J54	Z data ⁽¹⁾	-	-	D3 ⁽²⁾	D3 ⁽²⁾
ECL IN 4	J55	X data ⁽¹⁾	-	-	-	D4 ⁽²⁾
ECL IN 5	J56	Clock	Clock	Clock	Clock	Clock

Table 15: connectors allocation, single Ended ECL

(1): I/Q/Z/X data refers to direct symbol mapping. For example, I/Q/Z mode in 8PSK means that each bit fed to the input will be directly used to map the symbol using the mapping selected in the connected MDU. In BPSK, only I is relevant. In QPSK/OQPSK, I and Q are relevant. In 8PSK I, Q, and Z are relevant. In 16APSK and 16QAM, I, Q, Z and X are relevant.

(2): the convention for data merging is D1 D2 D3 D4, D1 being the MSB and first bit transmitted.

Here is the table of connector allocation depending on the data scheme, **for differential ECL**:

LABEL	ID	SERIAL
ECL IN 1	J52	
ECL IN 2	J53	Data +
ECL IN 3	J54	Data -
ECL IN 4	J55	Clock +
ECL IN 5	J56	Clock -

Table 16: connectors allocation, Differential ECL

IQZ and X bits are always read from left to right is the constellation mappings. Here is an example of IQZ mapping in 8PSK:

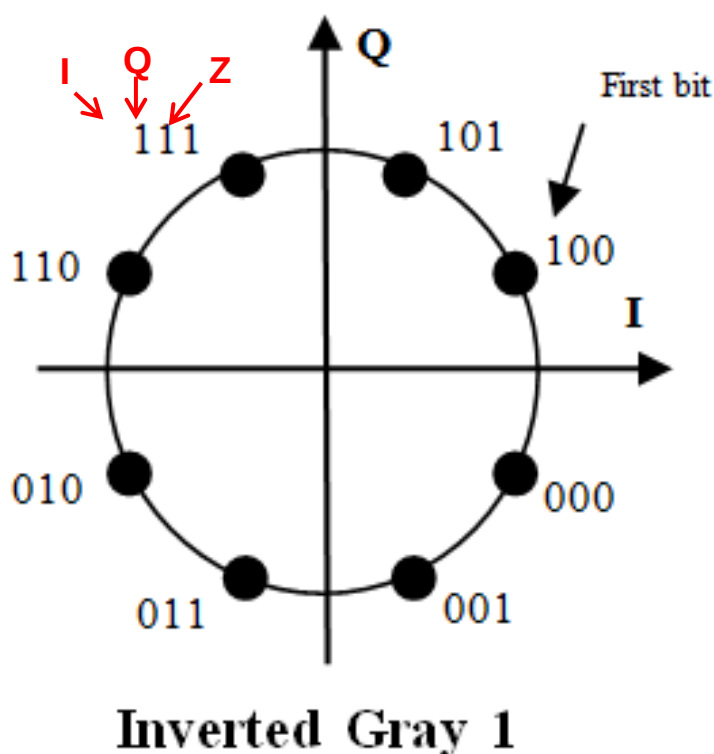


Figure 71: IQZ mapping example

3.6.4.1.2 Configuring the modulator and the format input

Direct input to the modulator can be obtained by following these guidelines:

- Use FPGA #1 for modulator
- Activate (mount) MDU1, MFI, and IFO in the global table
- In MDU 1:
 - Set MDU1 to the desired settings in terms of modulation, carrier frequency, RRC filtering, PCM code and phase rotation.
 - Set the Bitrate to the target overall bitrate:
 - In symbol mode, multiply the desired clock rate by the order of modulation
 - In data mode, multiply the desired clock rate by the input parallelism
 - Declare the Format Input as the input to the modulator
- In MFI:
 - Set the input type as standard
 - Select the data scheme (Hard Symbol or Data with required parallelism)
- Connect the required signals to the modulator board (please refer to Table 15: **connectors allocation, single Ended ECL**
-)

Below lies an example of a correct setting for direct input of serial data to a QPSK, 100 Mbps modulator.

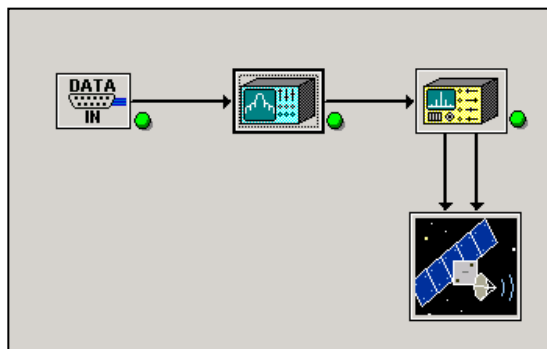


Figure 72: direct modulator input, as seen in the main modulator window

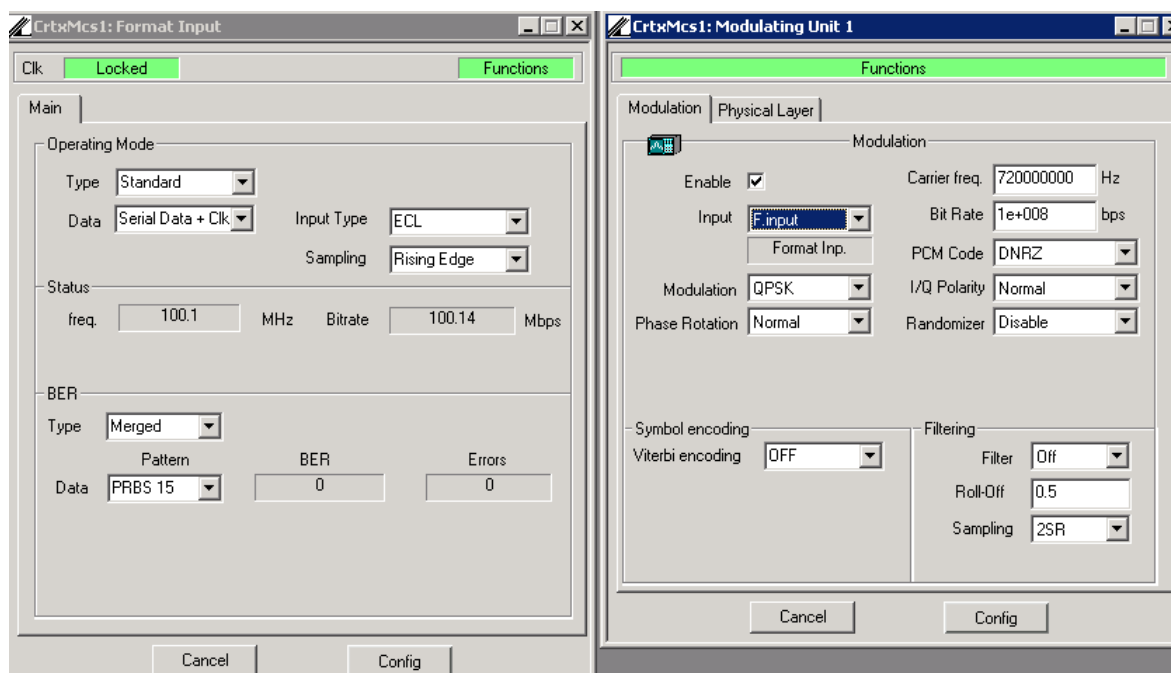


Figure 73: 100 Mbps, QPSK direct input setting

3.6.4.1.3 BER estimation

The format input enable BER estimation directly after sampling at the interface. Two BER modes can be selected:

- **Merged:** data of all parallel bits is merged into a single stream before PRBS comparison. In hard symbol mode, the merge logic is IQ, IQZ or IQZX, I being the first bit, X the last. In parallel data mode, the logic is D1 D2, D1 D2 D3 or D1 D2 D3 D4.
- **Split:** BER estimation is performed independently for each input connector. The number of connectors depends on the operating mode detailed earlier. Some examples are 8PSK, hard symbol mode (IQZ), or Any modulation with Data //4 (D1 D2 D3 D4).

The following PRBS testers are available: PRBS 7, 10, 11, 15, 23 and 31. The PRBS testers work in the same manner as the DMU and DPU BER estimators. They can be reset by issuing the correct extra command or by clicking the '0' button in the main GUI.

Note: only merged mode is available if modulator board is fitted with differential ECL.

3.6.4.1.4 MCS preset files

The following MCS files have been preset and can be used as a base for customization, in the D:\ZDS\Data\HDR\MCS\DefaultConfig\Test\IO Boards\Standard\Modulator folder

- ECL_serial_in_QPSK.mcs:
 - QPSK, 100 Mbps, 720 MHz carrier, DNRZ with 1 bit data + clk @ 100 MHz
- ECL_serial_in_8PSK.mcs:
 - 8PSK, 100 Mbps, 720 MHz carrier, NRZ-L with 1 bit data + clk @ 100 MHz
- ECL_IQ_in_QPSK.mcs:
 - QPSK, 200 Mbps, 720 MHz carrier, DNRZ with I/Q + clk @ 100 MHz
- ECL_IQZ_in_QPSK.mcs:
 - 8PSK, 300 Mbps, 720 MHz carrier, NRZ-L with I/Q/Z + clk @ 100 MHz

3.6.4.1.5 Constraints

Some constraints apply to this functionality. They are summarized hereafter.

- The maximal clock rate is 400 MHz
- The format input can be used only in FPGA menu #1, on the single MDU available.
- The data input to the modulator is **direct** meaning no encoding is available, apart from differential symbol encoding (DNRZ, NRZ-M...). As an example, RS223 and Viterbi are **not** available when using direct input to the modulator.
- The clock input is processed in real time to servo control the symbol rate of the modulator. This means that the clock should be stable, hole less. Good behavior in other contexts is not guaranteed.
- UQPSK and GMSK modulations are not supported.
- DVB-S2 and SCCC are also not supported.

3.7 Cable equalizer filter

The purpose of the **optional** cable equalizer filter is to compensate for cable degradation by means of a digital equalizer integrated to the receiver. Analog equalizers are commonly used to solve this problem; however a digital implementation offers the following advantages:

- ❑ More flexibility,
- ❑ Higher performances.

The primary cause of transmission loss below 1 GHz is skin effect, which is proportional to the square root of frequency:

$$\text{Amplitude Loss}_{(dB)} = A * \sqrt{\text{frequency}}$$

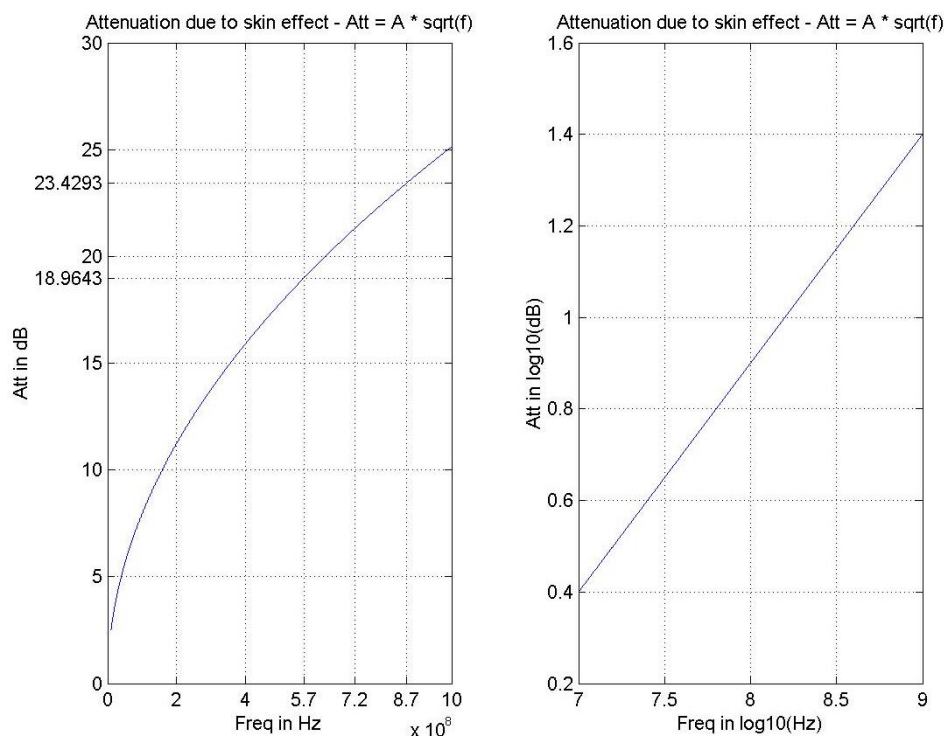


Figure 74: Example of Non-linearity in a RG188A/U Cable

Since this type of non-linearity is almost linear between 500 MHz and 1 GHz (above figure), it can be well compensated by a straight line with opposite slope of the tangent.

The sampled signal is filtered before baseband transposition. By this way, a symmetrical low-pass filter can be implemented:

Main features of this low-pass filter are:

- ☐ Fully programmable solution: the gain slope defined by the amplitude difference between the transmission at 570 MHz and 870 MHz. This gain slope can be locally or remotely programmed between 0 to 12 dB.
- ☐ Gain compensation performance in the bandwidth better than 0.1dB (see Figure 74)
- ☐ Intrinsic phase linearity performance.

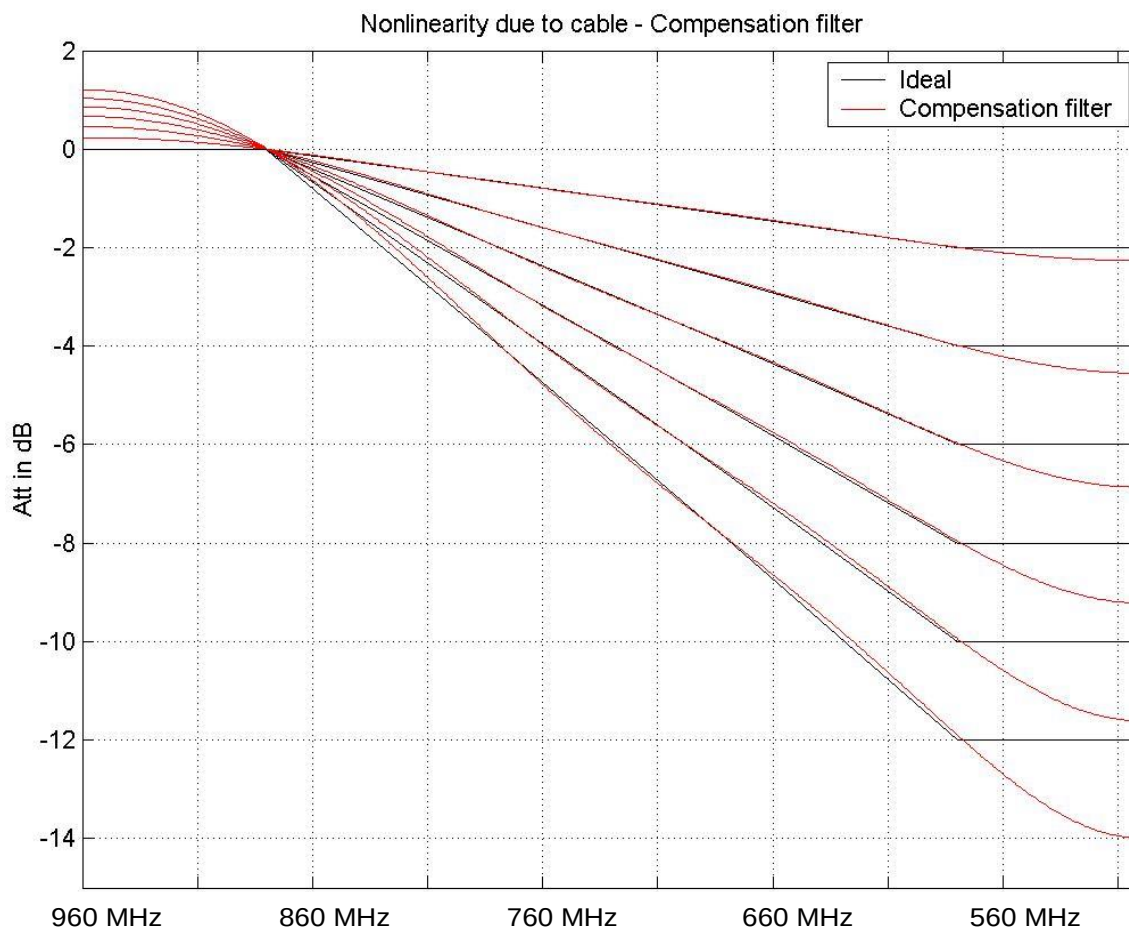


Figure 75: Example of cable compensation

3.8 Data logging

3.8.1 Standard Logging

The HDR provides capacities to record inside logging xml files. 8 logs are possible simultaneously.

For each log there is a list of pre-defined control parameters and/or status and their corresponding values. The recording can be periodical, on transition of a parameter/status value or on variation of a parameter/status value versus a pre-defined threshold. All the control parameters or status of the HDR can be recorded.

The logging files are recorded inside:

D:\ZDS\DATA\HDR\log\stl\<FILE_PATH>\<FILE_NAME>.xml

<FILE_PATH> and <FILE_NAME> are configured into text files named **spyconfN.txt**. They are located into:

D:\ZDS\DATA\HDR\log\stl\conf\logi, 1 ≤ i ≤ 8.

N.B.: If <TYPE_LOG> is specified in <FILE_PATH> and/or <FILE_NAME>, associated logging is split into periodic/transition/variation files.

By default, if <FILE_PATH> and <FILE_NAME> are not configured, logging files will be generated in:

D:\ZDS\DATA\HDR\log\stl\logi, 1 ≤ i ≤ 8.

With the following file names, YYYYMMDDhhmmss being the date:

**YYYYMMDDhhmmss_periodic.xml
YYYYMMDDhhmmss_transition.xml
YYYYMMDDhhmmss_variation.xml**

These files start with configuration information: spyconfN.txt file used for this file and format of the logs (label of the parameters to log). The following example is a generated 20150805151010_periodic.xml for log1:

```
.....
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<log1>
  <conf>
    <file>d:\ZDS\Data\HDR\log\stl\conf\log1\spyconf0.txt</file>
    <format>date,IFLevel_A,eb_n0_A,bitSync_A,frameSync_A</format>
  </conf>
.....
```

Then for each log node, the type and date of the log are specified in attributes, and the corresponding values.

```
.....  
<log type="P" date="18717013.041881">-20.0;25.8;1=Locked;2=Locked</log>  
<log type="P" date="18717013.548893">-20.1;25.8;1=Locked;2=Locked</log>  
<log type="P" date="18717014.068880">-20.0;25.7;1=Locked;2=Locked</log>  
<log type="P" date="18717014.575887">-20.0;25.7;1=Locked;2=Locked</log>  
<log type="P" date="18717015.082870">-20.1;25.7;1=Locked;2=Locked</log>  
</log1>  
.....
```

In this example, have been periodically recorded for log1 the values of the IF level (label: IFLevel_A), Eb/N0 (label: eb_n0_A), the status of the bit synchronizer (label: bitSync_A), the status of the frame synchronizer (label: frameSync_A).

The rules for the recording are defined inside text files named: **spyconfN.txt** recorded in the following directories:

D:\ZDS\DATA\HDR\log\stl\conf\logi, 1 ≤ i ≤ 8.

A specific window allows the user to build this control file: definition of the control parameters/status to record, labels of these parameters, occurrence of the recording or thresholds.

An automatic erase process exists on logging folder: **D:\ZDS\DATA\HDR\log\sti**

A maximum size is defined in registry (HKEY_LOCAL_MACHINE\SOFTWARE\IN-SNEC\CrtxHdr\SetDefault\Preferences\Global\PurgeSize_STILog). By default, install value is set to 100MBytes.

When this size is reached, oldest files are erased, exception made on "conf" folder.

3.8.2 Extended Logging

The extended logging is based on the same principles as the standard logging (i.e. the possibility to periodically record each parameter defined in the STI inside a file with the selection of the parameters defined inside a configuration text file), except that:

- Up to 4 different data flows can be simultaneously recorded inside different log files
- The start and stop of the recording for each flow can be done from the GUI or through special command via the Control TCP/IP port (refer to the document STI 100013_hdr).
- The log files are binary files. This allows to record more data simultaneously. These binary files can be exploited off line with the tool "ReadLog.exe" delivered within the HDR software.
- The names, paths, maximum sizes of the files are configurable through Windows Registry.

The Windows Registry is the following:

1 – To define the logging capabilities

PATH : [HKEY_LOCAL_MACHINE\SOFTWARE\In-sneec\Cortex Station\SetDefault\Log]

Registry	Type	Description / default value
<i>Storage</i>	DWORD	To enable/disable global logging data storage on the disk Default: 0 0 – Disable the storage capacity 1 – Enable the storage, start ON by default 2 – Enable the storage, start OFF by default Si la clé « Storage » par défaut est positionnée à 0, le maillon d'enregistrement ne sera pas créé et cela quel que soit la valeur des autres clés.
<i>File Dir</i>	Character string	To define the file default name and access path Default setting at delivery: D:\ZDS\Data\Storage\Menu1\Log\Logging\LOG
<i>StoreSize</i>	DWORD	To define the default maximum data file size Default setting at delivery: 4096 (bytes)
<i>StoreTotal</i>	DWORD	To define the default total data record size Default setting at delivery: 65536 (bytes)

2 - To create the different flows:

PATH : [HKEY_LOCAL_MACHINE\SOFTWARE\In-snec\Cortex Station\SetDefault\Log\10]
[HKEY_LOCAL_MACHINE\SOFTWARE\In-snec\Cortex Station\SetDefault\Log\11]
[HKEY_LOCAL_MACHINE\SOFTWARE\In-snec\Cortex Station\SetDefault\Log\12]
[HKEY_LOCAL_MACHINE\SOFTWARE\In-snec\Cortex Station\SetDefault\Log\13]

Registry	Type	Description / default value
<i>EnableLog</i>	DWORD	To enable/disable the specific additional data flow processing Default: 0 0 – Disable the Specifics additional flow 1 – Enable the Specifics additional flow
<i>FileConfig</i>	Character string	To define the configuration file name and access path Default setting at delivery: D:\ZDS\Data\CRT\SPS\Logging\Menu1\LogFlow10.txt

Example of a configuration text file for the logging of 4 parameters of the demodulator unit 1:

```
* -----
* Example of a Logging Configuration File
*
* The lines beginning by * or # are comments
*
* The first line is mandatory and gives the sampling period of the parameters recording
(minimum: * 100 ms). The value "1" is kept for the coherence of the file.
*
* The other lines are the logging elements and defined as:
* the Component table number (hexa) followed by the offset of the parameter in the table
(decimal)
*
* -----

* Sampling period - Mini = 100ms
1000 1

* Cortex Station - Start/Stop storage status
201 28

* DEMOD A - Carrier Offset
3010 40

* DEMOD A - IF Level
3010 41

* DEMOD A - Eb/No
3010 43

* DEMOD A - PLL Status
3010 42

* -----
```

3.8.3 System Logging

This logging traces events on Cortex HDR. A verbosity level is defined in registry, enabling to trace alarms (value 0), HDR state changes (1: default install value).

There is a daily log file: when Signal Processing software detects a day change on an event log, current file is closed and a new file opened.

N.B.: for this logging Windows time is used.

Generated files have XML format. They are located in:

D:\ZDS\DATA\HDR\log\system

An automatic erase process exists on logging folder: **D:\ZDS\DATA\HDR\log\system**

A maximum size is defined in registry (HKEY_LOCAL_MACHINE\SOFTWARE\IN-SNEC\CrtxHdr\SetDefault\Preferences\Global\PurgeSize_systLog). By default, install value is set to 100MBytes.

When this size is reached, oldest files are erased.

4 OPERATING MANUAL

4.1 Starting and Configuring the CORTEX HDR

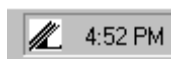
4.1.1 First Steps

4.1.1.1 Starting the PC Workstation

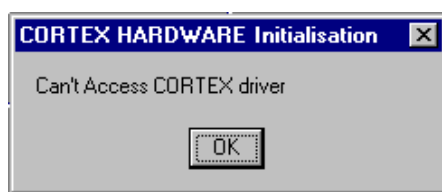
1. Verify that the CORTEX HDR has not been damaged during transportation. If needed, open the top cover and verify that the electronic sub-assemblies are in good condition.
2. Check that the CORTEX HDR chassis is properly grounded and connected to the AC power via the appropriate power cord.
3. Switch on the chassis (ON/OFF switch is at the rear panel). Allow 8 to 10 seconds for the power supply to start.
4. At the front panel display, follow the boot-up sequence. Depending on the **Autolog** setting (ON or OFF), manual intervention may be required (type Ctrl + Alt + Delete, then enter the login password). Default password at delivery is *cortex*. Default Autolog setting at delivery: ON.

4.1.1.2 Signal Processing Software. Start-up Menu

At the end of the boot-up sequence, the Signal Processing Software (SPS) is automatically started. If the Demodulator board is present, the following **SPS icon** appears at the lower right corner of the screen:



If the board is not mounted or in the case of a major hardware problem, the following message is displayed:



4.1.1.3 Starting the Monitoring & Control Software

Once the Signal Processing software is running, the Monitoring & Control software is automatically started. If needed, connect the MCS to the SPS for starting the GUI.

A detailed description of the Graphical User Interface is given in the next sections.

4.1.2 Factory Configuration

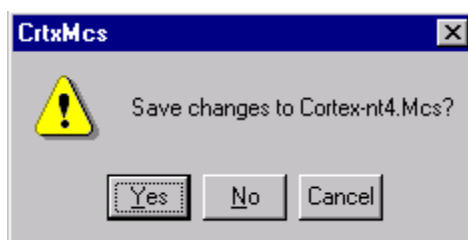
The CORTEX HDR has been configured in factory in accordance with the required hardware components (electronic cards, etc...) and software licenses (maximum bit rate supported, data decoding licenses, filter licenses, data processing licenses, etc...).

Each CORTEX HDR unit comes with a factory acceptance test book showing the detailed hardware and software configuration of the unit at delivery.

4.1.3 Software Exit & CORTEX HDR Shutdown

4.1.3.1 Exiting the Monitoring & Control Software

To exit the Monitoring & Control Software, click on **File** menu and select **Exit**. or click on  in the upper right corner of the MCS window. If any configuration parameter has been changed since the MCS Document was updated on the disk, the following message is displayed:



Click on **Yes** to update the MCS File on the disk. Otherwise, click on **No** or **Cancel** the command.

4.1.3.2 Exiting the Signal Processing Software

To exit the Signal Processing Software (SPS), click on the SPS icon  at the lower right part of the screen and select **Exit**.

4.1.3.3 Restarting the SPS & MCS

Assuming the application software (SPS + MCS) has been stopped, click on shortcut menu **Start / All Programs / Startup / crtstart** to restart it.

To restart the MCS only, click on shortcut menu **Start / All Programs / Cortex Application / MCS**.

To restart the SPS only, click on shortcut menu **Start / All Programs / Cortex Application / SPS**.

4.1.3.4 CORTEX HDR Shutdown

PRIOR TO SWITCHING OFF YOUR CORTEX PRODUCT

SHUT DOWN THE PC-COMPATIBLE WORKSTATION (CLICK ON *Start / Shutdown / Shut down the computer*).

FAILURE TO OBSERVE THIS PROCEDURE CAN AFFECT DATA ON THE DISK AND THE SYSTEM ITSELF.

4.1.4 Screen Saver

WARNING:

THE SCREEN SAVER INSTALLED AT DELIVERY ENABLES TO SWITCH THE LCD BACKLIGHT OFF.

TO AVOID PREMATURE AGEING / FAILURE OF THE LCD AND DATA TRANSFER PROBLEMS ON THE PCI BUS,

DO NOT CHANGE THE SCREEN SAVER SETTING.

4.2 Signal Processing software (SPS)

4.2.1 CTX Documents

CTX Documents are configuration files **created and managed by the CORTEX SPS**. A CTX Document contains the whole set of configuration parameters that have been previously stored on the disk under file extension .CTX.

A CTX Document is loaded by the SPS every time the CORTEX is started and the CORTEX is automatically configured from the contents of this CTX file.

If the machine has never been started since its software was installed, the default loaded CTX file is **menu1.ctx**. Else (usual case), the CTX document which is loaded is the **last active CTX file** (See 4.2.9.1.1 and 4.2.9.1.2).

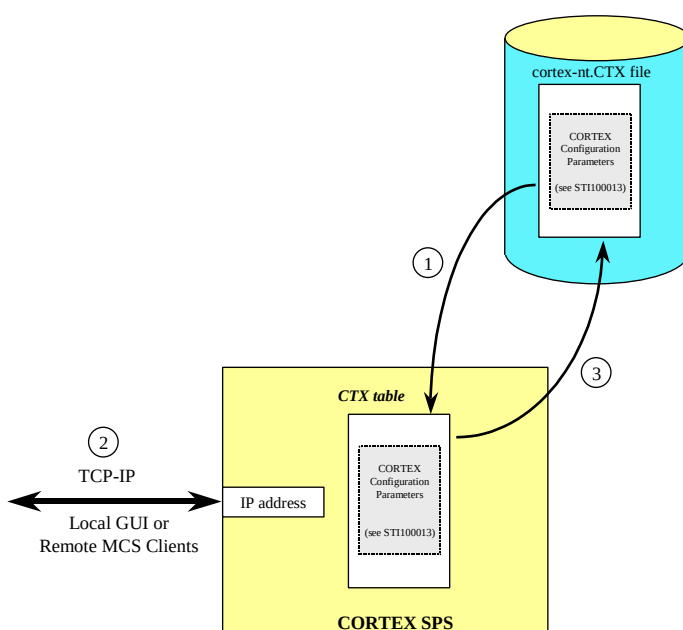
A copy of this file (CTX Table) is maintained in SRAM and is updated whenever the CORTEX configuration is changed by the operator. The CTX Table is automatically saved to the CTX Document on the hard disk every 10 seconds and at application ending.

Additional CTX Documents reflecting mission-specific configurations can be created at will by the operator.

CTX Documents are **binary files**; they cannot be displayed on the screen or printed out.

4.2.2 SPS Start Sequence

The last active CTX Document is loaded by the SPS when the CORTEX is started. The CORTEX is configured from the content of this file.



STEP 0: Windows boot up sequence.

STEP 1: The Signal Processing Software (SPS) is automatically started. The CORTEX is configured from parameters stored in the last active CTX file. A copy of the file is maintained in RAM (CTX Table).

STEP 2: The Monitoring & Control Software (MCS) (local GUI or remote MCS) opens a TCP-IP connection to the Monitoring port and starts monitoring the machine. The CTX Table is updated when configuration requests are received on the TCP-IP Control port.

STEP 3: The CTX file is automatically refreshed every 10 seconds and at application ending.

Figure 76: SPS Start Sequence

4.2.3 Configuring the CORTEX from CTX Documents

CTX Documents are stored on the disk in `C:\Program Files\IN-SNEC\crtxnt\Menus\CTX` directory.

A copy of the **original CTX files** (those installed from the reference CORTEX CD ROM) is available in `C:\Program Files\IN-SNEC\crtxnt\Menus\CTX_ref` directory. These files are set to **read-only**. They can be copied to `C:\Program Files\IN-SNEC\crtxnt\Menus\CTX`, if required, for being used by the SPS (remove the read-only protection on the copies).

4.2.3.1 CTX Document Name Table

Management of the CTX Documents is via the **CTX Document Name Table** (See 4.2.9.1.1). This table assigns a name to each CTX Document, to make file selection easier for the operator (see STI 100013 for table format).

The CTX Document Name Table is **ASCII-coded** and is transmitted over the LAN on reception of a monitoring request with component code 200_H.

The CTX Document Name Table is not included in the set of tables returned to the monitoring client on reception of a monitoring request for All component tables.

The CTX Document Name Table is automatically saved to the disk (`C:\Program Files\In-sneC\Crtxnt\CTXDocument\NameTable.TXT`) whenever the name of a CTX Document is changed or at application ending.

4.2.3.2 CTX Document Storage and Loading

CTX Documents can be created, modified, (re)named and loaded via two configuration commands (see Annex 1, section 2.3.2.3) available in the CTX Document Management window (see section 4.3.9):

- ❖ **Save command:** name and store the active CTX Table into one of the 10 CTX Documents on the disk.
- ❖ **Load command:** select and load one of the 10 CTX Documents and reconfigure the CORTEX.

The CTX Document name can also be modified by editing the **NameTable.TXT** file before starting the CORTEX SPS. If a CTX Document is not available (not existing or corrupted), the assigned name is set to an empty character string.

In the CORTEX Series monitoring table (offset 16) and in the Global CORTEX monitoring table (offset 22), the **CTX Document ID** status indicates which one of the 10 CTX Documents was previously loaded or stored and whether or not this CTX Document was modified since loaded or stored (**Update CTX Document** status). See STI 100013 and STI 100013_HDR in Annex 1 for more details.

Both statuses are set to 0 at boot.

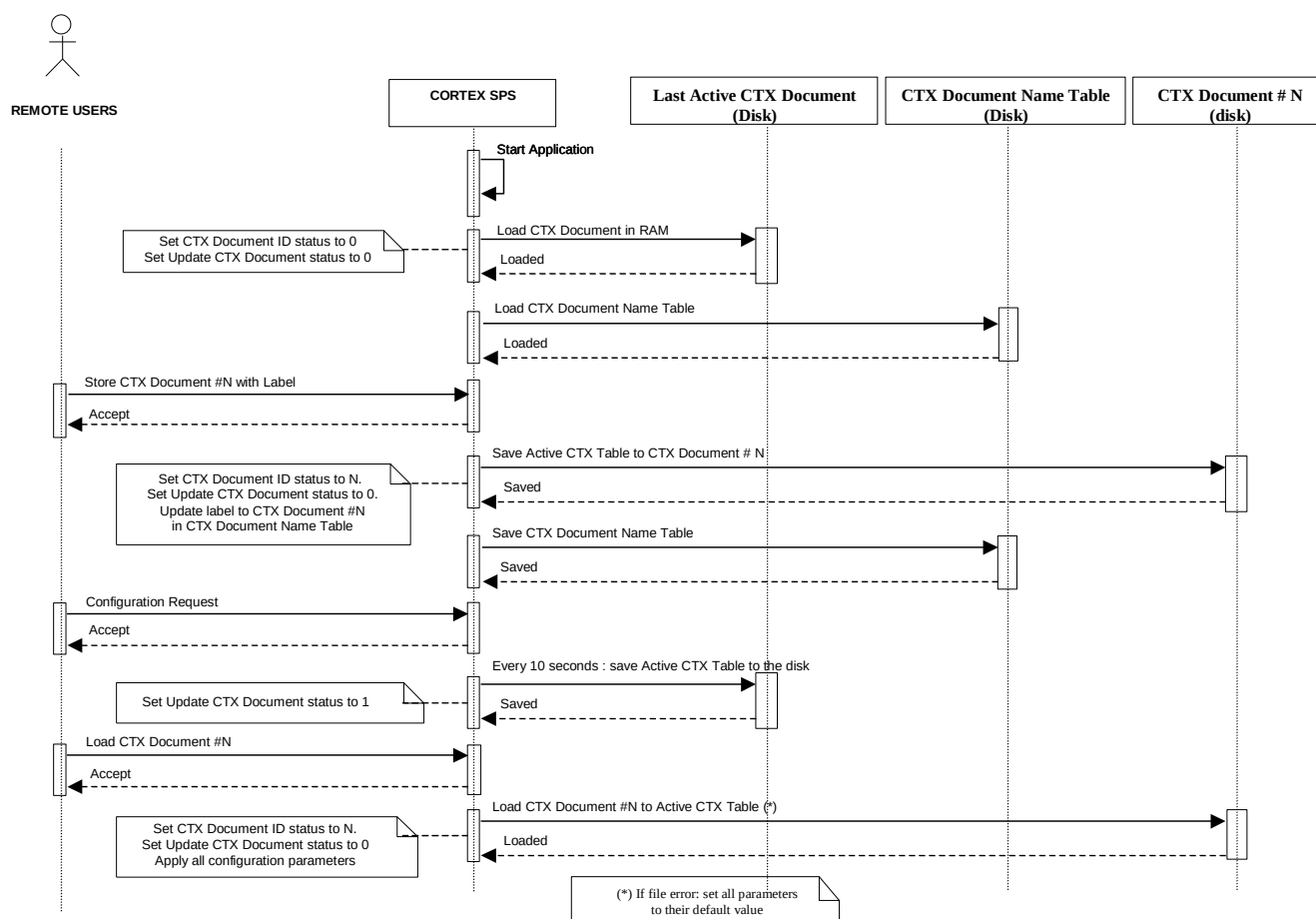


Figure 77: CTX Document Management

4.2.4 Start-up Menu Selection by Remote Access Control (RAC)

The RAC functionality is supported by all CORTEX products (CRT, HDR, ACS, ...) from **SPS Kernel version 2.7** and **MCS kernel version 1.9**.

4.2.4.1 General

On some CORTEX products (CORTEX CRT, CORTEX HDR, etc...) a start-up menu offers several high-level architectures (combination of hardware and software resources) corresponding to different mission requirements.

When starting up the CORTEX software, menu selection can be done either locally (user interface described hereafter) or remotely (Remote Access Control) by sending control commands to a specific RAC data server on the CORTEX machine (RAC port # 3003). Control commands are described in the Ethernet ICD STI 100013, section 2.6.

Starting a new menu while the CORTEX is running takes only a few seconds (2 seconds best case, up to 20 seconds when a new FPGA software needs to be downloaded from the hard disk to the target FPGA).

The following commands are supported:

- ☐ **Stop the active menu:** the CORTEX application software is not stopped but all IP connections (to the monitoring port, telemetry port, ranging ports, etc...), except the connections to the RAC port, are automatically closed. The connection status on the top level window of the GUI indicates DISCONNECTED.
- ☐ **Start a new menu:** select and start a new menu.
- ☐ **Check the active menu:** indicates the active menu.
- ☐ **Load the menu table:** on reception of this command, the CORTEX returns the list of available menus.

4.2.4.2 MCS Graphical User Interface

From the IN-SNEC MCS, the operator can access the RAC server by clicking successively, in the upper toolbar, on **Window** → **Menu_Selection**, or by clicking on the  icon.

Next picture shows the operational scenarios offered by the start-up menu. Use the **Stop** active menu and **Start** new menu pushbuttons to re-configure the CORTEX.

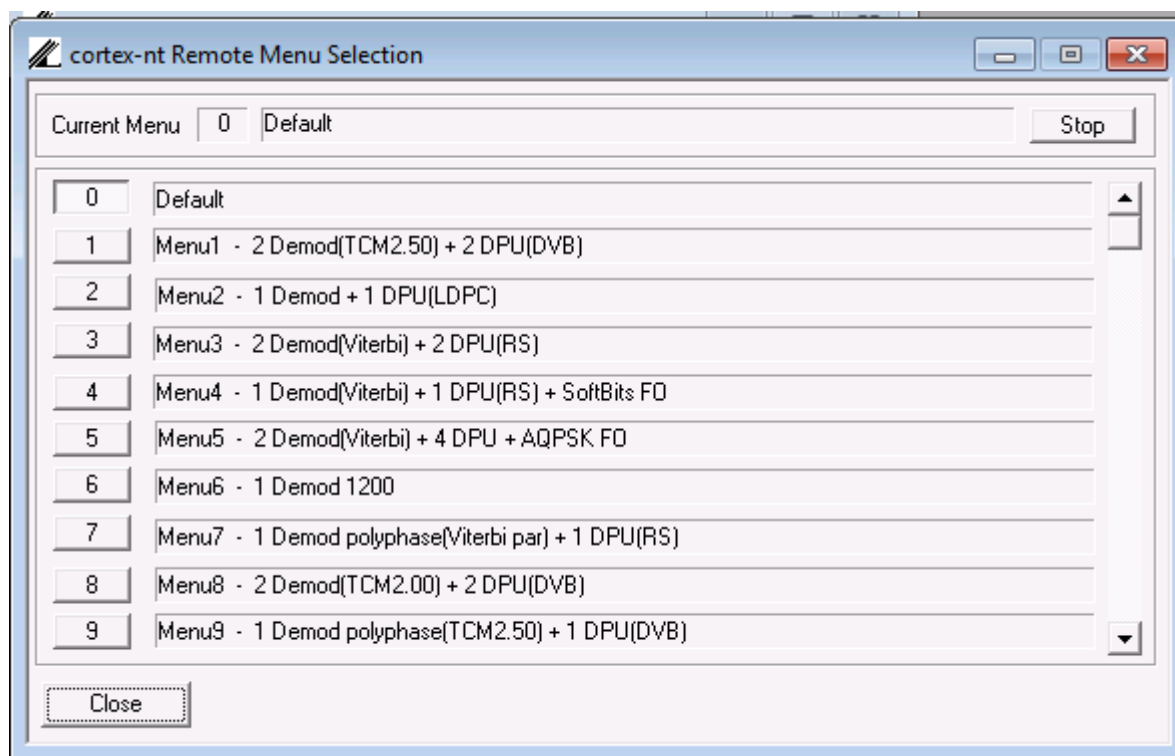


Figure 78: MCS Menu Management Window

4.2.4.3 *Menus Description File*

A detailed description of the start-up menu is available in:

C:\Program Files\IN-SNEC\crtxnt\Menus\menu.txt

How to read this file?:

Lines of comments (or disabled choices) starts with a # character (see the four first lines of the file). Other lines are menu lines.

A menu line consists of several arguments separated by a semi-comma character. Arguments are (from left to right):

- ☐ **Menu index:** the above menu allows for different configuration. Indexes 1 through 7 are for **basic menus** while others indexes are for **mission menus**.
- ☐ **Menu name:** basic menus are named *Menu 1* through *Menu 7* while missions menus refer mainly to the satellite to be tracked.
- ☐ **Registry base:** access path to the CTX file to be loaded.
- ☐ **CTX file:** CTX file to be loaded.

4.2.5 CORTEX HDR-4G Licenses and Basic Menus

The following table shows the hardware limitations for each basic menu of the demodulator:

MENU NBR.	IF INPUT	HARDWARE EXTENSION	NUMBER OF DEMOM	Max. SR ¹	A/UQPS K	Coding	NUMBER OF DPU	Coding
1	Wide Band	NO	1	1600	NO	NA	1	
2	Wide Band	(2)	1	500	NO	SCCC	1	
3	Wide Band	(2)	1	1200	NO		1	LDPC 7/8
4	2700/720	(1)	5	130	NO	6D TCM	5	
5	720/1200	NO	2	360/600	NO	TCM 2,50/3	2	RS 223/255
6	Wide Band	(2)	1	800	NO	DVB-S2	1	
7	720/1200	NO	2	360/600	NO	TCM 2,50/3	2	RS 239/255
8	720/1200	NO	2	360/600	NO	TCM 2,00/3	2	RS 223/255
9	720/1200	NO	2	360/600	NO	TCM 2,00/3	2	RS 239/255
10	720/1200	NO	2	360/600	NO	TCM 2,50/3	2	RS DVB
11	720/1200	NO	2	360/600	NO	TCM 2,00/3	2	RS DVB
12	720/1200	NO	2	360/600	NO	Viterbi	2	RS 223/255
13	720/1200	NO	2	360/600	NO	Viterbi	2	RS 239/255
14	720	NO	3	240	NO	TCM 2,50/3	3	RS DVB
15	720/1200	NO	2	360/600	NO	TCM 2,25/3	2	RS 239/255
16	720/1200	NO	2	360/600	NO	TCM 2,75/3	2	RS 239/255
17	720/1200	NO	2	360/600	NO		4	RS Worldview
18	720/1200	(2)	1	360/500	NO	DVB-S2	1	
19	720	NO	3	240	NO	TCM 2,50/3	3	RS 239/255
20	720/1200	NO	2	360/600	NO		2	RS 223/255

MENU NBR.	IF INPUT	HARDWARE EXTENSION	NUMBER OF DEMOS	Max. SR ¹	A/UQPS K	Coding	NUMBER OF DPU	Coding
21	720/1200	(2)	2	360/600	NO		2	LDPC 7/8 (I/Q interleave)
22	720/1200	(2)	2	360/600	NO		4	LDPC 7/8
23	720/1200	NO	2	360/600	NO	Stacked Viterbi	2	RS 223/255
24	720/1200	NO	2	360/600	NO	Stacked Viterbi	2	RS 239/255
25	Wide	NO	1	150	NO	NA	1	RS 223/255
26	Wide	(1)	1	150	NO	DVB-S2	1	
27	Wide	(1)	1	1600	NO	TCM 2,50/3	1	RS 239/255
28	Wide	(2)	1	1250	NO		1	LDPC 7/8
29	720/1200	(1)	2	480/410	NO		16	RS 223/255
30	720/1200	(2)	2	240/410	YES	Viterbi	4	RS 223/255
31	720/1200	NO	1	360/600	NO	TCM 2/3 TCM 2.5/3	1	RS 223/255 RS 239/255
32	720/1200	(1)	2	360/600	NO		2	LDPC 7/8
33	Wide	(1)	1	1600/6399	NO	Viterbi 1/2	1	RS 223/255
34	720/1200	(2)	1	360/500	NO	SCCC	1	
35	Wide	(2)	1	1600	NO		1	LDPC 7/8
36	Wide	(2)	1	600	NO	DVB-S2 ²	0	
37	720/1200	(2)	2	320/320	NO	DVB-S2	2	
38	720/1200	(2)	2	360/600	NO		2	LDPC 7/8
39	720/1200	(1)	2	360/600	NO		2	RS 223/255 RS 239/255
40	720/1200	(2)	1	360/600	NO		1	LDPC 1/2 4K
41	720/1200	(2)	1	360/600	NO		1	LDPC 1/2 1K

MENU NBR.	IF INPUT	HARDWARE EXTENSION	NUMBER OF DEMOD	Max. SR ¹	A/UQPS K	Coding	NUMBER OF DPU	Coding
42	720/1200	(2)	1	360/600	NO		1	LDPC 1/2 16K
43	720/1200	(2)	1	360/600	NO		1	LDPC 2/3 4K
44	720/1200	(2)	1	360/600	NO		1	LDPC 2/3 1K
45	Wide	(1)	1	460	NO		10	RS 223/255
46	720/1200	(1)	2	360/600	NO	Viterbi	4	RS 223/255 RS 239/255 RS 239/255 (conv 2)
47	720/1200	(2)	2	360/600	NO		2	LDPC 7/8
48	720/1200	(2)	2	360/600	NO		1	LDPC 7/8
49	720/1200	(1)	2	240/400	NO		2	LDPC 7/8
50	Wide	(1)	2	460	NO		2	RS 223/255 RS 239/255 RS 239/255 (conv 2)
51	720/1200	(1)	2	360/600	NO	TCM 2,50/3	6	RS 223/255
52	Wide	(1)	2	300	NO	TCM 2,50/3	2	RS 239/255
53	720/1200	NO	NO	-	NO		2	RS 223/255 RS 239/255 RS 239/255 (conv 2)
54	720/1200	NO	NO	-	NO		4	RS 223/255 RS 239/255 RS 239/255 (conv 2)
55	720/1200	(1)	1	360/600	NO	TCM 2/3 TCM 2.5/3	3	RS 223/255 RS 239/255
56	720/1200	(1)	1	360/600	NO	TCM 2.25/3 TCM 2.75/3	3	RS 223/255 RS 239/255

Table 17: The basic demodulator menus of the CORTEX HDR

The following table shows the hardware limitations for each basic menu of the test modulator:

Basic Menu	DGU	MDU	IFU	Data In/Out	Coding	DVBS2	SCCC	Channel Simulation	Comments
1	2	1	2	Standard ECL input	Viterbi, TCM RS (223, 239, 239 conv2), LDPC 7/8				Very high data rate single channel modulator (2.5 Gbauds)
2	2	2	2				YES	YES	Dual channel SCCC (625 Mbauds) with channel simulation
3	2	2	2		Viterbi, TCM RS (223, 239), LDPC 7/8			YES	High data rate dual channel modulator (1.25 Gbauds)
4	2	1	2	Wizard-link			YES		Single channel SCCC with Custom Wizard-link interface
5	2	1	2		Viterbi, TCM AR4JA(*) 1K & 4K				Single channel AR4JA 1K/4K (1.25 Gbps)
6	2	1	2		Viterbi, TCM AR4JA(*) 16K				Single channel AR4JA 16K (1/2 ; 2/3 ; 4/5) 580 ; 830 ; 1250 Mbps
7	2	1	2			YES			Single channel high speed DVB-S2 (1.25 Gbaud)
8	2	2	2			YES		YES	Dual channel DVB-S2 (625 Mbauds) with channel simulation
9	3	3	2		Viterbi, TCM RS (223, 239, 239 conv2) RS DVB				Three-channel RS-DVB modulator
10	2	1	2	Wizard-link		YES			Single channel DVB-S2 with Wizard-Link interface

Basic Menu	DGU	MDU	IFU	Data In/Out	Coding	DVBS2	SCCC	Channel Simulation	Comments
11	3	3	2		Customer specific 6D-TCM				Three-channel customer specific 6D-TCM
12	4	2	2		UQPSK 2 channels				Dual channel UQPSK modulator (625 Mbauds)
13	8	2	2		RS (223, 239)			YES	8 DGU, bit interleave, dual channel with channel simulation (625 Mbauds)
14	2	2	2		RS (223, 239, 239 conv2)			YES	Dual channel (1.25 Gbauds)
15	RESERVED								
16	3	1	2	LVDS input	Viterbi, TCM RS (223, 239), LDPC 7/8 AR4JA(*) 1K & 4K				LVDS input, single channel, 3 DGUs modulator with standard encoders on all DGUs and AR4JA on DGU1 only

Table 18: The basic test modulator menus of the CORTEX HDR

(*) Marked encoders are only available on DGU 1.

4.2.6 4G / XXL / XL demodulator menus equivalence

Here are tables which provide demodulator menu number equivalence with previous generation of Cortex HDR.

4.2.6.1 XL with XXL and 4G

XL Menu		XXL Menu	4G Menu	Comment
Description	Number	Number	Number	
1 DMU VIT + 1 DPU RS	1	5	30	
2 DMU VIT + 2 DPU RS	2	3	12	
2 DMU FIR + 2 DPU RS	3	3	12	
2 DMU VIT	4	3	12	
1 DMU VIT ¼ + 1 DPU RS	5	36	12	
1 DMU TCM2.50/3 + 1 DPU RS	6	62	5	
1 CLOCK RECOVERY VIT + 1 DPU RS	7	-	-	
1 DMU TCM2.50/3 + 1 DPU RS-DVB	8	1	10	
1 DMU + 1 DPU RS	9	3	12	
2 DMU + 2 DPU	10	3	12	Mission specific.
2 DMU + 2 DPU	11	3	12	Mission specific.
1 DMU VIT + 1 DPU RS	12	3	12	Mission specific.

4.2.6.2 XXL with 4G

XXL Menu			4G Menu	
Description	Demod Number	Ext. Number	Number	Comment
2 DMU TCM 2.50/3 + 2 DPU RS DVB	1	-	10	
1 DMU + 1 DPU LDPC 7/8	2	-	3	Up to 16 APSK
			32	Supported except for 32APSK/64APSK modulations
2 DMU VIT + 2 DPU RS 223/255	3	-	12	
2 DMU VIT + 2 DPU RS 239/255	4	-	13	
2 DMU VIT + 4 DPU + AQPSK FO	5	-	30	
1 DMU VIT 1/4 + 1 DPU RS 223/255	6	-	12	
1 DMU VIT + 1 DPU RS 223/255	7	-	12	
2 DMU TCM 2.00/3 + 2 DPU RS DVB	8	-	11	
1 DMU TCM 2.50/3 + 1 DPU RS DVB	9	-	10	
1 DMU TCM 2.00/3 + 1 DPU RS DVB	10	-	11	
2 DMU + 4 DPU RS WORLDVIEW	11	-	17	
1 DMU TCM 2.50/3 + 1 DPU RS 239/255	12	-	7	
1 DMU TCM 2.50/3 + 1 DPU RS 239/255	13	-	7	
1 DMU + 1 DPU LDPC 1/2 4k	15	-	40	LDPC 1/2 is not supported if one decoder only.
1 DMU + 8 DPU RS 223/255	16	-	29	
1 DMU + 1 DPU LDPC 7/8	17	1	3	
1 DMU + 1 DPU LDPC 1/2 4k		2	40	
1 DMU + 1 DPU LDPC 2/3 4k		3	43	
1 DMU + 1 DPU LDPC 1/2 4k		4	41	
1 DMU + 1 DPU LDPC 2/3 1k		5	44	
1 DMU + 1 DPU LDPC 1/2 16k		6	42	
1 DMU + 1 DPU LDPC 7/8	18	1	3	
1 DMU + 1 DPU LDPC 1/2 4k		2	40	
1 DMU + 1 DPU LDPC 2/3 4k		3	43	
1 DMU + 1 DPU LDPC 1/2 4k		4	41	
1 DMU + 1 DPU LDPC 2/3 1k		5	44	
1 DMU + 1 DPU LDPC 1/2 16k		6	42	
1 DMU + 1 DPU LDPC 7/8	19	-	32	
1 DMU + 1 DPU LDPC 1/2 4k	20	-	40	LDPC 1/2 is not supported if one decoder only.
1 DMU + 1 DPU LDPC 1/2 1k	21	-	41	LDPC 1/2 is not supported if one decoder only.
1 DMU + 1 DPU LDPC 1/2 1k	22	-	41	LDPC 1/2 is not supported if one decoder only.
1 DMU VIT + 1 DPU RS 223/255	23	-	12	

XXL Menu			4G Menu	
Description	Demod Number	Ext. Number	Number	Comment
2 DMU TCM 2.50/3 + 2 DPU RS 239/255 + RS 10.6	24	-	7	
1 DMU + 1 DPU SCCC	25	7	34	
1 DMU + 1 DPU DVB-S2		8	18	
1 DMU + 1 DPU SCCC	26	7	34	
1 DMU + 1 DPU DVB-S2		8	18	
1 DMU VIT + 2 DPU RS 223/255 + AQPSK FO	27	-	30	
1 DMU TCM 2.50/3 + 1 DPU RS DVB	28	-	10	
1 DMU VIT + 1 DPU RS 239/255	29	-	13	
2 ECL Input + 2 DPU RS 223/255	30	-	-	Not supported. Mission specific.
2 ECL Input + 2 DPU RS 239/255	31	-	-	Not supported. Mission specific.
2 DMU VIT + COMBI + 2 DPU RS 223/255	32	-	39	
2 DMU X-DEAF + 2 DPU RS 223/255	33	-	39	
2 DMU TCM 2.25/3 + 2 DPU RS 239/255 + RS 10.6	34	-	15	
2 DMU TCM 2.75/3 + 2 DPU RS 239/255 + RS 10.6	35	-	16	
1 DMU VIT 1/4 + 1 DPU RS 223/255	36	-	12	
2 DMU VIT + 4 DPU RS 223/255	37	-	46	
1 DMU VIT + 1 DPU RS 223/255	38	-	12	
1 DMU VIT + 1 DPU RS 239/255	39	-	13	
2 DMU + COMBI + 1 DPU LDPC 7/8	40	-	48	
2 DMU TCM 2.50/3 + COMBI + 1 DPU RS 239/255	41	-	7	
1 DMU FO + 1 DPU LDPC 7/8	42	-	-	Not supported. Mission specific.
1 DMU + 2 DPU LDPC 7/8	43	-	22	
1 DMU + ECL-in + ECL-out + 1 DPU LDPC 7/8	44	-	-	Not supported. Mission specific.
2 DMU + 2 DPU + SYMBOL FO	45	-	5	
2 DMU TCM 2.50/3 + 2 DPU	46	-	-	Not supported. Mission specific.
2 DMU VIT + 2 DPU RS 223/255 + COMP. RS	47	-	-	Not supported. Mission specific.
2 DMU TCM 2.50/3 + 2 DPU RS 239/255 + COMP. RS	48	-	-	Not supported. Mission specific.
2 DMU + FI + FO	49	-	-	Not supported. Mission specific.
1 DMU VIT + 2 DPU RS 223/255 + AQPSK FO	50	-	30	
2 DMU VIT + 4 DPU RS 239/255	51	-	46	
1 DMU + 8 DPU RS 223/255	52	-	29	
1 DMU X-DEAF TCM 2.50/3 + 1 DPU RS 239/255	53	-	7	
1 DMU X-DEAF TCM 2.00/3 + 1 DPU RS 239/255	54	-	9	
1 DMU X-DEAF TCM 2.00/3 + 1 DPU RS 223/255	55	-	8	
1 DMU X-DEAF TCM 2.50/3 + 1 DPU RS 223/255	56	-	5	

XXL Menu			4G Menu	
Description	Demod Number	Ext. Number	Number	Comment
2 DMU + 2 DPU RS 223/255 + 2 FI	57	-	-	Not supported. Mission specific.
2 DMU + 4 DPU RS 223/255 + 3 FI	58	-	-	Not supported. Mission specific.
2 DMU TCM DSNG + 2 DPU RS DVB DSNG	59	-	-	Not supported.
TEST MENU (3 MERGED PRN ON EXTERNAL CLOCK)	60	-	-	Not supported. Mission specific.
2 DMU + 2 DPU RS 223/255 + FI SPLIT3	61	-	-	Not supported. Mission specific.
2 DMU TCM 2.50/3 + 2 DPU RS 223/255 + RS 10.6	62	-	5	
2 DMU TCM 2.00/3 + 2 DPU RS 239/255 + RS 10.6	63	-	9	
2 DMU TCM 2.00/3 + 2 DPU RS 223/255 + RS 10.6	64	-	8	
1 DMU TCM 2.25/3 + 1 DPU RS 239/255	65	-	15	
1 DMU TCM 2.75/3 + 1 DPU RS 239/255	66	-	16	
1 DMU TCM 2.50/3 + 3 DPU RS 223/255 + SPLIT3	67	-	-	Not supported. Mission specific.
1 DMU + 1 DPU LDPC 7/8 I/Q	68	-	21	
1 DMU TCM 2.00/3 + 3 DPU RS 223/255 + SPLIT3	69	-	-	Not supported. Mission specific.
1 DMU TCM 2.50/3 + 3 DPU RS 239/255 + SPLIT3	70	-	-	Not supported. Mission specific.
1 DMU TCM 2.00/4 + 3 DPU RS 239/255 + SPLIT3	71	-	-	Not supported. Mission specific.
2 DMU + 2 DPU + 2 LDPC 7/8 EXT	72	9	32	
2 DMU + 4 DPU + 4 LDPC 7/8 EXT	73	10	22	
2 DMU + 2 DPU I/Q + 4 LDPC 7/8 EXT	74	10	21	
2 DMU + 2 DPU RS 223/255	75	-	-	Not supported.
2 DMU TCM 2.25/3 + 2 DPU RS 223/255 + RS 10.6	76	-	15	
2 DMU TCM 2.75/3 + 2 DPU RS 223/255 + RS 10.6	77	-	16	
1 DMU TCM 2.00/3 + 1 DPU RS DVB	78	-	11	

4.2.7 4G / XXL test modulator menus equivalence

Here is a table which provides test modulator menu number equivalence with previous generation of Cortex HDR.

XXL Menu		4G Menu	
Description	Number	Number	Comment
Default	1	3	
UQPSK	2	12	
High Speed	3	1	
Byte out	4	4	
Parallel DGU	5	13	
FIR 4SR	6	3	
Doppler	7	3	
DVBS2	8	2	
GOES-R	9	-	Not supported. Mission specific.
CSO/Pleiades	10	4	
Myriad	11	-	Not supported. Mission specific.
AIT	12	-	Not supported. Mission specific.
AIT	13	9	
Astra	14	-	Not supported. Mission specific.
FIR 2SR	15	3	
AIT	16	-	Not supported. Mission specific.
AIT	17	-	Not supported. Mission specific.
AIT	18	-	Not supported. Mission specific.
AIT	19	-	Not supported. Mission specific.
LDPC 1/2	20	5	
AIT	21	-	Not supported. Mission specific.
AIT	22	-	Not supported. Mission specific.
AIT	23	-	Not supported. Mission specific.
Pi/4-DQPSK	24	-	Not supported.

4.2.8 SPS Version Identification

4.2.8.1 Common Data

Software version at delivery is described in Annex 5 (Section 1) and, with more details, in the CORTEX Acceptance Test Report.

- ❑ To know the version of the PROMed software (SDSY electronic cards), refer to the Acceptance Test Report (see also the marking on the PROM chips).

- ❑ To know the Signal Processing Software (SPS) version and what version of software is

downloaded at boot to the SDSY electronic cards, click on  at the lower right corner of the display and select **About Cortex**:

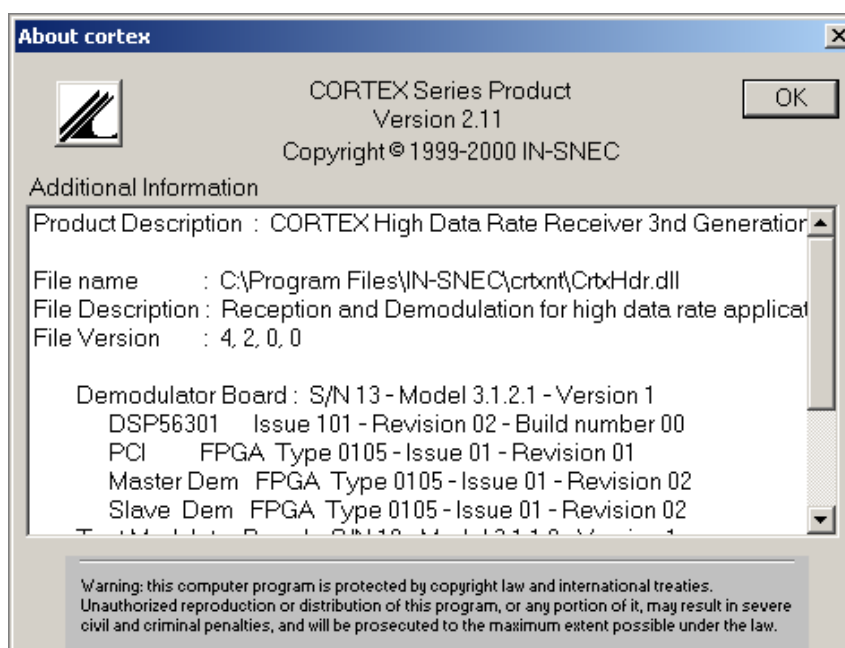


Figure 79: About CORTEX SPS Version

The SPS is built around an **SPS kernel** (supporting multiple CORTEX products such as the CORTEX CRT, CORTEX HDR, CORTEX ACS, CORTEX RTR, etc...) and a **product-specific SPS software layer**.

The header of the above window shows the version of the SPS kernel (example: *Version 2.4*).

4.2.8.2 CORTEX HDR Specific Data

The inner window shows the version of the SPS software layer specific to the CORTEX HDR product (example: *2.2.0.0*).

4.2.9 CORTEX HDR System Parameters in Registry

4.2.9.1 CORTEX Station

4.2.9.1.1 Menus Description

Access path to the Menu Description file (**menu.txt**).

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/Cortex Station/Remote Control**
Keyword: **MenuDescription**
Default setting: **"C:\\Program Files\\IN-SNEC\\crtxNt\\Menus\\menu.txt"**

4.2.9.1.2 Last Used Menu

This status indicates what was the working menu before shutting down the machine

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/Cortex Station/Remote Control**
Keyword: **LastConfiguration**

4.2.9.1.3 RAC Port Number

Allows changing the RAC port number:

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/Cortex Station/Remote Control**
Keyword: **PortCntx**
Default setting: **3003**

4.2.9.2 Demodulation Unit

4.2.9.2.1 Output Clock ON/OFF

Enabling or disabling the data clock from the Bit Synchronizer:

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Preferences**
/Dmu#n
Keyword: **ClkOut**

Enter **0** for permanent clock

Enter **1** for disabling the clock signal when the Bit Synchronizer is unlocked.

Enter **2** for limiting the clock signal frequency deviation to 0.1% of the programmed bit rate when the Bit Synchronizer is unlocked, and outputting the effective bit rate when the Bit Synchronizer is locked.

Enter **3** for clock setting accuracy better than 1 ppm when the Bit Synchronizer is unlocked and outputting the effective bit rate when the Bit Synchronizer is locked.

Enter **4** for clock disabling during frequency acquisition.

Default setting: **0** (clock always present)

4.2.9.2.2 Punctured Viterbi Setting

Setting of the punctured codes and swap function:

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Preferences/Dmu#n**

Keywords: **ViterbiPunctCode0, ViterbiPunctCode1 and ViterbiPunctSwapp**

For Code 0 and code 1: enter **0** for no puncturing and **1** for punctured Viterbi. Enter **0** for swap OFF, **1** for swap ON convention 1 and **2** for swap ON convention 2.

Default setting: **0**.

4.2.9.2.3 BER on Merged or Separate I/Q

Selecting the BER measurement technique in PRN BER mode (see 3.3.8) and BER display on the GUI:

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Dmu#n**

Keyword: **BerIQ**

Enter **0** for one BER value on merged I & Q (using method 1).

Enter **1** for two BER values (on I & on Q) (using method 1).

Enter **2** for one BER value on merged I & Q (using method 2).

Default setting: **0**.

4.2.9.2.4 High EbNo Tracking Mode

This key describes the mode of tracking of the high EbNo:

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Dmu#n**

Keyword: **TrackBERHigh**

Default setting: Enter **0** for EbNo computed from the I/Q constellation. Enter **1** for high EbNo (> 12 dB) update from IF level measurements (assuming the noise level is constant).

4.2.9.2.5 I/Q Synchronization in Capture Mode (menu # 7)

Expected timing for external I/Q input signals (external I/Q data recovery mode):

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Dmu#n**

Keyword: **BitSyncMode_IQ_sync**

Enter **0** for synchronized I/Q and **1** for Q data timing delayed by ½ symbol compared to I data.

Default setting: **0**

4.2.9.2.6 DNRZ PCM Decoding (deprecated)

Selecting the DNRZ code in QPSK and OQPSK demodulation mode:

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Dmu#n**
Keyword: **DNRZ_code**
Default setting: **0**

4.2.9.2.7 OQPSK (SQPSK) Algorithm

OQPSK demodulation algorithm definition

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Dmu#n**
Keyword: **Algo_OQPSK**

Enter **0** for standard algorithm and **1** for enhanced algorithm (better resistance to pattern, but higher synchronization threshold).

Default setting: **0**.

4.2.9.2.8 DVB-S2 randomization polynomial

Selection of the randomization polynomial for DVB-S2 standard

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Dmu#n**
Keyword: **DVBS2_PIScrambling_Num**

Enter **0** to **262141** (refer to the DVBS2 standard, document EN 302307 V1.1.1 § 5.55.4)

Default setting: **0**.

4.2.9.3 Transport Layer: Frame Synchronizer Setting

4.2.9.3.1 Check to Lock Strategy (CTL) for RS DVB

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**
Keyword: **RSDVB_Ctl**

Enter an integer value between **0** and **10**.

Default setting: **2**.

4.2.9.3.2 Lock to Search Strategy (LTS) for RS DVB

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**

Keyword: **RSDVB_Lts**

Enter an integer value between **0** and **10**.

Default setting: **2**.

4.2.9.3.3 Synchronization Threshold (SYN) for RS DVB

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**

Keyword: **RSDVB_Syn**

Enter an integer value between **0** and **10** (maximum number of corrupted bits in the 32-bit synchronization word)

Default setting: **2**.

4.2.9.3.4 Synchronization Word for RS DVB

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**

Keyword: **RSDVB_DATA_SW**

Enter an integer value between **0** and **255 (0xFF)**.

Default setting: **184 (0xB8)**.

4.2.9.3.5 Idle Synchronization Word for RS DVB

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**

Keyword: **RSDVB_IDLE_SW**

Enter an integer value between **0** and **255 (0xFF)**.

Default setting: **150 (0x96)**.

4.2.9.3.6 Frame Size for RS DVB

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**

Keyword: **RSDVB_FrameSize**

Enter the transport frame size in bytes (synchronization word included): **0** and **65535 (0xFFFF)**.

Default setting: **4080 (0xFF0)**.

4.2.9.3.7 Control of the R-S DVB Decoding Chain for RS DVB

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**
Keyword: **RSDVB_Ctrl**
Default setting: **15 (0x0F).**

4.2.9.3.8 Check to Lock Strategy (CTL) for LDPC

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**
Keyword: **LDPC#m_Ctl**

Enter an integer value between **0** and **10**.
With #m: 12 for LDPC 1/2, 23 for LDPC 2/3, 78 for LDPC 7/8

4.2.9.3.9 Lock to Search Strategy (LTS) for LDPC

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**
Keyword: **LDPC#m_Lts**

Enter an integer value between **0** and **10**.
With #m: 12 for LDPC 1/2, 23 for LDPC 2/3, 78 for LDPC 7/8

4.2.9.3.10 Synchronization Threshold (SYN) for LDPC

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**
Keyword: **LDPC#m_Syn**

Enter an integer value between **0** and **10** (maximum number of corrupted bits in the 32-bit synchronization word)
With #m: 12 for LDPC 1/2, 23 for LDPC 2/3, 78 for LDPC 7/8

4.2.9.3.11 Derandomization for LDPC

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**
Keyword: **LDPC#m_descrambling**

Enter **0** (disable) or **1** (enable)
With #m: 12 for LDPC 1/2, 23 for LDPC 2/3, 78 for LDPC 7/8

4.2.9.3.12 Synchronization Word LSB for LDPC

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**
Keywords: **LDPC#m_SW_Lsb**

Enter an integer value between **0x0** and **0xFFFFFFFF**
With #m: 12 for LDPC 1/2, 23 for LDPC 2/3, 78 for LDPC 7/8

4.2.9.3.13 *Synchronization Word MSB for LDPC*Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**Keyword: **LDPC#m_SW_msb**Enter an integer value between **0x0** and **0xFFFFFFFF**

With #m: 12 for LDPC 1/2, 23 for LDPC 2/3, 78 for LDPC 7/8

4.2.9.3.14 *Synchronization Mask LSB for LDPC*Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**Keyword: **LDPC#m_SW_MASK_lsb**Enter an integer value between **0x0** and **0xFFFFFFFF**

With #m: 12 for LDPC 1/2, 23 for LDPC 2/3, 78 for LDPC 7/8

4.2.9.3.15 *Synchronization Mask MSB for LDPC*Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**Keyword: **LDPC#m_SW_MASK_msb**Enter an integer value between **0x0** and **0xFFFFFFFF**

With #m: 12 for LDPC 1/2, 23 for LDPC 2/3, 78 for LDPC 7/8

4.2.9.3.16 *Synchronization Word Length for LDPC*Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**Keyword: **LDPC#m_SW_size**

Enter the transport frame sync word size in bits (synchronization word included): 0 and 64

With #m: 12 for LDPC 1/2, 23 for LDPC 2/3, 78 for LDPC 7/8

4.2.9.3.17 *Decoder refresh period*Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**Keyword: **DecoderRefreshPeriod**

Period of the update of the decoder in millisecond.

Default setting: **0x3e8 (1000 ms)**

4.2.9.3.18 *Transport layer status*

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/InSnec/CrtxHdr/SetDefault/Signal Processing/Dpu#n**

Keyword: **PermanentTransportStatus**

0: when the frame synchronizer of the CADU layer is unlock, the counters of the Transport layer are set to 0 and not updated

1: when the frame synchronizer of the CADU layer is unlock, the counters of the Transport layer are updated.

Default setting: **0**

4.2.9.4 *Test Modulation Unit*

4.2.9.4.1 *Selecting the DNRZ code in QPSK and OQPSK modulation mode:*

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /tstu**

Keyword: **DNRZ_code**

Default setting: **0**

4.2.9.5 *Global HDR*

4.2.9.5.1 *Selecting LVDS / ECL inputs of the test modulator:*

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Global**

Keyword: **LVDS_input**

0: selection of the ECL inputs

1: selection of the LVDS inputs

Default setting: **0**

4.2.9.5.2 *Recording in raw data mode:*

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Global**

Keyword: **UnlockedNoTM**

0: in raw data mode, data are recorded even if the demodulator is not locked

1: in raw data mode, data are recorded only if the bit synchronizer of the demodulator is locked

Default setting: **0**

4.2.9.5.3 *DMX table:*

Registry path: ***HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Global***

Keywords: ***DisableDmxAllTable***

1: disable the DMX table from the All Components table (refer to the document STI 100013 HDR)

0: enable the DMX table inside the All Components table

Default setting: **0** for a new software installation

Default setting: **1** for a software upgrade installation

4.2.9.5.4 *TCP/IP data transmission:*

Registry path: ***HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Global***

Keyword: ***TimeoutTcp***

Delay (in second) before transmitting of an uncompleted Telemetry Data Message over TCP/IP

Default setting: **0**

4.2.9.5.5 *TCP/IP Playback data reception*

Registry path: ***HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxHdr/SetDefault/Signal Processing /Global***

Keyword: ***TimeoutPbk***

Delay (in milliseconds) before disconnecting the user from Playback port (PBK-1 or PBK-2), stopping the playback LAN session and flushing all remaining data; if user is no longer sending data to Playback port.

Default setting: **5000**

4.3 Monitoring & Control Software (MCS)

4.3.1 General

Main MCS functions are:

- ☐ Display the current CORTEX configuration and status from the monitoring tables transmitted by the SPS,
- ☐ Modify the setting of the equipment. This function is **password-protected**,
- ☐ Reset the TCP-IP connections to the CORTEX (close all TCP-IP connections to a given port or close all connections to the CORTEX). This function is **password-protected**.
- ☐ **Product-specific functions:** telemetry Quick Look (CORTEX CRT), satellite commanding and ranging (CORTEX CRT), etc.... For more details, see next sections.

The MCS can be installed in any PC workstation using and having 32 Mbytes of RAM or more. The MCS and SPS communicate via an Ethernet link under TCP-IP.

Some of the TCP-IP ports of the CORTEX (monitoring, logging, etc...) accept simultaneous connections of several clients while other ports (Control, Reset, etc...) accept only one connection at a time. For more details concerning the Ethernet interface, refer to:

- ☐ CORTEX Series TCP-IP Ethernet interface (reference STI 100013): Ethernet ICD common to all CORTEX products),
- ☐ CORTEX XXX TCP-IP Ethernet interface: product-specific Ethernet interface extension, where XXX is the product extension name. Example: STI 100013_CRT is the CORTEX CRT Ethernet ICD, STI 100013_HDR is the CORTEX HDR Ethernet ICD, STI 100013_ACS is the CORTEX ACS Ethernet ICD, etc....

The MCS is built around an **MCS kernel** (supporting multiple CORTEX products such as the CORTEX CRT, CORTEX HDR, CORTEX ACS, CORTEX RTR, etc...) and a **product-specific SPS software layer** which is automatically identified when the MCS connects to the CORTEX SPS.

4.3.2 MCS Documents

MCS Documents are configuration files **created and managed by the CORTEX MCS**.

An MCS Document contains the whole set of configuration parameters that have been previously stored on the disk under file extension .MCS.

The MCS document also contains the **name or Ethernet address** of the CORTEX whose configuration has been saved in the MCS document. This name or address is displayed on top of the graphical window.

A default MCS Document (*cortex-nt.mcs*) is loaded by the MCS when the CORTEX is started; it can be then used to re-configure the machine. A copy of this file (MCS Table) is maintained in RAM and is updated whenever the CORTEX configuration is changed by the MCS operator. The MCS Table can be **manually saved** to the MCS Document on the hard disk by MCS Operator.

Additional MCS Documents can be created at will by the operator. A set of MCS Documents (common satellites configuration) is available on the hard disk C:\Program Files\IN-SNEC\crtxMcs\DefaultConfig).

A copy of the original MCS files (those installed from the CORTEX reference CD-ROM) is available in directory C:\Program Files\IN-SNEC\crtxMcs\DefaultConfig_ref. These files are in read-only mode. They can be copied to the working directory (C:\Program Files\IN-SNEC\crtxMcs\DefaultConfig) if necessary (remove the read-only protection on the copies).

MCS documents are **binary files**. They cannot be displayed on the screen or printed out, except if the MCS software package is installed in a remote PC with printing capability (PRINT PREVIEW & PRINT commands).

4.3.3 MCS Start Sequence. MCS Control Status

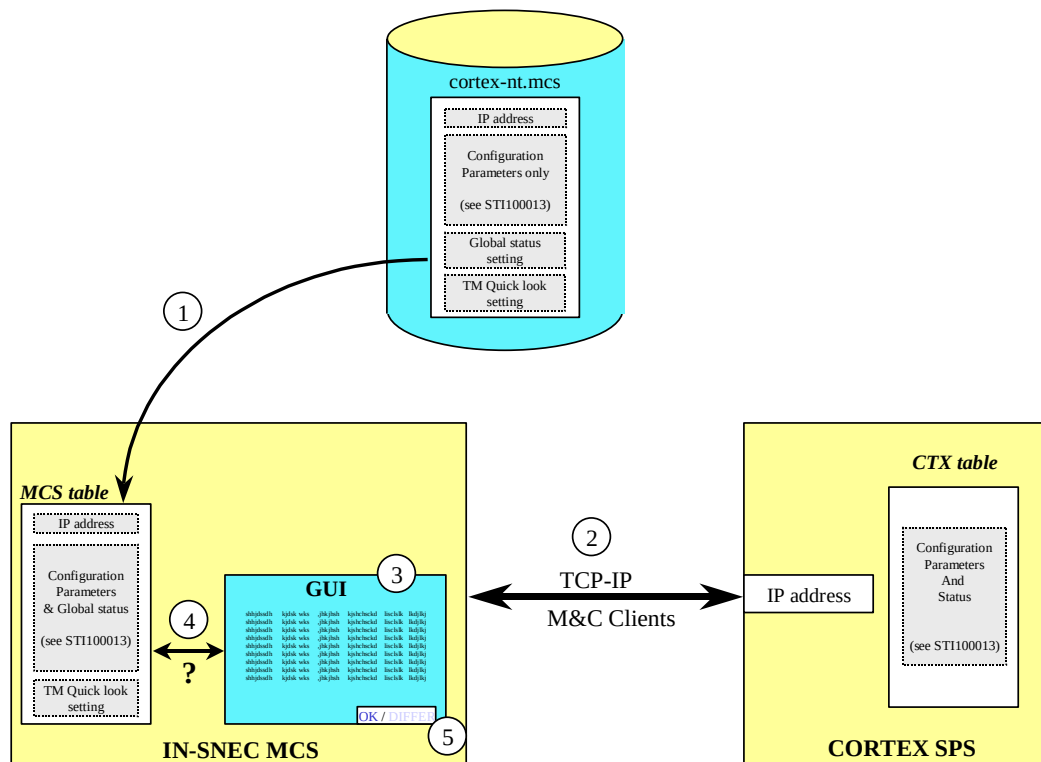
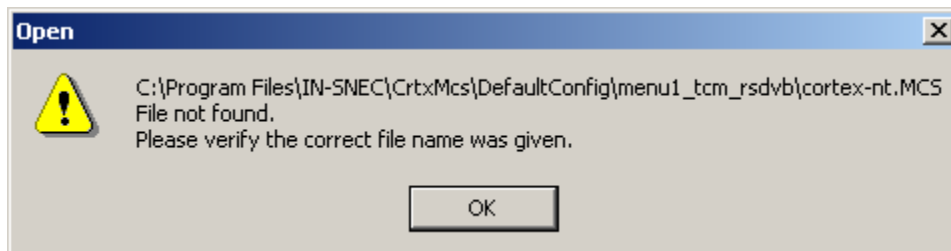


Figure 80: MCS Start Sequence

- ❑ **STEP 1:** When the Monitoring & Control Software (MCS) is started, it opens the default MCS Document (program argument) *c:\Program Files\In-snecl\ctx\Mcs\DefaultConfig\menu1_tcm_rsdvb\cortex-nt.MCS* on the disk and makes a copy in RAM (**MCS Table**). If the default MCS Document is not found the following warning message is displayed:



- ❑ **STEP 2:** The MCS connects to the Monitoring port of the CORTEX and starts monitoring the machine. If the MCS fails to open the connection, the MCS Login window indicates **Disconnected** and a warning message is displayed.



- ❑ **STEP 3:** Once the connection to the CORTEX has been established, the configuration parameters, the status and alarms received from the connected machine are refreshed in real time on the GUI. Configuration commands entered at the GUI are executed and the MCS Table refreshed accordingly.
- ❑ **STEP 4:** The MCS permanently compares the configuration parameters from the MCS Table to the ones displayed on the GUI.
- ❑ **STEP 5:** If the GUI strictly reflects the contents of the MCS Table, the **MCS control status** at the bottom of the Top Level window displays:



Else (the CORTEX has been reconfigured by a remote M&C client or the MCS document has not been saved properly prior shutting down the machine) the **MCS control status** is changed to:



- ❖ Click on icon  to reconfigure the CORTEX from the MCS Table, or
- ❖ Click on icon  to update the MCS Table from the configuration parameters on the GUI (CTX table).

4.3.4 M&C Session. MCS Documents Management

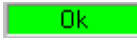
4.3.4.1 Close the Active M&C Session

- Click on FILE → CLOSE. If required, save the active MCS Table to the disk.

4.3.4.2 Open an M&C session with a CORTEX

- Click on FILE → NEW. An empty MCS Login window appears on the screen. The MCS creates an empty MCS Table in RAM.
- Enter the IP address or name of the CORTEX unit and click on  (or click on CONNECT). Once the monitoring connection is established, the MCS starts monitoring the unit and initializes the contents of the MCS Table (IP address or name of the unit included). The MCS control status is set to , indicating that the contents of the MCS Table strictly matches the configuration of the unit. The MCS Table is automatically updated every time the Operator modifies the setting of the unit from the MCS on which the M&C session has been started (the MCS control status remains to ).
- The MCS control status automatically turns to  if any remote Monitoring & Control system modifies the configuration of the unit.
- For saving the MCS Table to an MCS Document on the disk, see next section. Note: **multiple M&C sessions** can be simultaneously started from the MCS.

4.3.4.3 Save the CORTEX Configuration to an MCS Document on the Disk

- Verify that the MCS control status is set to . If it is set to  click first on .
- If the MCS Table has never been saved to the disk, click on FILE → SAVE or FILE → SAVE AS and either confirm the name proposed in the dialog box or enter another name.
- If the MCS Table has already been previously saved to the disk, click on FILE → SAVE to save it again to the same MCS Document or click on FILE → SAVE AS to save it to another new or existing MCS Document.

4.3.4.4 Reconfigure the CORTEX from an Existing MCS Document

- Close the active M&C session (click on FILE → CLOSE).
- Click on FILE → OPEN and select an MCS Document in the proposed list. The selected MCS Document is copied to an MCS Table in RAM.
- Click on  to reconfigure the CORTEX from the new MCS Table. The MCS control status is changed to , indicating that the unit has been successfully reconfigured.

4.3.5 Graphical User Interface

The following window appears on the screen when the MCS is started:

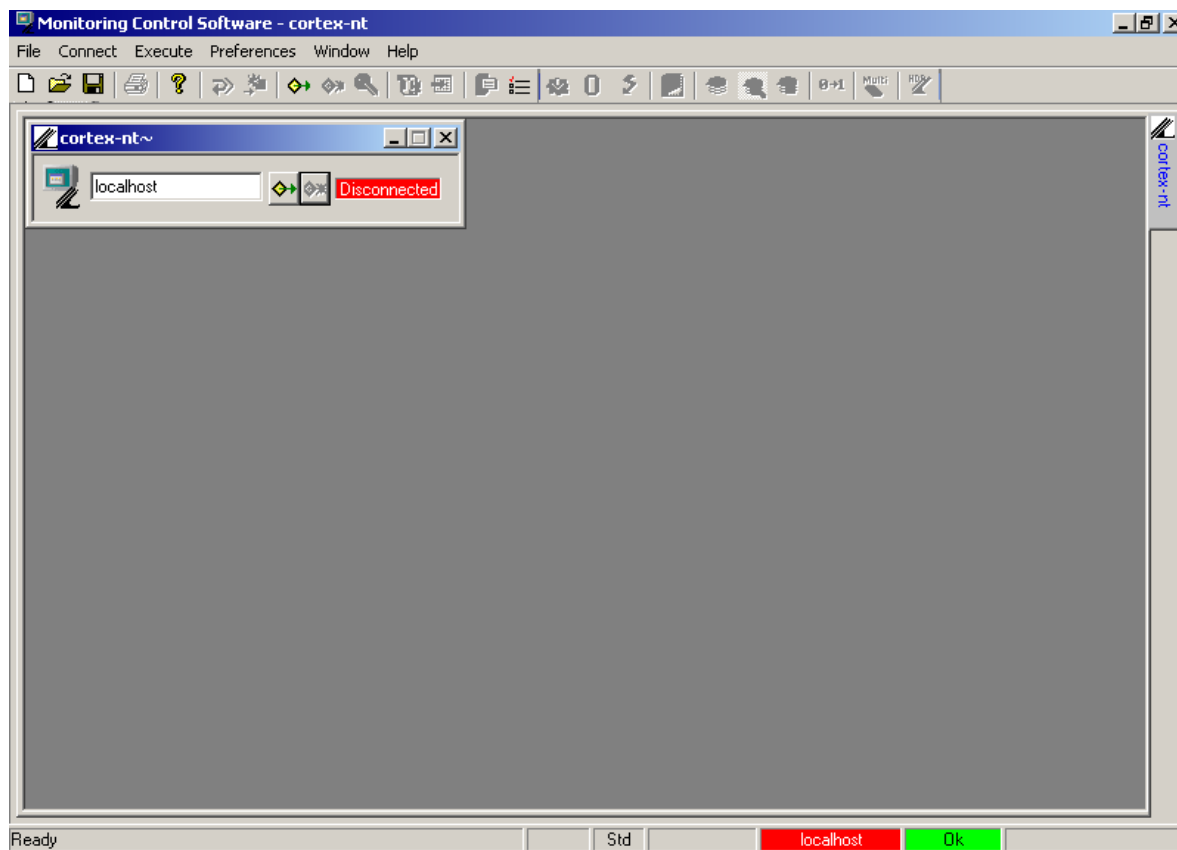


Figure 81: MCS Top-Level Window

The window name (**Monitoring Control Software**) is followed by the name of the MCS Document opened by the Monitoring & Control Software.

4.3.5.1 MCS Login Window

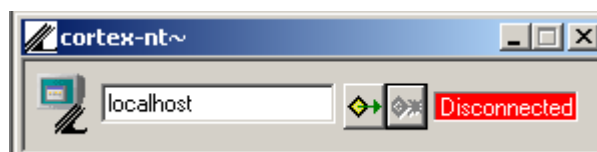


Figure 82: MCS Login Window

The MCS login window enables to connect or disconnect to the Cortex SPS and check the connection status. The connection buttons are disabled if no machine name or IP address is provided in the host identification zone. Start entering the machine name or IP address and the connection button will automatically be enabled.

Connect the MCS to the SPS by clicking on  or by pressing the RETURN or ENTER key. If the host machine accepts the connection, the disconnection button  is enabled and the connection status turns to **Connected**. Otherwise, a connection error message appears on the screen.

After the connection is established the host identification zone becomes gray. Changing the IP address is no more possible unless the  disconnect button is pressed.

Next sections describe the commands and statuses available from the MCS window after the connection is established.

4.3.5.2 MTDI (Multiple Tabbed Document Interface)

The MCS software is based on MTDI (Multiple Tabbed Document Interface) space. The MTDI enables to separate in different working spaces the MCS Documents. Every document is identified by the host name and shows all the windows corresponding to this connection. The tabs enable to go from a working space to another. The tabs can be placed either at the bottom or right side (default setting) of the MCS window (menu PREFERENCES).

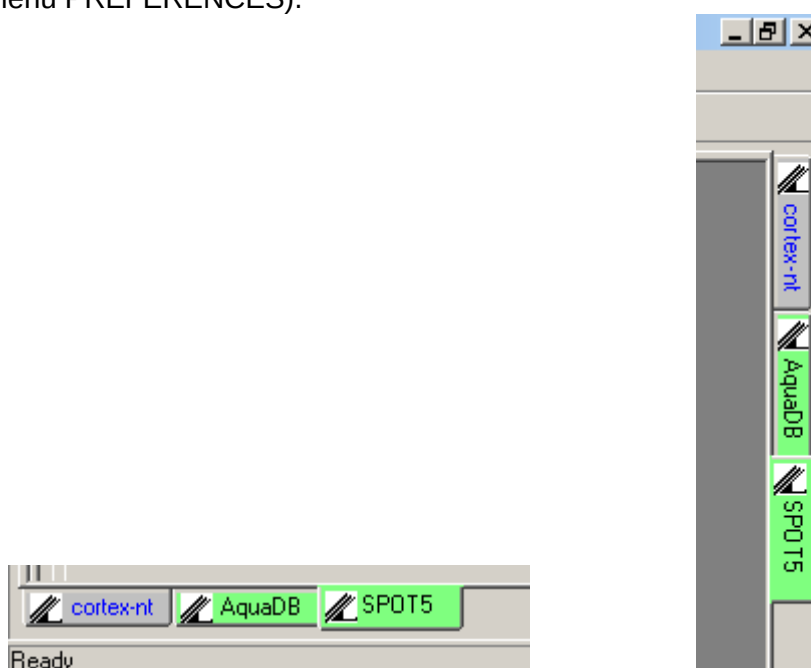


Figure 83: Multiple Tabbed Document Interface

The tabs enable to know the status of the connection and host CORTEX:

- in black on green back : MCS is connected and host CORTEX is OK,
- in white on red back : MCS is connected and host CORTEX is in alarm,
- in blue on gray back : MCS is disconnected.

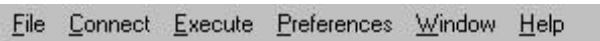
In case of host CORTEX alarm the tabs turn to red.

Another option enables to display on the tab the name of the host machine or MCS Document (menu PREFERENCE).

4.3.5.3 Upper Toolbar

4.3.5.3.1 Common Data

The following commands are common to all of the CORTEX products. For product specific commands, see next section.



FILE

NEW: Create a new MCS Document to open a new M&C session.

OPEN: Open an M&C session from an existing MCS Document on the disk

CLOSE: Close the current M&C session and document. If the document is not saved, a messages proposes to save it. The workspace is closed too.

SAVE: Save the active MCS Document to the original document on the disk.

SAVE AS: Save and rename the active MCS Document to the disk.

PRINT: Print out the contents of the active MCS Table.

PRINT PREVIEW: Display the contents of the active MCS Table.

PAGE SETUP: Change the printer and printing options.

EXIT: Exit the MCS.

<file_list >: Select an existing MCS document from the list.

CONNECT

CONNECT TO: Connect the MCS to the monitoring port of a CORTEX. Command available only if an IP address is provided in the host identification zone.

DISCONNECT: Close the monitoring connection to the CORTEX.

EXECUTE

CONTROL ACCESS: Activate or inhibit the configuration mode.

OPEN CONTROL PORT: Open a TCP-IP connection to the Control port. Opening the connection will inhibit access to the Control port by other remote Control Clients.

CLOSE CONTROL PORT: Close the TCP-IP connection to the Control port.

SPECIAL COMMANDS

INITIALISATION: Re-program the CORTEX to boot up state.

RESTART DEMODULATOR: Restart the acquisition of the demodulator.

CLOSE TCP: Close the TCP-IP connection for the active window.

INITIALISATION: Re-program the CORTEX to boot up state.

LOAD CONFIGURATION: Load a User configuration

PREFERENCES	TOOL BAR: Display or hide the tool bar.
	STATUS BAR: Display or hide the status bar.
	BOTTOM TABS: Move the document tabs to the bottom or to the right side of the MCS window (MTDI).
	VIEW HOSTNAMES: Display on the tabs the name of the host machine or the name of the active MCS Document.
	OK / DIFFER MANAGEMENT: Management of the CONFIG/APPLY mode: STANDARD or PROTECTED or AUTOMATIC See Section 4.3.6.4 for a detailed description of the CONFIG/APPLY modes
WINDOW	NEW MAIN WINDOW: Create a new product main window.
	MENU_SELECTION: Start-up menus management (see section 4.2.4)
	CASCADE: Reorganize the windows in cascade.
	TILE: Reorganize the windows in order to display everything.
	ARRANGE ICONS: Represent the windows in icons.
	<list_window >: List the opening windows in the workspace.
HELP	ABOUT: Display the versions of the MCS kernel, products and plugins.

4.3.5.4 Lower Toolbar

Hide or display the lower toolbar from menu PREFERENCE / TOOL BAR. The lower toolbar appears on the screen only after the MCS is started.

4.3.5.4.1 Common Data

The following commands are common to all of the CORTEX products. For product specific commands, see next section.



	Same as command FILE - NEW.
	Same as command FILE - OPEN.
	Same as command FILE - SAVE.
	Same as command FILE - PRINT.
	Same as command HELP - ABOUT.
	Update the MCS Table from the real configuration of the connected CORTEX.
	Reconfigure the connected CORTEX from the active MCS Table.
	Connect to the monitoring port of the CORTEX.
	Close the connection to the monitoring port of the CORTEX.
	Activate/inhibit the configuration mode. Password-protected function.
	Open the TCP window.
	Toggle command: open/close the connection to the Control port (see section 4.3.6.3).
	Open the CTX Document management window (see section 4.3.9).
	Open the MCS Menu Management window (see section 4.2.4.2).

4.3.5.4.2 CORTEX HDR Specific Data



Restart signal acquisition by the Demodulator. Turns to yellow when the Demodulator is locked. Turns to gray when the Demodulator is unlocked (see SPECIAL COMMAND - RESTART IFR).



Reset the BER counter for the active Demodulator Unit window or
Restart the status counter for the active Data Processing Unit window.



Close the TCP-IP connection for the active window. Turns to red when the connection is active and to gray when the connection is closed (see SPECIAL COMMAND - CLOSE TCP)



Open the data logging set-up window (refer to Logging window)



Open the Disk Partitioning & Memory Allocation window.



Toggle command: select the Configuration or Acquisition mode.



In Acquisition mode, close all open data acquisition processes to toggle in Configuration mode.



Invert one bit in the simulated data pattern (pseudo-random pattern or file). Bit inversion takes place before convolutional encoding (if enabled).



Allows the setting of a same parameter in all the active channels from a single window.



Recall the MCS Login window.



Load a User Configuration MCS file

4.3.5.5 Bottom Status Toolbar



From left to right:

- **OK/DIFFER mode:** Standard (**Std**), Protected (**Prot**) or Automatic (**Auto**). See section 4.3.6.4)
- **Alarm status:**
OK (gray background).
Alarm (red background) if any CORTEX component is in alarm. Detailed information on the alarm is available at the GUI.
Temperature (red background) if any temperature monitored by the IPMIMonitor tool exceeds the alarm threshold (see section 5.3.2).
- **Host name & connection status:** cortex-hdr01 (connected) or cortex-hdr01 (disconnected)
- **MCS control status:** Ok or Differ (see section 4.3.3).
- **Date and time** from the built-in time code reader: 1 - 01:51:42. Yellow background if the time decoder is unlocked or in flywheel mode; gray background if the decoder is locked.

4.3.6 CORTEX Monitoring & Configuration

4.3.6.1 Default Operating Mode

At start up, the default CORTEX MCS operating mode is **Monitoring only**. In all configuration windows, control pushbuttons are inhibited:

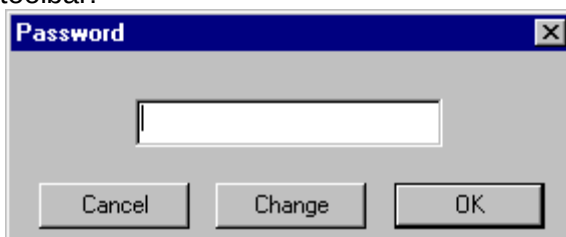


Configuration changes (by the MCS Operator or by any remote Control client), status and alarms are refreshed in real time on the GUI.

4.3.6.2 CORTEX Configuration

To reconfigure the CORTEX in a **single-operator** environment (no configuration conflict due to simultaneous access of several Control clients to the machine):

- Click on  in the lower toolbar:

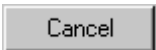


- Enter the **Control password**. Default setting at delivery: no password. Control pushbuttons are changed to:



- Click on . The pushbutton is changed to .

- Change the parameter setting and click on . The MCS:
 1. opens a connection to the Control port,
 2. transmits the configuration commands to the Signal Processing Software,
 3. closes the connection to the Control port and returns to the monitoring mode. The pushbutton is changed to .

To cancel the modifications, click on  instead of  or close the window.

To reconfigure the CORTEX in a **multi-operator** environment, read section 4.3.6.4 .

4.3.6.3 Forced Connection to the Control Port (Pseudo Local Mode)

After entering the Control password, the CORTEX can be set to a pseudo local mode by keeping open a TCP-IP connection to the Control port: reconfiguration of the unit by remote Control Clients remains impossible until the connection is released by the local Operator or a Reset Control port command is received on the Reset port.

To open a connection to the Control port, click on EXECUTE → OPEN CONTROL PORT. To close the connection to the Control port, click on EXECUTE → CLOSE CONTROL PORT.

4.3.6.4 OK/DIFFER Management Mode

The OK/DIFFER management mode is used to customize the configuration process. Three modes are proposed: STANDARD, PROTECTED and AUTOMATIC. Mode selection is from menu PREFERENCES.

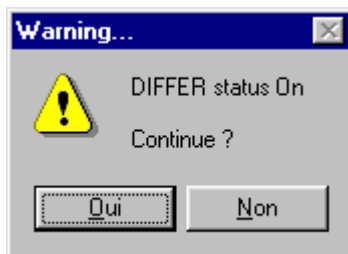
4.3.6.4.1 Standard Mode

When the CONFIG button is pressed, the status part of the GUI keeps on being refreshed in real time but the parameter part shows the contents of the active MCS Table (and not the parameter values returned in the monitoring tables). At this point, it should be noted that if the **MCS control status** is set to , **clicking on APPLY will reconfigure the equipment from the contents of the MCS Table !**

New parameter values entered by the Operator are stored in the MCS Table and when the APPLY button is pressed, the CORTEX is re-configured from the MCS Table and the MCS control status indicates .

4.3.6.4.2 Protected Mode

This mode is identical to the Standard mode above except that, if the MCS control status were **Differ** before clicking on CONFIG, a confirmation message is displayed on the screen when the APPLY button is pressed:



This message is to warn the Operator that the CORTEX will be re-configured from the MCS Table and that one or several parameters, else than those modified by the Operator, will be changed, thus affecting other remote CORTEX Operators. Click on NO to abort or YES to re-configure the machine.

4.3.6.4.3 Automatic Mode

In this mode, the MCS Table is automatically updated whenever a change in the configuration is made by the operator. Consequently, the MCS control status remains permanently to **Ok**, indicating that the contents of the MCS Table and the contents of the monitoring tables always match.

4.3.7 MCS Version Identification

To know the Monitoring & Control Software (MCS) version, click on  or click on **Help** in the top-level window, then select **About**.

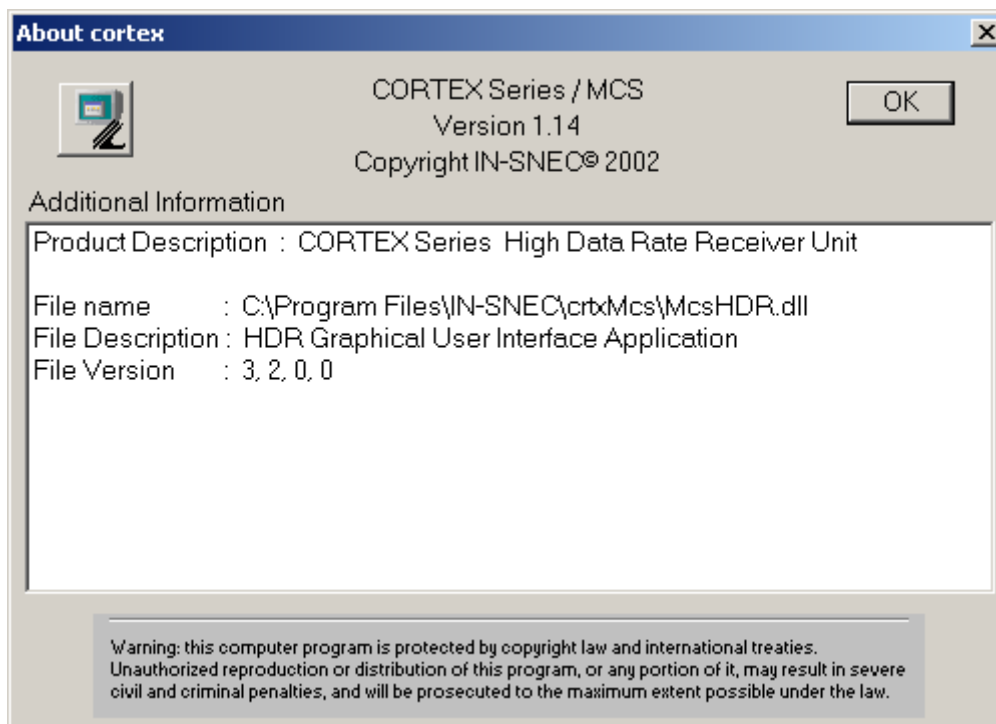


Figure 84: Cortex MCS About: Software Version

The MCS is built around an **MCS kernel** (supporting multiple CORTEX products such as the CORTEX CRT, CORTEX HDR, CORTEX ACS, CORTEX RTR, etc...) and a **product-specific MCS software layer**.

The header of the above window shows the version of the MCS kernel.

The inner window shows the version of the MCS software layer specific to the CORTEX HDR product.

4.3.8 Changing a Password

The following procedure applies to all passwords except the user login password.

To change a password:

1. Click on icon , or select button '**Change**'
2. Click on **CHANGE**,
3. Enter and confirm new password.

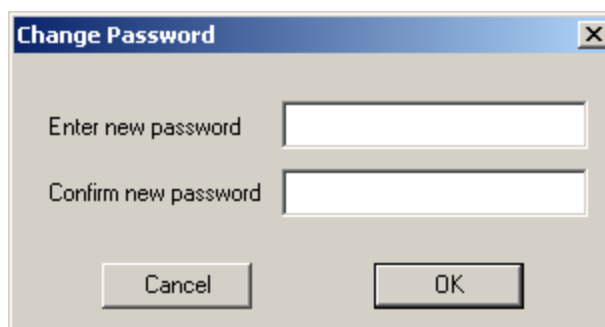


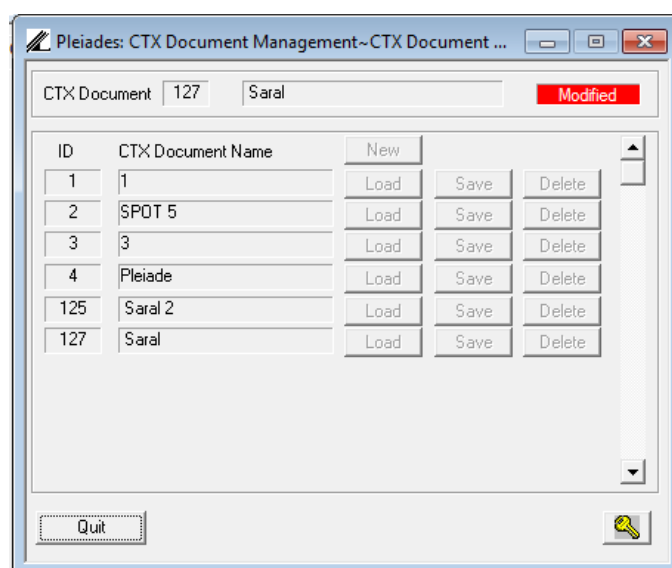
Figure 85: Changing a Password

Notes:

1. The Control password does not protect the CORTEX from re-configuration by remote Control Clients.
2. The TCP-IP password does not protect from reset commands issued by remote Reset Clients.

4.3.9 MCS Interface for CTX document management

Click on  to access the CTX Document management window, which enables to save up to 127 different configurations:



- Click on  and enter the password enabling the **Load**, **Save** and **Delete** commands
- To create a new CTX Document, click on **New** and enter the name (maximum name size: 16 characters), and the number (0 to 127) of the new CTX document
- Click on **Save** to capture the current CORTEX configuration and save it to an existing CTX Document on the disk
- Click on **Load** to load a CTX document in the current CORTEX configuration.

The last loaded or saved CTX Document is identified in the upper part of the window by its **ID number** and **name**. A configuration status indicates whether the current configuration of the unit matches the contents of the last loaded or saved CTX Document (Ok) or not (Differ).

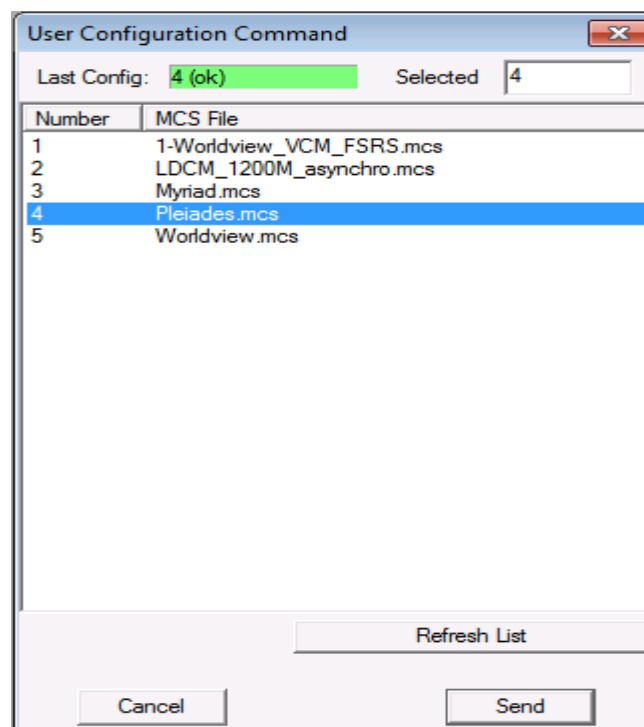
The CTX Document name can be modified only with the **Save** Command or by editing the **NameTable.TXT** file before starting the Signal Processing Software (SPS). If a CTX Document is not available (not created or damaged), the assigned name is blank. The Configuration Names table is transmitted over the LAN on reception of a monitoring request with Component code 200_H. This table is not part of the tables that can be monitored under an **All components tables** monitoring cycle.

4.3.10 MCS Interface for User MCS files loading

This window enables to display and to load a configuration file (MCS file) previously recorded inside the directory: c:\Program File\In-Snec\CrtxMcs\UserConfig or d:\ZDS\Data\HDR\CrtxMcs\UserConfig (for Windows 7) by the user.

Note that these MCS configuration files can be also remotely loaded using a special TCP/IP command.

Click on  (Upper toolbar of the GUI) to access the User Configuration Command window:



- Click on “Refresh List” to refresh the list of the available MCS files previously recorded by the user inside the directory: ... \CrtxMcs\UserConfig.
- Click on the desired line to select the configuration file,
- Click on” Send” to load the configuration inside the HDR.

The number of the selected file appears inside the "Selected" box. The status "Last Conf" is equivalent to the OK/Differ status of GUI.

4.3.11 MCS File System Architecture

IN-SNEC MCS software can be found in the following directories. Do not move or modify these directories:

ACCESS PATH			DESCRIPTION
C:\Program Files\In-snec Or D:\ZDS\Data\HDR			Root directory Upgrade.txt: list of s/w packages (ID + version)
\CrtxMcs			Monitoring & Control Software Graphical User Interface
			DefaultConfig
			MCS files
			DefaultConfig_ref
			MCS files (read only reference files)
			UserConfig
			MCS files (read only reference files)

4.3.12 MCS System Parameters in Registry

MCS system parameters are high-level parameters in the registry. Changing them requires restarting the CORTEX MCS application software:

4.3.12.1 Monitoring

4.3.12.1.1 Monitoring Rate

To change the time interval between two consecutive monitoring cycles by the MCS software:

Registry path: **HKEY_LOCAL_MACHINE / SOFTWARE\In-Snec\CrtxMcs\Tcp**
Keyword: **Monitoring**
Setting: **≥ 500D. Default: 500. Expressed in ms**

Example: for a monitoring request every four seconds, enter **FA0** (in hexadecimal) or **4000** (in decimal).
Minimum value: 1F4_H = 500_D (monitor the equipment twice per second).

4.3.12.2 MCS Auto-Connect

The following registry keys enables to force the MCS to regularly attempt to reconnect to the monitoring port of the CORTEX. The auto-connect mode is particularly useful on multi-menu CORTEX.

4.3.12.2.1 Auto Connect ON/OFF

Registry path: **HKEY_LOCAL_MACHINE / SOFTWARE\In-Snec\CrtxMcs\Tcp**
Keyword: **AutoReconnect**
Setting: **0: auto-connect mode disabled. 1: auto-connect mode enabled**

4.3.12.2.2 *Auto Connect Time-out*

Registry path: **HKEY_LOCAL_MACHINE / SOFTWARE\In-Snec\CrtxMcs\Tcp**
Keyword: **AutoReconnect** (time interval between retry to connect)
Setting: **Any integer value ≥ 1 . Default: 5000. Expressed in ms**

4.3.12.2.3 *Auto Connect Retry*

Registry path: **HKEY_LOCAL_MACHINE / SOFTWARE\In-Snec\CrtxMcs\Tcp**
Keyword: **NbRetryAutoReconnect** (maximum number of retry to connect)
Setting: **Any integer value ≥ 1 . Default: 3**

4.3.12.3 *MCS Preferences*

4.3.12.3.1 *Synchronization Word: Bit Masking Convention*

Synchronization Word bit masking convention for frame synchronization.

Registry path: **HKEY_LOCAL_MACHINE/SOFTWARE/In-Snec/CrtxMcs/Product/Product4**
/Preferences
Keyword: **SW Mask**

Enter **0** (ignore SW bits corresponding to bits set to 1 in the mask) or **1** (ignore SW bits corresponding to bits set to 0 in the mask).

Default setting: **0**.

4.4 Operating System and files architecture

4.4.1 Introduction

From the release E9R0, the Cortex HRD units come with Windows 7 embedded operating system in place of the former Windows 2000 and Windows XP.
Depending on the Operating System, the Cortex HDR software and files can be found in different directories. The paths of these different directories are given in the following paragraphs.
To improve security, with Windows 7, the Cortex user does not have access to C drive anymore. The configurations files have been moved to D drive.

4.4.2 Files Architecture

4.4.2.1 Files associated with Monitoring & Control Software

FILE	WINDOWS 7	WINDOWS 2000 AND XPE
MCS Application	C:\Program Files\ZDS\CrtxMcs	C:\Program Files\IN-SNEC\CrtxMcs
MCS Files	D:\ZDS\Data\HDR\CrtxMcs\DefaultConfig D:\ZDS\Data\HDR\CrtxMcs\DefaultConfig_ref (and subdirectories)	C:\Program Files\IN-SNEC\CrtxMcs\DefaultConfig C:\Program Files\IN-SNEC\CrtxMcs\DefaultConfig_ref (and subdirectories)
User's MCS Files	D:\ZDS\Data\HDR\CrtxMcs\UserConfig	C:\Program Files\IN-SNEC\CrtxMcs\UserConfig

4.4.2.2 Files associated with the Signal Processing Software

FILE	WINDOWS 7	WINDOWS 2000 AND XPE
SPS Application	C:\Program Files\ZDS\crtxnt	C:\Program Files\IN-SNEC\crtxnt
Menu List	D:\ZDS\Data\HDR\crtxnt\Menus	C:\Program Files\IN-SNEC\crtxnt\Menus
Menu CTX Files	D:\ZDS\Data\HDR\crtxnt\Menus\CTX D:\ZDS\Data\HDR\crtxnt\Menus\CTX_ref	C:\Program Files\IN-SNEC\crtxnt\Menus\CTX C:\Program Files\IN-SNEC\crtxnt\Menus\CTX_ref
User's CTX Files	D:\ZDS\Data\HDR\crtxnt\NameTable.txt D:\ZDS\Data\HDR\crtxnt\SatCnfXX.ctx (with XX = 01 to 10)	C:\Program Files\IN-SNEC\crtxnt\NameTable.txt C:\Program Files\IN-SNEC\crtxnt\SatCnfXX.ctx (XX = 01 to 10)
BER Files	D:\ZDS\Data\HDR\crtxnt\BER	C:\Program Files\IN-SNEC\crtxnt\BER
VCM Files	D:\ZDS\Data\HDR\crtxnt\VCM	C:\Program Files\IN-SNEC\crtxnt\VCM
DVB S2 & SCCC Gamma Files	D:\ZDS\Data\HDR\crtxnt\VCM\gamma	C:\Program Files\IN-SNEC\crtxnt\VCM\gamma
FIR Files	D:\ZDS\Data\HDR\crtxnt\fir	C:\Program Files\IN-SNEC\crtxnt\fir
Test Definition Files	D:\ZDS\Data\HDR\crtxnt\TEST	C:\Program Files\IN-SNEC\crtxnt\TEST
Script Files	D:\ZDS\Data\HDR\crtxnt\Qual D:\ZDS\Data\HDR\crtxnt\Scripts (and subdirectories)	C:\Program Files\IN-SNEC\crtxnt\Qual C:\Program Files\IN-SNEC\crtxnt\Scripts (and subdirectories)
Storage Definition Files	D:\ZDS\Data\HDR\Storage	D:\ZDS\Data\HDR\Storage
FTP Sorting Files	D:\ZDS\Data\HDR\ftp (and subdirectories)	C:\Program Files\IN-SNEC\ftp (and subdirectories)
CFDP Sorting Files	D:\ZDS\Data\HDR\cfdp (and subdirectories)	C:\Program Files\IN-SNEC\cfdp (and subdirectories)
Log Files	D:\ZDS\Data\HDR\log	C:\Program Files\IN-SNEC\log
STI Logging Files	D:\ZDS\Data\HDR\log\param_surv	C:\Program Files\IN-SNEC\log\param_surv

4.4.2.3 Files associated with the Cortex HDR Hardware

FILE	WINDOWS 7	WINDOWS 2000 AND XPE
Drivers	C:\Program Files\ZDS\Drivers (and subdirectories)	C:\Program Files\IN-SNEC\Drivers (and subdirectories)
Firmware	C:\Program Files\ZDS\hdr (and subdirectories)	C:\Program Files\IN-SNEC\hdr (and subdirectories)
IPMI Monitor	C:\Program Files\ZDS\Tools\Monitor	C:\Program Files\IN-SNEC\Tools\Monitor

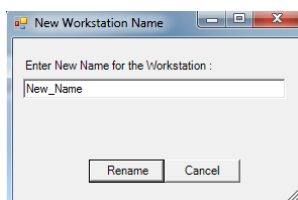
4.4.2.4 Other Files

FILE	WINDOWS 7	WINDOWS 2000 AND XPE
Documentation	D:\ZDS\Data\HDR\DOC (and subdirectories)	C:\Program Files\IN-SNEC\DOC (and subdirectories)
CreateCFDP App	C:\Program Files\ZDS\Tools\createCFDP	C:\Program Files\IN-SNEC\Tools\createCFDP
CreateCFDP Files	D:\ZDS\Data\HDR\Tools\createCFDP\Files (and subdirectories)	C:\Program Files\IN-SNEC\Tools\createCFDP\Files (and subdirectories)
Magic Test Tool	C:\Program Files\ZDS\Tools\magicTEST	C:\Program Files\IN-SNEC\Tools\magicTEST
UMMI Tool	C:\Program Files\ZDS\uihm	C:\Program Files\IN-SNEC\uihm
FileZilla Installer	C:\Program Files\ZDS\Tools\FileZillaServer	C:\Program Files\IN-SNEC\Tools\FileZillaServer
DosHdr Tool	C:\Program Files\ZDS\Tools\doshdr	C:\Program Files\IN-SNEC\Tools\doshdr
Disks Utilities	C:\Program Files\ZDS\Tools\disks	C:\Program Files\IN-SNEC\Tools\disks
Upgrade File	D:\ZDS\Data\HDR\Upgrade.txt C:\Program Files\ZDS\Upgrade.txt	C:\Program Files\IN-SNEC\Upgrade.txt

4.4.3 Windows 7 Tools & Functionalities

With Windows 7, the following tools and functionalities are available to the Cortex User:

- Network Connections (Desktop – Network Connections): Offers the possibility to modify the IP address, subnet, gateway of the Cortex Unit.
- Rename Workstation Utility (Start Menu – All Programs – Rename Workstation): Offers the possibility to change network name of the Cortex Unit.



Procedure: Enter the new network name and click on rename (Cortex unit will restart automatically with its new name).

- Notepad (Start Menu – All Programs – Accessories – Notepad).
- Calculator (Start Menu – All Programs – Tools – Calculator).

-
- IPMI Monitor (Start Menu – All Programs – Tools – IPMIMonitor): Offers the possibility to view motherboard integrated sensors.
 - Paint (Start Menu – All Programs – Tools – Paint).
 - Problem Step Recorder (Start Menu – All Programs – Tools – psr): Tool to makes a visual and text-based documentation of everything that happens on the Cortex Unit while you're recording.
 - Regedit (Start Menu – All Programs – Tools – Regedit): Offers the possibility to modify the Cortex application registry (HKEY_LOCAL_MACHINE\SOFTWARE\IN-SNEC).
 - Touch Screen Utility (Start Menu – All Programs – Tools – TouchUtility).
 - Wordpad (Start Menu – All Programs – Tools – Wordpad).

4.5 CORTEX HDR GUI

4.5.1 Conventions & Warnings

IN THE FOLLOWING SECTIONS, EACH CONFIGURATION PARAMETER, STATUS OR ALARM IS RED-MARKED WITH THE NAME OF A CORTEX « UNIT ». THIS MARK ALLOWS TO LOCATE THE PARAMETER, STATUS OR ALARM IN THE ETHERNET INTERFACE CONTROL DOCUMENT IN ANNEX 1.

The following marks are used:

- ❖ **(SWID)** : Refer to the Software & ICD Version ID table
- ❖ **(GLOB)** : Refer to the Global CORTEX table
- ❖ **(DMU)** : Refer to the Demodulator Unit table
- ❖ **(DPU)** : Refer to the Data Processing Unit table
- ❖ **(DCU)** : Refer to Diversity Combining Unit table
- ❖ **(MGLO)** : Refer to the Modulator Global table
- ❖ **(DGU)** : Refer to the Data Generating Unit table
- ❖ **(MDU)** : Refer to the Modulating Unit table
- ❖ **(IFO)** : Refer to the IF Output table
- ❖ **(CHM)** : Refer to the Channel Modeling table
- ❖ **(MFI)** : Refer to the Modulator Format Input table
- ❖ **(MFO)** : Refer to the Modulator Format Output table
- ❖ **(DRG)** : Refer to the Data Recording Global table
- ❖ **(DRU)** : Refer to the Data Recording Unit table
- ❖ **(PLBK)** : Refer to the Playback Unit table
- ❖ **(PROJ)** : Refer to the Project-specific table

PROGRAMMING PARAMETERS DEFINED AS FLOAT VALUES IN THE INTERFACE CONTROL DOCUMENT CAN BE ENTERED IN FLOAT OR REAL FORMAT (SEE ANNEX 1).

PARAMETERS DEFINED AS INTEGER VALUES MUST BE ENTERED IN REAL FORMAT.

PARAMETERS MARKED « HEX » IN THE GRAPHICAL USER INTERFACE MUST BE ENTERED IN HEXADECIMAL FORMAT. OTHER PARAMETERS MUST BE ENTERED IN DECIMAL FORMAT.

4.5.2 Graphical User Interface. Windows Breakdown

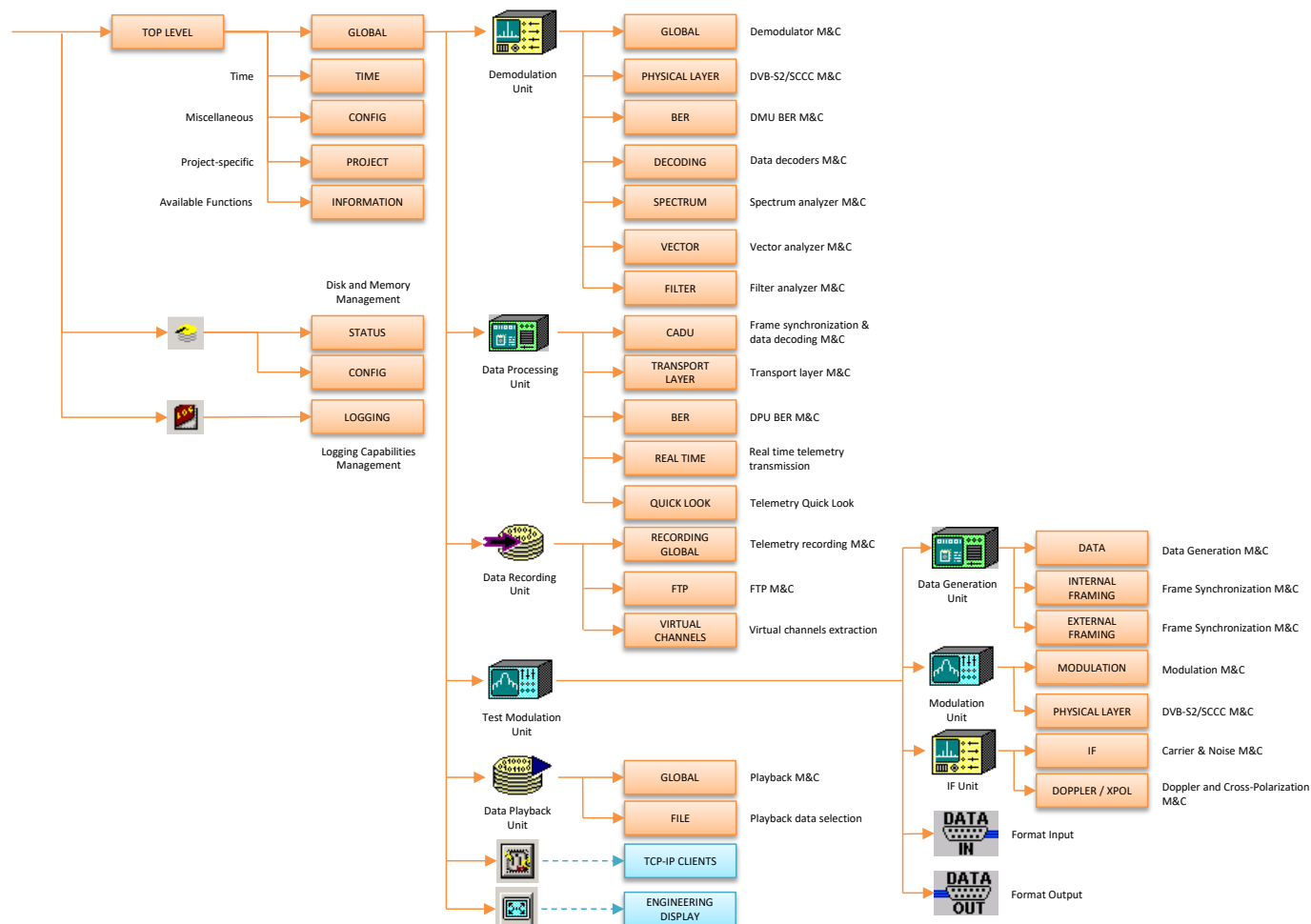


Figure 86: Graphical User Interface: Windows Breakdown

4.5.3 CORTEX HDR Top-level Windows

4.5.3.1 "Global" Window

4.5.3.1.1 CORTEX HDR Architecture & Global Status

The following window appears on the screen after the MCS is connected to the SPS (note that the number of functional units depends on the selected configuration menu and available licenses):

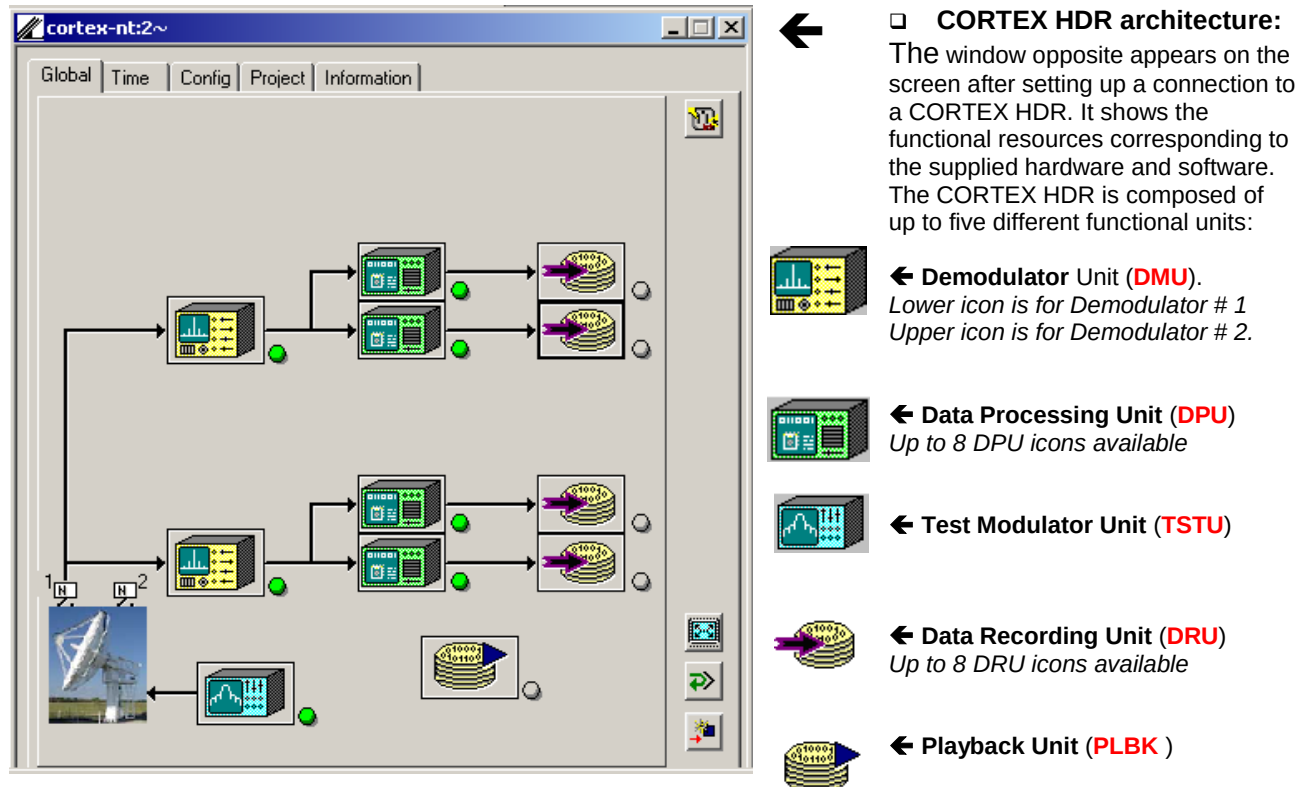
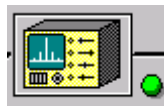


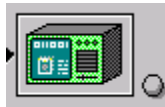
Figure 87: CORTEX HDR: Global Window

Demodulator Unit global status:



Demodulator locked if the LED is green (Carrier PLL and Bit Synchronizer locked - **DMU**) or unlocked if the LED is unlit (gray background). The LED turns to red if a serious hardware or software (license) problem is detected on the demodulation chain.

❑ **Data Processing Unit global status:**



Frame synchronizer status: green LED when locked, amber LED in flywheel mode, unlit LED when unlocked (searching). The LED turns to red if a serious hardware or software problem is detected on the demodulation chain.

❑ **Data Recording Unit global status:**



Recorder status: green LED when recording in progress, unlit LED when recording off. The LED turns to red if a problem is detected on the recording chain.

❑ **Playback Unit global status:**



Playback status: green LED when playback is operating, unlit LED when playback is off. The LED turns to red if a problem is detected on the playback function.

❑ **TCP-IP data flow status:**



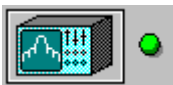
Number of clients connected to the Spectral port (SPA) and Vector port (VEC): none (left), one (middle) or more than one (right).

❑ **IF inputs status:**



Reflects the IF input selection for both demodulators with N for normal input, and A for Alternate input.

❑ **Test Modulator global status:**



In normal conditions, this LED must be green. The LED turns to red if a serious hardware problem (Test Modulator failure) or software problem is detected.

❑ **MCS Table update:** click on



❑ **CORTEX re-configuration from the MCS Table:** click on



❑ **Open the Global Engineering window:** click on



❑ **Open the TCP-IP secondary window:** click on



4.5.3.1.2 Global Engineering Window

To open the Global Engineering window, click on icon



To go back to the standard windows display, click on the **Exit** key.

The Global Engineering window displays dynamically the current configuration and status of the Cortex HDR in a single window. The following figure gives an example.

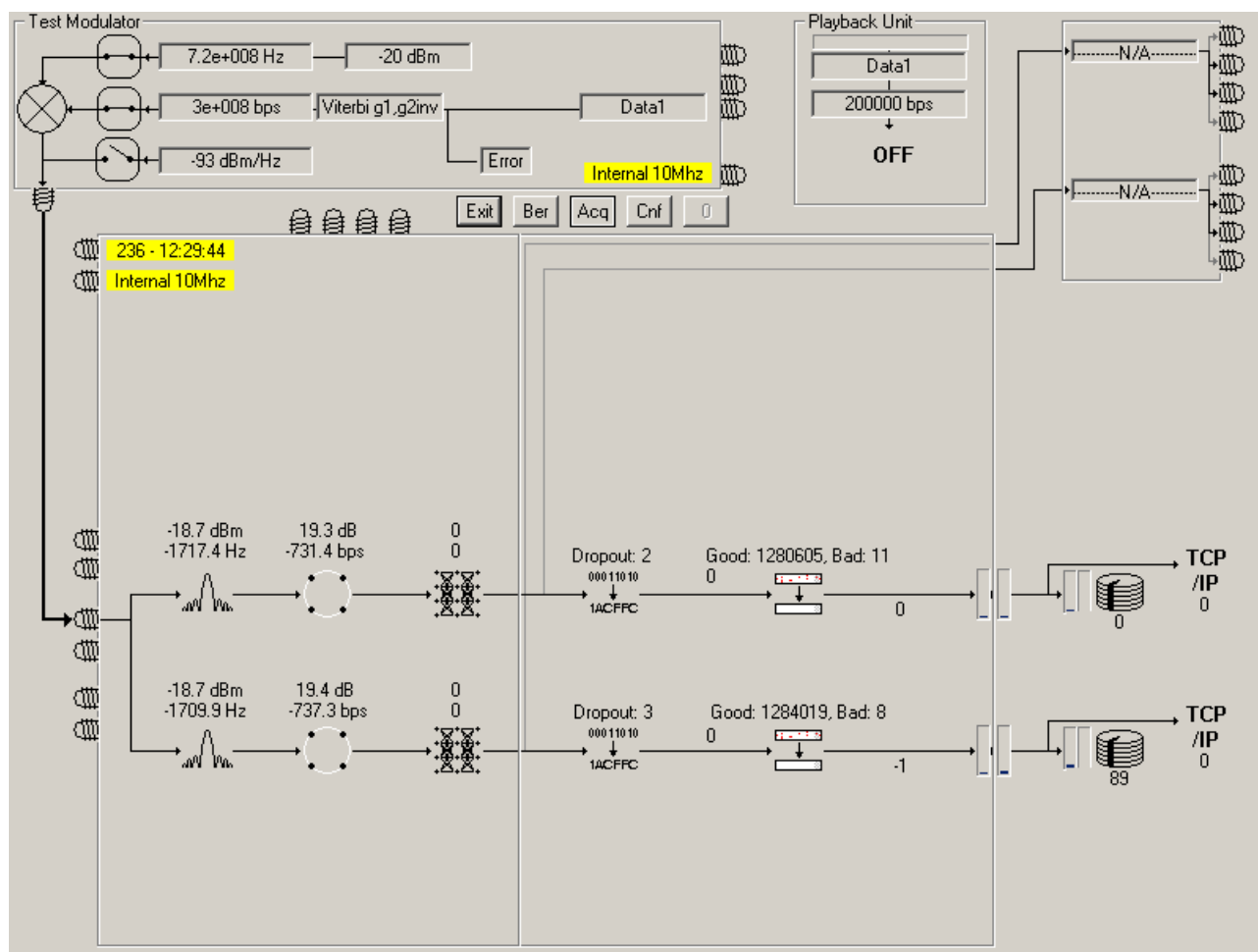


Figure 88: Global Engineering Window

4.5.3.1.3 TCP-IP Secondary Window

For detailed information on the TCP-IP clients, click on icon :

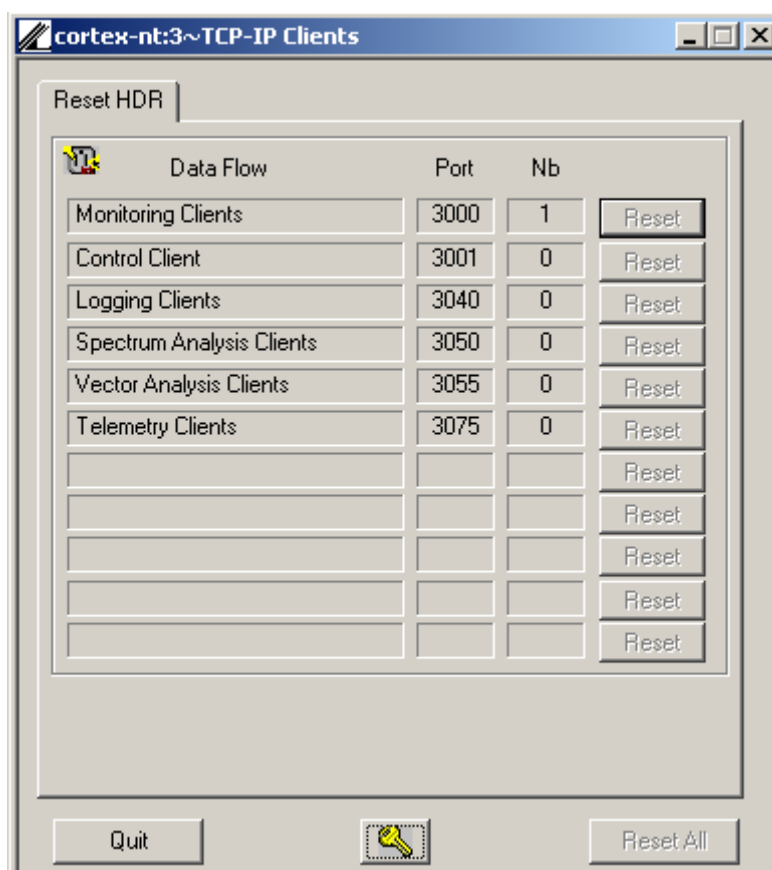
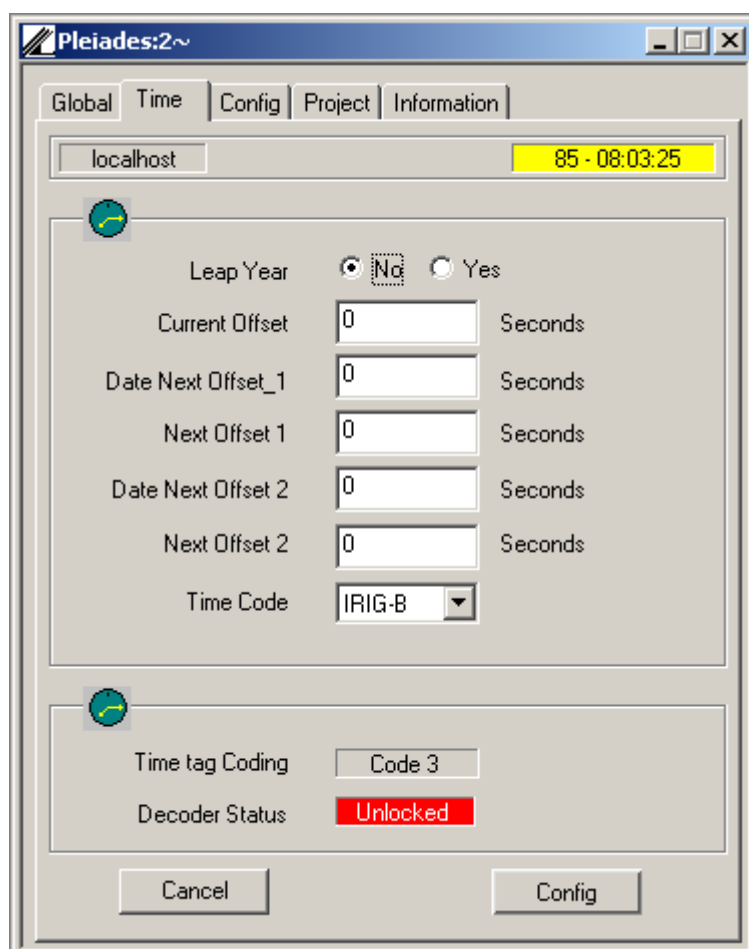


Figure 89: TCP-IP Secondary Window

The number of clients connected to each TCP-IP port is displayed in the above window. Each Reset button allows closure of all the TCP-IP sockets connected to the associated port. The Reset All pushbutton allows closure of all connections to all ports.

The TCP-IP reset function is password protected: click on  and enter the **TCP-IP password**. Default setting at delivery: no password. To change the TCP-IP password, see Section 4.3.8).

4.5.3.2 "Time" Window



From the opposite window, the User can program time offsets to cope with time-tagging requirements with local time or UTC time or GPS time, etc....or to have the day counter not reset at the beginning of each year. Leap years should be declared in this window to avoid potential loss of lock on the internal Time Code Reader at the end of the year. Expected time code must be entered for proper synchronization of the Time Code Reader. Parameters, status and alarms in this window are described in **GLOB**.

Figure 90: CORTEX « Time » Window

4.5.3.2.1 Control

LEAP YEAR	Leap year programming (365 or 366 days). Select Yes or No (ensure the external time code generator setting is identical).
CURRENT OFFSET	Configuration parameter and status. Allows to enter a time offset for immediate application and to read the current time offset. Expressed in seconds (signed value). The current time offset is added to the time delivered by the internal Time Code Reader, for time tagging telemetry frames, range measurements, etc...
DATE OF NEXT OFFSET 1	At this date, Next Offset 1 will replace the current offset. Expressed in seconds with January 1 st 00:00:00 = 0.
NEXT OFFSET 1	This time offset, expressed in seconds (signed value), will become effective when the current date matches Date of Next Offset 1 . See note 1.
DATE OF NEXT OFFSET 2	At this date, Next Offset 2 will replace the current offset. Expressed in seconds with January 1 st 00:00:00 = 0.
NEXT OFFSET 2	This time offset, expressed in seconds (signed value), will become effective when the current date matches Date of Next Offset 2 . See note 1. Note that if Date of Next Offset 1 = Date of Next Offset 2 , then Date of Next Offset 2 will be ignored (no offset update: priority to Offset 1).
TIME CODE	Expected time code (IRIG-B or NASA-36) or NTP if expected time from an external NTP server (refer to § 4.4.3.2.3)

Note 1: At boot up, the current offset value is set to the latest value programmed by the Operator (automatic adjustment of **Current Offset** from **Next Offset 1** or **Next Offset 2** is not memorized by the machine).

4.5.3.2.2 Monitoring

TIME-TAG CODING	Time-tag coding in TCP-IP messages: Code 3 Time-tags are expressed in seconds and microseconds of the second (integer format).
DECODER STATUS	Time Code Reader status: Locked or Unlocked or Flywheel .

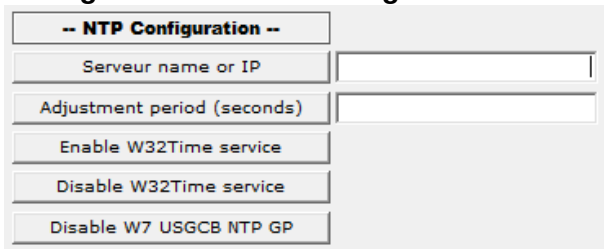
4.5.3.2.3 NTP client set-up

The HDR is fitted with an NTP client. The activation of the NTP client is done by setting the parameter "**TIME CODE**" to "**NTP**".

Two parameters are available for the set-up: the address of the server and the time refresh period (Server name or IP address, Adjustment period in seconds). **The time is periodically updated when the HDR is set in Configuration mode.**

These set-up parameters are not directly accessible through the GUI. The setup is done using the tool "**uihm.exe**". This tool is available in the directory **C:\Program Files\ZDS\uihm** for Windows 7 otherwise **C:\Program Files\IN-SNEC\uihm**.

After having started uihm.exe and connected it with the SPS software, the NTP server parameters are available and can be set following, in the “uihm” window, the path **Commands -> Hardware Configuration -> NTP Configuration**:



4.5.3.3 Time Management Procedures

4.5.3.3.1 Loss of Time Code or Time Code Missing

In case of wrong code selection or degraded or missing time code signal, the Time Code Reader remains unlocked and generates its own time code in flywheel mode. When operating in flywheel mode, time-tagging accuracy is set by the reference clock (internal or external) performances (see Section 1.6.8.2). Should the time code be missing at boot up, the time will be **initialized from the Windows time**.

4.5.3.3.2 Leap Second Management

As the earth's rotation varies in relationship to atomically maintained time (UTC), it is often necessary to officially add or delete a second from the UTC time scale. Insertion or removal of leap seconds can occur as often as twice per year. Pending leap second changes are notified by the Paris-based Bureau International de l'Heure (France).

If data time-tagging on the CORTEX require to be aligned on UTC, then the leap second changes can be made either on the external Time Code Generator or on the CORTEX.

☐ Leap second change on the external Time Code Generator:

If the change is made on the Time Code Generator, the Time Code Reader in the CORTEX will lose lock for a short period of time (typically: 10 seconds) at the very moment the time information, which modulates the time code signal, is updated.

Ensure that parameters **Current Offset**, **Next Offset 1** and **Next Offset 2** are set to 0 to avoid taking twice into account the leap second change.

☐ Leap second change:

Should it be mandatory to keep the Time Code Reader permanently synchronized (or if the external Time Code Generator does not feature leap second entry), the leap seconds entry must be done on the CORTEX instead of the external Time Code Generator.

Example: let us assume that:

- ❖ the current leap second value is 10. Non leap year.
- ❖ a positive leap second change (new value = 11) has been notified for July 1st, 00 h, 00 mn, 00 s
- ❖ the CORTEX is started before June 30st, with **all time offsets set to 0**.

The procedure is as follows (see next figure):

1. Start the CORTEX and go to the « Time » window.
2. Program the following values:

Current Offset = 10
Date of Next Offset 1 = Date of Next Offset 2 = 15 638 400
Next Offset 1 = 11

3. On July 1st, 00 h, 00 mn, 00 s, check that the Current Offset is automatically set to 11 and that the time-tags are updated accordingly. Note: **15 638 400** = July 1st, 00 h, 00 mn, 00 s = 181 days x 86400.
4. Before December 31st, program the following values:

Date of Next Offset 1 = 31 535 989
Next Offset 1 = - 31 535 989
Date of Next Offset 1 = 0
Next Offset 1 = 11

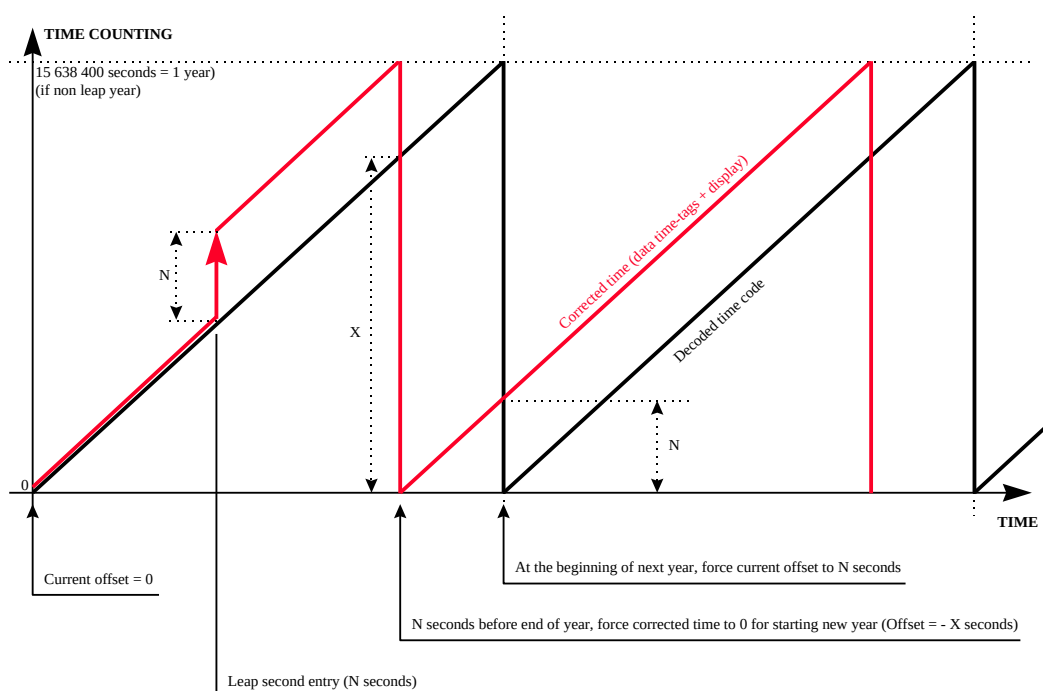


Figure 91: Leap Second Management

4.5.3.3.3 Absolute Time Reference

The IRIG-B and NASA-36 time codes are serial time format with the time-of-year coded in Binary Coded Decimal (BCD) notation as days, hours, minutes and seconds. Years are not coded and the day counter is reset every January 1st, 00 h, 00 mn, 00 s. Time parameters can be programmed so that the day counter will not be restarted every year.

Example:

- ❖ The absolute time reference is January 1st, 1998 at 0 AM.
- ❖ The CORTEX is started in the course of 1998 with **Current Offset = 0**.

The procedure is as follows:

1. Enter: **Date of Next Offset 1 = Date of Next Offset 2 = 0**
Next Offset 1 = 31 536 000 (year 1998 = 365 days = 31 536 000 seconds)
2. On January 1st, 1999 at 0 AM, check that the Current Offset is set to 31 536 000 and that the time-tags are updated accordingly (365 days are automatically added to the time-tags). The day counter is equal to 366.
3. In the course of 1999, enter **Next Offset 1 = 63 072 000** (year 1999 = 365 days = 31 536 000 seconds).
4. On January 1st, 2000 at 0 AM, check that the Current Offset is set to 63 072 000 and that the time-tags are updated accordingly. The day counter is now equal to 731.
5. Each following year, increment parameter **Next Offset 1** by the number of seconds of the current year.

4.5.3.3.4 Absolute Reference Time & Leap Second Management

The following scenario is a combination of the two previous scenarios.

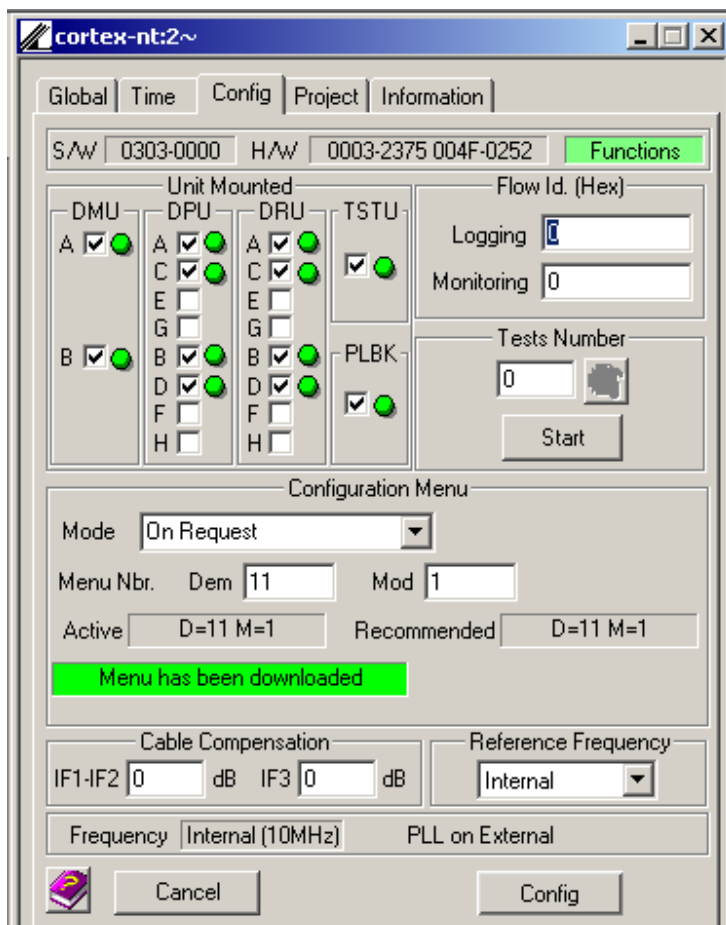
Example:

- ❖ The absolute time reference is January 1st, 1998 at 0 AM.
- ❖ The CORTEX is started on January 1st 1999 at 0 AM with **Current Offset = 0**.
- ❖ The current leap second value is 10.

The procedure is as follows:

1. Enter: **Date of Next Offset 1 = 3600**
Next Offset 1 = 31 536 010 (one non leap year + 10 seconds)
2. On January 1st, 1999 at 1 AM, (3600 seconds after boot up) check that the Current Offset is set to 31 536 010 and that the time-tags are updated accordingly. The day counter is now equal to 366.
3. In the course of 1999, enter:
Date of Next Offset 1 = 0 (January 1st 2000)
Next Offset 1 = 63 072 010 (two non-leap years + 10 seconds)
Next Offset 2 = Date of Next Offset 2 = 0
4. On January 1st, 2000 at 0 AM, check that the Current Offset is set to 63 072 010 and that the time-tags are updated accordingly. The day counter is now equal to 731.
5. Repeat step 3 every following year.

4.5.3.4 "Config" Window



From the window opposite, the User can monitor & control miscellaneous system parameters. Parameters, status and alarms in this window are described in **GLOB**.

Figure 92: CORTEX HDR: Config. Window

4.5.3.4.1 Control

UNIT MOUNTED

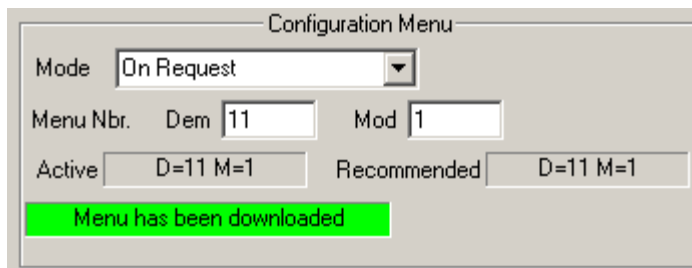
Identification of the mounted units in the current configuration. A green LED indicates that the unit is mounted and enabled. A unit can be enabled or disabled by clicking in the associated selection box:

- **DMU:** Demodulator Unit
- **DPU:** Data Processing Unit
- **DRU:** Data Recording Unit
- **TSTU:** Test Modulator Unit
- **PLBK:** Playback Unit

TESTS NUMBER

Reserved for internal tests.

MONITORING FLOW ID	Identification of the monitoring data flow. This value is reported in all TCP-IP messages transmitted over the Monitoring port (MON). Enter any hexadecimal value from 0 to FFFFFFF . The Flow ID value in the incoming monitoring requests must match the programmed value.
LOGGING FLOW ID	Identification of the logging flow. This value is reported in all TCP-IP messages transmitted over the Logging port (LOG). Enter any hexadecimal value from 0 to FFFFFFF .
CABLE COMPENSATION	Determines the evaluated loss (in dB) due to cables among the propagating path. This value is reported in the calculation of the compensating filter's coefficients. Enter any value from 0 to 12 (Option)
REFERENCE FREQUENCY	Selection of the frequency reference <i>internal</i> or <i>external</i> .
CONFIGURATION MENU	The easier way to configure the unit is to load the selected MCS file (refer to paragraph 4.3). As soon as the MCS file is loaded, the unit can automatically load the firmware corresponding to this MCS file. The Configuration Window allows the User to force the selection of a basic menu and gives also indications about the status of the selection an loading operations. In manual mode (Mode: On Request or On Accept) the operator can input in the field: Menu Nbr the numbers of a basic menus (Demodulator, test Modulator and the optional Extension board when present). If the operation mode is set to On Accept, the loading will be effective only after having clicked the button Accept Request Menu In automatic mode (Automatic or Automatic with Accept) the basic menu corresponding to the current configuration is selected automatically and its number is displayed in the Recommended status field. If the operational mode is set to Automatic with Accept, the loading will be effective after having clicked the button Accept Requested Menu.



In all cases the number of the current and active basic menu is displayed in the **Active** status field.

A status field indicates if the menu is being loaded or has been loaded and is ready to operate. This status is displayed in the Configuration Window and in the upper right corner of the Global Window:



For detailed information on the current configuration and the possible basic menus, click on icon  to access the help windows:

Menu Help Window																					
Demodulator																					
Capability	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
DMU-1 Available	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
DMU-2 Available	+	-	+	+	+	-	+	+	-	+	+	-	-	-	-	-	-	-	-	-	-
DPU-1 Available	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
DPU-2 Available	+	-	+	+	+	-	-	+	-	-	+	-	-	-	-	+	-	-	-	-	-
DPU-3 Available	-	-	-	-	+	-	-	-	-	-	+	-	-	-	-	+	-	-	-	-	-
DPU-4 Available	-	-	-	-	+	-	-	-	-	-	+	-	-	-	-	+	-	-	-	-	-
DPU-5 Available	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	-	-	-	-	-
DPU-6 Available	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	-	-	-	-	-
DPU-7 Available	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	-	-	-	-	-
DPU-8 Available	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	-	-	-	-	-
DMU Combi.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
DMU : Ext. Bit sync.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
: TCM 2.00/3 Decoder	-	-	-	-	-	-	-	+	-	+	-	-	-	-	-	-	-	-	-	-	-
: TCM 2.25/3 Decoder	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
: TCM 2.50/3 Decoder	+	-	-	-	-	-	-	-	+	-	-	+	+	-	-	-	-	-	-	-	-
: TCM 2.75/3 Decoder	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
: Viterbi 1/2 Decoder	-	-	+	+	+	-	+	-	-	-	+	-	-	-	-	-	-	-	-	-	-
: Viterbi 1/4 Decoder	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
: Viterbi Par Decoder	-	-	-	-	-	-	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-
: Fast Decoder	-	-	-	-	-	-	+	-	+	+	-	+	+	-	-	-	-	-	-	-	-
: Derandomizer	-	-	√35	√35	-	-	-	-	-	-	√√	-	-	-	-	-	-	-	-	-	-
: X-DEAF	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
: VCM QQPSK/8PSK	-	-	-	-	-	-	-	-	-	-	+	-	-	-	-	-	-	-	-	-	-
: VHR E/O DVBS2 - ...	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Figure 93: Configuration help Window for the Demodulator

Menu Help Window									
Modulator									
Capability	1	2	3	4	5	6	7	8	9
Modulation uqpsk	-	+	-	-	-	-	-	-	-
Modulation 16qam	+	-	+	+	+	+	+	+	+
VHR E/O DVBS2 - ...	-	-	-	-	-	-	-	+	+
High Symbol Rate	-	-	+	-	-	-	-	+	-
Parallel data generat...	-	-	-	-	+	-	-	-	-
Byte output	-	-	-	+	-	-	-	-	-
FIR Capable	-	-	-	-	-	+	-	-	+
Doppler Capable	-	-	-	-	-	-	+	-	-

Figure 94: Configuration help Window for the Test Modulator

The help windows recall inside a single table the capabilities of each basic hardware menu available for the demodulator, the test modulator and the optional extension board.

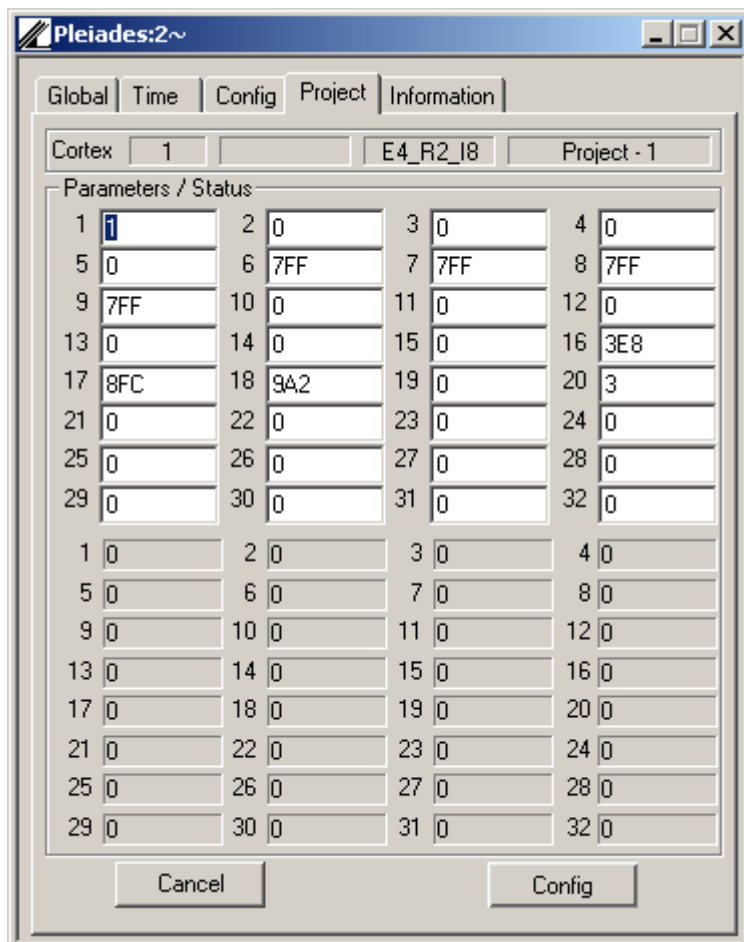
The tables indicate also which basic menu(s) is or are authorized for the current loaded configuration: green or yellow columns (green: full compliance between the menu and the loaded configuration, yellow: full compliance only for the demodulation channel #1). For the other non-authorized menus, a red stamp is applied on the function(s) or performance(s) which do not match what is required by the loaded configuration.

4.5.3.4.2 *Monitoring*

Software Id	0101-0023	Hardware Id	0101-0000
-------------	-----------	-------------	-----------

- ❖ Demodulator board: software ID. From left to right:
 DSP software version: Edition (2 digits) + Revision (2 digits).
 DSP software: build number (4 digits).
- ❖ Demodulator board: Hardware ID.

4.5.3.5 "Project" Window



Parameters / Status			
1	0	3	0
5	0	7	7FF
9	7FF	11	0
13	0	15	0
17	8FC	19	0
21	0	23	0
25	0	27	0
29	0	31	0
32	0		

From the window opposite, the User can monitor & control parameters, status and alarms that have been implemented from **Customer-specific requirements**. Should be all zeros for the standard CORTEX HDR product.

Unless otherwise specified, parameters, status and alarms in this window are described in **PROJ**.

Figure 95: CORTEX HDR: Project Window

4.5.3.5.1 Control

PARAMETER 1 to 32 32 project-specific parameters (project-specific parameters are described in Annex 5).

4.5.3.5.2 Monitoring

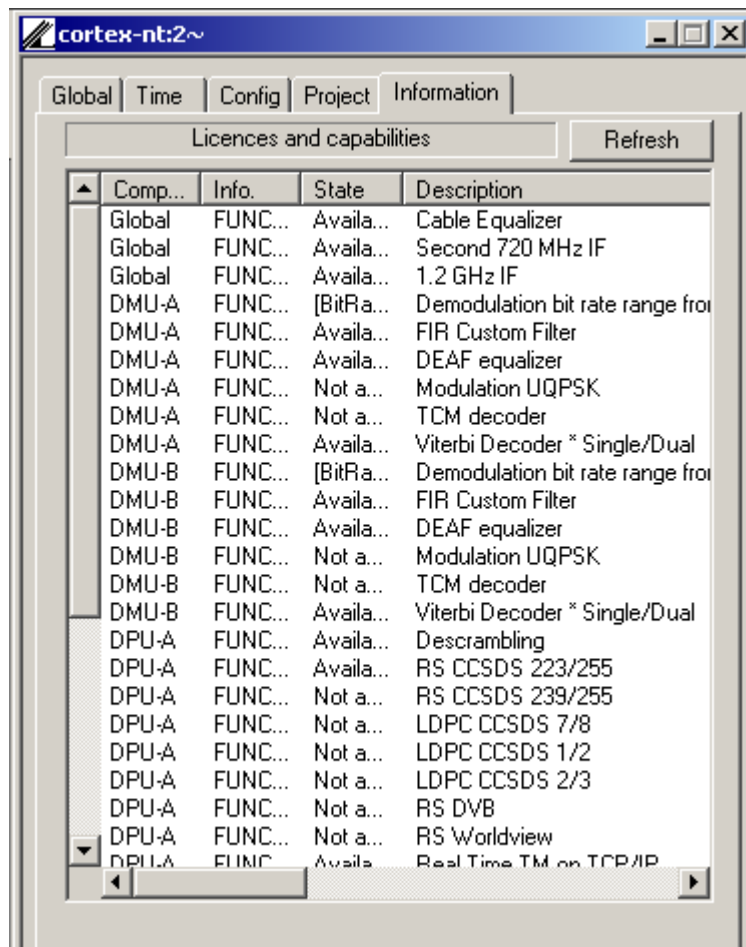


From left to right:

- ❖ CORTEX HDR number (**GLOB**).
- ❖ Software and Ethernet ICD version ID (**SWID**).
- ❖ Project ID. 0 for generic CORTEX HDR. Other values reserved for custom design (see Annex 5).

STATUS 1 to 32 32 project-specific status & alarms. (project-specific status and alarms are described in Annex 5).

4.5.3.6 "Information" Window



From the window opposite, the User can check the availability of the different functionalities of the CORTEX HDR.

Some functionalities are menus and licenses dependent refer to paragraph Licenses and Basic Menus.

Figure 96: CORTEX HDR: Information Window

Another way to check the availability of the functionalities is to position the mouse cursor on the **Function** status in the different windows.

4.5.3.7 "Mod" Window

The following window appears on the screen when clicking on the TSTU in the "Global" Window (note that the number of functional units depends on the selected configuration menu and available licenses):

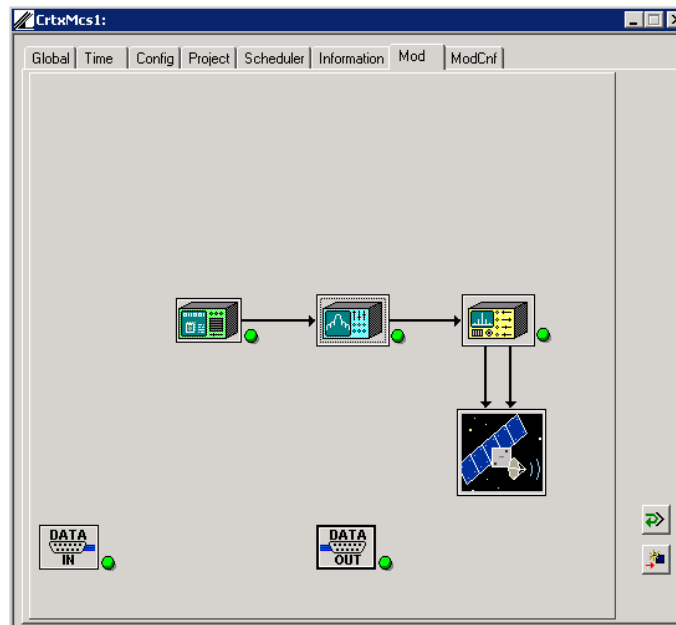
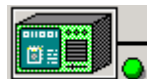


Figure 97: "Mod" Window

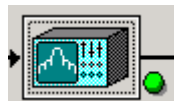
The modulator can be composed of different elements.

☐ Data Generating Unit:



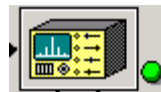
The LED turns to red if a serious hardware or software (license) problem is detected on the **DGU**.

☐ Modulating Unit:



The LED turns to red if a serious hardware or software (license) problem is detected on the **MDU**.

☐ IF Unit:



The LED turns to red if a serious hardware or software (license) problem is detected on the **IFU**.

☐ Format Input:



The LED turns to red if a serious hardware or software (license) problem is detected on the **FI**.

❑ **Format Output:**



The LED turns to red if a serious hardware or software (license) problem is detected on the **FO**.

4.5.3.8 "ModCnf" Window

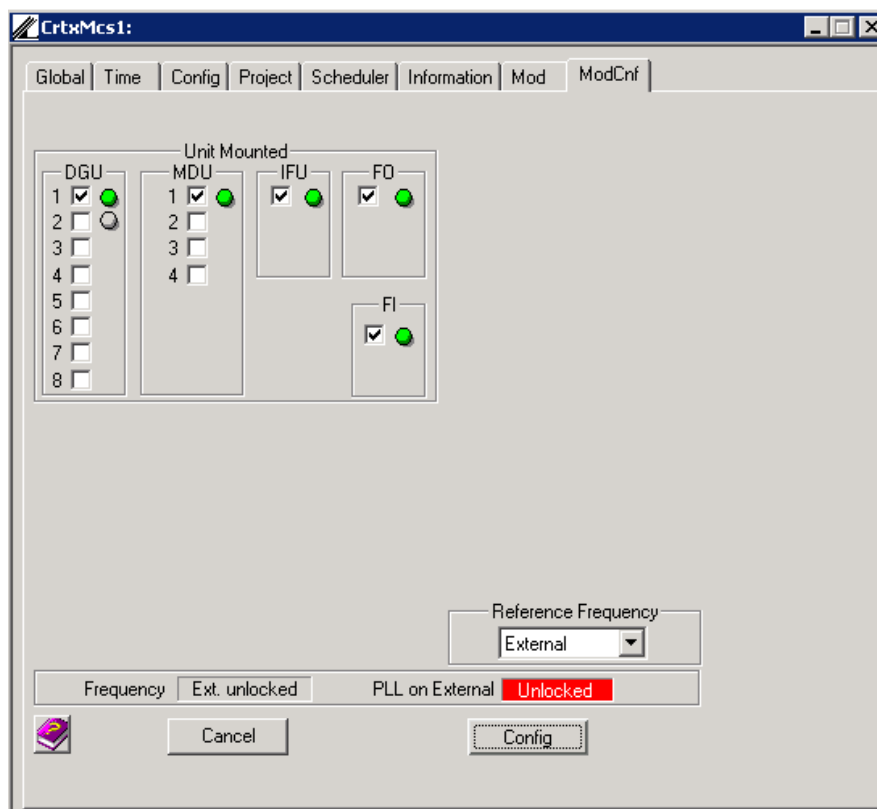


Figure 98: "ModCnf" Window

For each type of unit, a grey LED indicates that the unit is available but not used. A green LED indicates that the Unit is mounted.

The type and number of available units depends on the configuration selected on the "Config" Window.

Reference Frequency can be chosen external or internal.

A status indicates the source of Reference Frequency.

When the Reference Frequency is external, a status indicates if it is locked (Green) or unlocked (Red).

4.5.4 Demodulator Windows

4.5.4.1 "Global" Window

❑ BPSK, QPSK Modulation, I&D filter:

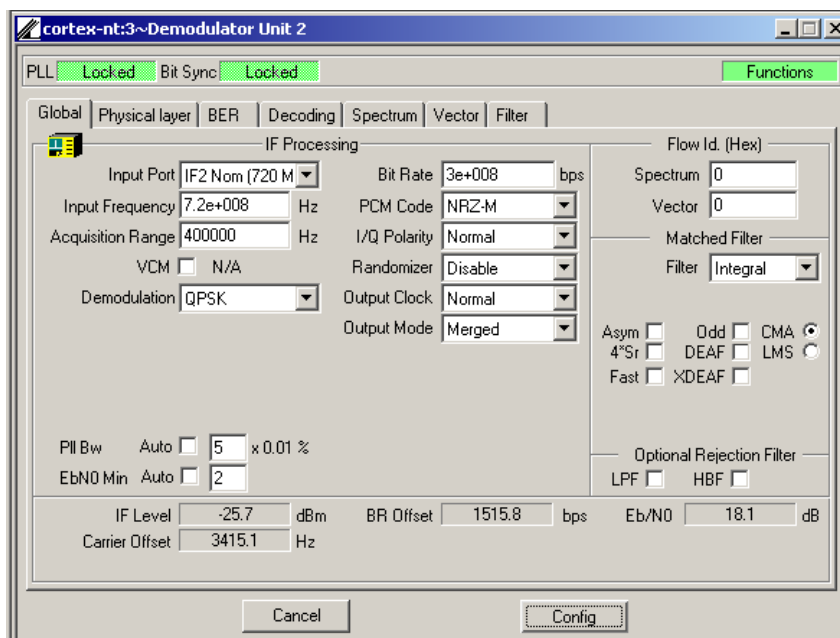


Figure 99: Demodulator: Global Window for QPSK Modulation + I & D Filter

❑ BPSK, QPSK Modulation, Raised cosine filter:

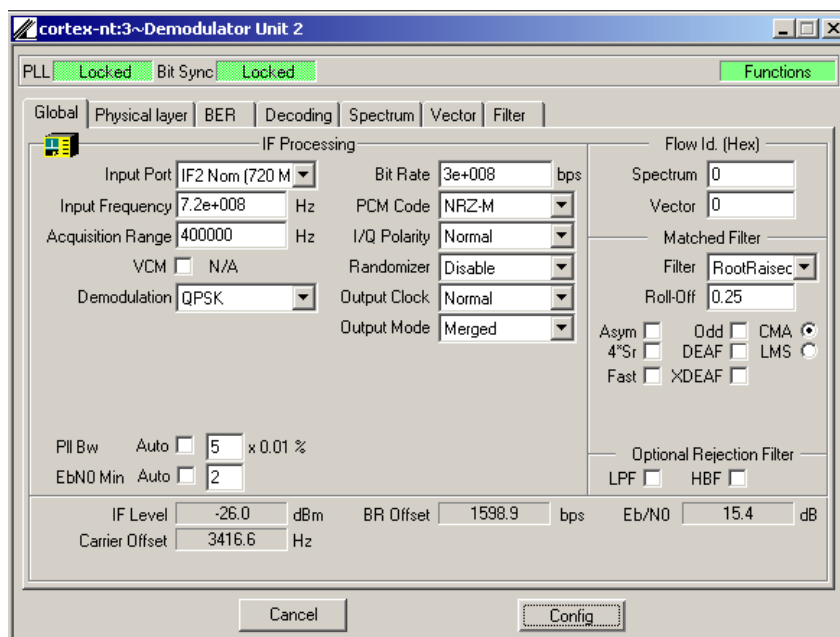


Figure 100: Demodulator: Global Window for QPSK Modulation + Raised Cosine Filter

❑ **BPSK, QPSK Modulation, custom filter:**

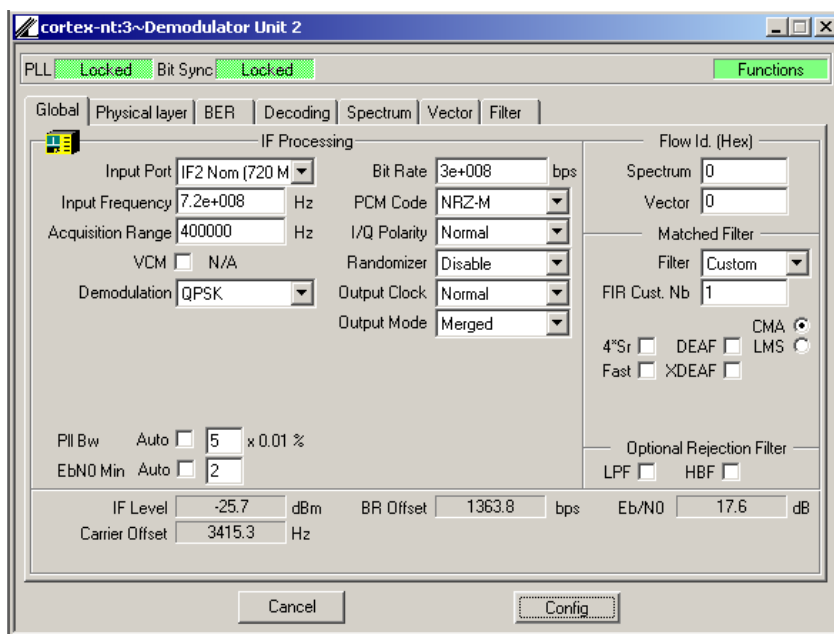


Figure 101: Demodulator: Global Window for QPSK Modulation + Custom Filter

❑ **OQPSK (SQPSK) Modulation:**

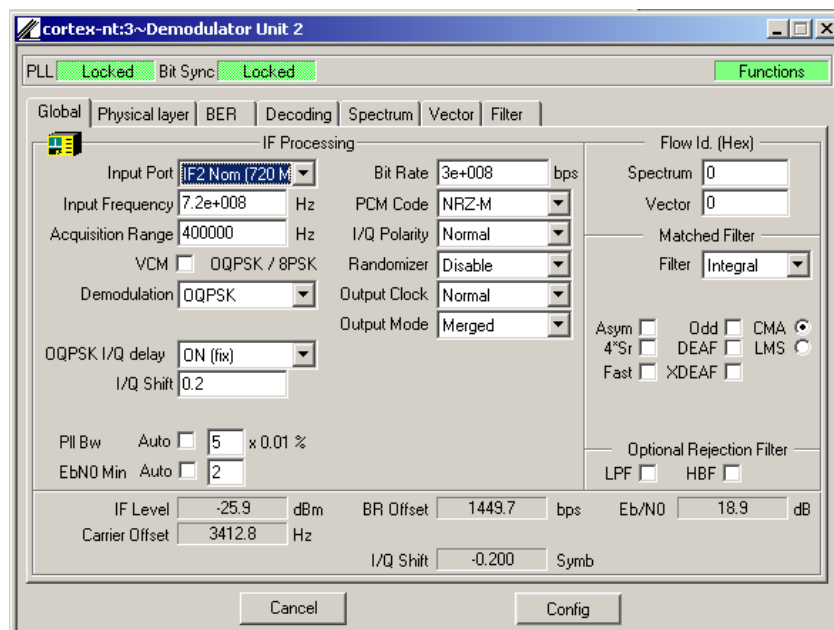


Figure 102: Demodulator: Global Window for OQPSK Modulation

❑ GMSK Modulation:

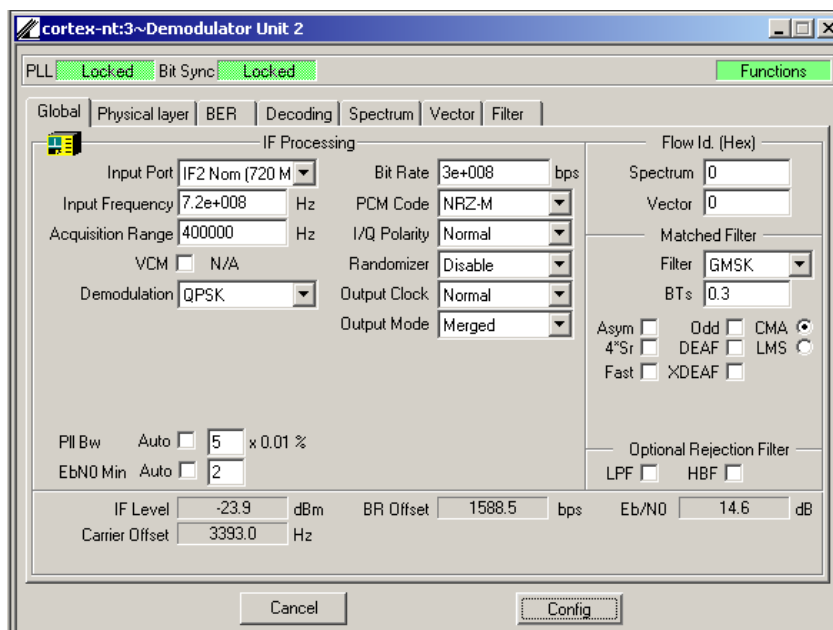


Figure 103: Demodulator: Global Window for GMSK Modulation

❑ SOQPSK Modulation:

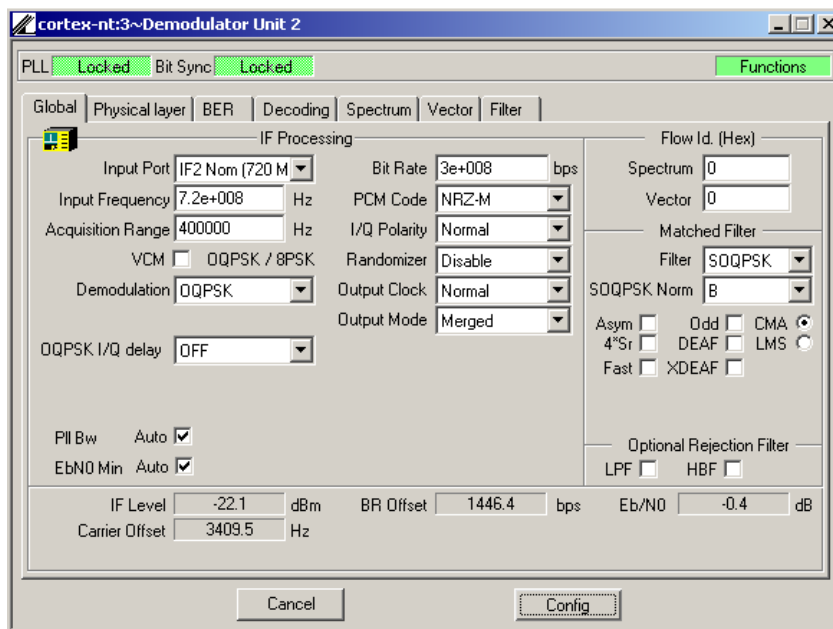
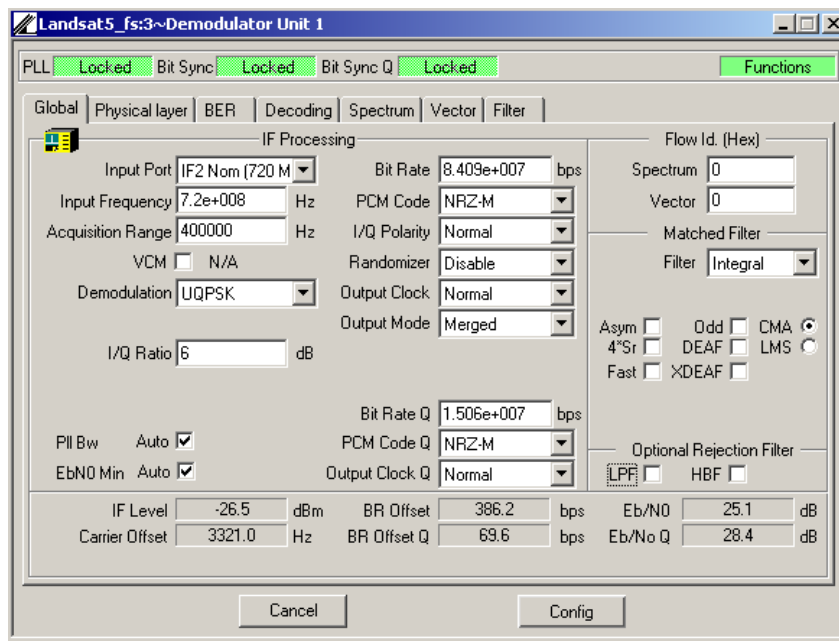


Figure 104: Demodulator: Global Window for SOQPSK Modulation

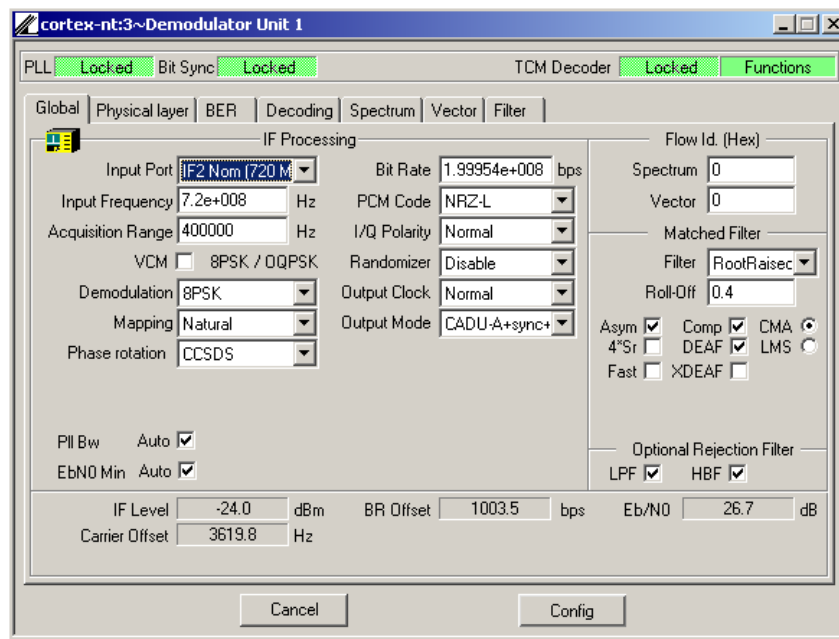
❑ UQPSK Modulation:



The screenshot shows the 'Global' tab of the 'Landsat5 fs:3~Demodulator Unit 1' window. The 'IF Processing' section is active. The 'Input Port' is set to 'IF2 Nom (720 M)'. The 'Input Frequency' is '7.2e+008 Hz'. The 'Acquisition Range' is '400000 Hz'. The 'VCM' is 'N/A'. The 'Demodulation' is set to 'UQPSK'. The 'I/Q Ratio' is '6 dB'. The 'Bit Rate' is '8.409e+007 bps'. The 'PCM Code' is 'NRZ-M'. The 'I/Q Polarity' is 'Normal'. The 'Randomizer' is 'Disable'. The 'Output Clock' is 'Normal'. The 'Output Mode' is 'Merged'. The 'Flow Id. (Hex)' is '0'. The 'Spectrum' is '0'. The 'Vector' is '0'. The 'Matched Filter' is 'Integral'. The 'Filter' is 'Integral'. The 'Asym' checkbox is checked. The 'Odd' checkbox is checked. The 'CMA' radio button is selected. The '4*sr' checkbox is checked. The 'DEAF' checkbox is checked. The 'LMS' radio button is selected. The 'Fast' checkbox is checked. The 'XDEAF' checkbox is checked. The 'Optional Rejection Filter' section has 'LPE' and 'HBF' checkboxes, both of which are checked. The 'IF Level' is '-26.5 dBm'. The 'BR Offset' is '386.2 bps'. The 'Eb/N0' is '25.1 dB'. The 'Carrier Offset' is '3321.0 Hz'. The 'BR Offset Q' is '69.6 bps'. The 'Eb/N0 Q' is '28.4 dB'. The 'Cancel' and 'Config' buttons are at the bottom.

Figure 105: Demodulator: Global Window for UQPSK Modulation

❑ 8PSK Modulation:



The screenshot shows the 'Global' tab of the 'cortex-nt:3~Demodulator Unit 1' window. The 'IF Processing' section is active. The 'Input Port' is set to 'IF2 Nom (720 M)'. The 'Input Frequency' is '7.2e+008 Hz'. The 'Acquisition Range' is '400000 Hz'. The 'VCM' is '8PSK / DQPSK'. The 'Demodulation' is set to '8PSK'. The 'Mapping' is 'Natural'. The 'Phase rotation' is 'CCSDS'. The 'Bit Rate' is '1.99954e+008 bps'. The 'PCM Code' is 'NRZ-L'. The 'I/Q Polarity' is 'Normal'. The 'Randomizer' is 'Disable'. The 'Output Clock' is 'Normal'. The 'Output Mode' is 'CADU-A+sync+'. The 'Flow Id. (Hex)' is '0'. The 'Spectrum' is '0'. The 'Vector' is '0'. The 'Matched Filter' is 'RootRaisec'. The 'Roll-Off' is '0.4'. The 'Asym' checkbox is checked. The 'Comp' checkbox is checked. The 'CMA' radio button is selected. The '4*sr' checkbox is checked. The 'DEAF' checkbox is checked. The 'LMS' radio button is selected. The 'Fast' checkbox is checked. The 'XDEAF' checkbox is checked. The 'Optional Rejection Filter' section has 'LPE' and 'HBF' checkboxes, both of which are checked. The 'IF Level' is '-24.0 dBm'. The 'BR Offset' is '1003.5 bps'. The 'Eb/N0' is '26.7 dB'. The 'Carrier Offset' is '3619.8 Hz'. The 'Cancel' and 'Config' buttons are at the bottom.

Figure 106: Demodulator: Global Window for 8PSK Modulation

❑ **16QAM Modulation:**

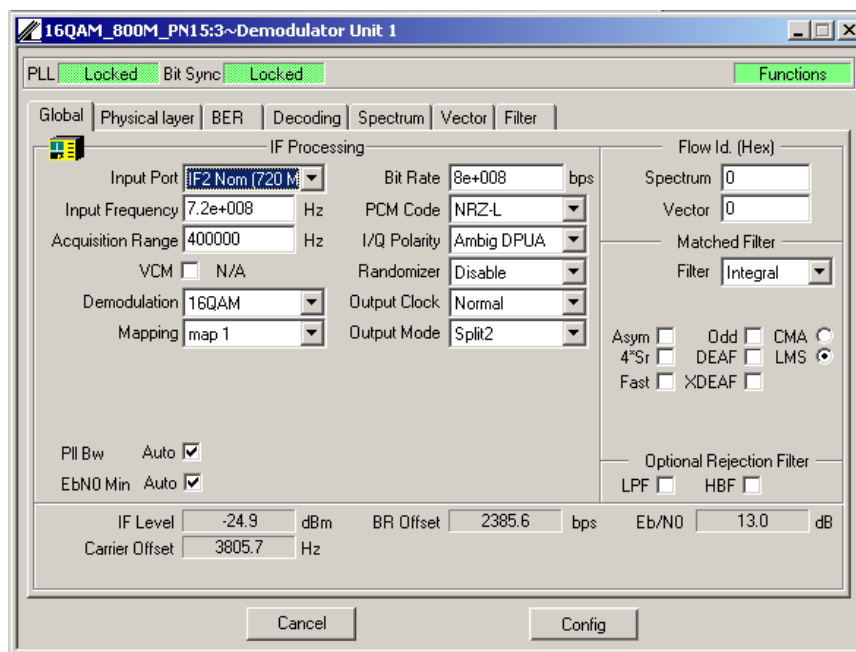


Figure 107: Demodulator: Global Window for 16QAM Modulation

❑ **16APSK Modulation:**

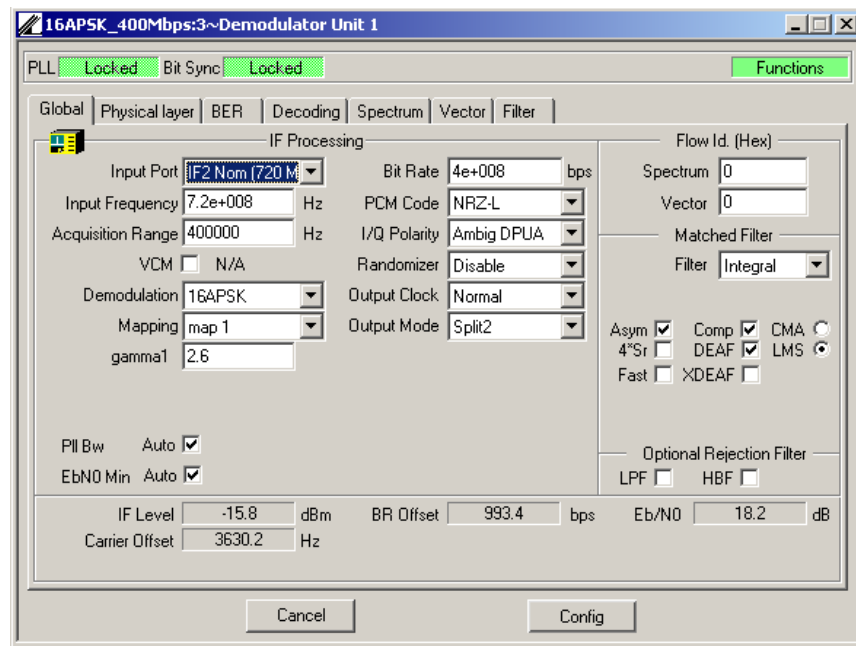
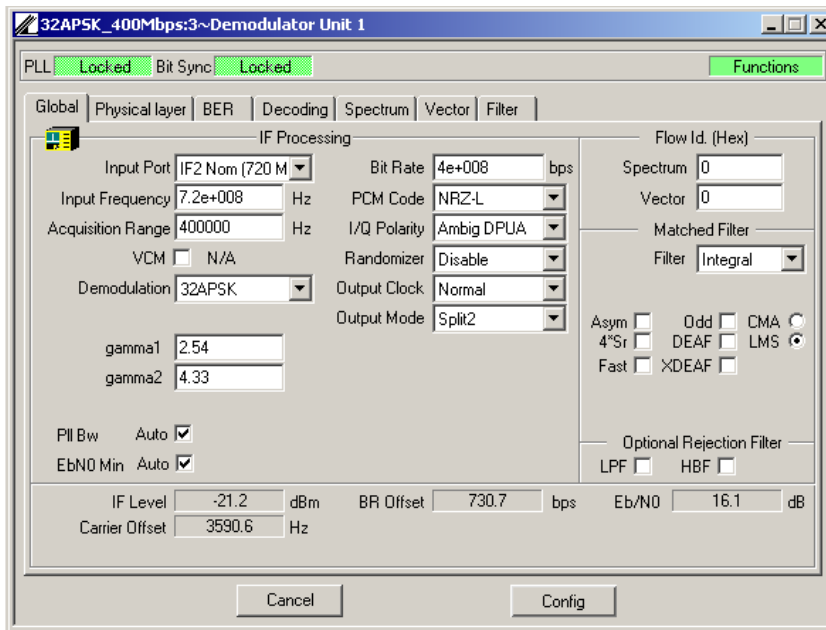


Figure 108: Demodulator: Global Window for 16APSK Modulation

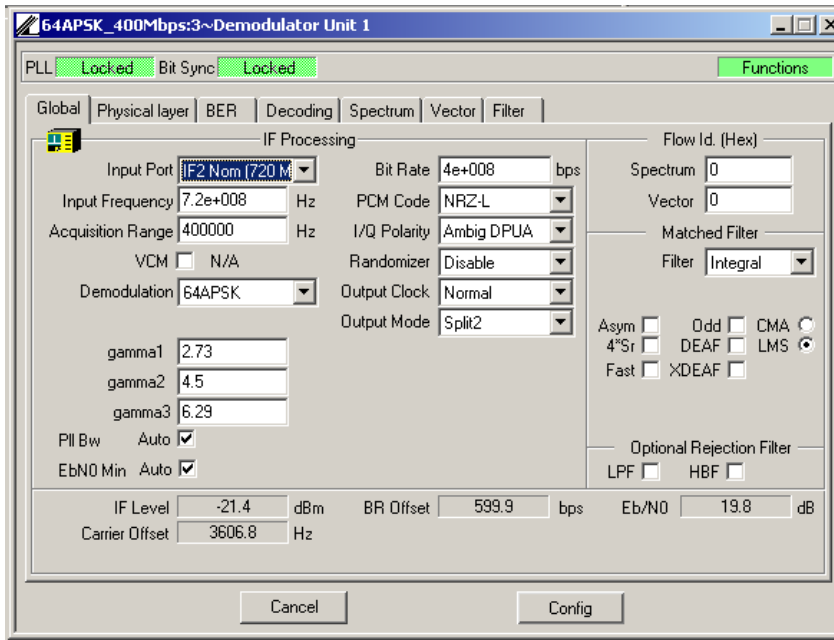
❑ **32APSK Modulation:**



The screenshot shows the '32APSK_400Mbps:3~Demodulator Unit 1' window. The 'Global' tab is active. The 'IF Processing' section includes: Input Port (IF2 Nom (720 M)), Input Frequency (7.2e+008 Hz), Acquisition Range (400000 Hz), VCM (N/A), Demodulation (32APSK), gamma1 (2.54), gamma2 (4.33), Pll Bw (Auto), and EbN0 Min (Auto). The 'IF Processing' section also includes: Bit Rate (4e+008 bps), PCM Code (NRZ-L), I/Q Polarity (Ambig DPUA), Randomizer (Disable), Output Clock (Normal), and Output Mode (Split2). The 'Flow Id. (Hex)' section includes: Spectrum (0) and Vector (0). The 'Matched Filter' section includes: Filter (Integral). The 'Optional Rejection Filter' section includes: LPF and HBF. The 'Functions' section includes: Asym, 4'Sr, Fast, Odd, DEAF, XDEAF, CMA, and LMS. The bottom status bar shows: IF Level (-21.2 dBm), BR Offset (730.7 bps), Eb/N0 (16.1 dB), and Carrier Offset (3590.6 Hz).

Figure 109: Demodulator: Global Window for 32APSK Modulation

❑ **64APSK Modulation:**



The screenshot shows the '64APSK_400Mbps:3~Demodulator Unit 1' window. The 'Global' tab is active. The 'IF Processing' section includes: Input Port (IF2 Nom (720 M)), Input Frequency (7.2e+008 Hz), Acquisition Range (400000 Hz), VCM (N/A), Demodulation (64APSK), gamma1 (2.73), gamma2 (4.5), gamma3 (6.29), Pll Bw (Auto), and EbN0 Min (Auto). The 'IF Processing' section also includes: Bit Rate (4e+008 bps), PCM Code (NRZ-L), I/Q Polarity (Ambig DPUA), Randomizer (Disable), Output Clock (Normal), and Output Mode (Split2). The 'Flow Id. (Hex)' section includes: Spectrum (0) and Vector (0). The 'Matched Filter' section includes: Filter (Integral). The 'Optional Rejection Filter' section includes: LPF and HBF. The 'Functions' section includes: Asym, 4'Sr, Fast, Odd, DEAF, XDEAF, CMA, and LMS. The bottom status bar shows: IF Level (-21.4 dBm), BR Offset (599.9 bps), Eb/N0 (19.8 dB), and Carrier Offset (3606.8 Hz).

Figure 110: Demodulator: Global Window for 64APSK Modulation

❑ VCM QPSK / 8PSK Modulation:

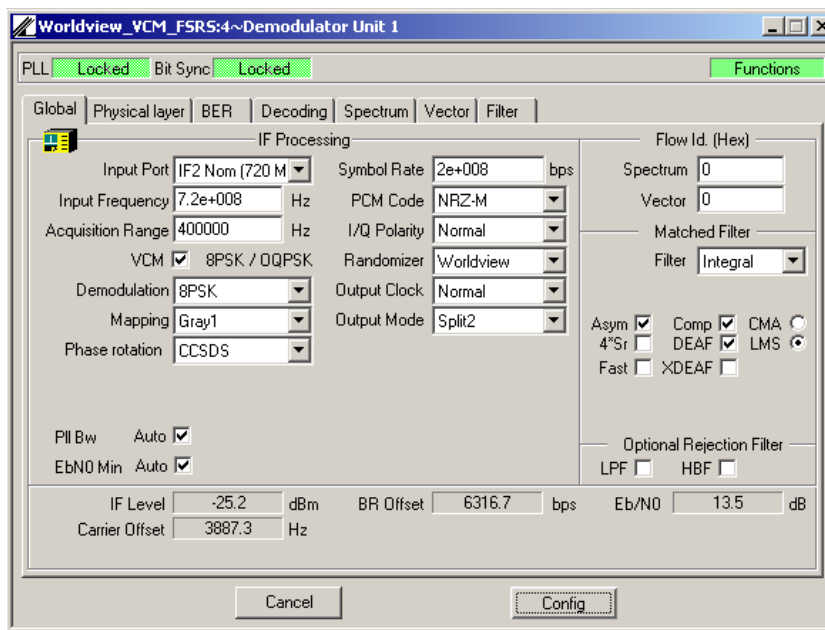


Figure 111: Demodulator: Global Window for VCM QPSK / 8PSK

❑ DVB-S2 / SCCC Modulation:

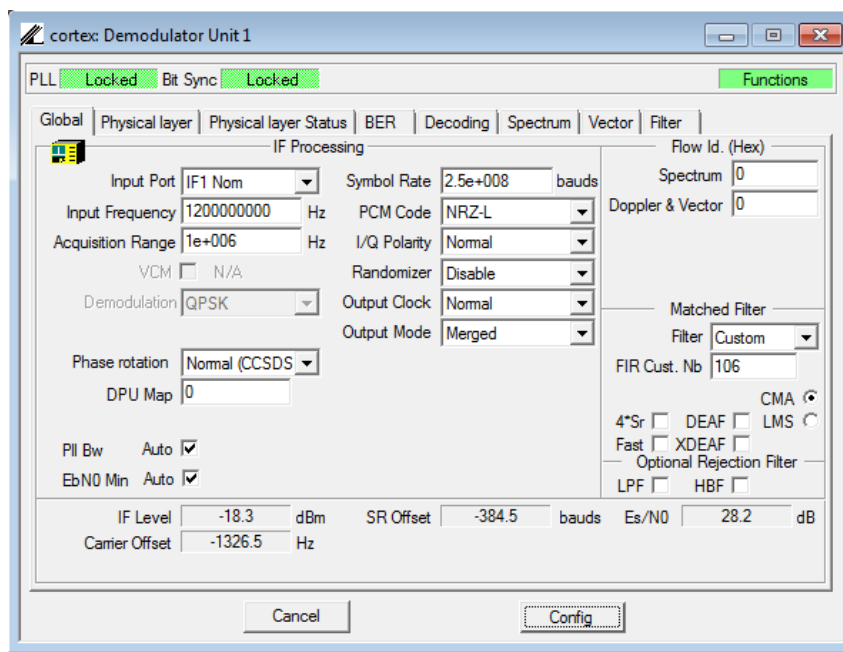
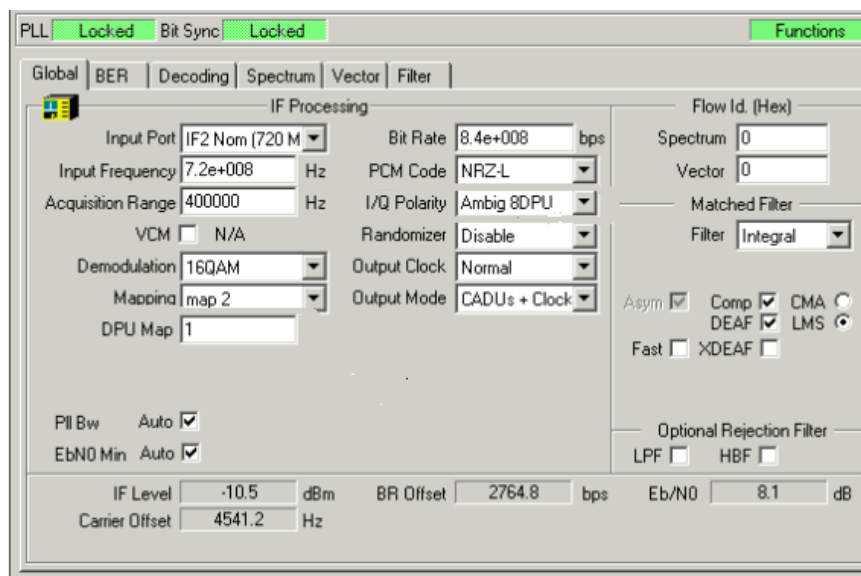


Figure 112: Demodulator: Global Window for DVB-S2 / SCCC

❑ **Multi-DPU Modulation (16QAM or QPSK):**



PLL: Locked Bit Sync: Locked Functions

Global | BER | Decoding | Spectrum | Vector | Filter

IF Processing

Input Port: IF2 Nom (720 M) Bit Rate: 8.4e+008 bps

Input Frequency: 7.2e+008 Hz PCM Code: NRZ-L

Acquisition Range: 400000 Hz I/Q Polarity: Ambig 8DPU

VCM: ☐ N/A Randomizer: Disable

Demodulation: 16QAM Output Clock: Normal

Mapping: map 2 Output Mode: CADUs + Clock

DPU Map: 1

Flow Id. (Hex)

Spectrum: 0

Vector: 0

Matched Filter

Filter: Integral

Asym: ☒ Comp: ☒ CMA: ☐

DEAF: ☒ LMS: ☒

Fast: ☐ XDEAF: ☐

Optional Rejection Filter

LPF: ☐ HBF: ☐

PIBw: Auto ☒

EbN0 Min: Auto ☒

IF Level: -10.5 dBm BR Offset: 2764.8 bps Eb/N0: 8.1 dB

Carrier Offset: 4541.2 Hz

Figure 113: Demodulator: Global Window for Multi-DPU Operation

4.5.4.1.1 Control

Parameters in this window are described in **DMU**.

4.5.4.1.1.1 IF Processing

INPUT PORT IF input port selection (for 720 MHz or 1.2 GHz IF)

IF1 Nom , IF1 Alt

IF2 Nom, IF2 Alt

IF input port selection (for Wide Band)

IF1 Nom , IF1 Alt

INPUT FREQUENCY Expected IF **520** to **920** MHz (nominal 720 MHz IF input) or **960** to **1600** MHz (nominal 1.2 GHz IF input). expressed in Hz. Enter any value in float or real format. Example: for 720 MHz, type: **72e+7** or **720000000**. The programmed value is always displayed in float format. Resolution step: 32-bit float format.

ACQUISITION RANGE Carrier acquisition peak range (from the expected carrier frequency) and PLL tracking range. Expressed in Hz. Enter any value from **10,000** to **2e6**. Select a value that takes into account the frequency offset in the worst operational conditions. (Doppler, frequency uncertainties, etc.).

Carrier acquisition can be restarted at any moment by clicking on .

DEMODULATION

- **BPSK** BPSK demodulation.
- **QPSK** QPSK demodulation.
- **OQPSK** OQPSK or SQPSK or SOQPSK demodulation.
- **UQPSK** UQPSK or AQPSK, AUQPSK, AUSQPSK demodulation
- **8PSK** 8PSK demodulation
- **16QAM** 16 QAM demodulation
- **16APSK** 16 APSK demodulation
- **32 APSK** 32 APSK demodulation
- **64 APSK** 64 APSK demodulation
- For GMSK** Select OQPSK + Matched filter = GMSK and then enter BTs value
- For SOQPSK** Select OQPSK + Matched filter = SOQPSK and then select the dedicated filter

VCM - 8PSK/OQPSK

Enable the automatic switch function between 8PSK and OQPSK demodulation. This function is only available with OQPSK and 8PSK modulation. In this mode, the programmed symbol rate is constant.

BIT RATE OR SYMBOL RATE	Telemetry bit rate. Expressed in bps. Enter any value from 1e6 to the maximum admissible bit rate (license, modulation and firmware are menu dependent)
	Important: if the data are convolutionally encoded, enter the bit rate at the Viterbi decoder input (bit synchronizer output). In DVB-S2, SCCC and VCM modes, the Bit Rate is replaced by the Symbol Rate
PCM CODE	Telemetry PCM coding technique: NRZ-L/M/S , DNRZ , Bi-P-L/M/S , NRZ-M Merge Important: if NRZ-M Merge is selected, the NRZ-M to NRZ-L decoding is performed after the merging of the data coming from I and Q. If NRZ-M is selected the decoding is performed separately on I and Q before data merging.
RANDOMIZER	Auto-synchronized de-scrambler: ■ Disable de-randomizer disabled ■ Worldview Worldview type (menu 11 only) ■ CCITT V.35 Menu 3 only ■ Intelsat V.35 Menu 3 only
I/Q POLARITY	For QPSK, 16QAM and n-APSK modulations: ■ Normal Direct I & Q ■ I Inverted I inverted, direct Q ■ Q Inverted direct I, Q inverted ■ I/Q Inverted I & Q inverted ■ Swap Normal I&Q swapped & direct ■ Swap I Inv. I&Q swapped, I inverted & direct Q ■ Swap Q Inv. I&Q swapped, direct I & Q inverted ■ Swap I/Q Inv. I&Q swapped & inverted ■ Ambig DPUA (B) Ambiguity resolution by DPU A (B) ■ Ambig DPUC (D) Ambiguity resolution by DPU C (D) ■ Ambig DPUA+C (B+D) Ambiguity resolution by DPU A&C (B&D) ■ Ambig BER Ambiguity resolution by the BER tester ■ Ambig 4DPU Ambiguity resolution by 4 DPU (specific menu 16) ■ Ambig 8DPU Ambiguity resolution by 8 DPU (specific menu 16) For 8PSK modulation: ■ Normal Direct output (no rotation) ■ Rot 45° 45° rotation of the 8PSK phase constellation ■ Rot 90° 90° rotation ■ Rot 135° 135° rotation ■ Rot 180° 180° rotation ■ Rot 225° 225° rotation

- **Rot 270°** 270° rotation
- **Rot 315°** 315° rotation
- **Ambig DPUA (B)** Ambiguity resolution by DPU A (B)
- **Ambig DPUC (D)** Ambiguity resolution by DPU C (D)
- **Ambig DPUA+C (B+D)** Ambiguity resolution by DPU A&C (B&D)
- **Ambig BER** Ambiguity resolution by the BER tester
- **Ambig 4DPU** Ambiguity resolution by 4 DPU (specific menu 16)
- **Ambig 8DPU** Ambiguity resolution by 8 DPU (specific menu 16)

Note: **DPU A** and **DPU C** are connected to **DMU 1**. **DPU B** and **DPU D** are connected to **DMU 2**

Note: with the specific basic menu # 16, **all DPU** (up to 8) are connected to the only available demodulator **DMU 1**

OUTPUT CLOCK

Clock polarity: Normal, Inverted or Dual Edge

OUTPUT MODE

Data output mode:

- **Merged** I & Q Channels are interleaved on a single output
- **Split 2** I & Q Channels each on 2 separate outputs
- **Split 4** Data are distributed over 4 separate outputs
- **Normal Soft bits**
- **Merge Soft bits**
- **CADU DPU A (B) + sync + user**
- **CADU DPU A (B) + sync + good**
- **CADU DPU A (B) + split2 + sync**
- **CADU DPU C (D) + sync + user**
- **CADU DPU C (D) + sync + good**
- **CADU DPU C (D) + split2 + sync**
- **CADU DPU A (D) + CADU C (D)**
- **CADUs + Clock** All CADUs data at the output of DPU and Clock are distributed on HDR outputs (specific menu 16)
- **CADUs + User Clock** All CADUs data at the output of DPU and Clock are distributed on HDR outputs the clock is present only with the useful data (specific menu 16)
- **Good CADUs + User Clock** All good CADUs data at the output of DPU and Clock are distributed on HDR outputs the clock is present only with the useful data (specific menu 16)
- **Split 8** Data at the output of DMU are distributed over 4 separate HDR outputs (specific menu 16)
- **Playback Merge** To be used jointly with the Playback function

- **Playback Split2** To be used jointly with the Playback function

Note: DPU A and DPU C are connected to DMU 1. DPU B and DPU D are connected to DMU 2

- PLL BW AUTO** Selection of the PLL BW automatic adjustment.
If not selected , a value must be set from 1 to 255, with unit: 0.01 % of the bit rate
- EBN0 MIN AUTO** Selection of the automatic adjustment of the acquisition threshold.
If not selected , a value must be set from -3 to 10 dB, with unit: 1 dB

In OQPSK mode, the following additional parameters appear at the GUI for controlling the I/Q delay compensation:

- OQPSK I/Q DELAY** OQPSK I/Q delay compensation:
- **OFF** No compensation
 - **ON (Auto)** Automatic compensation
 - **ON (Fixed)** Fixed compensation (see *I / Q shift* parameter)
- I/Q SHIFT** When **OQPSK I/Q DELAY** is set to **ON (Fixed)**:
- I/Q delay compensation: enter any value from **-0.25** to **+0.25** (symbol)
- Enter **0** for no compensation (optimal OQPSK)

Important: in **UQPSK (AQPSK, AUQPSK, AUSQPSK) mode**, parameters **BIT RATE**, **PCM CODE** and **OUTPUT CLOCK** described here above are for the I Channel. The following additional parameters appear at the GUI for controlling the Q Channel:

- BIT RATE Q** Telemetry bit rate for the Q Channel. Expressed in bps. Enter any value from **1e6** to **480.e6** (model dependent)
- In AQPSK mode, the I Channel bit rate is supposed to be higher or equal to the Q Channel bit rate.
- PCM CODE Q** PCM code on the Q Channel: **NRZ-L/M/S**, **DNRZ**, **Bi-P-L/M/S**
- OUTPUT CLOCK Q** Clock polarity for the Q Channel: **Normal** or **Inverted**
- I/Q RATIO** Power ratio between I Channel & Q Channel in dB: from **0** to **100**

In **8PSK mode**, the following additional parameters appear at the GUI:

PHASE ROTATION	Select CCSDS (Counter Clock Wise) or Inverted (Clock Wise). (see section 3.6.2.2)
8PSK MAPPING	Select Natural , Natural2 , Gray1 , Gray2 , Gray3 , Gray4 , Gray5 or Gray6 mapping (see section 3.6.2.2)

In **16QAM mode**, the following additional parameter appears at the GUI:

16QAM MAPPING	Select Map 1 , Map 2 or Map 3 (see section 3.6.2.2)
----------------------	---

In **16APSK mode**, the following additional parameter appears at the GUI:

GAMMA1	Allow to adjust R2/R1 of the 16APSK constellation
16APSK MAPPING	Select Map 1 or Map 2 (see section 3.6.2.2)

In **32APSK mode**, the following additional parameter appears at the GUI:

GAMMA1	Allow to adjust R2/R1 of the 32APSK constellation
GAMMA2	Allow to adjust R3/R1 of the 32APSK constellation

In **64APSK mode**, the following additional parameter appears at the GUI:

GAMMA1	Allow to adjust R2/R1 of the 64APSK constellation
GAMMA2	Allow to adjust R3/R1 of the 64APSK constellation
GAMMA3	Allow to adjust R4/R1 of the 64APSK constellation

In **DVB-S2 and SCCC modes**, the demodulation mode is controlled from the Physical Layer window.

In **Multi DPU mode**, the following additional parameter appears at the GUI:

DPU MAP	Select 0 or 1 for th emulti-DPU processing mapping (refer to paragrah 3.4.13)
----------------	---

4.5.4.1.1.2 Flow Id. (Hex)

SPECTRUM	Identification of the Spectrum data flow. This value is reported in all TCP-IP messages transmitted over the Spectrum Analysis Data port (SPA). Enter any hexadecimal value from 0 to FFFFFFF
VECTOR	Identification of the Vector Analysis data flow. This value is reported in all TCP-IP messages transmitted over the Vector Analysis Data port (VEC). Enter any hexadecimal value from 0 to FFFFFFF

4.5.4.1.1.3 Matched Filter

FILTER	Bit synchronizer filter. Satellite-dependent parameter: ❖ Integral Integral & Dump ❖ Raised Raised Cosine (See Roll-off) ❖ Root Raised Root Raised Cosine (See Roll-off) ❖ Custom Custom filters (See Fir Cust. Nb) ❖ GMSK GMSK filters (See BT) ❖ SOQPSK SOQPSK filters (See SOQPSK Norm)
ROLL-OFF	Filter roll-off factor: enter any value from 0.1 to 1.0 .
FIR CUSTOM NB	FIR custom Number: enter any value from 1 to 1000 .
BT FACTOR	GMSK Filter BT factor: 0.25 or 0.5.
SOQPSK NORM	❖ MIL MIL standard filter ❖ TG IRIG standard filter (IRIG standard 106-04) ❖ A CCSDS A-type filter (CCSDS 413.0-G-1) ❖ B CCSDS B-type filter
ASYM	Select asymmetrical FIR coefficients, else symmetrical FIR coefficients are applied.
COMP	Select complex FIR coefficients, else real FIR coefficients are applied (dependent of the other FIR characteristics selections).
ODD	Select odd FIR coefficients, else even FIR coefficients are applied.
4*SR	Select the sample rate of the FIR filter as 4 time the symbol rate, else 2 time the symbol rate is applied.
DEAF	Select the DEAF mode (Digital Equalization Automatic Filter).
CMA / LMS	Select the algorithm of the DEAF function. It is recommended to use the LMS algorithm for the reception of frequency non symmetrical signals.
FAST	Select fast DEAF computation.
XDEAF	Select the automatic cross-polarization compensation function.

4.5.4.1.1.4 *Optional rejection Filter*

LPF	Select additional Low Pass Filter +/- 128 MHz bandwidth
HBF	Select additional rejection filter

4.5.4.1.2 *Monitoring*

Status and alarms in this window are described in **DMU**.

4.5.4.1.2.1 *Upper Banner*

PLL	Demodulator global status: ■ Locked green ■ Unlocked red
BIT SYNC	Bit Synchronizer status: ■ Locked green ■ Unlocked red
BIT SYNC Q	Bit Synchronizer status for Q Channel (for UQPSK configuration): ■ Locked green ■ Unlocked red
VITERBI DECODER	Viterbi decoder status for I & Q Channels (logical AND of the VITERBI DECODER I and VITERBI DECODER Q status in case of dual Viterbi configuration): ■ Locked ■ Unlocked
TCM DECODER	TCM decoder status (for 4D-8PSK-TCM configuration): ■ Locked ■ Unlocked
FUNCTIONS	One or more parameter values do not match the installed licenses, or firmware capabilities: ■ Functions (green): OK Move the cursor to the status field to see the lists of available capabilities ■ Warning (red): the requested configuration is not supported. Move the cursor to the status field to check which parameter is incorrect (Tool Tip).

4.5.4.1.2.2 Other Status & Alarms

IF LEVEL	IF carrier level in dBm. The nominal AGC range is from -10 to -50 dBm. Degraded measurement if the level of the received signal is not within these limits.
CARRIER OFFSET	Offset from the expected IF carrier frequency (<i>real frequency - expected frequency</i>). Expressed in Hz. This value depends both on the oscillator drift (on-board and ground equipment) and on the Doppler conditions.
BR OFFSET OR SR OFFSET	Offset from the expected bit rate (<i>real BR - expected BR</i>). Expressed in bit per second. In DVB-S2 and SCCC modes, the Bit Rate Offset is replaced by the Symbol Rate Offset
EB/NO OR ES/NO	Energy per bit in a 1-Hz noise bandwidth, measured at the bit synchronizer input. Expressed in dB. In DVB-S2 and SCCC modes, the Eb/No is replaced by the Es/No (Es: Energy per symbol)

In **OQPSK mode**, the following additional status appears at the GUI for monitoring the I/Q delay compensation:

I/Q SHIFT	When OQPSK I/Q delay compensation is:
■ OFF	No compensation
■ ON (Auto)	Detected compensation value (Note 1)
■ ON (Fixed)	Fixed compensation value (Note 1 and 2)

Note 1: The I/Q shift compensation applies to Q channel (after IF to base-band down-conversion). As the OQPSK demodulation is ambiguous, the sign of the status changes in accordance with the demodulation ambiguity.

Note 2: In fixed mode, as the demodulation is ambiguous, both signs are tested and an error estimation is done in both cases. If the best error value is inferior to a certain threshold, the automate will be locked on this sign. If the two errors are not acceptable (i.e. if the programmed compensation does not reflect the real delay), the automate will indefinitely change the sign of the compensation, which is reflected by the sign change of this status. In this unstable configuration, the I/Q shift value must be refined.

Important: in **UQPSK (AQPSK, AUQPSK, AUSQPSK) mode**, status **BR OFFSET**, **Eb/No** and **BIT SYNC** described here above are for the I Channel. The following additional status appears at the GUI for monitoring the Q Channel:

BR OFFSET Q	Offset from the expected bit rate for the Q Channel (<i>real Q Channel BR - expected Q Channel BR</i>). Expressed in bit per second.
Eb/No Q	Energy per bit in a 1-Hz noise bandwidth, measured at the bit synchronizer input for the Q Channel. Expressed in dB.

4.5.4.2 "Physical layer" Window

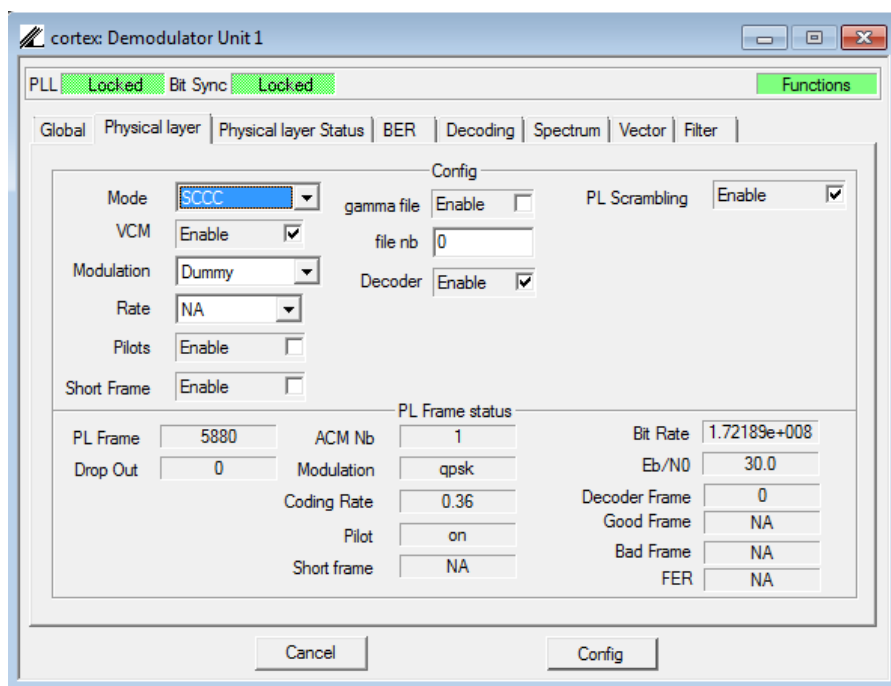


Figure 114: Demodulator: Physical layer window for SCCC and DVB-S2 standard

4.5.4.2.1 Control

Parameters in this window are described in **DMU**.

MODE	Selection of the demodulation mode: ■ OFF Standar fixed demodulation mode ■ VHR E/O DVB-S2 Very High Rate DVB-S2 standard ■ SCCC SCCC standard ■ DVBS2X DVB-S2X standard ■ SCCC-X SCCC-X standard
VCM	Enable or not the VCM mode: Automatic detection of the current modulation (if the option not accessible, VCM mode is automatically enabled).
MODULATION	If VCM mode is disabled, select the demodulation mode ■ QPSK ■ 8PSK ■ 16APSK ■ 32APSK ■ 64APSK
RATE	N/A
PILOTS	If enabled, indicates that the received physical frames are with pilots
SHORT FRAME	If enabled, indicates that the received physical frames are short frames (only with DVB-S2 standard).
PL SCRAMBLING	If enabled, indicates that the received physical frames are randomized.
DECODER ENABLE	Enable or Disable the decoding
GAMMA FILE (ENABLE) / FILE NB	Allows to enable and select a specific or customized set of the radius ratios of the mapping for the different available APSK modes (16APSK, 32APSK and 64APSK). The files are available in D:\ZDS\Data\HDR\crtxnt\VCM\gamma.

4.5.4.2.2 *Monitoring*

Status and alarms in this window are described in **DMU**.

PL FRAME	Counter of the received frames of the physical layer.
DROP OUT	Number of drop out events at the physical layer level.
ACM NB	Detected current ACM number: Modulation and Coding format (refer to DVB-S2 and SCCC standard).
MODULATION	Detected current modulation mode (refer to DVB-S2 and SCCC standard).
CODING RATE	Detected current coding rate (refer to DVB-S2 and SCCC standard).
PILOT	Presence or not of pilot inside the current received frame (refer to DVB-S2 and SCCC standard).
SHORT FRAME	The received frames are short frames or not (refer to DVB-S2).
BIT RATE	Bit Rate in bps of the current received frame.
EB/NO	Measured Eb/No in dB.
DECODER FRAME	Counter of the received frames of the decoder.
GOOD FRAME	Number of good frames
BAD FRAME	Number of bad frames
FER	Frame error Rate

4.5.4.3 "Physical layer status" Window

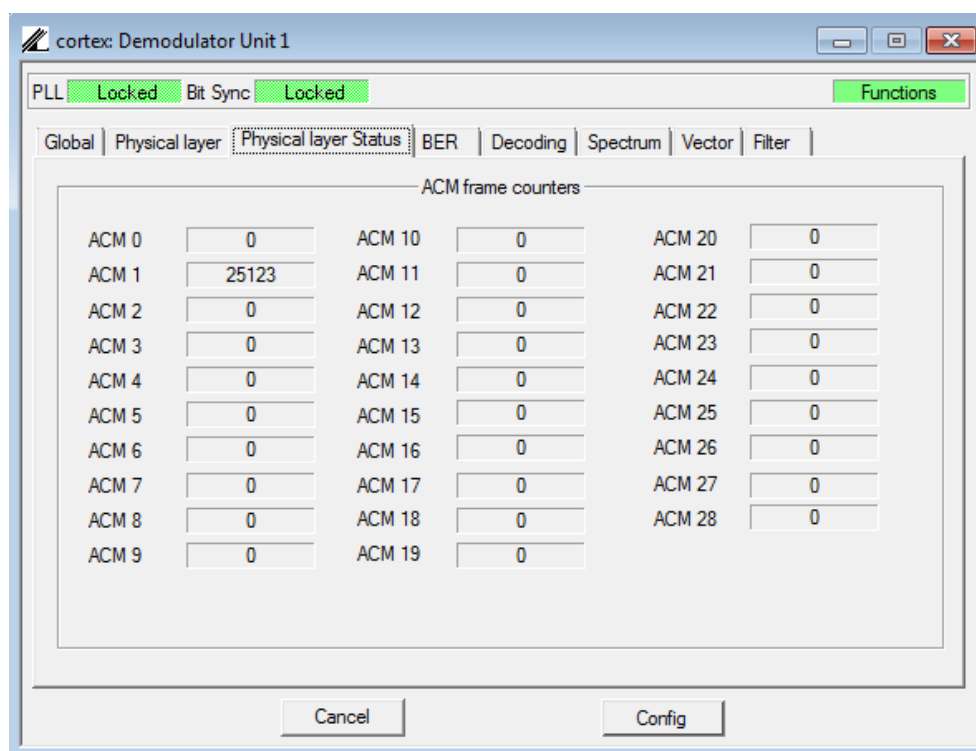


Figure 115: Demodulator: Physical layer status window for SCCC and DVB-S2 standard

4.5.4.3.1 Monitoring

Statuses in this window are described in **DMU**.

ACM N Counters of received frames for the different ACM.

OTHER ACM Counters of received frames for ACM numbers above 28.

4.5.4.4 "BER" Window

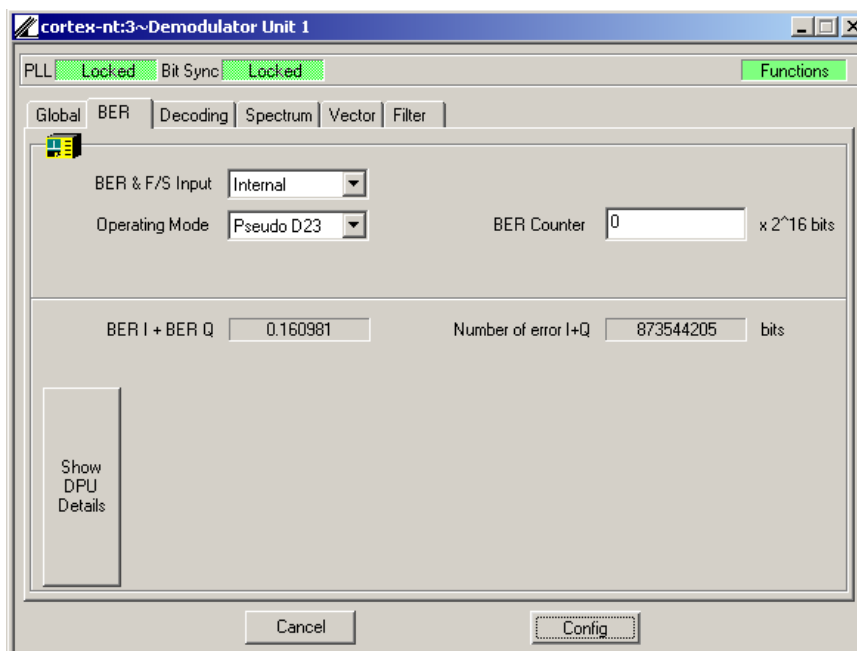


Figure 116: Demodulator: BER Window PRN mode

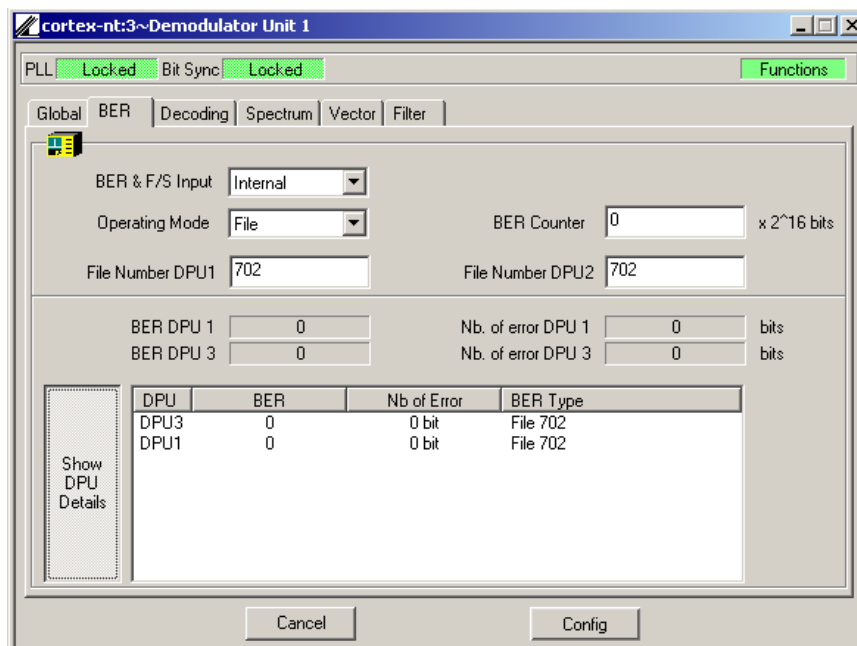


Figure 117: Demodulator: BER Window File mode

4.5.4.4.1 Control

Parameters in this window are described in **DMU**.

BER & F/S INPUT	■ Reserved (Unused). Set to Internal
OPERATING MODE	BER measurement mode (simulated data for BER analysis):
	■ Off BER measurement stopped (for reset purposes)
	■ File Simulated data from a file on the disk (note 1)
	■ PseudoDxx Pseudo random data, using from PN generator xx (where xx = 7, 10, 11, 15, 23 or 31. See Section 3.6.1.1.1)

Note: In File mode, the BER is computed by comparing the original frame to the reconstructed frame from the frame synchronizer. The BER value is not significant if the frame synchronizer is not correctly configured and locked.

FILE NUMBER File number for BER in file mode: enter any value from **1** to **N** for each associated DPU.

BER COUNTER Total number of bits to process for BER measurement in Pseudo mode
Resolution step: 65536 bits. In File Mode, Infinite accumulation.



(upper banner)

Reset the BER counter for the active Demodulation window.

4.5.4.4.2 Monitoring

Status and alarms in this window are described in **DMU**.

4.5.4.4.3 Upper Banner

See section 4.5.4.1.2.1 .

4.5.4.4.3.1 Other Status & Alarms

For BER measurement in PRN mode, on I & Q merged and if the demodulation mode is not UQPSK:

BER I + BER Q	Display a BER value between 0 and 1 , or No data if the demodulation chain is unlocked or if the total number of bits regenerated by the bit synchronizer is less than the programmed value for parameter BER Counter .
NUMBER OF ERROR I + Q	Number of bits in error since the BER counter was reset.

If Separate I/Q BER display requested or in UQPSK demodulation:

BER I Display a BER value between **0** and **1**, or
No data if the demodulation chain is unlocked or if the total number of bits regenerated by the bit synchronizer is less than the programmed value for parameter **BER Counter**.

NUMBER OF ERROR I Number of bits in error since the BER counter was reset.

BER Q Display a BER value between **0** and **1**, or
No data if the demodulation chain is unlocked or if the total number of bits regenerated by the bit synchronizer is less than the programmed value for parameter **BER Counter**.

NUMBER OF ERROR Q Number of bits in error since the BER counter was reset.

For BER measurement in File mode, for each associated DPU:

BER DPU N Display a BER value between **0** and **1**, for DPU #N

NUMBER OF ERROR DPU N Number of bits in error since the BER counter was reset for DPU #N

Note: From this window, by clicking on “**Show DPU Details**”, it is possible to view the BER values computed at the output of each DPU associated with the DMU and the used BER mode.

4.5.4.5 "Decoding" Window

For BPSK, QPSK, OQPSK, UQPSK modulations, $\frac{1}{2}$ and $\frac{1}{4}$ Viterbi decoder are available (if licence purchased):

$\frac{1}{2}$ -based, punctured Viterbi are available too:

❑ BPSK Modulation:

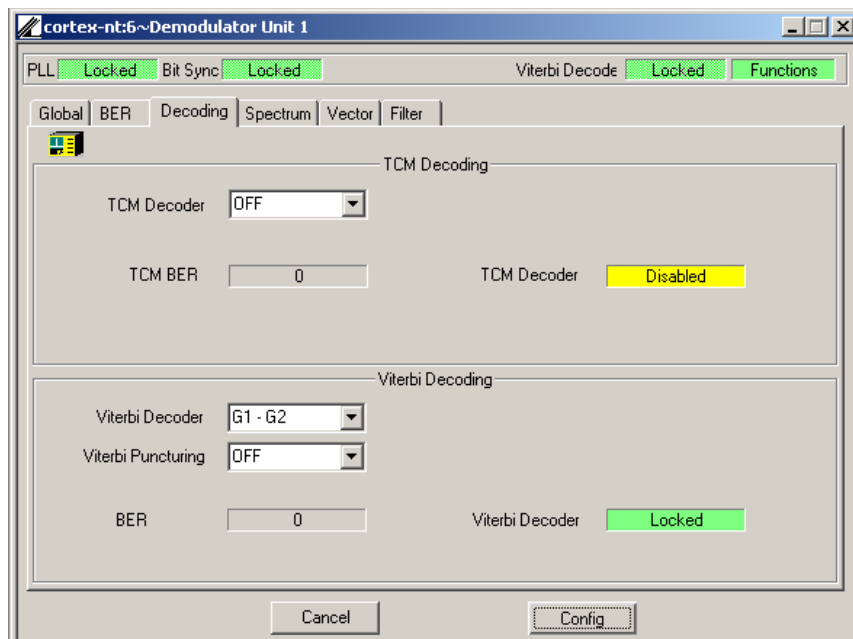


Figure 118: Demodulator: Decoding Window for BPSK Modulation

❑ QPSK or OQPSK (SQPSK) Modulation: Viterbi single

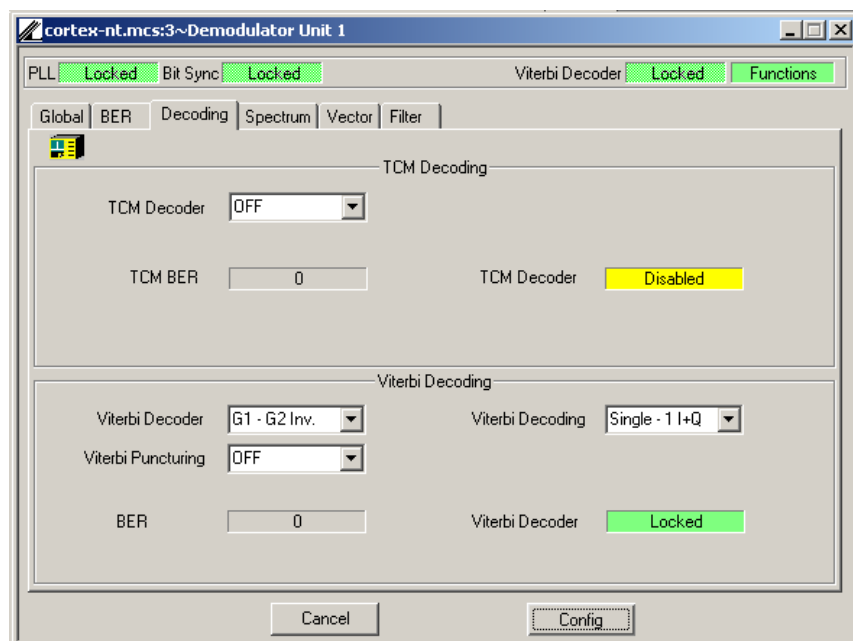


Figure 119: Demodulator: Decoding Window for QPSK & OQPSK Modulation (Viterbi Single)

❑ **QPSK or OQPSK (SQPSK) Modulation: Viterbi dual**

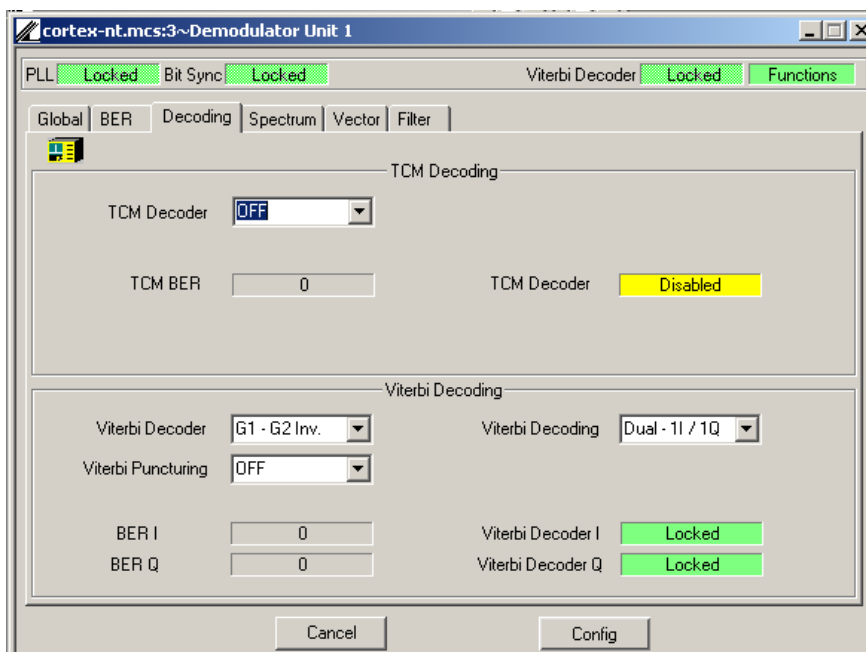


Figure 120: Demodulator: Decoding Window for QPSK & OQPSK Modulation (Viterbi Dual)

❑ **UQPSK (AQPSK, AUQPSK, AUSQPSK) Modulation:**

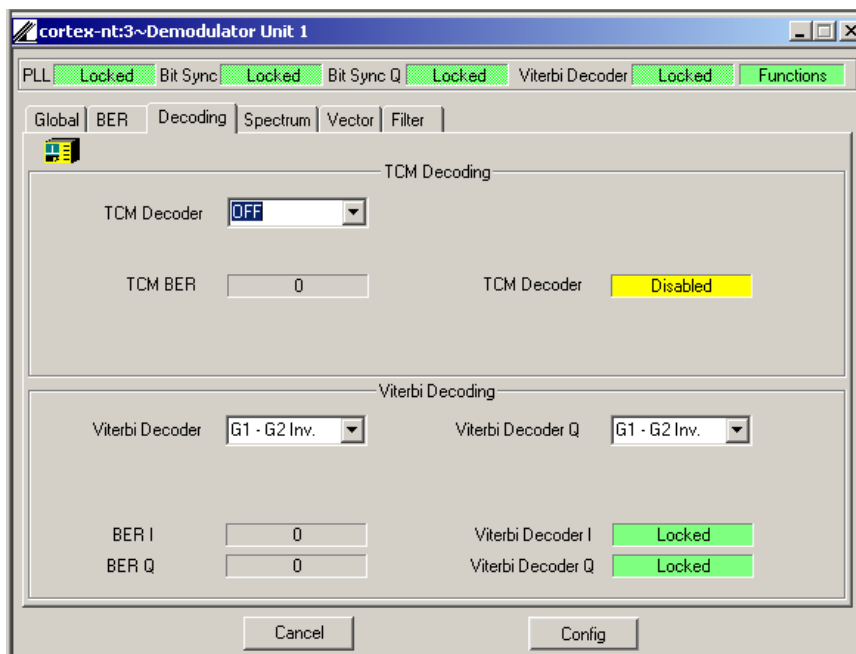


Figure 121: Demodulator: Decoding Window for UQPSK Modulation

For 8PSK modulation, TCM decoder is available (license dependent):

□ **8PSK Modulation:**

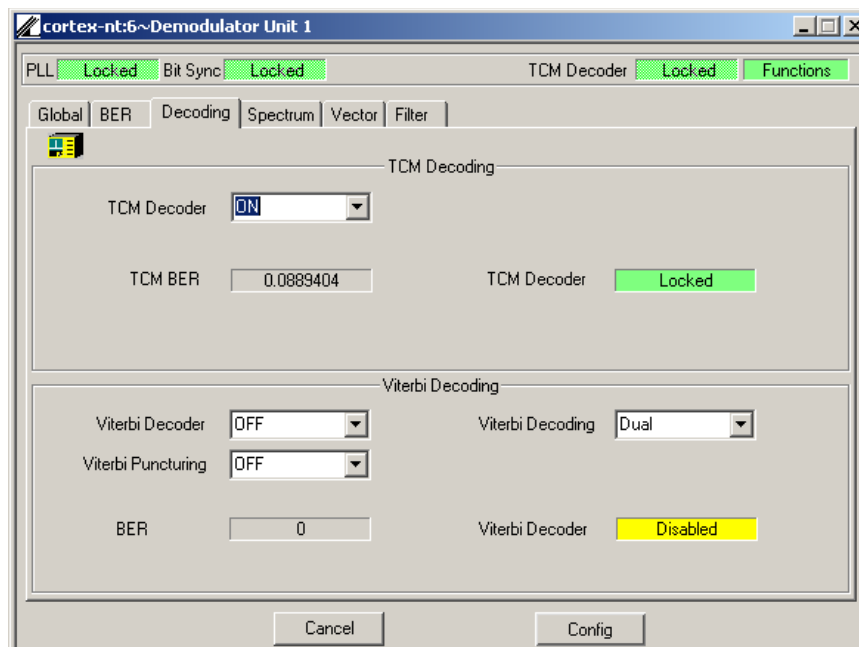


Figure 122: Demodulator: Decoding Window for 8PSK Modulation

4.5.4.5.1 Control

Parameters in this window are described in **DMU**.

TCM DECODER

TCM decoder setting , in 8PSK mode:

- **Off** TCM disabled
- **2.00/3** TCM 2.00 enabled
- **2.25/3** Not implemented today
- **2.50/3** TCM 2.50 enabled
- **2.75/3** TCM 2.75 enabled

Important: TCM and Viterbi decoding are mutually exclusive.

VITERBI DECODER

Viterbi decoder setting for the I Channel in BPSK or UQPSK (AQPSK, AUQPSK, AUSQPSK) mode or for both I & Q Channels in QPSK, OQPSK (SQPSK) mode:

- **Off** Viterbi decoder disabled
- **G1 →G2 Inv** first Viterbi decoder enabled with G2 inverted. G1 transmitted (CCSDS recommendation)
- **G1 →G2** first Viterbi decoder enabled with G2 normal. G1 transmitted first.
- **G2 Inv →G1** first Viterbi decoder enabled with G2 inverted. G2 transmitted first
- **G2 → G1** Viterbi decoder enabled with G2 normal. G2 transmitted first

Refer to the CCSDS recommendations for “G2 Normal/ G2 Inverted” definition.

VITERBI DECODING

Viterbi decoder setting in QPSK, OQPSK (SQPSK) mode:

- Single- 1 I+Q Single Viterbi decoder common to I & Q Channels
- Dual – nI / nQ Separate Viterbi decoders for I & Q Channels, n is the number of parallel decoders and can be selected between 1 and 16.

See Section 0 for more details

VITERBI PUNCTURING

Viterbi puncturing setting for I Channel in BPSK or UQPSK (AQPSK, AUQPSK, AUSQPSK) mode or for both I & Q Channels in QPSK, OQPSK (SQPSK) mode:

- Off Viterbi puncturing disabled
- ON (registry) Puncturing scheme defined in registry.
- 2/3 2/3 Viterbi
- ¾ (conv.1) ¾ Viterbi (convention 1)
- ¾ (conv.2) ¾ Viterbi (convention 2)
- 4/5 4/5 Viterbi
- 5/6(conv.1) 5/6 Viterbi (convention 1)
- 5/6(conv.2) 5/6 Viterbi (convention 2)
- 6/7 6/7 Viterbi
- 7/8 (conv.1) 7/8 Viterbi (convention 1)
- 7/8 (conv.2) 7/8 Viterbi (convention 2)
- 1/4 Esrin 1/4 Viterbi (ESA / ESRIN type)

VITERBI DECODER Q

Viterbi decoder setting for Q Channel in UQPSK (AQPSK, AUQPSK, AUSQPSK) mode:

- Off Viterbi decoder disabled
- G1 →G2 Inv first Viterbi decoder enabled with G2 inverted. G1transmitted (CCSDS recommendation)
- G1→G2 first. Viterbi decoder enabled with G2 normal. G1 transmitted first.
- G2 Inv→G1 first Viterbi decoder enabled with G2 inverted. G2 transmitted first
- G2→ G1 Viterbi decoder enabled with G2 normal. G2 transmitted first

Refer to the CCSDS recommendations for “G2 Normal/ G2 Inverted” definition.

VITERBI PUNCTURING Q

Viterbi puncturing setting for Q Channel in UQPSK (AQPSK, AUQPSK, AUSQPSK) mode:

- Off Viterbi puncturing disabled
- ON (registry) Puncturing scheme defined in registry.
- 2/3 2/3 Viterbi
- ¾ (conv.1) ¾ Viterbi (convention 1)
- ¾ (conv.2) ¾ Viterbi (convention 2)
- 4/5 4/5 Viterbi

- 5/6(conv.1) 5/6 Viterbi (convention 1)
- 5/6(conv.2) 5/6 Viterbi (convention 2)
- 6/7 6/7 Viterbi
- 7/8 (conv.1) 7/8 Viterbi (convention 1)
- 7/8 (conv.2) 7/8 Viterbi (convention 2)
- 1/4 Esrin 1/4 Viterbi (ESA / ESRIN type)

4.5.4.5.2 Monitoring

4.5.4.5.2.1 Upper Banner

See section 4.5.4.1.2.1 .

4.5.4.5.2.2 Other Status & Alarms

□ **For TCM configuration:**

BER TCM Bit Error Rate (must be 0 if no error detected). **Important:** the BER is analyzed at the bit synchronizer output (before TCM decoding)

TCM DECODER TCM decoder status:

- *Locked*
- *Unlocked*
- *Disabled*

□ **For single Viterbi configuration:**

BER Viterbi Bit Error Rate (must be 0 if no error detected). **Important:** the BER is analyzed at the bit synchronizer output (before Viterbi decoding)

VITERBI DECODER Viterbi decoder status:

- *Locked*
- *Unlocked*
- *Disabled*

❑ For dual Viterbi configuration:

BER I Viterbi Bit Error Rate on I Channel (must be 0 if no error)

BER Q Viterbi Bit Error Rate on Q Channel (must be 0 if no error)

VITERBI DECODER I Viterbi decoder status for I Channel:

- *Locked*
- *Unlocked*
- *Disabled*

VITERBI DECODER Q Viterbi decoder status for Q Channel:

- *Locked*
- *Unlocked*
- *Disabled*

4.5.4.6 "Spectrum" Window

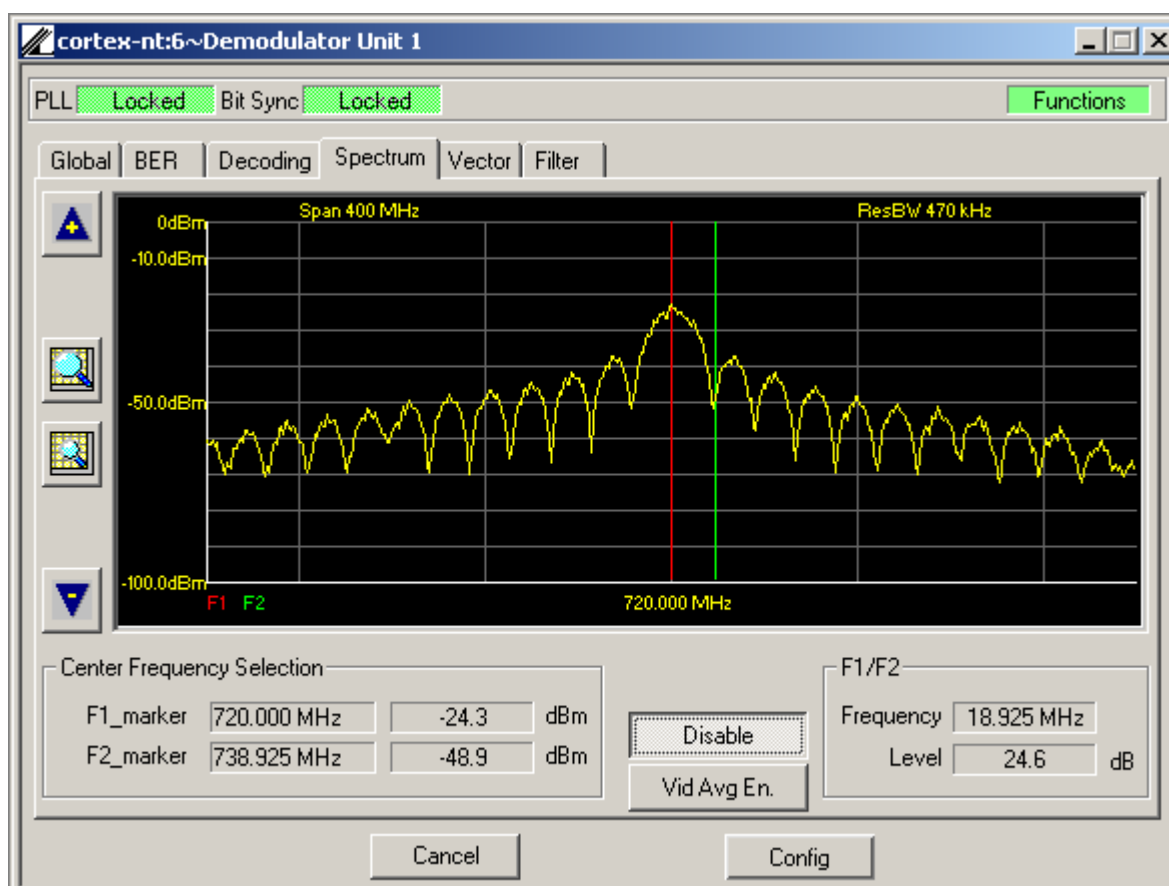


Figure 123: Demodulator: Spectrum Window

❑ Starting the Spectrum Analyzer:

- ❖ In the Demodulator Spectrum window, click on *Enable*. A TCP-IP connection is opened to the Spectrum port (SPA). To close the connection, click on *Disable* or click on icon  or exit the Spectrum window.

❑ Selecting the video average display mode:

- ❖ In the Demodulator Spectrum window, click on *Vid Avg En..* The averaging mode display is selected. To return to the non averaging display mode , click on *Vid Avg Dis..*

❑ Level Scale Adjust:



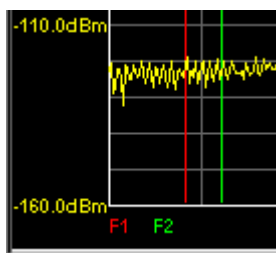
- ❖ Adjust the signal level scale (vertical toolbar) from 0.1 to 10 dB/division.

❑ Reference Level Adjust:



- ❖ Adjust the reference signal level from 0 dBm to –60 dBm.

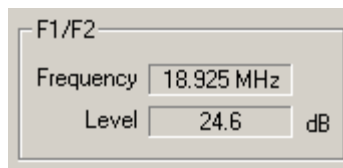
❑ Marker Setting & Center Frequency Selection:



- ❖ Clicking in the spectrum, with the left key of the mouse, will set a red marker. The signal frequency and level at this point is displayed in the window entitled Center Frequency Selection (F1 marker).
To center the spectrum on frequency F1:
 - Click on CONFIG,
 - Set the red marker with the left key of the mouse,
 - Click on F1 marker,
 - Click on APPLY.
- ❖ Clicking in the spectrum, with the right key of the mouse, will set a green marker. The signal frequency and level at this point are displayed in the next window under F2 marker. To center the spectrum on frequency F2, follow the above procedure using F2 marker.

Center Frequency Selection			
F1_marker	720.000 MHz	-24.3	dBm
F2_marker	738.925 MHz	-48.9	dBm

- ❖ The frequency and level offsets between markers F1 and F2 are displayed in a separate window:



4.5.4.7 "Vector" Window

❑ BPSK Modulation:

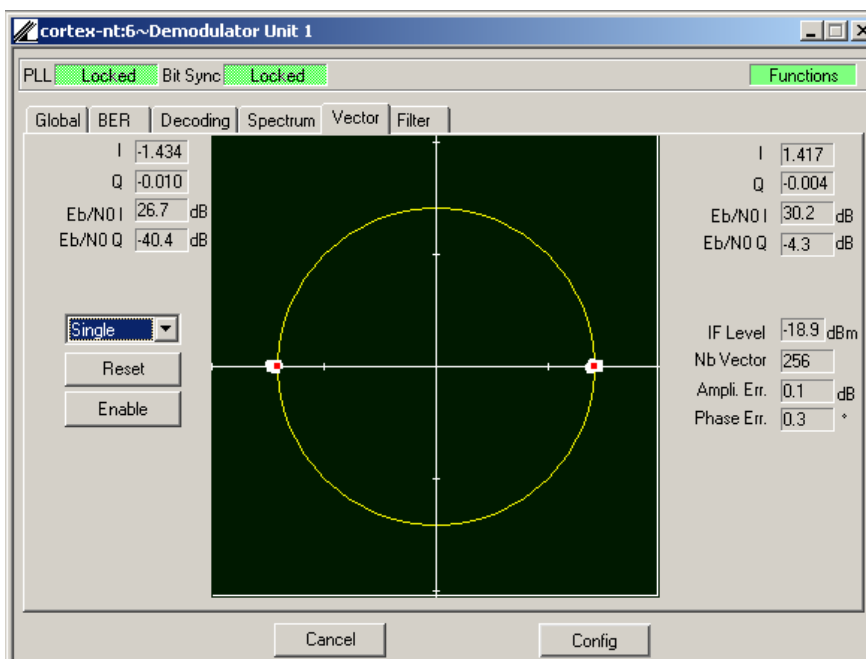


Figure 124: Demodulator: BPSK Vector Window

□ QPSK, OQPSK (SQPSK) or UQPSK Modulation:

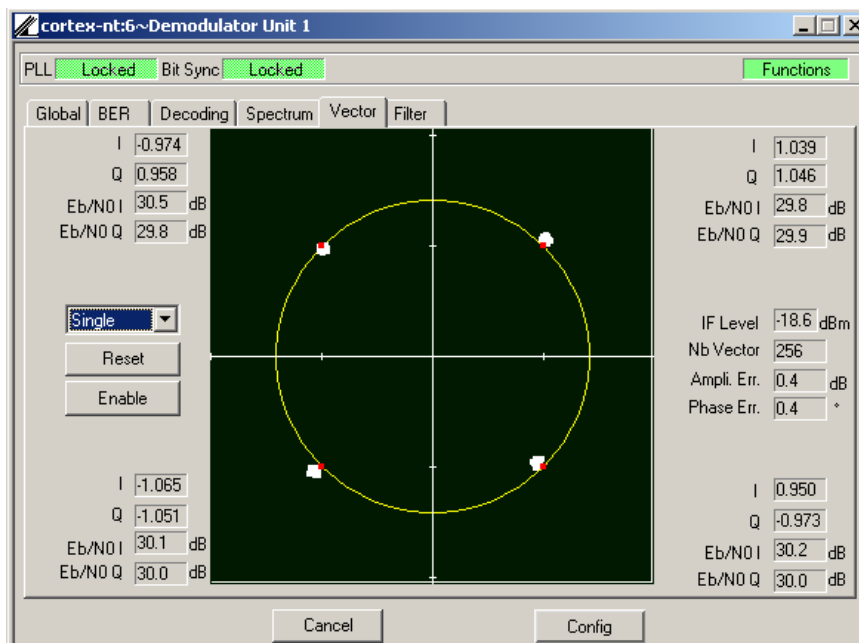


Figure 125: Demodulator: QPSK(OQPSK) Vector Window

□ 8PSK Modulation:

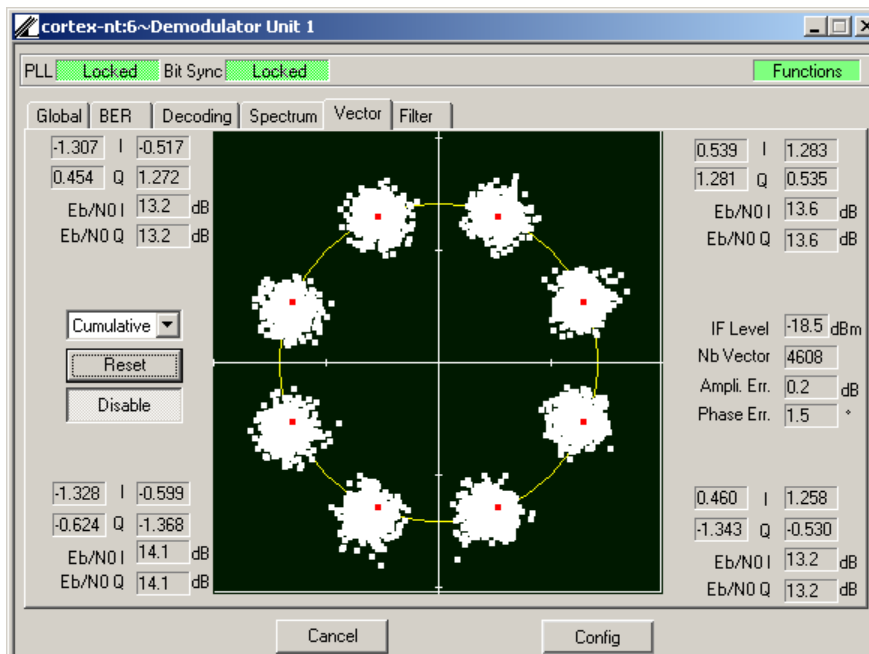


Figure 126: Demodulator: 8PSK Vector Window

□ **16QAM Modulation:**

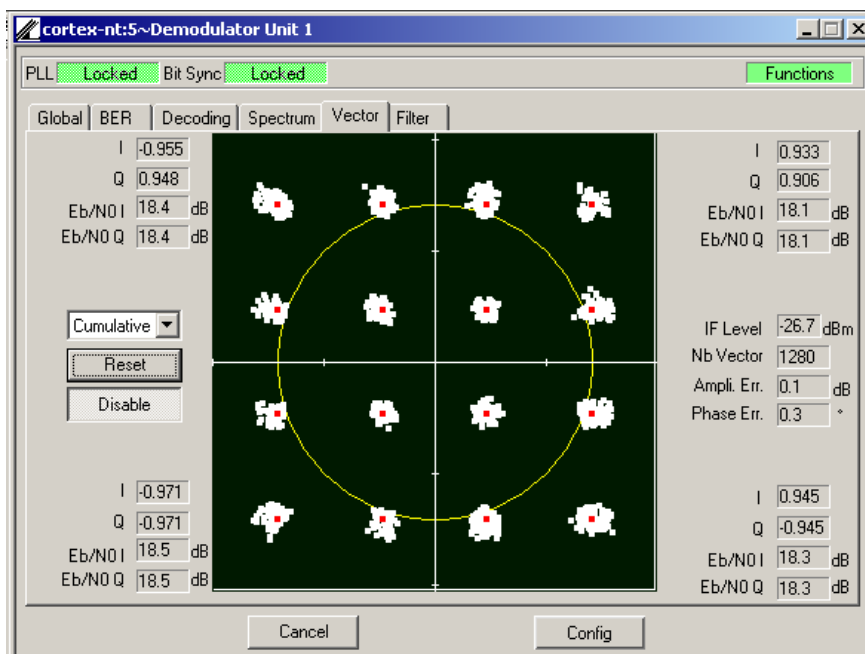


Figure 127: Demodulator: 16QAM Vector Window

□ **16APSK Modulation:**

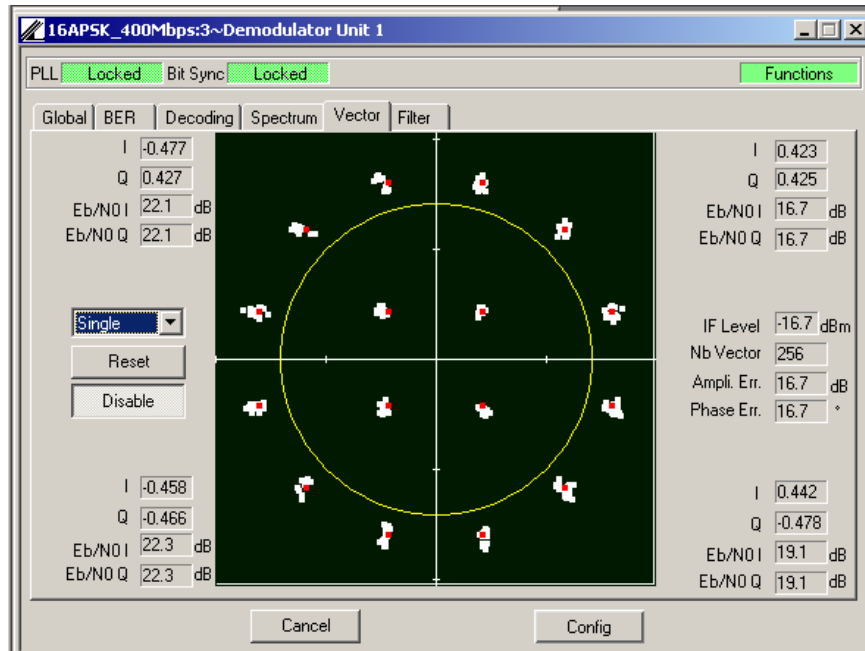


Figure 128: Demodulator: 16APSK Vector Window

❑ 32APSK Modulation:

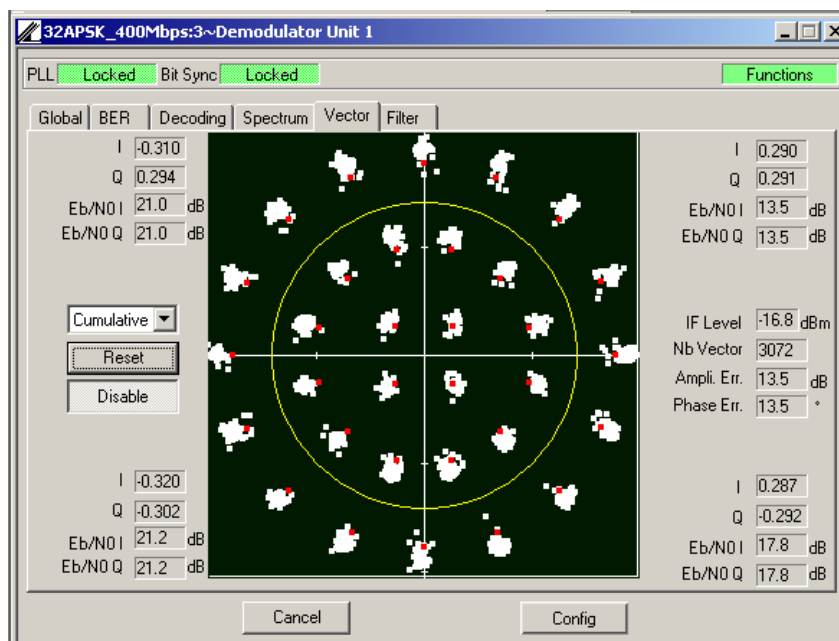


Figure 129: Demodulator: 32APSK Vector Window

❑ Starting the Vector Analyzer:

- ❖ In the Demodulator Global window (see section 4.5.4.1.1.1), set parameter **Mode** to **Demod**.
- ❖ In the Demodulator Vector window, click on **Enable**. A TCP-IP connection is opened to the Vector port (VEC). To close the connection, click on **Disable** or click on icon  or exit the Vector window.

❑ Vector display:



- ❖ Vectors are transmitted to the local MCS in bursts (TCP-IP packets) of 256 vectors.
- ❖ In **Single** mode, the display is refreshed burst by burst, with all vectors displayed.
- ❖ In **Cumulative** mode, the display is refreshed every 20 bursts only.
- ❖ In **Average S** mode, the MCS displays the average vector value for the received burst of vectors in each I/Q quadrant.
- ❖ In **Average C** mode, the MCS displays the average vector value for the 20 last bursts of vectors in each I/Q quadrant.
- ❖ Click on the **Reset** button to reset the display.

❑ **Vector analysis:**

I	1.037
Q	0.970
Eb/NO I	22.7 dB
Eb/NO Q	21.6 dB

- ❖ I & Q values and associated signal-to-noise ratio (Eb/No) are displayed for each quadrant.

❑ **Signal analysis:**

IF Level	-22.5 dBm
Nb Vector	5120
Ampli. Err.	0.3 dB
Phase Err.	2.1 °

- ❖ IF level gives the level of the input signal.
- ❖ Nb Vector gives the number of displayed bursts of vector values.
- ❖ Ampli. Err. And Phase Err. give respectively the RMS errors on the Amplitude and phase measurements.

4.5.4.8 "Filter" Window

❑ **Time Display Mode:**

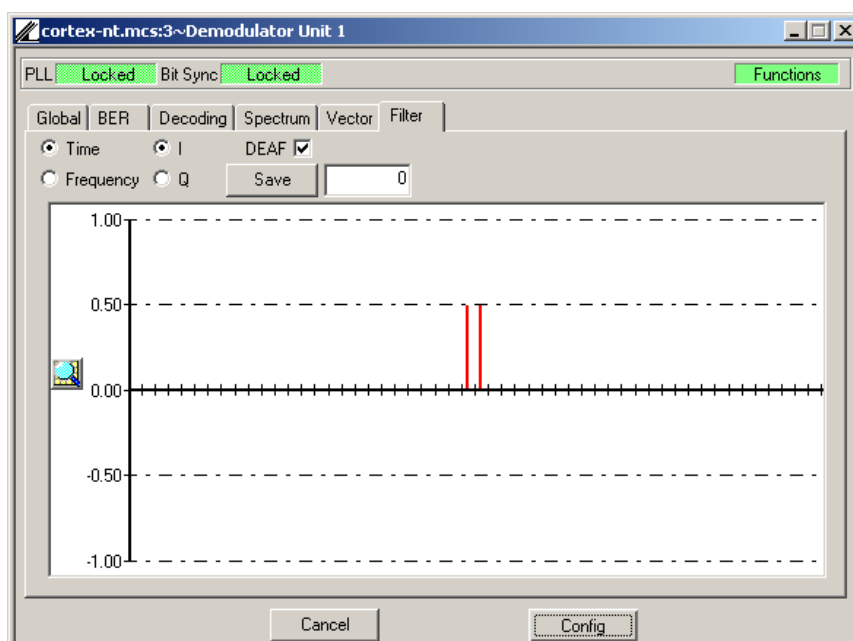


Figure 130: Filter: Time Display Selection

❑ Frequency Display Mode:

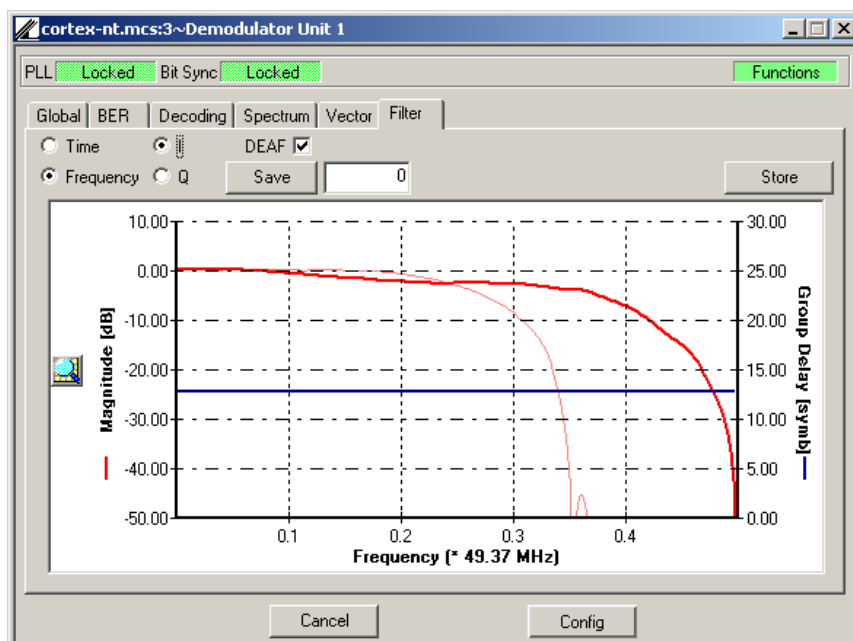


Figure 131: Filter: Frequency Display Selection

Note: The “Filter” Window enables to give a representation of the characteristics of the digital matched filter used in the demodulation process just before the Bit Decision. Depending on the User’s selection, this representation can be done in the time domain (pulse response, i.e. taps of the digital filter) or in the frequency domain (amplitude and phase responses). If the **DEAF** filter mode is selected, the display is continuously updated with new computed filter characteristics. The selection of the filter type and consequently its characteristics is done in the “Global” Window (Matched Filter box).

❑ Selection of the display mode:

- ❖ To select the time domain display mode, click on **Time**
- ❖ To select the frequency domain display mode, click on **Frequency**

❑ Control of the DEAF mode:

- ❖ To select the auto-adaptive mode and to get a dynamical update display of the filtering characteristics, click on **DEAF**

❑ Saving the filter parameters in a custom filter file (save and learn mode):

- ❖ In DEAF mode Click on **Store** to freeze the display and to temporary save the filter characteristics in a buffer.
- ❖ Select a filter file number from **0 to 1000**, this number will be used to recall the custom filter in the DEMOD global window.
- ❖ Click on **Save** to store the filter parameters in the selected custom file.

❑ **Level Scale Adjust:**



- ❖ Adjust the signal level scale (vertical toolbar).

❑ **Reference Level Adjust:**



- ❖ Adjust the reference signal level.

4.5.5 Data Processing Unit Windows

4.5.5.1 "CADU" Window

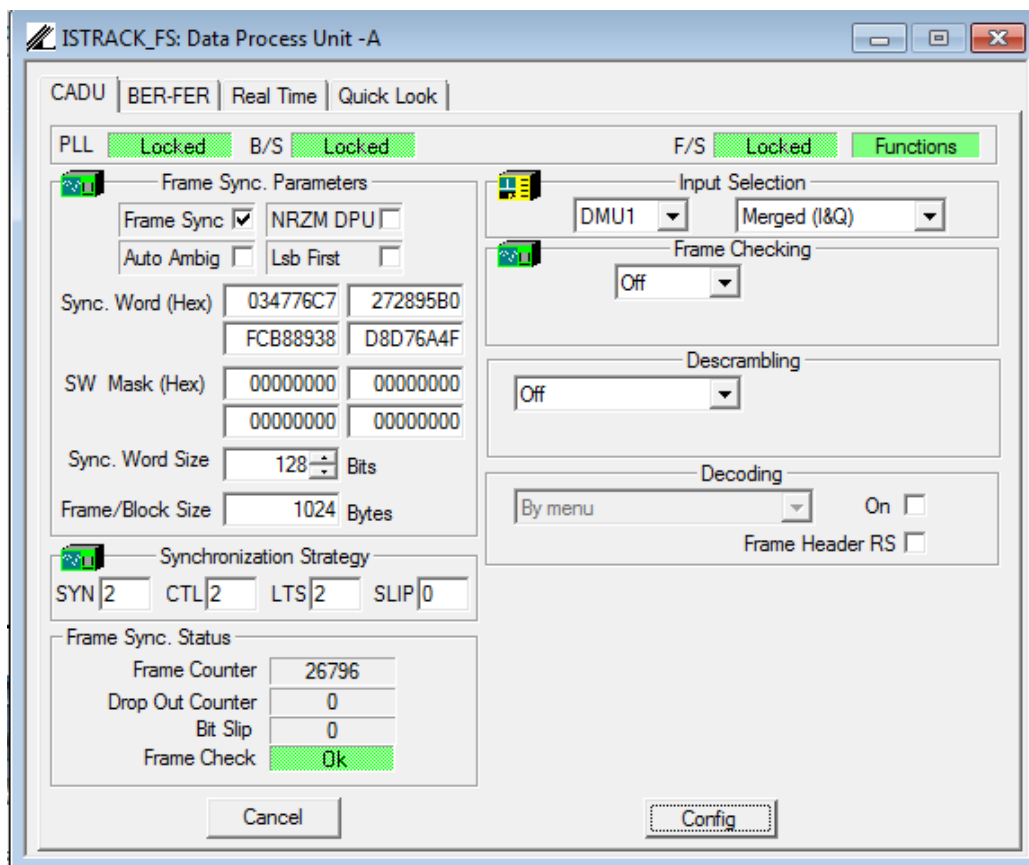


Figure 132: DPU: CADU Window

Position the mouse cursor on **Function** to see which DPU-related licenses are available. Depending on the available option (DVB Transport Layer, CCSDS Reed Solomon, or LDPC decoding), the right low part of the window lets appear the status of the corresponding function: RS DVB Status, Reed Solomon Status or LDPC Status.

4.5.5.1.1 Control

Parameters in this window are described in **DPU**.

FRAME SYNC.

Click on the check box to enable or disable the frame synchronizer:

Enable

Frame synchronizer ON. Minor frames reconstruction according to the programmed strategy.

Disable

Frame synchronizer OFF. Transmission of raw data frames to the Telemetry Clients.

NRZM DPU	Click on the check box to enable or disable the function: Enable The received data in the DPU are decoded from NRZ-M to NRZ-L before being sent to the frame synchronizer Disable No PCM decoding is performed
AUTO AMBIG.	Click on the check box to enable or disable the ambiguity resolution operation by the frame synchronizer: Enable Automatic Ambiguity Resolution ON. Disable Automatic Ambiguity Resolution OFF.
LSB FIRST	Enable The frame synchronizer works with LSB first byte format Disable The frame synchronizer works with MSB first byte format
SYNC. WORD	Synchronization Word. Enter any 64-bit hexadecimal value (left field = 32 MS bits, right field = 32 LS bits). MSB-aligned value. Example: If the Synchronization Word is 1ACFFC1D _H , enter 1ACFFC1D00000000_H .
SW MASK	Synchronization Word mask. Enter any 128-bit hexadecimal value. MSB-aligned value. « 1 » bits in the mask indicate which bits in the Synchronization Word must be ignored by the frame synchronizer. Enter 0 for no mask. Example: for a 32-bit Synchronization word, enter 00000000XXXXXXXX_H (no masked bits).
SYNC. WORD SIZE	Synchronization Word size, in bits. Must be an integer number of bytes. Enter 8, 16, 24, 32, 40, 48, 56, 64, ... or 128 . Example: if the Synchronization Word is 1ACFFC1D _H , enter 32 . For an non integer number of bytes specification, use the upper integer number of bytes value and use the mask to meet specification.
FRAME/BLOCK SIZE	Telemetry frame length, in bytes. Enter any integer from 16 to 1048580. Includes the Synchronization Word.
SYNCHRONIZATION STRATEGY	Frame synchronizer strategy: SYN Number of errors tolerated in the Synchronization Word. Enter any integer from 0 to 15. CTL Check-to-Lock threshold. Number of consecutive occurrences of correct Synchronization Words before going from the CHECK phase to the LOCK phase. Enter any integer from 0 to 15. LTS Lock-to-Search threshold. Number of consecutive occurrences of incorrect Synchronization Words before going from the LOCK phase to the SEARCH phase. Enter any integer from 0 to 15. Slip Number of bit slip tolerated for the search of the synchronization word in LOCK phase. Enter 0, 1 or 2.

INPUT SELECTION ¹

Allows to connect the DPU with the DMU:

DMU 1	Connect the DPU with DMU 1
DMU 2	Connect the DPU with DMU 2
ECL IN	Input external data (Data + Clock) in DPU only with menu 30 and 31
Merged (I&Q)	Connect the DPU input with the merged I&Q output of the selected DMU
Split 2 (I)	Connect the DPU input with the I of the selected DMU
Split 2 (Q)	Connect the DPU input with the Q of the selected DMU
Split 2 (I 1/2)	Connect the DPU input with the I of the selected DMU and take one data on two
Split 4 (Q 1/2)	Connect the DPU input with the Q of the selected DMU and take one data on two
Split 8 (I 1/4)	Connect the DPU input with the I of the selected DMU and take one data on four
Split 8 (Q 1/4)	Connect the DPU input with the Q of the selected DMU and take one data on four
In1+In2 (rising edge)	Both ECL data inputs are merged and sampled on the rising edge of the clock
In1 (rising edge)	Select ECL data input 1 and sample on the rising edge of the clock
In2 (rising edge)	Select ECL data input 2 and sample on the rising edge of the clock
In1+In2 (falling edge)	Both ECL data inputs are merged and sampled on the falling edge of the clock
In1 (falling edge)	Select ECL data input 1 and sample on the falling edge of the clock
In2 (falling edge)	Select ECL data input 2 and sample on the falling edge of the clock

FRAME HEADER RS

Header RS(6,10) **ENABLE** or **DISABLE**

Reed Solomon decoding of the header of the transfer frames in AOS CCSDS

DESCRAMBLING

OFF

CCSDS CCSDS derandomization

Custom Customized derandomization applied before decoding

Custom after decoding Customized derandomization applied after decoding

DESCR. POLY. (HEX)

Derandomizer polynomial value, in hexadecimal (16 bits max)

INIT. (HEX)

Preset value for the derandomizer, in hexadecimal

DECODING

By menu	Automatic selection by menu
RS 223 (CCSDS)	CCSDS RS (223,255) decoding standard basis
DVB Transport Layer	RS decoding for DVB Transfer Layer
LDPC 7/8	LDPC 7/8 decoding
Worldview RS	Specific RS decoding for Worldview satellite
RS 239 (CCSDS)	CCSDS RS (239,255) decoding
Turbo decoder	Turbo decoder (future release)
LDPC 1/2	LDPC 1/2 decoding
LDPC 2/3	LDPC 2/3 decoding
RS 223 (Dual)	CCSDS RS (223,255) decoding dual basis
RS 239 (Dual)	CCSDS RS (239,255) decoding dual basis
LDPC 7/8 Transport Layer	LDPC 7/8 decoding with Transport Layer
LDPC 1/2 1K Transport Layer	LDPC 1/2 (1K) decoding with Transport Layer
LDPC 1/2 4K Transport Layer	LDPC 1/2 (4K) decoding with Transport Layer
LDPC 1/2 16K Transport Layer	LDPC 1/2 (16K) decoding with Transport Layer
LDPC 2/3 1K Transport Layer	LDPC 2/3 (1K) decoding with Transport Layer
LDPC 2/3 4K Transport Layer	LDPC 2/3 (4K) decoding with Transport Layer

FRAME CHECKING

CRC	16 Bit CRC enabled
OFF	16 Bit CRC disabled

CRC POLY. (HEX) CRC polynomial value, in hexadecimal. Enter **8005** for a CRC16 ($X^{16} + X^{15} + X^2 + 1$) or **1021** for a CCITT compliant CRC ($X^{16} + X^{12} + X^5 + 1$)

CRC PRESET (HEX) Preset value for CRC calculation, in hexadecimal. Enter **0000** for a CRC16 or **FFFF** for a CCITT compliant CRC

Note 1: An input selection not available is indicated by the mention “N/A”.

Note 2: Frame synchronizer input selection vs demodulation and Viterbi mode:

QPSK/OQPSK: No Viterbi or dual Viterbi mode	Merged I & Q: select Merged (I&Q) Separate I & Q: select Split 2 (I) or (Q)
QPSK/OQPSK: Single Viterbi mode	Select Split 2 (I)
BPSK with or without Viterbi	Select Split 2 (I)
UQPSK with or without Viterbi	Select Split 2 (I) or (Q)

4.5.5.1.2 Monitoring

Status and alarms in this window are described in **DPU**.

4.5.5.1.2.1 Upper Banner

See section 4.5.4.1.2.1 .

4.5.5.1.2.2 Frame Synchronizer Status

FRAME COUNTER	Telemetry frame counter. Reset to 0 when the frame synchronizer unlocks or by clicking on  in the upper tool bar (the DPU window must be active).
BIT SLIP	Bit slippage measured by the frame synchronizer.
DROP OUT COUNTER	Number of loss of lock on the frame synchronizer. Reset to 0 by clicking on  in the upper tool bar (the DPU window must be active).
FRAME CHECK	CRC or Check-Sum test result: Alarm or OK . Forced to OK if parameter Frame Checking is set to Off .

4.5.5.1.2.3 Reed Solomon decoders Status

CORRECTED ERRORS	Number of symbols corrected by the Reed-Solomon decoder (average value computed over a one-second period of time). Displays Exceeded if the maximum correction capability is reached (number of corrected symbols greater than [16 x Interleave Factor]).
GOOD FRAMES	Number of good transfer frames received from the Reed Solomon decoder. Reset to 0 when the Frame Synchronizer or the Reed Solomon decoder is set to OFF or by clicking on  in the upper tool bar (the DPU window must be active).
BAD FRAMES	Number of bad transfer frames received from the Reed Solomon decoder. Reset to 0 when the Frame Synchronizer or the Reed Solomon decoder is set to OFF or by clicking on  in the upper tool bar (the DPU window must be active).
BIT ERROR RATE	Ratio between the number of corrected bits and the total number of bits processed by the Reed Solomon decoder. Reset to 0 when the Frame Synchronizer or the Reed Solomon decoder is set to OFF or by clicking on  in the upper tool bar (the DPU window must be active). Important: the BER is calculated before Reed Solomon decoding.

4.5.5.1.2.4 LDPC Status

CORRECTED ERRORS	Number of symbols corrected by the LDPC decoder (average value computed over a one-second period of time).
GOOD FRAMES	Number of good transfer frames received from the LDPC decoder. Reset to 0 when the Frame Synchronizer or the LDPC decoder is set to OFF or by clicking on  in the upper tool bar (the DPU window must be active).
BAD FRAMES	Number of bad transfer frames received from the LDPC decoder. Reset to 0 when the Frame Synchronizer or the LDPC decoder is set to OFF or by clicking on  in the upper tool bar (the DPU window must be active).
BIT ERROR RATE ON INPUT DATA	Estimation of the BER at the input of the LDPC decoding. Reset to 0 when the Frame Synchronizer or the LPC decoder is set to OFF or by clicking on  in the upper tool bar (the DPU window must be active).
BIT ERROR RATE ON OUTPUT DATA	Estimation of the BER at the output of the LDPC decoding. Reset to 0 when the Frame Synchronizer or the LDPC decoder is set to OFF or by clicking on  in the upper tool bar (the DPU window must be active).

4.5.5.1.2.5 CRC Status

GOOD FRAMES	Number of good transfer frames received from the CRC decoder. Reset to 0 when the Frame Synchronizer or the Reed Solomon decoder is set to OFF or by clicking on  in the upper tool bar (the DPU window must be active).
BAD FRAMES	Number of bad transfer frames received from the CRC decoder. Reset to 0 when the Frame Synchronizer or the Reed Solomon decoder is set to OFF or by clicking on  in the upper tool bar (the DPU window must be active).

4.5.5.2 "Transport Layer" Windows

This window is only available with RS DVB and LDPC decoding. It gives the status of the Transport layer.

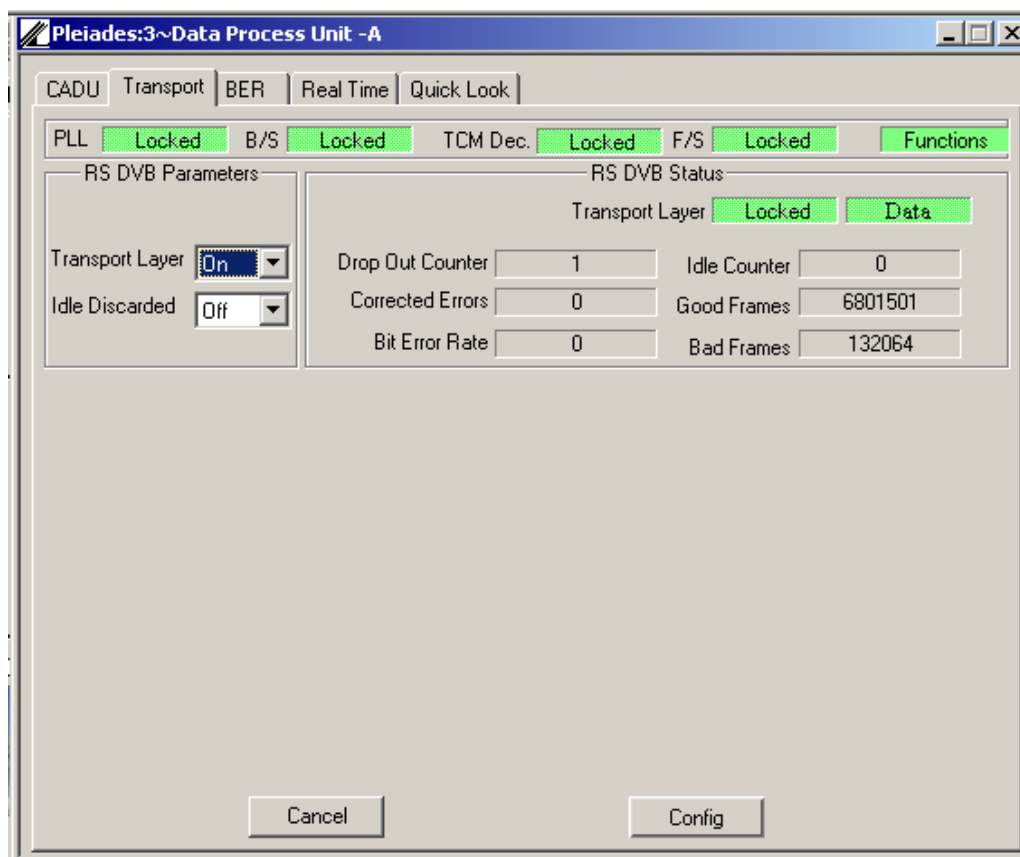


Figure 133: DPU: Transport Layer Window (R-S DVB Mode)

4.5.5.2.1 Control (Transport Layer Window)

Unless otherwise specified, parameters in this window are described in **DPU**.

TRANSPORT LAYER	Set the Transport Layer ON or OFF
IDLE DISCARDED	Discard (ON) or transmit the idle frames to the CADU frame synchronizer (OFF)

4.5.5.2.2 Monitoring (Transport Layer Window)

Unless otherwise specified, status & alarms in this window are described in **DPU**.

DROP OUT COUNTER	Number of frame drop-out events at transport layer level
CORRECTED ERRORS	Number of bits corrected by the decoder at transport layer level
BIT ERROR RATE	Bit error rate at transport layer level
IDLE COUNTER	Number of received idle frames at transport layer level
GOOD FRAMES	Number of good frames
BAD FRAMES	Number of bad frames (could not be corrected)
TRANSPORT LAYER	Frame Synchronizer status. From left to right: <i>Search, Check, Locked</i> or <i>Flywheel</i> <i>Data</i> or <i>Idle</i> frames

Note: When the frame synchronizer of the CADU layer is lock, the status of the counters in the Transport layer can be updated or not. That is controlled by a Registry key (*PermanentTransportStatus*).

4.5.5.3 “BER” Windows

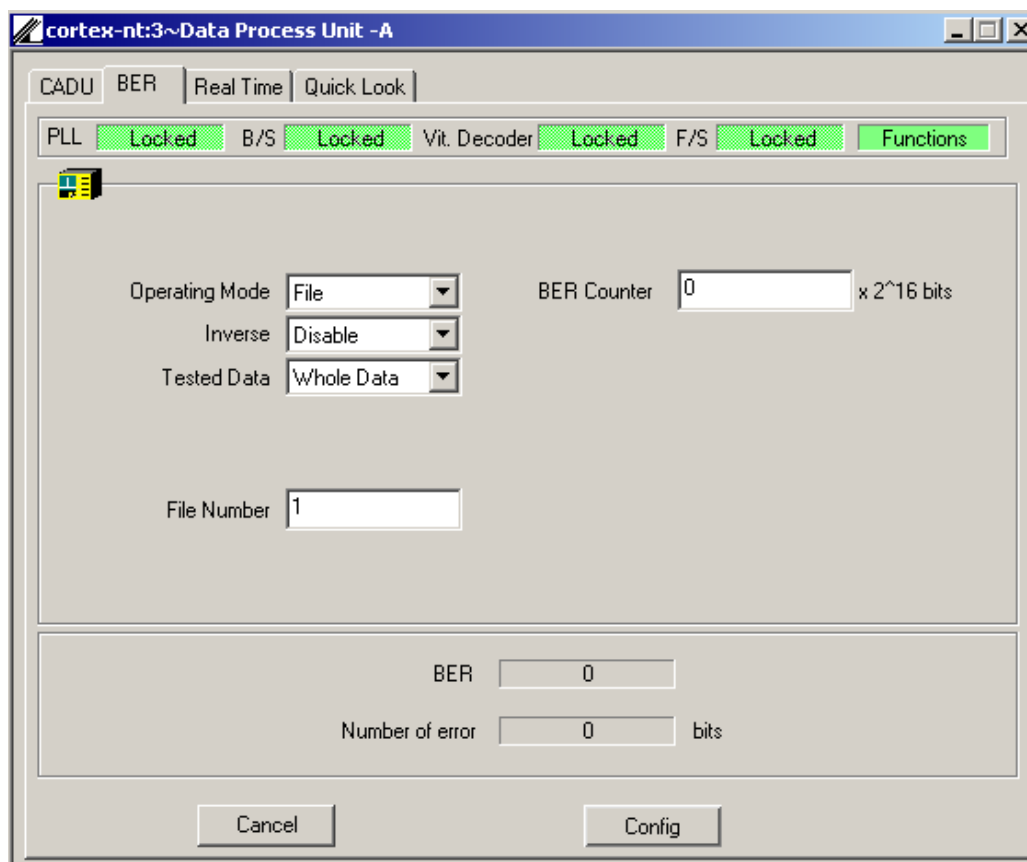


Figure 134: DPU: BER Window

4.5.5.3.1 Control

Unless otherwise specified, parameters in this window are described in **DPU**.

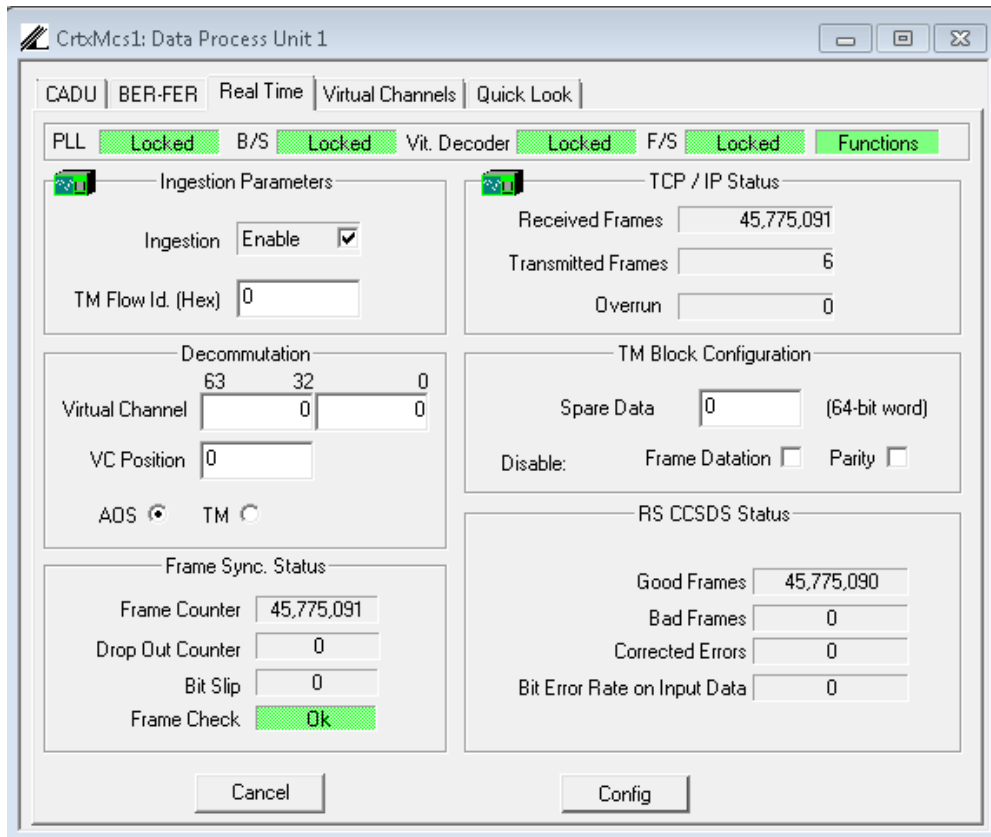
OPERATING MODE	BER measurement mode (simulated data for BER analysis): Off BER measurement stopped File Simulated data from a file on the PseudoDxx Pseudo random data, using from PN generator xx (where xx = 7, 10, 11, 15, 23 or 31)
INVERSE	Inverse (Enable) or not (Disable) the data used for BER computation
TESTED DATA	User Data Compute the BER only on the User data of the frames (excluded added check symbols) Whole Data Compute the BER on all the received data of the frames Custom Pattern Compute the BER on a user-defined portion of the frame. The custom BER pattern is defined inside an ASCII file stored on the hard drive.
FILE NUMBER	File number for BER in file mode: enter any value from 1 to N.
BER PATTERN NUMBER	File number for user-defined BER pattern in Custom Pattern mode: enter any value from 1 to N
BER COUNTER	Total number of bits to process for BER measurement in Pseudo mode Resolution step: 65536 bits. In File Mode, Infinite accumulation.

4.5.5.3.2 Monitoring

Unless otherwise specified, status & alarms in this window are described in **DPU**.

BER	Computed Bit Error Rate
NUMBER OF ERROR	Cumulated number of errors

4.5.5.4 “Real Time Frame” Window



CrtxMcs1: Data Process Unit 1

Tabs: CADU | BER-FER | **Real Time** | Virtual Channels | Quick Look

PLL **Locked** B/S **Locked** Vit. Decoder **Locked** F/S **Locked** Functions

Ingestion Parameters

Ingestion ☒ Enable

TM Flow Id. (Hex)

Decommutation

Virtual Channel

VC Position

AOS ☒ TM ☐

Frame Sync. Status

Frame Counter

Drop Out Counter

Bit Slip

Frame Check **Ok**

TCP / IP Status

Received Frames

Transmitted Frames

Overrun

TM Block Configuration

Spare Data (64-bit word)

Disable: Frame Datation ☐ Parity ☐

RS CCSDS Status

Good Frames

Bad Frames

Corrected Errors

Bit Error Rate on Input Data

Buttons: Cancel Config

Figure 135: Data Processing Unit: Real Time Frame Window

4.5.5.4.1 Control

Parameters in this window are described in **DPU**.

❑ Ingestion parameters:

FRAME SYNC. Enable or disable the frame synchronizer

TM FLOW ID. Identification of the telemetry flow. This value is reported in all telemetry messages over the Telemetry port (TM). Enter any hexadecimal value from **0** to **FFFFFFF**.

❑ **Decommutation:**

VIRTUAL CHANNEL	Virtual Channels 64-bit mask selection (0 = select, 1 = discard)
VC POSITION	Offset, in bits, between the last bit of the synchronization word and the first bit of the Virtual Channel ID field. Example: enter 0 if the VC ID field starts right after the synchronization word.
SPARE DATA	Select number of spare fields in the TM Block. Enter 0 to 8 . See Figure 42: Data Transmission over TCP-IP. Message Formatting

❑ **TM block configuration:**

SPARE DATA	Enables to add spare data (64 bits word) at the end of the TM data blocks (recorded and transmitted over TCP/IP). Possible values: 0 to 8 .
DISABLE FRAME DATATION	Enables to remove the insertion of the 8 time tag bytes at the end of the TM data blocks (recorded and transmitted over TCP/IP)
DISABLE PARITY	Enables to remove the parity in TM data

4.5.5.4.2 Monitoring

Status & alarms in this window are described in **DPU**.

4.5.5.4.2.1 TCP-IP Status

RECEIVED FRAMES	Number of demodulated TM Blocks. Reset to 0 when the CORTEX HDR is restarted.
TRANSMITTED FRAMES	Total number of TM Blocks (for all TM Clients) over TCP-IP. Reset to 0 when the CORTEX HDR is restarted or when switching from Acquisition mode to Configuration mode
OVERRUN	Amount of data lost during overrun conditions. Expressed in number of buffers of TM Packets (one buffer = 8 TM Packets).

4.5.5.4.2.2 Frame Sync. Status

See section 4.5.5.1.2.2 .

4.5.5.4.2.3 Upper Banner

See section 4.5.4.1.2.1 .

4.5.5.5 "Quick Look" Window

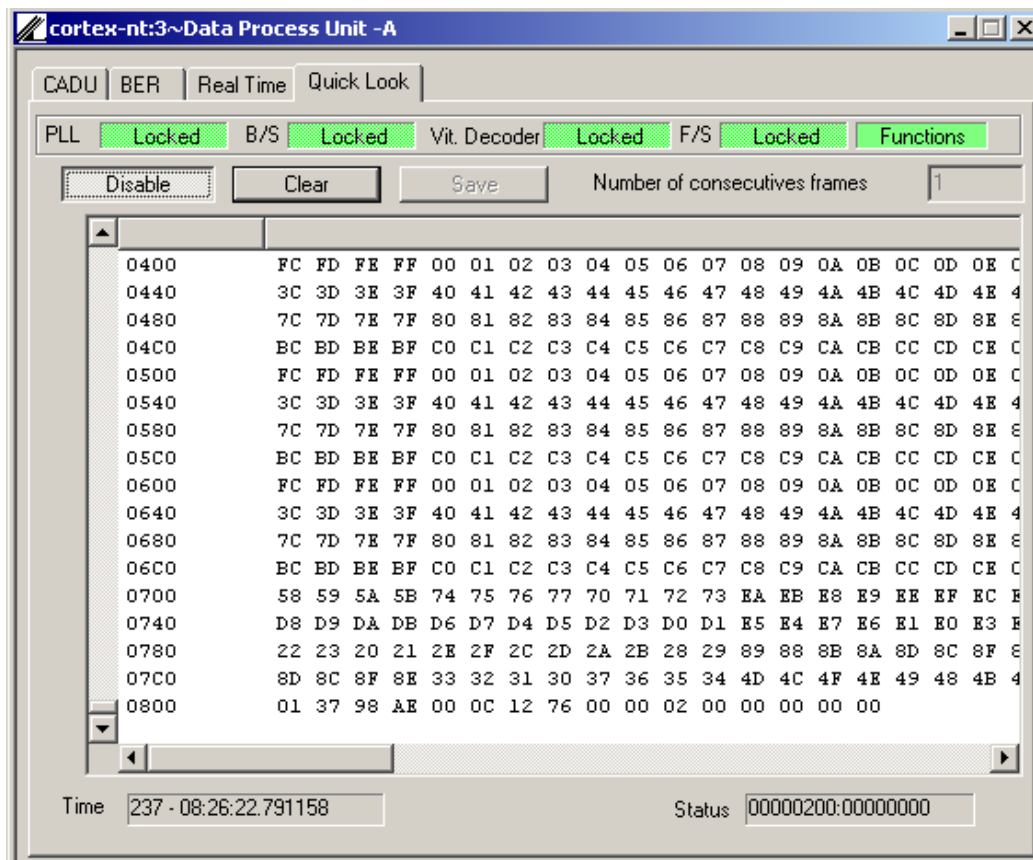


Figure 136: Data Processing Unit: Quick Look Window

4.5.5.5.1 Control

Parameters in this window are described in **DPU**.

ENABLE/DISABLE

Enable (the MCS opens a connection to the telemetry port) or Disable the Quick Look display.

Important: the Quick Look can only be enabled if the Acquisition mode is selected (click on icon ). Switching back to the Configuration mode is impossible as long as the Quick Look is enabled.

CLEAR

Clear the Quick Look window

SAVE

Save the data displayed on the Quick Look to a text file on the disk (click on **SAVE** only after the Quick Look has been disabled).

NUMBER OF CONSECUTIVE FRAMES

Frame display rate. Enter any value from **1** to **10**:

1: display one frame every second

2: display two consecutive frames every two seconds

etc...

4.5.5.5.2 Telemetry Blocks Quick Look

Telemetry Blocks are displayed back to back on the Quick Look window.

Offset **0000** marks the beginning of a Telemetry Block.

Telemetry Blocks are structured as follows (see section 3.4.11 : for more details):

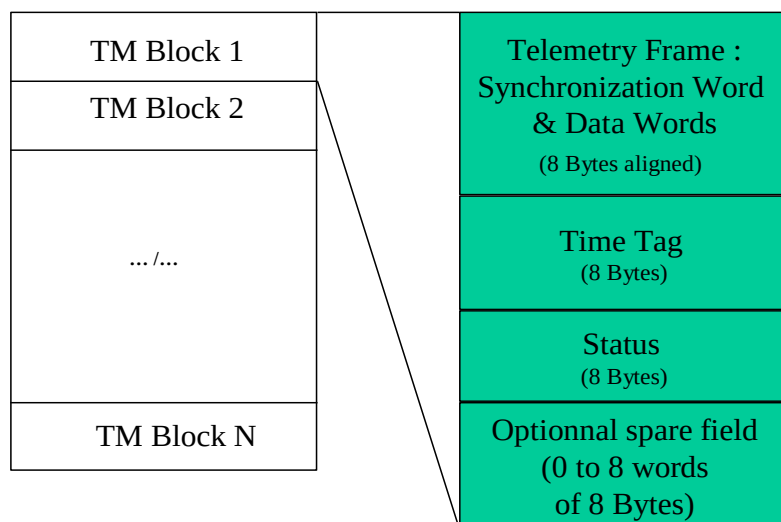


Figure 137: Telemetry Block Description

Telemetry frame:

Synchronization Word + telemetry frame (if frame synchronizer ON) or raw data (if frame synchronizer OFF). At the end of the frame, spare bytes for 8 bytes alignment (if required). These added spare bytes can have any values.

TM Block time tag (8 bytes):

Bits D0-31 (offset N): number of seconds elapsed since January 1st, 00:00:00 of current year
 Bits D32-63 (offset N+1): number of microseconds in the current second
 Both 32-bit fields are integers coded in hexadecimal form.

The time tag can be disabled by set-up (refer to “Real Time Frame” window: disable frame datation).

Telemetry Block status (8 bytes): the Telemetry Block status is captured and displayed in this field only if the frame synchronizer is ON.

Optional spare field (from 0 to 8 words of 8 bytes): the number of spare data words (64 bit words) is selected in the “Real Time Frame” Window. These data words are filled with “00000000” values.

4.5.5.5.3 Monitoring

Status & alarms in this window are described in **DPU**.

TIME	TM Block time-tag, with the display format: Day - Hour:Minute:Second.Microsecond
STATUS	TM Block status, expressed in hexadecimal. The left-hand digits of the status corresponds to offset N+4 of the Telemetry Block Description (see STI 100013_HDR, Table 21). The right-hand digits of the status corresponds to offset N+5 of the Telemetry Block Description.

4.5.5.5.3.1 Upper Banner

See section 4.5.4.1.2.1 .

4.5.6 Polarization Combining Windows

4.5.6.1 "Global" Window

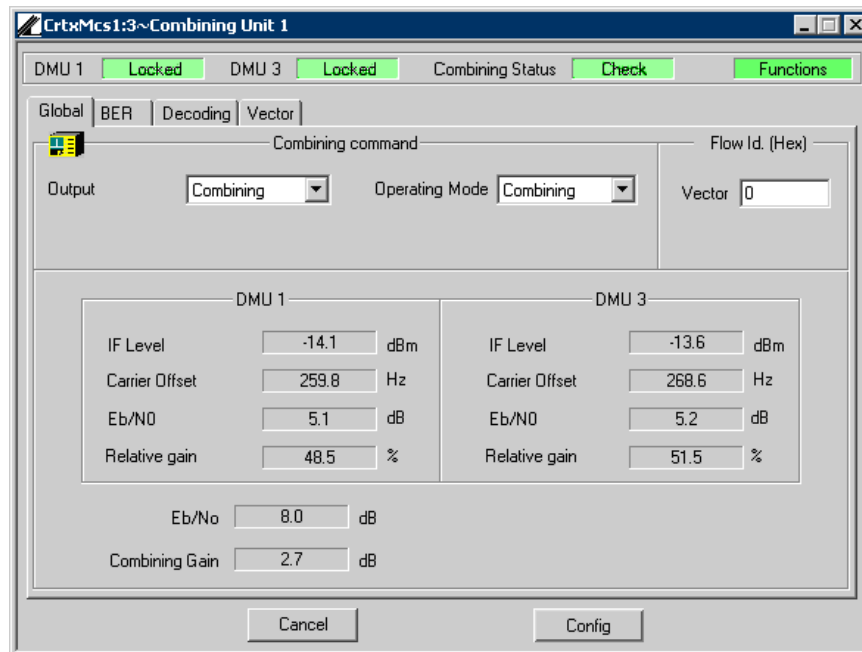


Figure 138: Polarization Combining: Global Window

4.5.6.1.1 Control

Parameters in this window are described in **DCU**.

OUTPUT

Affects the output of the DCU. Possible values are: **DMU1**, **DMU3** or **Combining**.

OPERATING MODE

Best channel only the best channel is fed to the output

Combining both channels are combined using optimal weights

4.5.6.1.2 *Monitoring*

Status of the channels:

DMU 1/3 Locking state of each channel. Displays “*Locked*” (green) when the PLL and the Bit Synchronization loops are locked, “*Unlocked*” (red) otherwise.

IF LEVEL IF carrier level. Expressed in dBm.

CARRIER OFFSET Offset from the specified IF carrier frequency (real frequency – expected frequency). Expressed in Hz.

Eb/N0 Energy per bit in a 1-Hz noise bandwidth. Expressed in dB.

Status of the combining:

COMBINING STATUS Locked (green), Check (green), Acquisition (amber), 1-channel (green), Unlocked (red), N/A (gray).

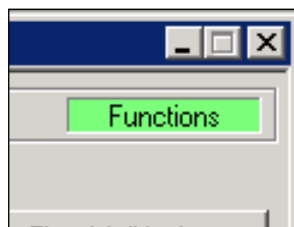
RELATIVE GAINS Combining weights, expressed in %. In *Best Channel* mode one weights is equal to 0, % the other one equals 100 %. The same goes when the unit is in *Single channel* state.

EB/N0 Energy per bit in a 1-Hz noise bandwidth of the combining unit output. Expressed in dB.

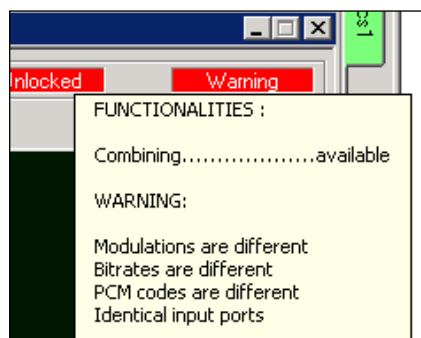
COMBINING GAINS Gain in energy per bit in a 1-Hz noise bandwidth of the combining unit output. Expressed in dB.

4.5.6.1.3 Other status and alarms

The upper right part of the window gives general information concerning the DCU. When everything is configured right, it displays.



When an incorrect configuration is detected (different bitrates of both demodulators, same IF input...), the box displays ***Warning*** (red). A tool tip is displayed when the pointer stops on the box. Below stands an example of an incorrect configuration:



4.5.6.2 “BER” Window

This window enables the user to monitor the BER at the output of the combining system. Please refer to the Demodulator Unit section for further details on this feature.

4.5.6.3 “Decoding” Window

This window enables the user to control and monitor the decoding features (Viterbi, TCM) at the output of the combining system. Please refer to the Demodulator Unit section for further details on this feature.

4.5.6.4 “Vector” Window

This window enables the user to monitor the constellation at the output of the combining unit. It is the same as the standard demodulator Vector windows, with the exception of three status (IF level, Amplitude Error, Phase Error) that do not apply to the combining feature. Please refer to the Demodulator Unit section for further details on this window.

4.5.7 Data Recording Unit Windows

4.5.7.1 "Global Disk and Memory Management" Window

In the upper toolbar, click on  to open the Disk Partitioning & Memory Allocation window:

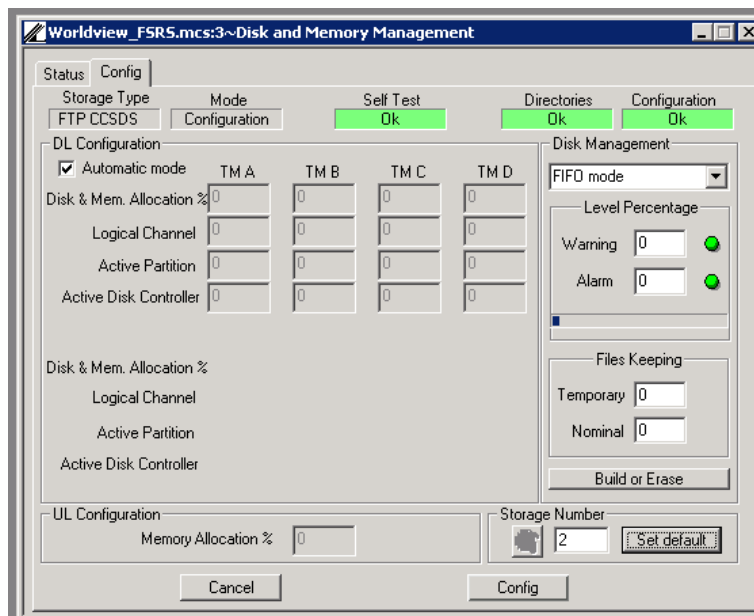


Figure 139: Global Disk & Memory Management Window (Configuration)

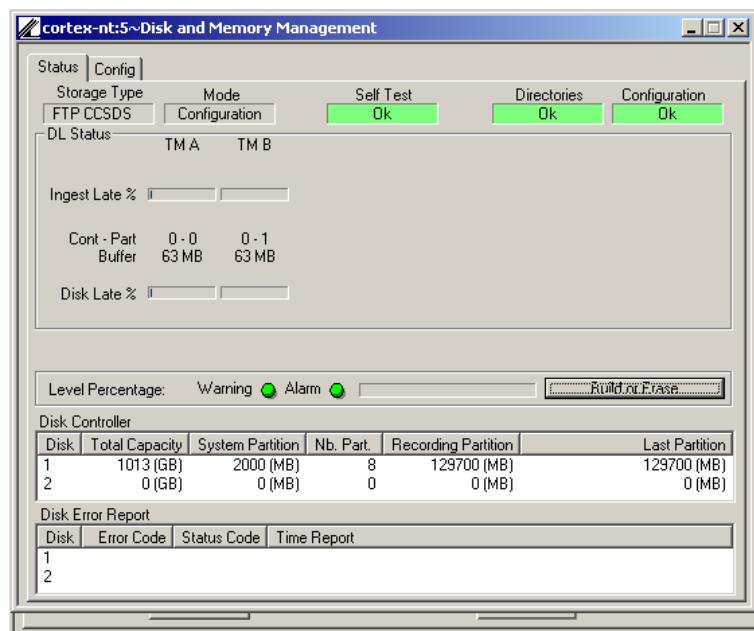


Figure 140: Global Disk & Memory Management Window (Status)

4.5.7.1.1 Control

Parameters in this window are described in **DRG**.

☐ Channels configuration:

DISK AND MEMORY ALLOCATION % FOR EACH MOUNTED DPU	Disk and PC RAM buffer partitioning. For TCP-IP and FTP recording modes, allows to allocate a percentage of the available memory for each mounted TM: 0% to 100 %
LOGICAL CHANNEL FOR EACH MOUNTED DPU	For FTP mode only, allows for the current pass to select a logical channel (first level FTP directory) 0 to 255 .
ACTIVE PARTITION FOR EACH MOUNTED DPU	For FTP mode only, allows for the current pass to select the active partition 0 to 64 .
ACTIVE DISK CONTROLLER FOR EACH MOUNTED DPU	For FTP mode only, allows for the current pass to select the active disk controller 0 or 1 . In case of one available controller, the value must be set to 0.
DISK LATE %	Status of the delay during the recording of the data on the disk.
DISK OVERRUN	Overflow during the recording of the received data (disk).
MEMORY LATE %	Status of the delay during the writing of the received data in the buffer memory.
MEMORY OVERRUN	Overflow during the writing of the received data (buffer memory).

☐ Disk Management:

MODE	In TCP-IP mode only. Select FIFO Mode (permanent recording. The storage area is structured as a FIFO) or Manual Erase (record data until the storage area is full, then stop and wait for manual erase of the recorded session). Note: it is recommended to use the FIFO mode (with auto-erase) only for single recording session (auto-erase of the least recent TM Files).	
LEVEL PERCENTAGE	Warning	In FTP mode only. If different from 0, allows to define a storage level threshold above which a warning is reported. Value expressed in percentage: 1% to 80%
	Alarm	In FTP mode only. If different from 0, allows to define a storage level threshold above which an alarm is reported. Value expressed in percentage: 1% to 80% . In case of alarm, the oldest files are deleted until to obtain a level equivalent to the warning storage level

FILES KEEPING

Temporary

In FTP mode only. Duration of the temporary storage for the directories keeping the files declared as NOK by the filtering operations.

The unity is in days for values between **1** and **128**

The unity is in minutes for values between **-1** and **-128**

Nominal

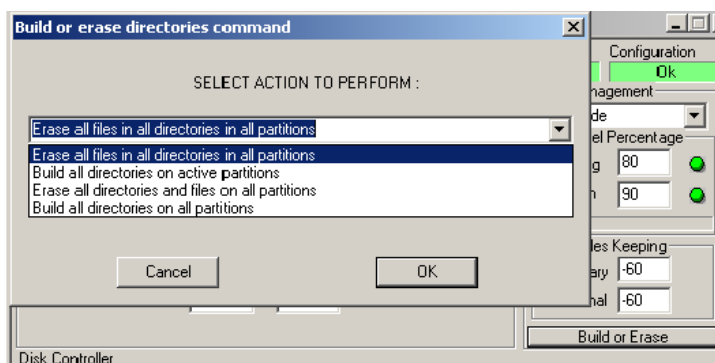
In FTP mode only. Duration of the nominal storage for the directories keeping the files not declared as NOK by the filtering operation.

The unity is in **days** for values between **+1** and **+127**

The unity is in **minutes** for values between **-1** and **-128**

Build or Erase

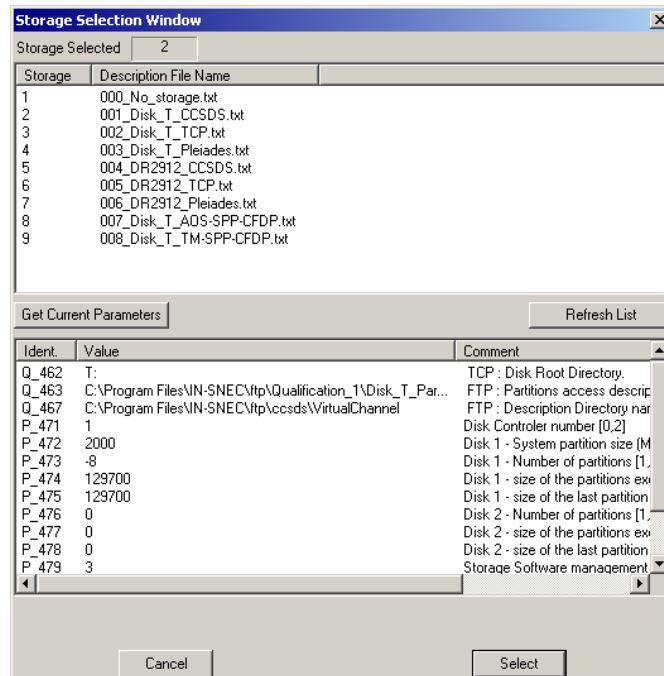
Allows to build or erase the files and/or the directories in FTP mode. This button appears in the window only when the HDR is set in “**Configuration**” mode. When pressed a pop-up menu appears with all the possible operations.



❑ Storage Selection:



Click on  to open the Storage Selection window:



To select a storage type click on the corresponding line in the upper window and then click on “Select”.

The proposed standard internal storage is the storage number 2 (002_Disk_T_CCSDS).

Storage types with the label “Disk_T” are used for internal hard disk storage. Storage type with the label “DR2912” are used for external storage on a RAID 5 DR2912 or ER2912 recorder.

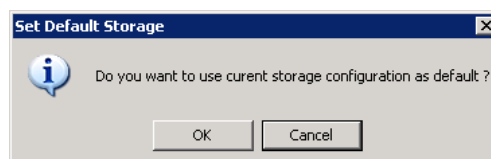
Storage types with the label “Pleiades” are used for specific projects.

Storage number 8 and 9 are reserved for CCSDS Packet data and CCSDS CFDP recording in AOS mode and Telemetry mode.

❑ Set default:



Click on  to select the current storage number as the default storage number. The following window will be displayed to confirm:



4.5.7.1.2 Monitoring

Status & alarms in this window are described in **DRG**.

STORAGE TYPE	Indicates which mode of storage is active example: FTP CCSDS (<i>standard</i>)
MODE	CORTEX HDR mode: Configuration or Acquisition. Important: in the Acquisition mode, the Data Processing Unit(s) cannot be re-configured. The maximum number of TM Packets per TM file (Data Recording Unit) cannot be changed. Data Recording Global (DRG) parameters cannot be changed. Click on  for changing the mode. Switching to Configuration mode is impossible if a TM Client is connected to the telemetry port (3075) or if telemetry storage is in progress. Switching to Acquisition mode is impossible if the requested configuration is incorrect.
SELF TEST	In configuration mode, a periodical auto-test is performed on the defined partitions. This test consists in writing a file in the directories and trying to read it. The result is OK or Non OK as soon as one partition is in error.
DIRECTORIES	Status of the FTP directories.
CONFIGURATION	General status of the actual configuration.
OTHER FIELDS	Disk partitioning information and status.

4.5.7.2 "Recording Global" Window

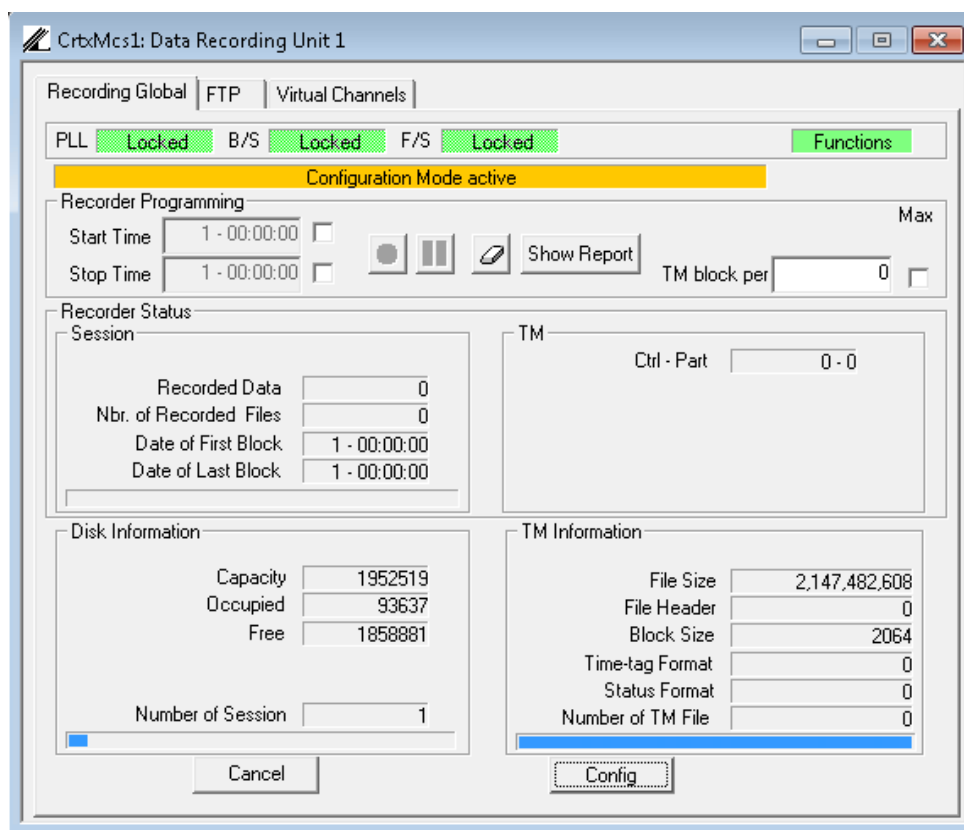


Figure 141: DRU: Recording Global Window

4.5.7.2.1 Control

Unless otherwise specified, parameters in this window are described in **DRU**.

START TIME Storage start time: Day-Hour:Minute:Second

STOP TIME Storage stop time: Day-Hour:Minute:Second

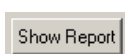
TM BLOCK PER FILE Maximum number of TM blocks per recorded File (defines the recorded files maximum size). If set to 0, the recorded files maximum size will be 2 Gbytes. If checkbox "Max" is checked, file maximum size is almost infinite, and one TM file is recorded for the recording session.



Start writing data to the disk.



Stop writing data.

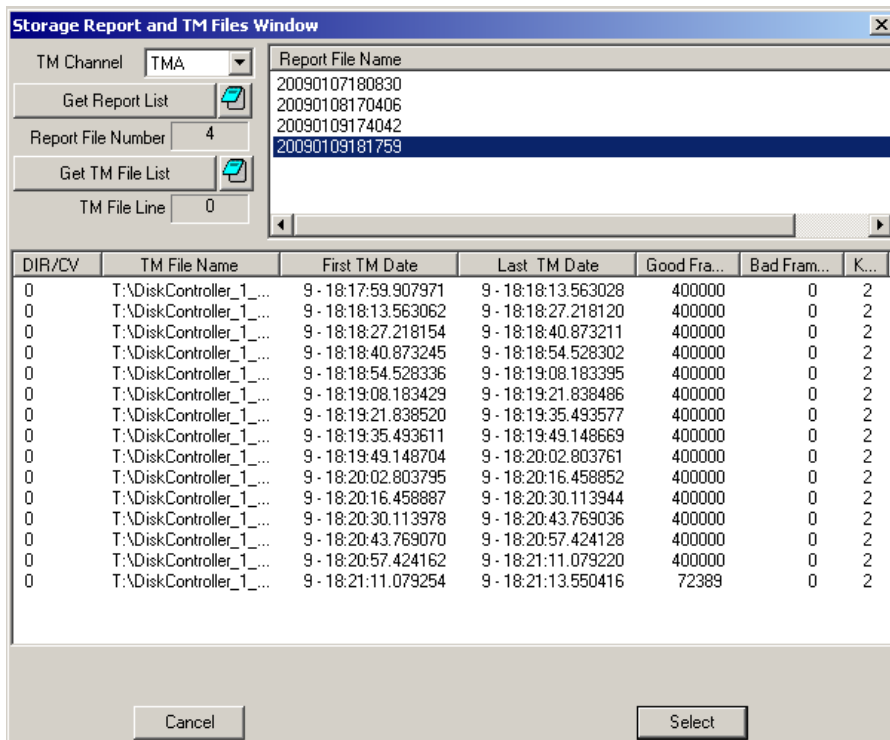


Show recorded reports (see below).



To access to the erase functions by DRU (Refer to the paragraph "Global Disk and Memory Management window).

When clicking on  the following window appears on the screen:



Storage Report and TM Files Window

TM Channel: TMA

Get Report List 

Report File Number: 4

Get TM File List 

TM File Line: 0

Report File Name:
20090107180830
20090108170406
20090109174042
20090109181759

DIR/CV	TM File Name	First TM Date	Last TM Date	Good Fra...	Bad Fram...	K...
0	T:\DiskController_1_...	9 - 18:17:59.907971	9 - 18:18:13.563028	400000	0	2
0	T:\DiskController_1_...	9 - 18:18:13.563062	9 - 18:18:27.218120	400000	0	2
0	T:\DiskController_1_...	9 - 18:18:27.218154	9 - 18:18:40.873211	400000	0	2
0	T:\DiskController_1_...	9 - 18:18:40.873245	9 - 18:18:54.528302	400000	0	2
0	T:\DiskController_1_...	9 - 18:18:54.528336	9 - 18:19:08.183395	400000	0	2
0	T:\DiskController_1_...	9 - 18:19:08.183429	9 - 18:19:21.838486	400000	0	2
0	T:\DiskController_1_...	9 - 18:19:21.838520	9 - 18:19:35.493577	400000	0	2
0	T:\DiskController_1_...	9 - 18:19:35.493611	9 - 18:19:49.148669	400000	0	2
0	T:\DiskController_1_...	9 - 18:19:49.148704	9 - 18:20:02.803761	400000	0	2
0	T:\DiskController_1_...	9 - 18:20:02.803795	9 - 18:20:16.458852	400000	0	2
0	T:\DiskController_1_...	9 - 18:20:16.458887	9 - 18:20:30.113944	400000	0	2
0	T:\DiskController_1_...	9 - 18:20:30.113978	9 - 18:20:43.769036	400000	0	2
0	T:\DiskController_1_...	9 - 18:20:43.769070	9 - 18:20:57.424128	400000	0	2
0	T:\DiskController_1_...	9 - 18:20:57.424162	9 - 18:21:11.079220	400000	0	2
0	T:\DiskController_1_...	9 - 18:21:11.079254	9 - 18:21:13.550416	72389	0	2

Cancel Select

Select a report by clicking on the corresponding line in the upper window to display the recorded files listing corresponding to the report.

Once the storage is started, the following message appears on the screen:



cortex-nt:3~Data Recording Unit -A

Recording Global FTP Virtual Channels

PLL Locked B/S Locked F/S Locked Functions

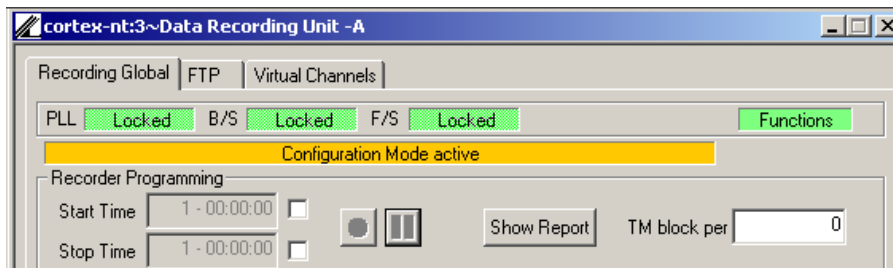
Recording in progress. Awaiting for Stop Command

Recorder Programming

Start Time 1 - 00:00:00   Show Report TM block per 0

Stop Time 1 - 00:00:00  

If the HDR is set in "Configuration" mode, the following message is display:



cortex-nt:3~Data Recording Unit -A

Recording Global FTP Virtual Channels

PLL Locked B/S Locked F/S Locked Functions

Configuration Mode active

Recorder Programming

Start Time 1 - 00:00:00   Show Report TM block per 0

Stop Time 1 - 00:00:00  

The recording cannot be started as long as the HDR is not set in "Acquisition" mode; use the button to switch from "Configuration" mode to the "Acquisition" mode.



4.5.7.2.2 Monitoring

Status & alarms in this window are described in **DRU**.

4.5.7.2.2.1 Upper Banner

See section 4.5.4.1.2.1 .

4.5.7.2.2.2 Recorder Status

☐ Session Status:

RECORDED DATA	Amount of recorded data for the current recording session. Expressed in Mbytes
NBR. OF RECORDED FILES	Recorded TM file number for the current recording session. 0 to N
DATE OF FIRST BLOCK	Time-tag of the first recorded TM Block for the current recording session
DATE OF LAST BLOCK	Time-tag of the last recorded TM Block for the current recording session

☐ TM Status:

CTRL/PART	Active hard disk controller (0 or 1) / Active hard disk partition (1 to 64)
------------------	---

☐ Disk Information:

CAPACITY	Total data recording capacity for both hard disk controllers (in Mbytes)
OCCUPIED	Occupied capacity for both hard disk controllers (in Mbytes)
FREE	Free capacity for both hard disk controllers (in Mbytes)
NUMBER OF SESSION	Number of recording sessions on the disk

☐ TM Information:

FILE SIZE	Size of the TM File (in bytes). This value corresponds to the TM block size multiplied by the value set in "TM block per".
FILE HEADER	TM File header
BLOCK SIZE	TM Block size (in bytes). Up to 1,048, 656 bytes. This value corresponds to the defined frame (CADU) size plus a few bytes of time tagging and frame sync. Status.

TIME-TAG FORMAT

Time tag format.

3 is the standard code format: field N is the number of seconds elapsed since January 1st, 0:00:00:000 of current year. Field N+1 is the number of microseconds in the current second. Both fields are 32-bit integers coded in hexadecimal form. Other codes reserved.

STATUS FORMAT

TM Block status format

NUMBER OF TM FILES

Number of TM Files on the disk

4.5.7.3 "FTP" window

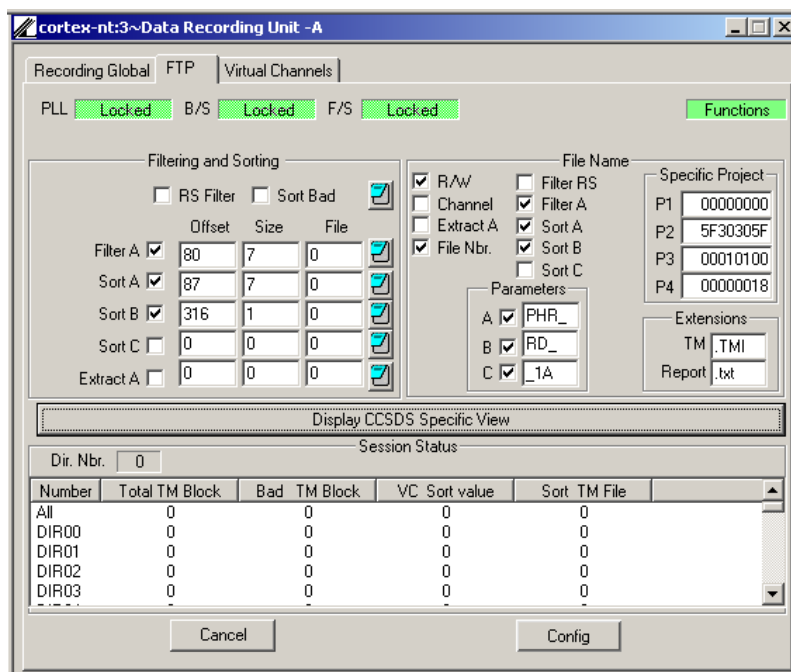


Figure 142: DRU: FTP Window (Generic)

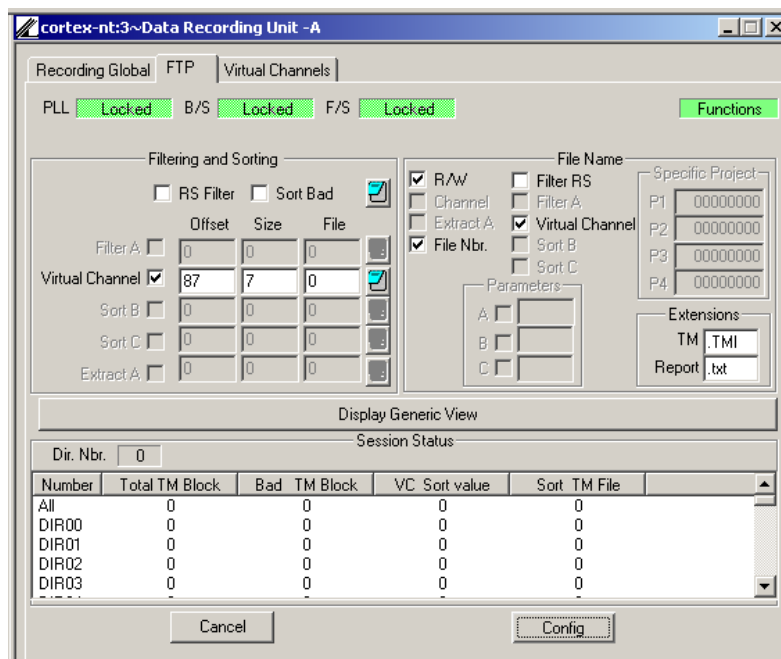


Figure 143: DRU: FTP Window (CCSDS standard)

4.5.7.3.1 Control

Parameters in this window are described in **DRU**.

☐ Filtering and sorting:

RS FILTER **Enable** or **disable** the filtering of Reed Solomon code blocks in errors.

SORT BAD **Enable** or **disable** the sorting of the RS code blocks considered as not corrected

FILTER A **Enable** or **disable** the Filter A operation

Offset: Offset in bits in the frame of the position of the word to extract to compare with the filter A value (0 is the first bit of the synchronization marker)

Size: Size in bits of the word to extract, from 0 to 8

File: Index of the Filter A configuration file, from 0 to 255

SORT A OR VIRTUAL CHANNEL **Enable** or **disable** the Sort A operation (Generic mode) or the Virtual Channels sorting (CCSDS mode)

Offset: Offset in bits in the frame of the position of the word to extract to be used for the sorting A operation (0 is the first bit of the synchronization marker)

Size: Size in bits of the word to extract, from 0 to 8

File: Index of the Sort A configuration file, from 0 to 255

SORT B	Enable or disable the Sort B operation Offset: Offset in bits in the frame of the position of the word to extract to be used for the sorting B operation (0 is the first bit of the synchronization marker) Size: Size in bits of the word to extract, from 0 to 8 File: Index of the Sort B configuration file, from 0 to 255
SORT C	Enable or disable the Sort C operation Offset: Offset in bits in the frame of the position of the word to extract to be used for the sorting C operation (0 is the first bit of the synchronization marker) Size: Size in bits of the word to extract, from 0 to 8 File: Index of the Sort C configuration file, from 0 to 255
EXTRACT A	Enable or disable the operation of Extract A, the operation is the same as for Sort A,B and C operations except that it impacts only the files names and not the directories. Offset: Offset in bits in the frame of the position of the word to extract to be used for the Extract A operation (0 is the first bit of the synchronization marker) Size: Size in bits of the word to extract, from 0 to 8 File: Index of the Extract A configuration file, from 0 to 255

Note: the data values used in the filtering, sorting and extract operations are defined in their corresponding configuration files selected by the index values.

☐ **Filename:**

The basic name of a recorded file is the date of its creation (default name). From this basic name a number of programmable markers can be added:

R/W	Enable or disable the insertion of the R/W marker in the recorded files names
CHANNEL	Enable or disable the insertion of the selected Logical Channel marker in the recorded files names (The selection of the logical channel for a recording session is done in the "Global Disk and Memory Management Window")
EXTRACT A	Enable or disable the insertion of the Extract A marker in the recorded files names
FILE NBR	Enable or disable the insertion of the File Number marker in the recorded files names (counter of the number of generated files during a recording session inside a dedicated directory)
FILTER RS	Enable or disable the insertion of the RS Filter marker in the recorded files names
FILTER A	Enable or disable the insertion of the Filter A marker in the recorded files names
SORT A OR VIRTUAL CHANNEL	Enable or disable the insertion of the Sort A marker (Generic mode) or the Virtual Channels marker (CCSDS mode) in the recorded files names

SORT B	Enable or disable the insertion of the Sort B marker in the recorded files names
SORT C	Enable or disable the insertion of the Sort C marker in the recorded files names
SPECIFIC PROJECT	P1, P2, P3, and P4 , hexadecimal values from 0 to FFFFFFFF . <i>Theses parameters are specific to the project.</i>
PARAMETERS	Specific user defined markers. For A, B and C, the corresponding fields allow to enable or disable the insertion of the marker and to enter the characters string of the marker (up to 12 characters for each field)
EXTENSIONS	Allows for the recorded Data files and the session Report files to define an extension suffix which will be inserted in the files names

Note: the characters strings markers of the filtering, sorting and extract operation are defined in their corresponding configuration files.

4.5.7.3.2 Monitoring

Status & alarms in this window are described in **DPU**.

4.5.7.3.2.1 Session Status

DIR. NBR.	Number of created directories (up to 64)
NUMBER	Directories indexes (DIR00 to DIR63). DIR00 corresponds to the first directory where sorted TM files are recorded, DR01 to the second directory and so on..
TOTAL TM BLOCK	Total number of TM Blocks for all directories (All line) and for each directory.(DR00 to DR63 lines)
BAD TM BLOCK	Total number of bad TM Blocks for all directories and for each directory
SORT TM FILE	Total number of Sort TM files created for all directories and for each directory
SORT VALUE	First four bytes of the last frame data which created the corresponding last sorting.

4.5.7.3.2.2 Upper Banner

See section 4.5.4.1.2.1 .

Detailed information for the set-up and operation of the functionality is given in the Recording and FTP Recovery User Manual DTU 10038.

4.5.7.4 "Virtual Channels" Window

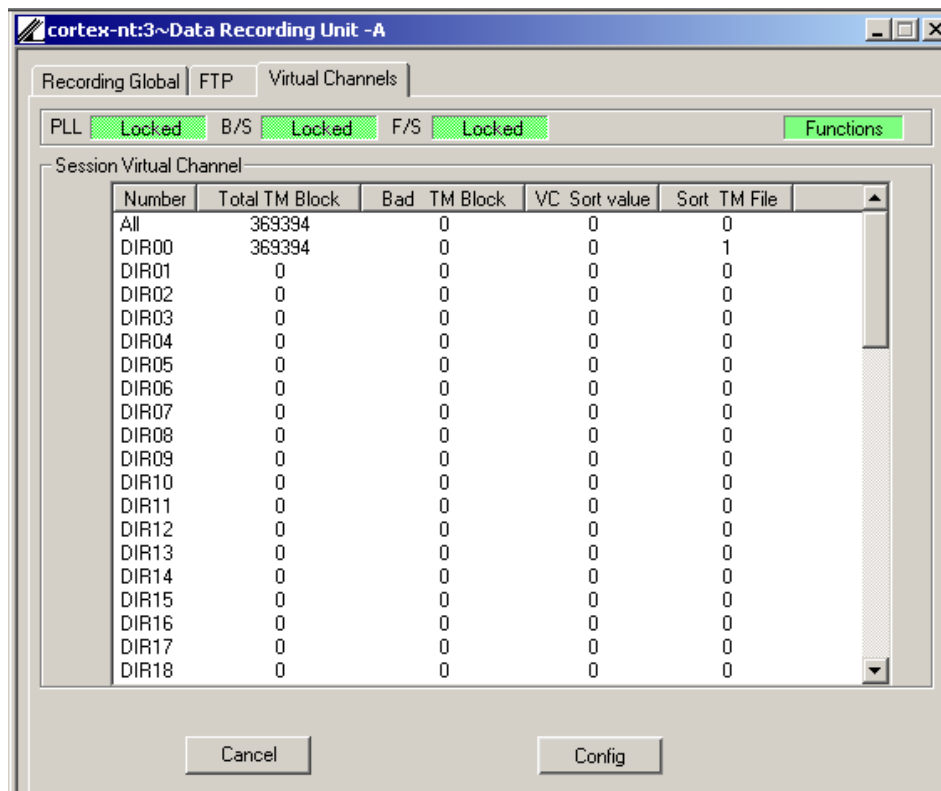


Figure 144: DRU: Virtual Channels Window

4.5.7.4.1 Control

None.

4.5.7.4.2 Monitoring

4.5.7.4.2.1 Virtual Channels Information

From left to right:

- Number of good TM Blocks (including corrected ones) for each virtual channel
- Number of bad TM Blocks (could not be corrected)

Click right in the window to select either **Session Virtual Channel** (display VC information for the current recording) or **TM File Virtual Channel** (display VC information for the current TM File).

4.5.7.4.2.2 Upper Banner

See section 4.5.4.1.2.1 .

4.5.8 Test modulator Windows

4.5.8.1 "Data Generation Unit" Windows

4.5.8.1.1 "Data" Window

4.5.8.1.1.1 PRN Selection

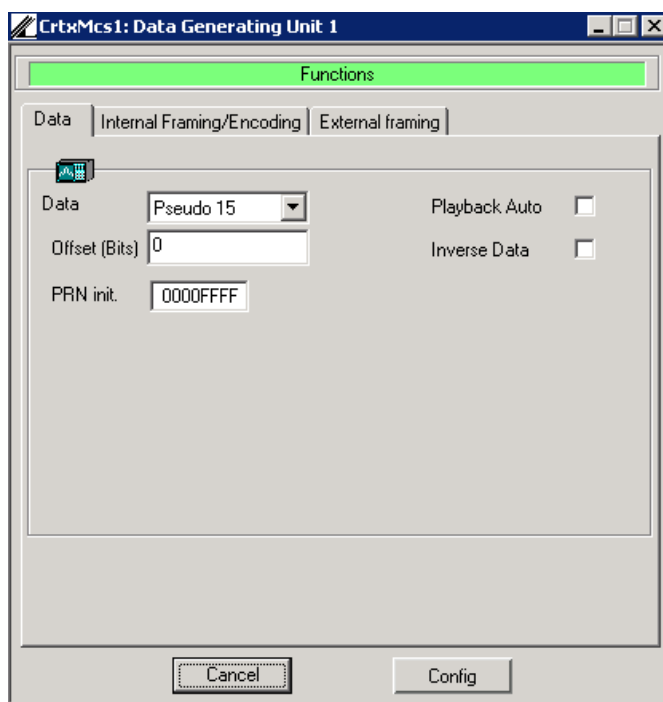


Figure 145: PRN Selection

4.5.8.1.1.2 File Selection

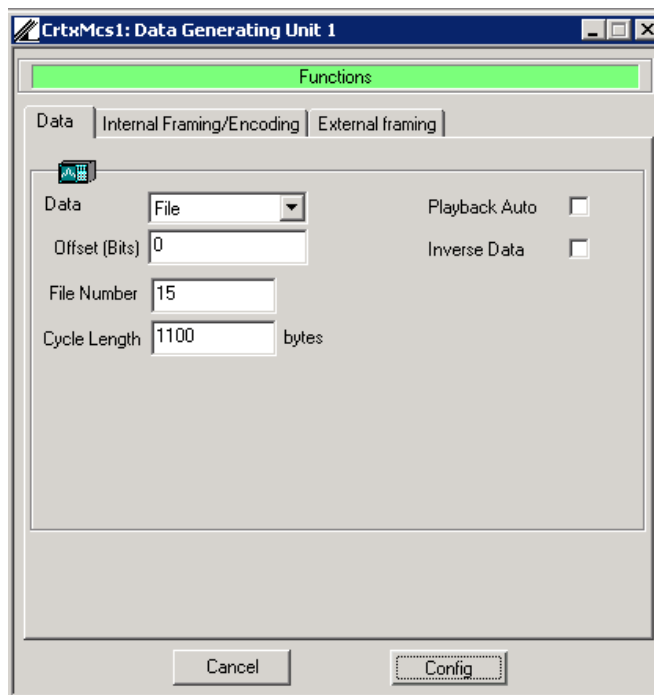


Figure 146: File Selection

4.5.8.1.1.3 File I/Q Selection

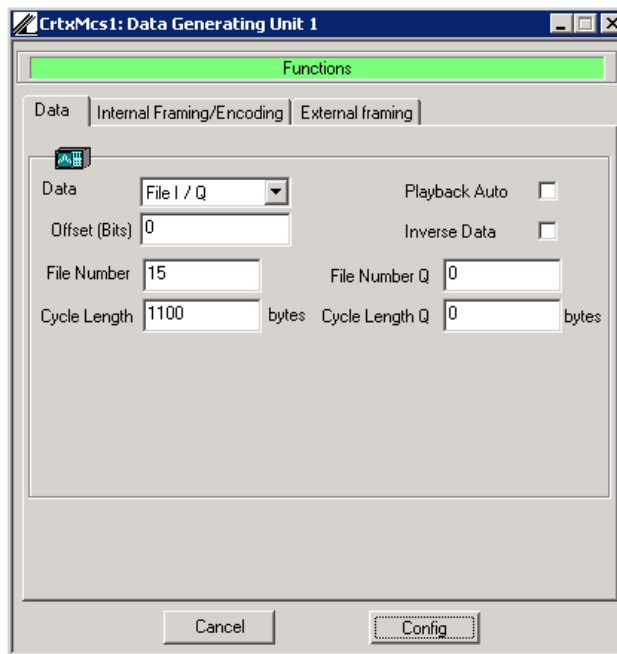


Figure 147: File I/Q Selection

4.5.8.1.1.4 Control

DATA	Simulated telemetry data source: • Off: For reset purposes • File: Read data from a file • File I/Q: Select two different files for I and Q • Playback: Connect the test modulator unit with the Playback • Pseudo XX: Pseudo random data from PRN generator XX. See Section 3.6.1.1.1 • External: External source of data
OFFSET (BITS)	In multi DGU mode, adjustable offset between DGU
FILE NUMBER FILE NUMBER Q	File number for BER in File mode: enter any value from 0 to 4,294,967,295
CYCLE LENGTH CYCLE LENGTH Q	Cycle length for File mode only (in bytes). Enter a value from 4,194,303 bytes
PLAYBACK AUTO	If selected insert PRN data before and after the actual recorded data in Playback
INVERSE DATA	If selected invert the data to be modulated
PRN INIT.	Initializing vector for PRN data generation

4.5.8.1.2 "Internal Framing" Window

4.5.8.1.2.1 General selection

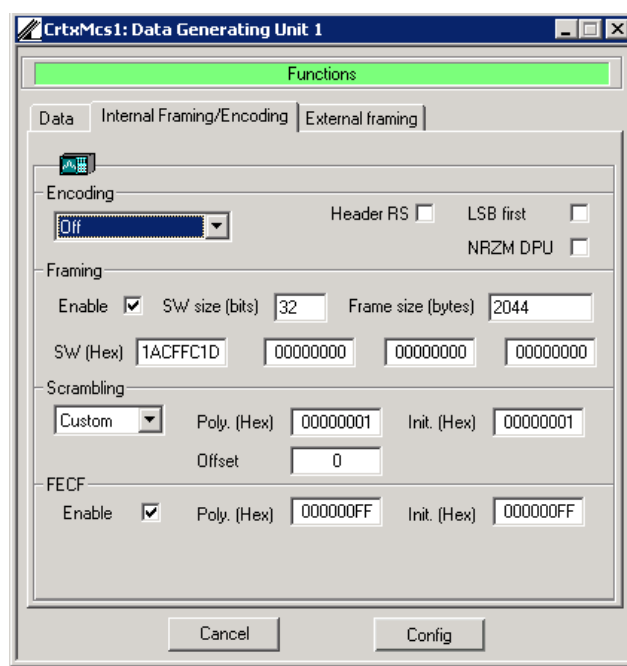


Figure 148: Internal Framing Window

4.5.8.1.2.2 Control

ENCODING	Selection of the DGU encoding: • Off: No encoding • CCSDS RS223: RS (223, 255) • CCSDS RS239: RS (239, 255) • CCSDS LDPC7/8: NASA-Shortened LDPC 7/8 • DVB: DVB Transport Layer • DVB DSN • CCSDS LDPC 1/2 • CCSDS LDPC 2/3 • CCSDS LDPC 4/5
HEADER RS	Internal Frame Header encoding
LSB FIRST	Enable or Disable the LSB-First byte formatting in the simulated data
NRZM DPU	Enable or Disable the NRZ-M encoding in the simulated data
FRAMING ENABLE	Enable or Disable the synchronization word inside the data. Essentially used with simulated Pseudo Random Data
SW SIZE	<i>Only if Framing enabled:</i> Selection of the size of the synchronization word
FRAME SIZE	<i>Only if Framing enabled:</i> Selection of the size of the Frame
SW	<i>Only if Framing enabled:</i> Value of the Synchronization Word (4 fields of 8 HEX characters)
SCRAMBLING SELECTION	• Off: No scrambling • CCSDS: CCSDS Scrambling • Custom: Customized Scrambling
SCRAMBLING POLY	<i>Only if Custom Scrambling</i>
SCRAMBLING INIT	<i>Only if Custom Scrambling</i>
SCRAMBLING OFFSET	<i>Only if Custom Scrambling</i>
FECF ENABLE	Enable or Disable FECF insertion
FECF POLY	<i>Only if FECF enabled</i>
FECF INIT	<i>Only if FECF enabled</i>

4.5.8.1.3 "External Framing" Window

4.5.8.1.3.1 General Selection

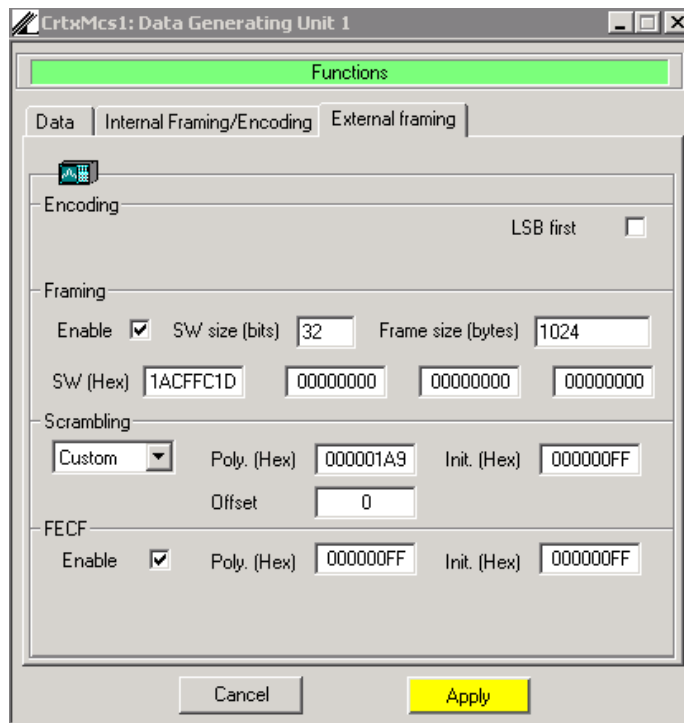


Figure 149: External Framing Window

4.5.8.1.3.2 Control

LSB FIRST	Enable or Disable the LSB-First byte formatting in the simulated data
FRAMING ENABLE	Enable or Disable the synchronization word inside the data. Essentially used with simulated Pseudo Random Data
SW SIZE	<i>Only if Framing enabled:</i> Selection of the size of the synchronization word
FRAME SIZE	<i>Only if Framing enabled:</i> Selection of the size of the Frame
SW	<i>Only if Framing enabled:</i> Value of the Synchronization Word (4 fields of 8 HEX characters)
SCRAMBLING SELECTION	- Off: No scrambling - CCSDS: CCSDS Scrambling - Custom: Customized Scrambling
SCRAMBLING POLY	<i>Only if Custom Scrambling</i>
SCRAMBLING INIT	<i>Only if Custom Scrambling</i>
SCRAMBLING OFFSET	<i>Only if Custom Scrambling</i>
FECF ENABLE	Enable or Disable FECF insertion
FECF POLY	<i>Only if FECF enabled</i>
FECF INIT	<i>Only if FECF enabled</i>

4.5.8.1.4 Status & Warnings

The upper banner indicates the Warning status: OK (green display) or ERROR (red display).
When pointing at the banner, more information about the Warning is displayed.

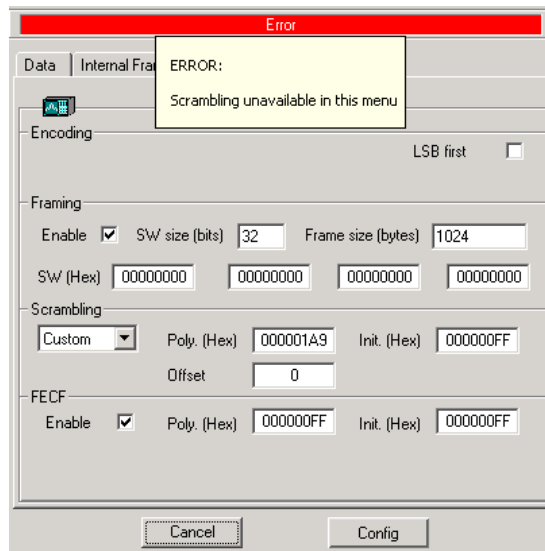


Figure 150: Pointing at an error

4.5.8.2 "Modulation Unit" Windows

4.5.8.2.1 "Modulation" Window

4.5.8.2.1.1 BPSK/QPSK/GMSK

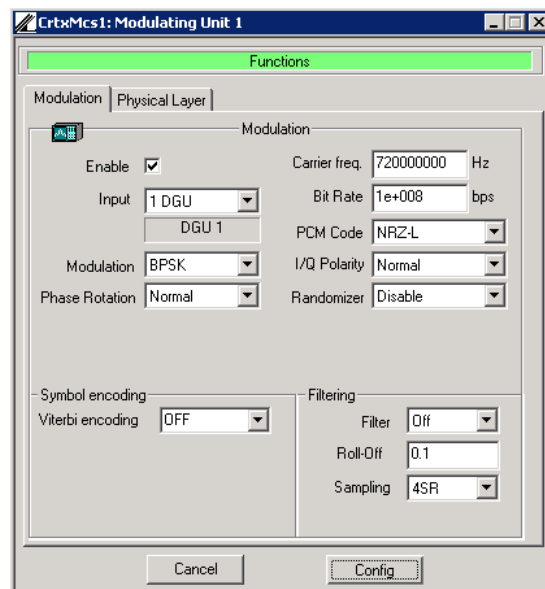


Figure 151: BPSK/QPSK/GMSK Configuration

4.5.8.2.1.2 OQPSK

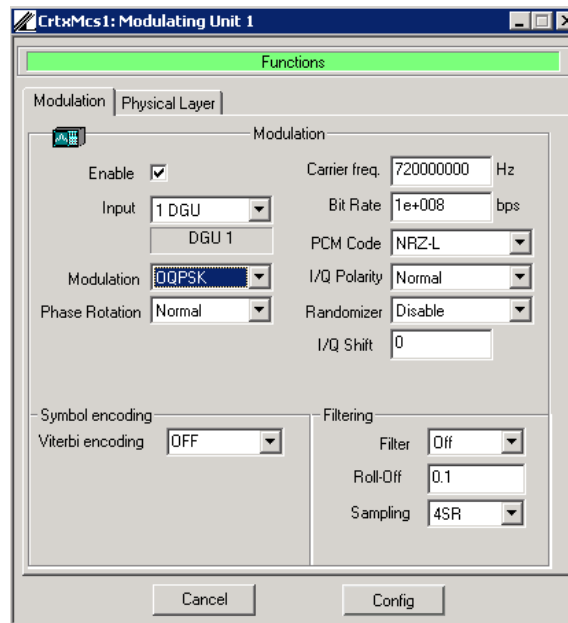


Figure 152: OQPSK Configuration

4.5.8.2.1.3 UQPSK

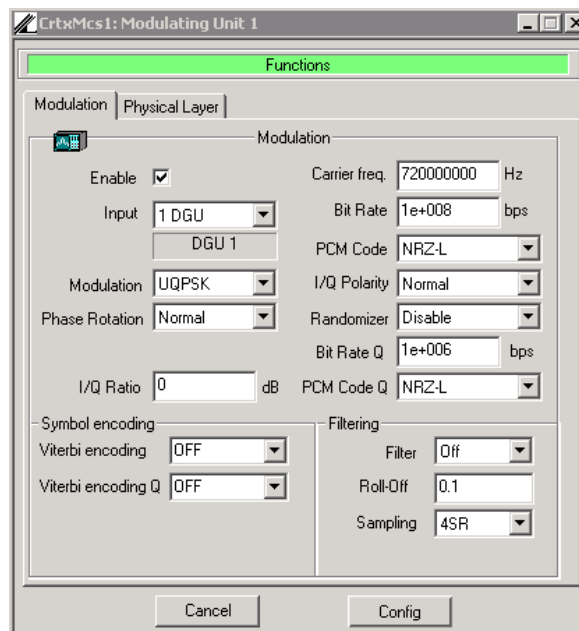


Figure 153: UQPSK Configuration

4.5.8.2.1.4 8PSK

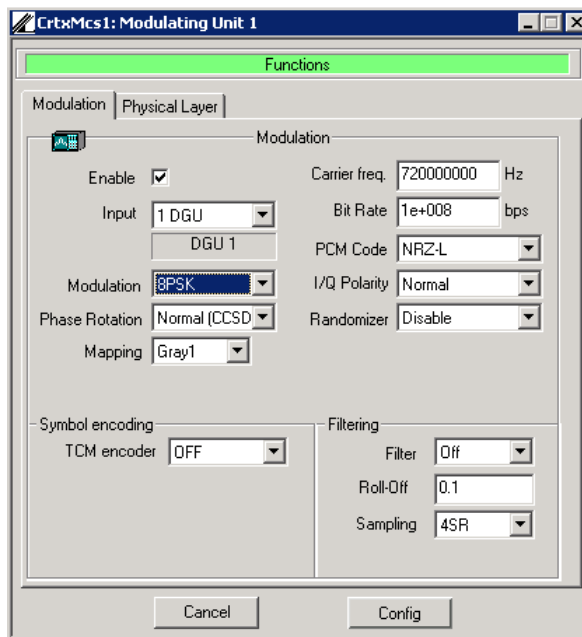


Figure 154: 8PSK Configuration

4.5.8.2.1.5 16QAM

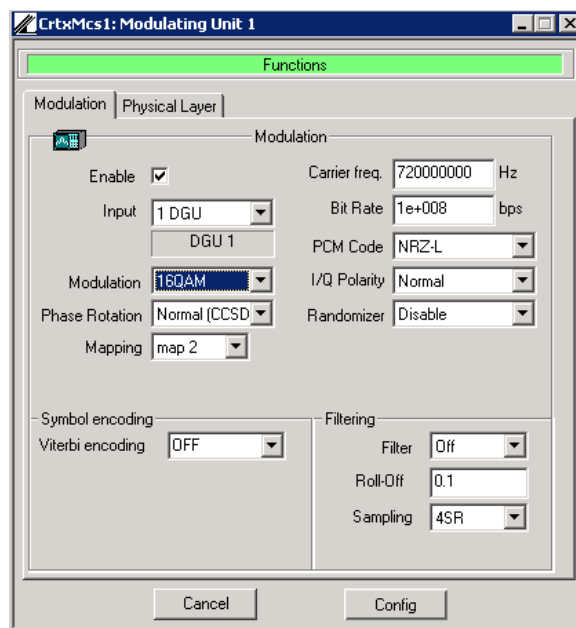


Figure 155: 16QAM Configuration

4.5.8.2.1.6 16APSK

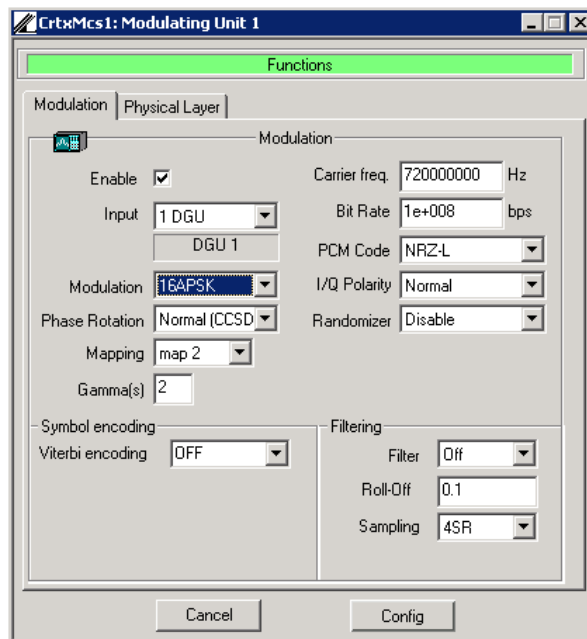


Figure 156: 16APSK Configuration

4.5.8.2.1.7 32APSK

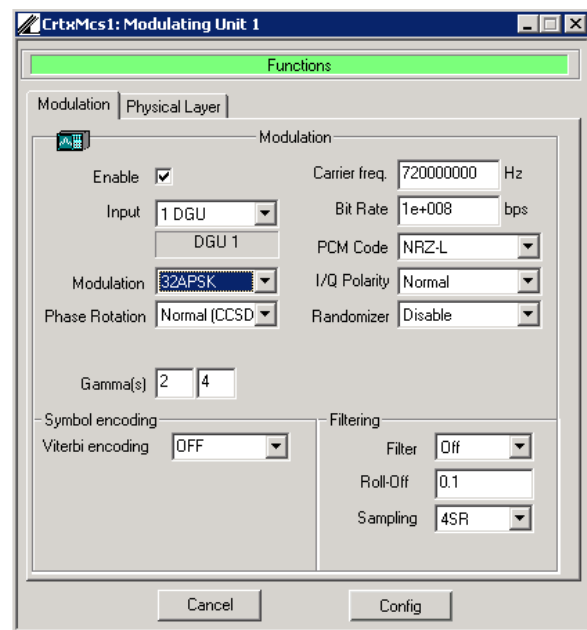


Figure 157: 32APSK Configuration

4.5.8.2.1.8 64APSK

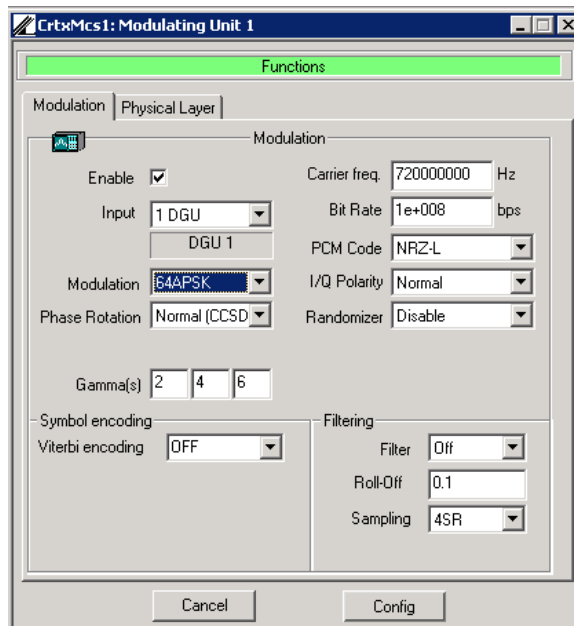


Figure 158: 64APSK Configuration

4.5.8.2.1.9 Viterbi Mode

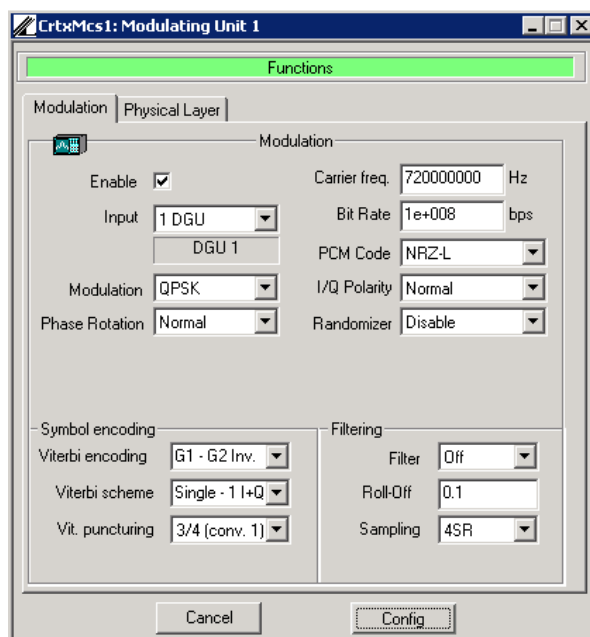


Figure 159: Viterbi Configuration

4.5.8.2.1.10 TCM Mode

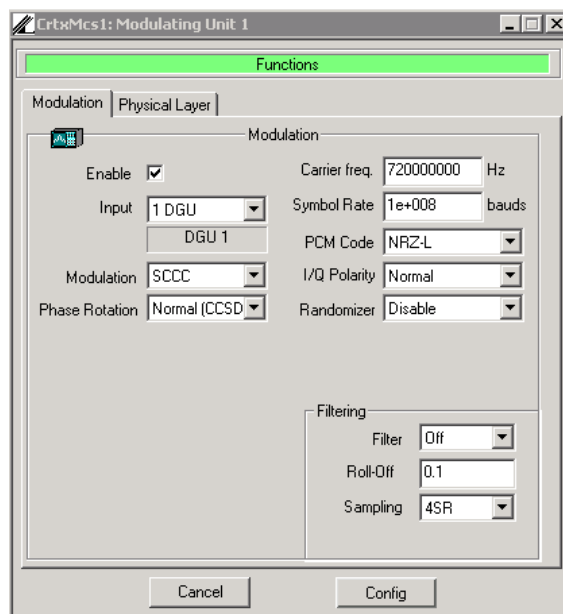


Figure 160: TCM Configuration

4.5.8.2.1.11 Control

MODULATION ENABLE	Enable or disable carrier modulation
CARRIER FREQUENCY	IF Carrier Frequency Enter any value in float or real format. Example: for 720 MHz, type 7.2e+8 or 720000000.
INPUT	Input of the MDU • X DGU (s): Merge of X DGUs • F. Input: Format Input (external source)
BIT RATE/SYMBOL RATE BIT RATE Q	Bit Rate or Symbol Rate expressed in bps or Bauds. Independent rates can be defined for I and Q in UQPSK.
PCM CODE PCM CODE Q	Telemetry PCM Coding technique: • NRZ-L/M/S: Non Return to Zero • DNRZ • BPL/M/S: Bi-Phase • NRZM Merge
DNRZ CODE	Type of DNRZ Code (only in QPSK or in OQPSK)
I/Q RATIO	In UQPSK: Ratio between I Channel and Q Channel in dB
MODULATION	Type of Modulation: BPSK, QPSK, OQPSK, UQPSK, 8PSK, GMSK, 16QAM, 16APSK, 32APSK, 64APSK, SCCC or DVB-S2
MAPPING	Select Gray X or Natural X in 8PSK Select Map X in 16QAM or 16APSK
GAMMA(S)	Allow to adjust R_2/R_1 for 16APSK, 32APSK, 64APSK Allow to adjust R_3/R_1 for 32APSK, 64APSK Allow to adjust R_4/R_1 for 64APSK
I/Q POLARITY	Allows to invert and/or swap I and/or Q Channels

PHASE ROTATION	Select CCSDS (Counter Clock Wise) or Inverted (Clock Wise)
RANDOMIZER	Type of Auto synchronized Randomizer • Disable • WorldView • CCITT V.35 • IntelSat V.35
I/Q SHIFT	Allows adjusting the I/Q offset in QPSK mode for test purposes Enter any value from +0.5 to -0.5 (in symbol). Select 0 for optimal QPSK (half-symbol offset). <i>This parameter is not yet available</i>
VITERBI ENCODING VITERBI ENCODING Q	Convolutional encoder setting for the I and/or Q Channel • OFF: disabled • G1 – G2 Inv: G2 inverted (G1 first) • G1 – G2: G2 normal (G1 first) • G2 Inv – G1: G2 inverted (G2 first) • G2 – G1: G2 normal (G2 first) <i>Refer to the CCSDS recommendations for “G2 normal/inverted” definition</i>
VITERBI SCHEME	• Single – 1 I+Q: Single encoder common to I & Q Channels • Dual – 1I / 1Q: Separate encoders for I & Q Channels • XI / XQ: Separate encoders for I & Q Channels. X is the number of parallel encoders
VITERBI PUNCTURING VITERBI PUNCTURING Q	Puncturing setting for Viterbi encoding • OFF • ON (registry): defined in registry • a/b (Conv. X): Puncturing a/b (convention 1 or 2) • 1/4 ESRIN
TCM ENCODER	TCM encoder for 8PSK modulation mode: • OFF: disabled • 2.XX/3: 4D-TCM rate 2.XX/3 • DSNG 5/6
FILTER	Modulation Filter: • Off • RC • RRC • GMSK • Custom
FILTER FILE	<i>Only when Custom Filter is selected</i>
ROLL-OFF	Roll-Off factor for the RC/RRC filter
BT	BT factor for the GMSK filter
SAMPLING	XSR: Enable the high rate filter at F_e equal X times the Symbol Rate

4.5.8.2.2 "Physical Layer" Window

4.5.8.2.2.1 SCCC Mode

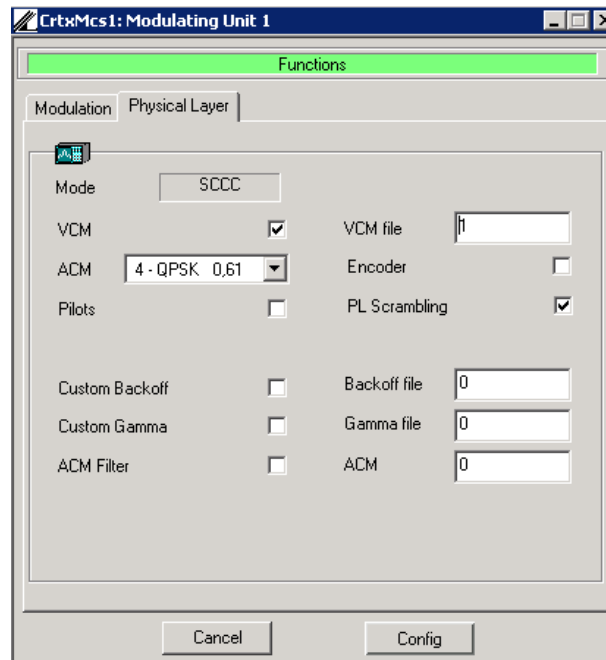


Figure 161: SCCC Configuration

4.5.8.2.2.2 DVB-S2 Mode

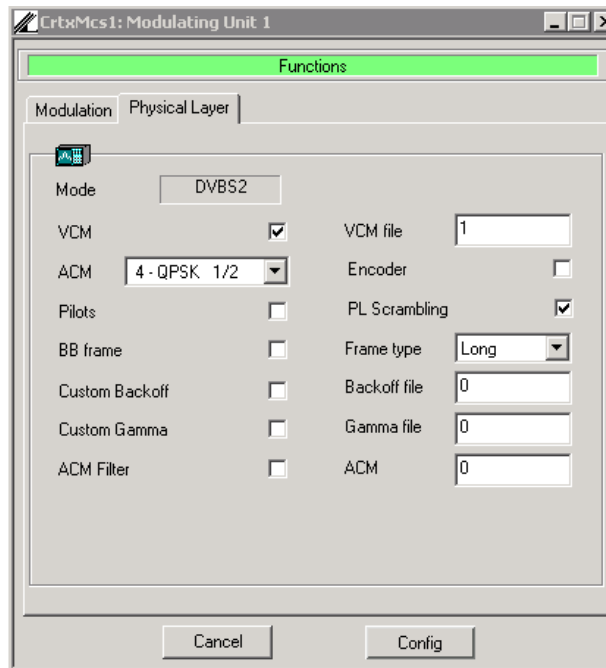


Figure 162: DVB-S2 Configuration

4.5.8.2.2.3 Control

VCM	Enable or not the VCM mode for the modulation
VCM FILE	If VCM mode is enabled, the modulation and coding modes sequence are described inside a simulation ASCII File. This field allows selecting the File Number. The file are available in D:\ZDS\Data\HDR\SPS\VCM The file contains 1 line per ACM. It begins with #frame# when the VCM is frame-based. If not, it is time-based. Each line must have the following syntax: Nb Frames ; modulation OFF ; stop ; ACM nb ; pilot ; short frame (Frame-based mode) time (seconds) ; noise level (dBm/Hz) ; ACM nb ; pilot ; short frame (Time-based mode) Lines beginning with '#' character and blank are ignored.
ACM	ACM number (describing modulation and rate, or DUMMY frames)
ENCODER	Enable or disable the encoding
PILOTS	Enable or disable the insertion of pilots inside the frames
PL SCRAMBLING	Enable or disable the randomization of the frames
CUSTOM BACKOFF	Enable or Disable custom values of Back-Off
BACKOFF FILE	Allows to select the file (ASCII file) defining the back off setting. The files are available in D:\ZDS\Data\HDR\SPS\VCM\backoff Please refer to provided example file for reference. Lines beginning with '#' character and blank are ignored.
CUSTOM GAMMA	Enable or disable custom values of Gamma(s)
GAMMA FILE	Allows to select the file (ASCII file) defining the gamma(s) setting. The files are available in D:\ZDS\Data\HDR\SPS\VCM\gamma Please refer to provided example file for reference. Lines beginning with '#' character and blank are ignored.
ACM FILTER	Enable or disable the "filtering" of frames. For BER test purposes BER can be evaluated in VCM mode on a single ACM. The filter ensures continuity of the data generated from the DGU
ACM FILTERED	When ACM Filter is enabled, this field indicates the ACM which is actually transmitted (with continuity of data). For any other ACM, padding frames are inserted.
BB FRAME	Enable or disable the insertion of BB Headers inside DVB-S2 Frames
FRAME TYPE	Long or Short Frames can be selected for DVB-S2 Mode

4.5.8.2.3 Status & Warnings

When the input of the MDU is selected, a status indicates which DGUs are actually connected to the MDU or if the MDU is connected to the Format Input.



Figure 163: Status: Input of the MDU

A status in the “Physical Layer” Window indicates which type of modulation is activated:

- SCCC
- SCCC-X
- DVB-S2
- DVB-S2X
- Standard (any other mode)



Figure 164: Modulation Type

The upper banner indicates the Warning status: OK (green display) or ERROR (red display). When pointing at the banner, more information about the Warning is displayed.

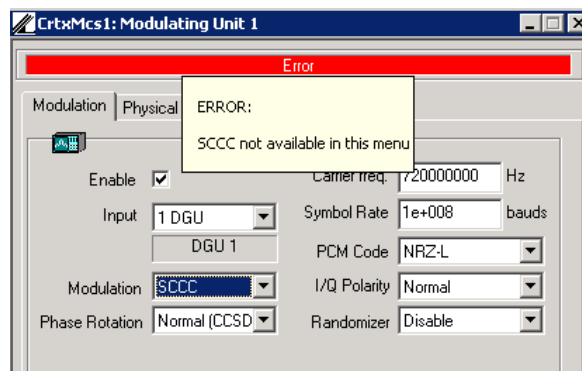


Figure 165: Pointing at an error

4.5.8.3 "IF Unit" Windows

4.5.8.3.1 "IF" Window

4.5.8.3.1.1 General Selection

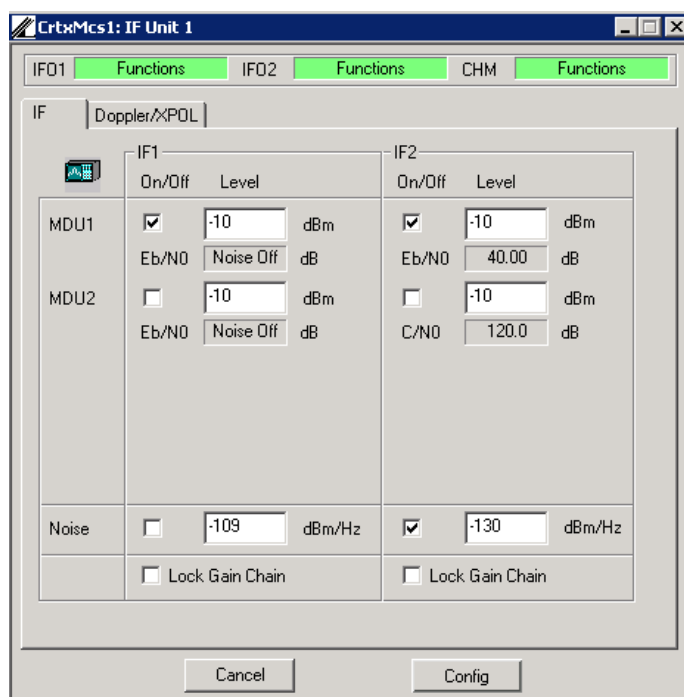


Figure 166: IF Window

4.5.8.3.1.2 Control

MDU X – IF Y – ON/OFF	Enable or disable the carrier (contribution of MDU X for IF Y)
MDU X – IF Y – LEVEL	Carrier Level in dBm
IF Y – NOISE	Enable or disable the noise generator
IF Y – NOISE LEVEL	Noise Level in dBm/Hz
IF Y – LOCK GAIN CHAIN	Setting reserved for factory puproses

4.5.8.3.2 "Doppler/XPOL" Window

4.5.8.3.2.1 Doppler Configuration

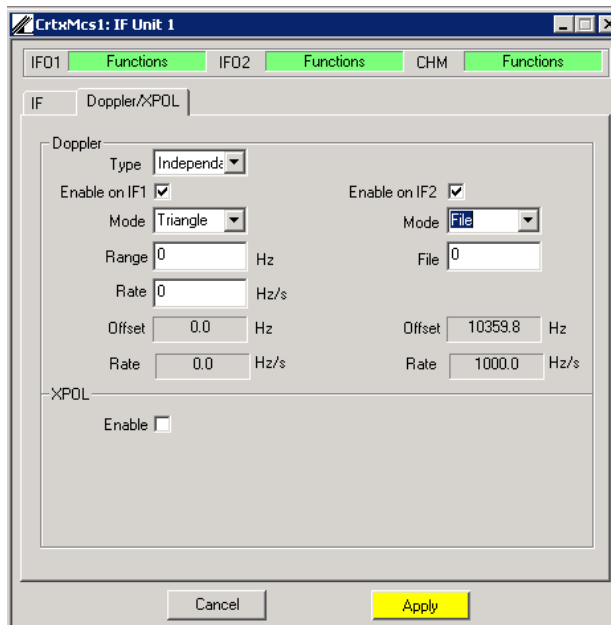


Figure 167: Doppler Configuration

4.5.8.3.2.2 XPOL Configuration: Still

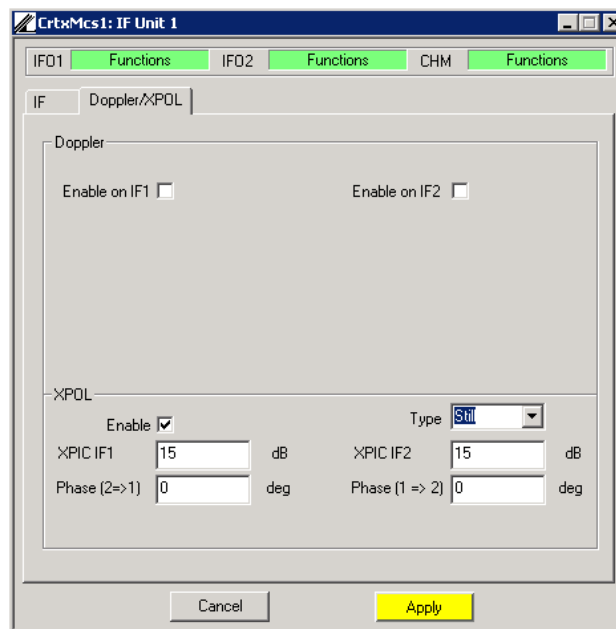


Figure 168: XPOL Configuration: Still

4.5.8.3.2.3 XPOL Configuration: File

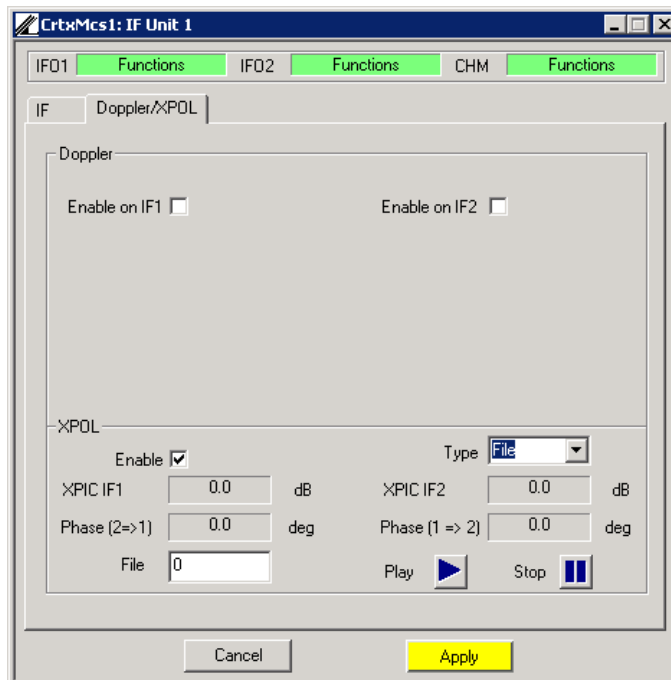


Figure 169: XPOL Configuration: File

4.5.8.3.2.4 Control

DOPPLER	
ENABLE (ON IF X)	Enable or disable Doppler (Frequency Sweeping) on IF 1 and/or IF 2
TYPE	Only when Doppler is available on both IF 1 & 2 • Same: Same Doppler scheme on both IF • Independent: Independent Doppler schemes
MODE (IF X)	• Triangle • Sinusoidal • Atan • File
RANGE (IF X)	Doppler Range in Hz
RATE (IF X)	Doppler Rate in Hz/s
FILE (IF X)	When File Mode is enabled, this is the number of the selected File The files can be found in D:\ZDS\DATA\HDR\SPS\DOPPLER\ and are named as follows: - File 1 will be named doppler1.csv The files contain two columns: "time" and "offset" - "time" is the time offset in seconds - "offset" is the frequency offset You can add "#REWIND#" at the end to loop on the file
XPOL	
ENABLE	Enable or disable Cross-Polarization
TYPE	• Still • File

FILE	When File Mode is enabled, this is the number of the selected File The files can be found in D:\ZDS\DATA\HDR\SPS\XPOL\ and are named as follows: - File 1 will be named xpol1.csv
XPIC IF X	In dB
PHASE (A => B)	In °

4.5.8.3.3 Status and Warnings

The upper banner indicates the Warning status: OK (green display) or ERROR (red display) for IF 1, IF 2 and CHM.

When pointing at the banner, more information about the Warning is displayed.

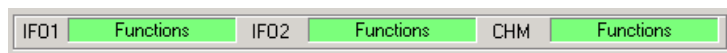


Figure 170: IF Warnings

In the “IF” Window, a status is placed under each contribution of each MDU for each IF. It indicates either “Noise Off” when Noise Generator is disabled for this IF, either an estimation of the E_b/N_0 based on the IF Level, the Noise Level and the Bit Rate.

E_b/N_0	Noise Off	dB
E_b/N_0	40.00	dB

Figure 171: E_b/N_0 status

In the “Doppler/XPOL” Window, when Doppler is active, several status indicate the actual rate and offset in real time.

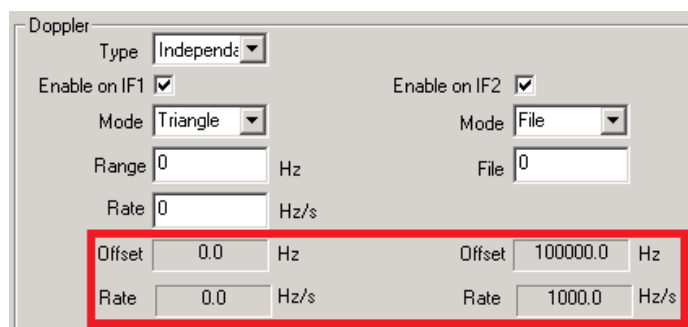


Figure 172: Doppler status

In “Doppler/XPOL” Window, when XPOL is active in File Mode, several status indicate actual XPIC and Phase in real time.

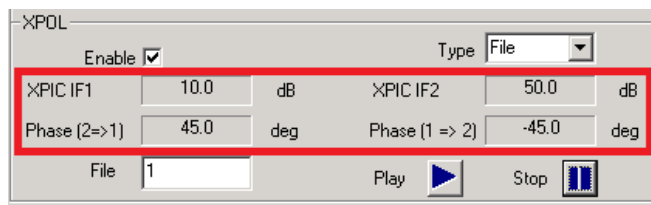


Figure 173: XPOL status

The “PLAY” button  starts playing the file.

The “STOP” button  stops playing the file.

4.5.8.4 "Format Input" Windows

4.5.8.4.1 "Format Input" Window – Merged BER

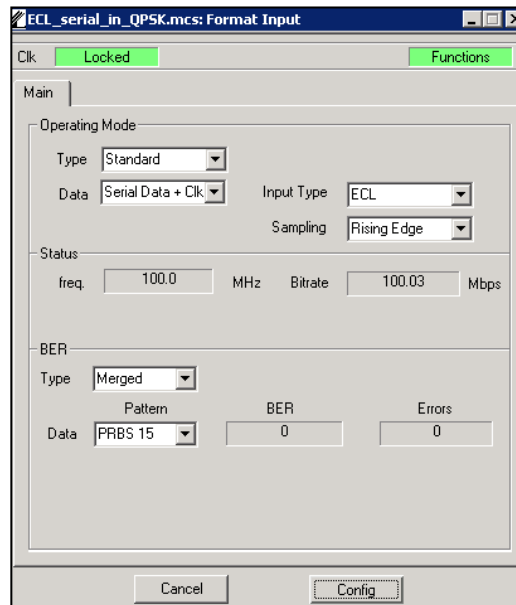


Figure 174: Serial data

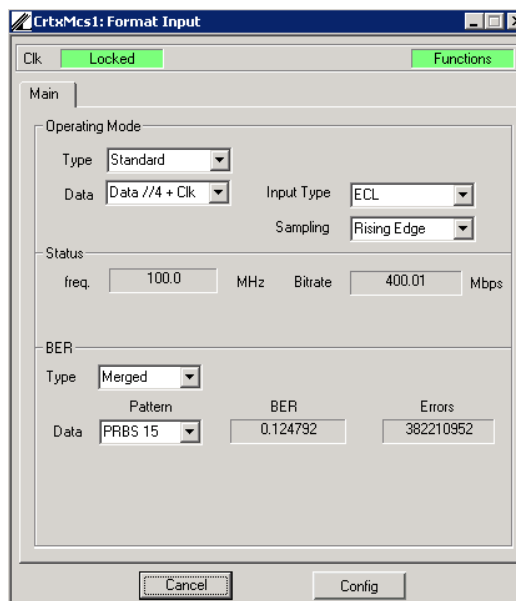
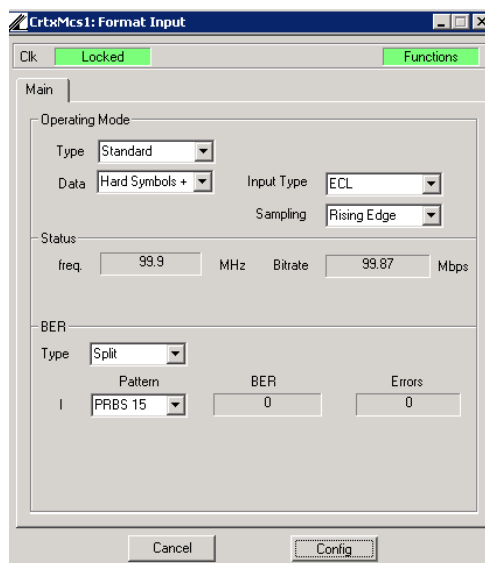


Figure 175: Data //4

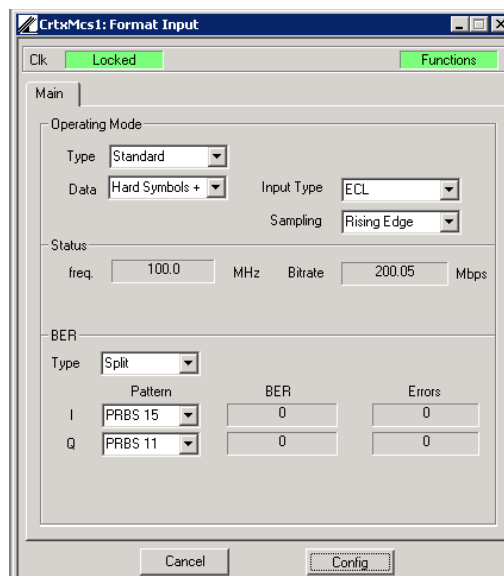
4.5.8.4.2 Format Input" Window –SPLIT BER



The screenshot shows the 'Format Input' window with the following settings:

- Operating Mode:** Type: Standard, Data: Hard Symbols +, Input Type: ECL, Sampling: Rising Edge.
- Status:** freq: 99.9 MHz, Bitrate: 99.87 Mbps.
- BER:** Type: Split, Pattern: PRBS 15, BER: 0, Errors: 0.

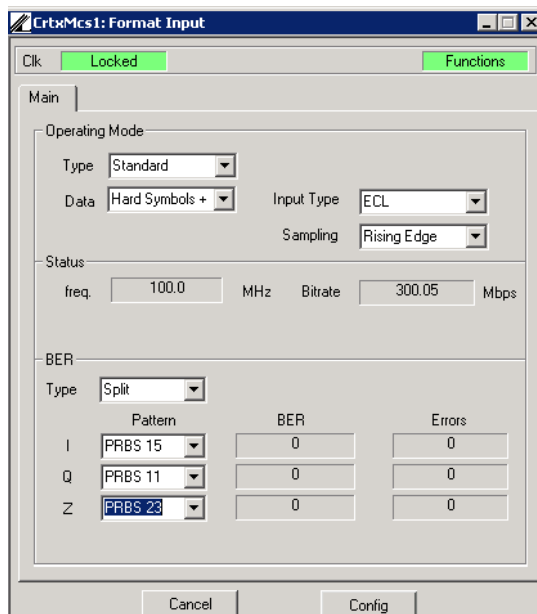
Figure 176: Hard Symbol, BPSK



The screenshot shows the 'Format Input' window with the following settings:

- Operating Mode:** Type: Standard, Data: Hard Symbols +, Input Type: ECL, Sampling: Rising Edge.
- Status:** freq: 100.0 MHz, Bitrate: 200.05 Mbps.
- BER:** Type: Split, Pattern: PRBS 15 (I) and PRBS 11 (Q), BER: 0, Errors: 0.

Figure 177: Hard Symbol, QPSK/QPSK

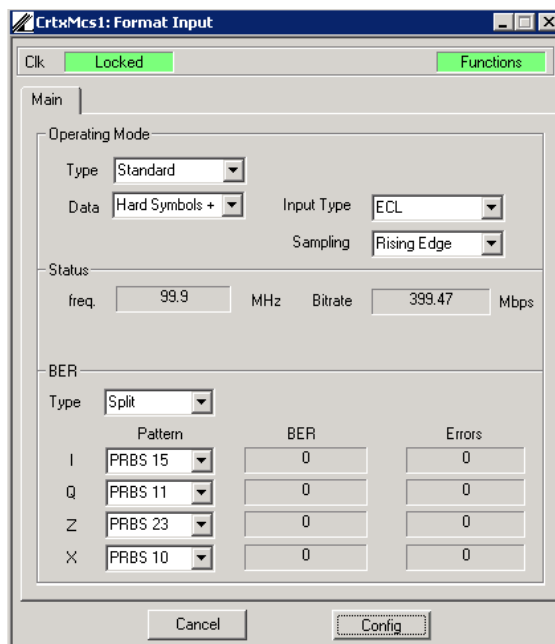


The screenshot shows the 'CrtxMcs1: Format Input' dialog box with the 'Main' tab selected. The 'Operating Mode' section has 'Type' set to 'Standard', 'Data' set to 'Hard Symbols +', 'Input Type' set to 'ECL', and 'Sampling' set to 'Rising Edge'. The 'Status' section shows 'freq.' as 100.0 MHz and 'Bitrate' as 300.05 Mbps. The 'BER' section has 'Type' set to 'Split'. Below this, there is a table for BER patterns:

	Pattern	BER	Errors
I	PRBS 15	0	0
Q	PRBS 11	0	0
Z	PRBS 23	0	0

At the bottom, there are 'Cancel' and 'Config' buttons.

Figure 178: Hard Symbol, 8PSK



The screenshot shows the 'CrtxMcs1: Format Input' dialog box with the 'Main' tab selected. The 'Operating Mode' section has 'Type' set to 'Standard', 'Data' set to 'Hard Symbols +', 'Input Type' set to 'ECL', and 'Sampling' set to 'Rising Edge'. The 'Status' section shows 'freq.' as 99.9 MHz and 'Bitrate' as 399.47 Mbps. The 'BER' section has 'Type' set to 'Split'. Below this, there is a table for BER patterns:

	Pattern	BER	Errors
I	PRBS 15	0	0
Q	PRBS 11	0	0
Z	PRBS 23	0	0
X	PRBS 10	0	0

At the bottom, there are 'Cancel' and 'Config' buttons.

Figure 179: Hard Symbol, 16APSK/16QAM

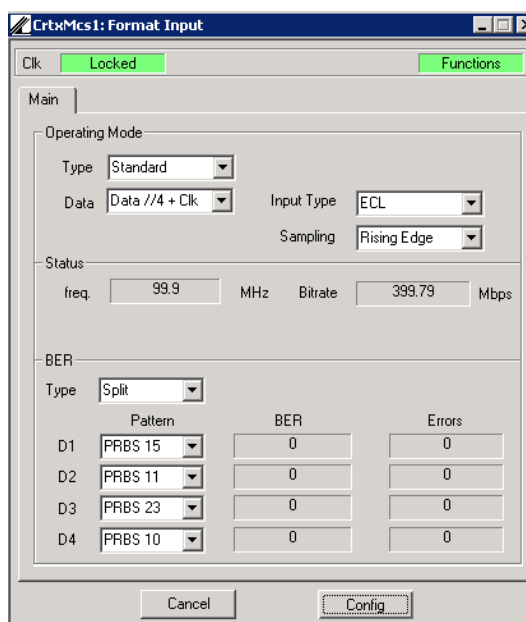


Figure 180: Data //4

4.5.8.4.2.1 Control

TYPE	Specifies the type of Input scheme: - Standard (HDR4G baseline) - Custom (Customer specific, may require dedicated HW and/or licence)
DATA	For standard input mode, specifies the input format to the modulator: - Serial data + clk: 1 bit data + clk - Data //2 + clk: 2 bits data + clk with clock rate at half the bitrate - Data //3 + clk: 3 bits data + clk - Data //4 +clk: 4 bits data +clk - Data //8 +clk: 8 bits data + clk (coming soon) - Hard Symbol +clk: n bits + clk, with n being the order of the modulation selected in the connected MDU.
INPUT TYPE	Specifies the electrical standard for the input: - ECL (on the modulator board) - LVDS (on the I/O panel – coming soon)
SAMPLING	Enable selection of either rising or falling edge with respect to clk on the interface
BER – TYPE	Specifies the BER scheme on the input: - Merged: all parallel bits are merged on a single stream before entering the BER estimator - Split: parallel bits are treated separately, with independent Pattern selection. Parallel bits can results from either “Hard symbols” or “data” in //2, 3, 4.
BER – DATA, OR D1, D2, D3, D4 OR I, Q, Z, X	Enable selection of the BER pattern to test among PRBS 7, 10, 11, 15, 23 or 31 for each possible stream to test (merged, or one for each input bit). “OFF” disables PRBS testing.

4.5.8.4.3 Status & Warnings

The upper left part of the window indicates whether an incoming clock is detected or not.

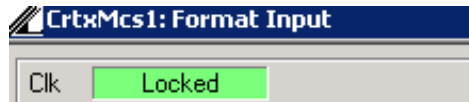


Figure 181: Status: clock status, Locked and Unlocked

The clock frequency and the raw bitrate are also indicated in the middle of the window:

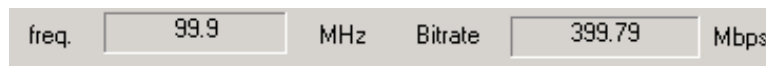


Figure 182: clock and raw bitrate estimates

The raw bitrate is estimated by multiplying the instant frequency estimate by the number of bits in parallel on the interface. This depends on the selected input scheme (parallel 2, 3 4 or Hard Symbol + modulation).

For each possible data stream (1 to 4), BER status and an errors status is indicated. Both indicators can be reset by clicking the '0' button in the main Cortex GUI.



Figure 183: BER and Errors status

The upper banner indicates the Warning status: OK (green display) or ERROR (red display). When pointing at the banner, more information about the Warning is displayed.

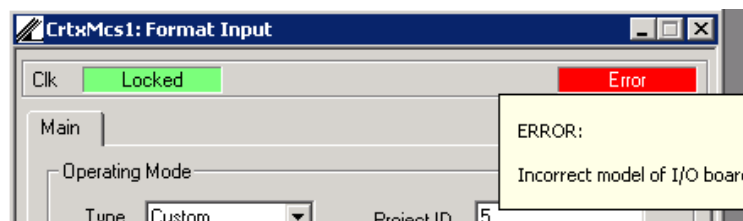


Figure 184: Pointing at an error

4.5.8.5 Error insertion

A click on the button  inverts one bit in the simulated data-pattern. Bit inversion takes place before convolutional encoding (if enabled).

- When a "DGU" Window is active, the bit is inverted in this DGU
- When a "MDU" Window is active, the bit is inverted in one of the DGUs connected to this MDU (randomly chosen). *This mode is not yet available.*
- When no window is active, the bit is inverted in one of the mounted DGUs (randomly chosen)

4.5.9 Playback Windows

4.5.9.1 "Global" Window

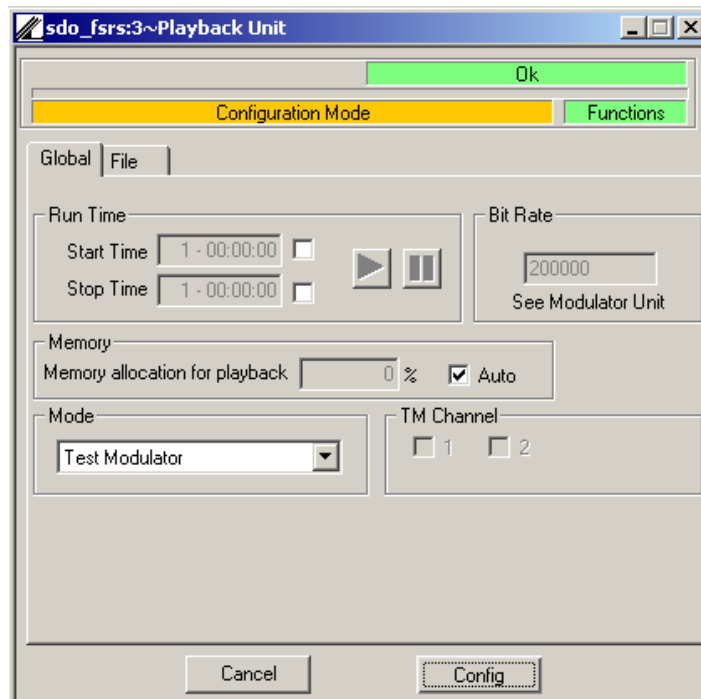


Figure 185: Playback: Global Window

4.5.9.1.1 Configuration and control

Parameters in this window are described in **PLBK**

To start the replay, the HDR must be set in "Acquisition" mode.

START TIME

Enables to define an automatic starting date for the replay.
Must be selected to be active.

STOP TIME

Enables to define an automatic stop date for the replay.
Must be selected to be active.



Manual start of the replay.



Manual stop of the replay.

BIT RATE

Selects the bit rate of the playback only used in "Data+Clock" mode.

Important: In "Test modulator" mode, the bit rate of the playback is the bit rate set in the test modulator window.

MEMORY ALLOCATION FOR PLAYBACK

Enables to allocate a fixed part of the available memory to the playback function when the selection box Auto is disabled. **The recommended operating mode is to enable the Auto mode.**

MODE

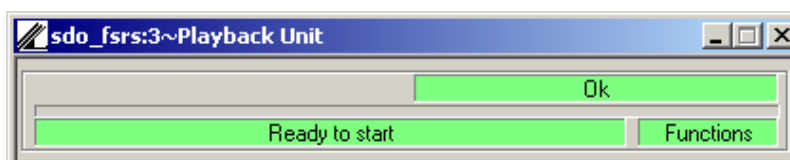
Select the playback mode:

- **Off** Playback OFF
- **Soft Mode** *Specific Project*
- **Data + Clock** Playback at the ECL outputs of the demodulators
- **Test Modulator** Playback at IF level using the test modulator

Important: Before selecting a playback mode ("Test Modulator" or "Data + Clock") the playback function must be connected to the test modulator (test modulator Global Window – Field: Operating Mode = Playback), or to the Demodulator outputs (Demodulator Global Window – Field: Output Mode = Playback Merge or Playback Split 2).

4.5.9.1.2 Status

Status and alarms in this window are described in **PLBK**.



The status field is divided into two parts:

A general status line gives the status of the operation: "Ready to start" or "Play in progress, waiting for stop".

A bar-graph gives in real-time the level of the current playback as a percentage of the consumed data from the selected file.

4.5.9.1.3 Specific operating mode: Playback LAN

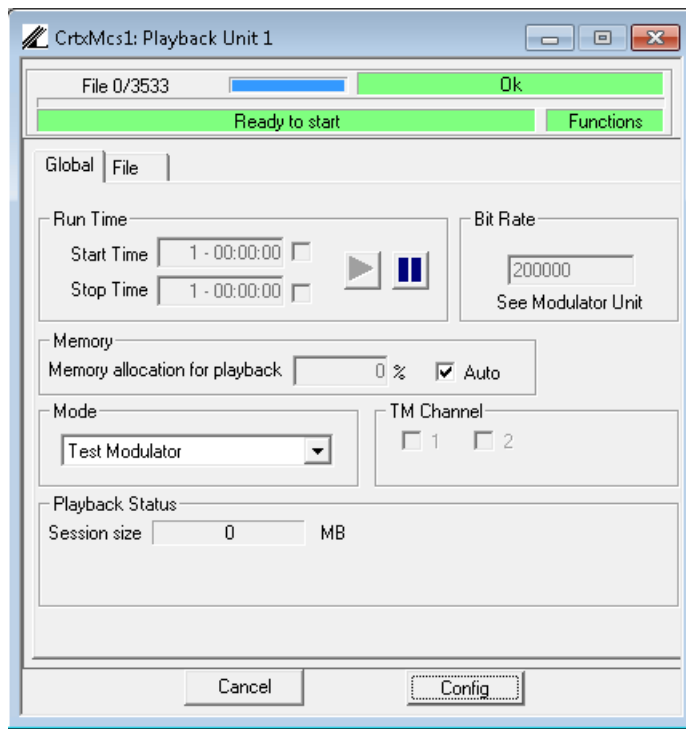


Figure 186: Playback LAN: Global Window

In “Playback LAN” mode, the manual start of playback, start time and stop time are disabled. The session starts when user transmits data over the Playback port (PBK-1 or PBK-2). The session stops when:

- User disconnects from the Playback port (PBK-1 or PBK-2); session ends after modulation of all received data.
- User clicks on the manual stop of replay; user is disconnected from Playback port (PBK-1 or PBK-2), session is immediately stopped and all remaining data are flushed.

Note:

Session stops also when:

- User clicks on “Reset playback client” button in “TCP-IP Clients” window; user is disconnected from Playback port (PBK-1 or PBK-2), session ends after modulation of all received data.
- User clicks on “Configuration” button then on “Stop All” button in the toolbar; user is disconnected from Playback port (PBK-1 or PBK-2), session is immediately stopped and all remaining data are flushed.

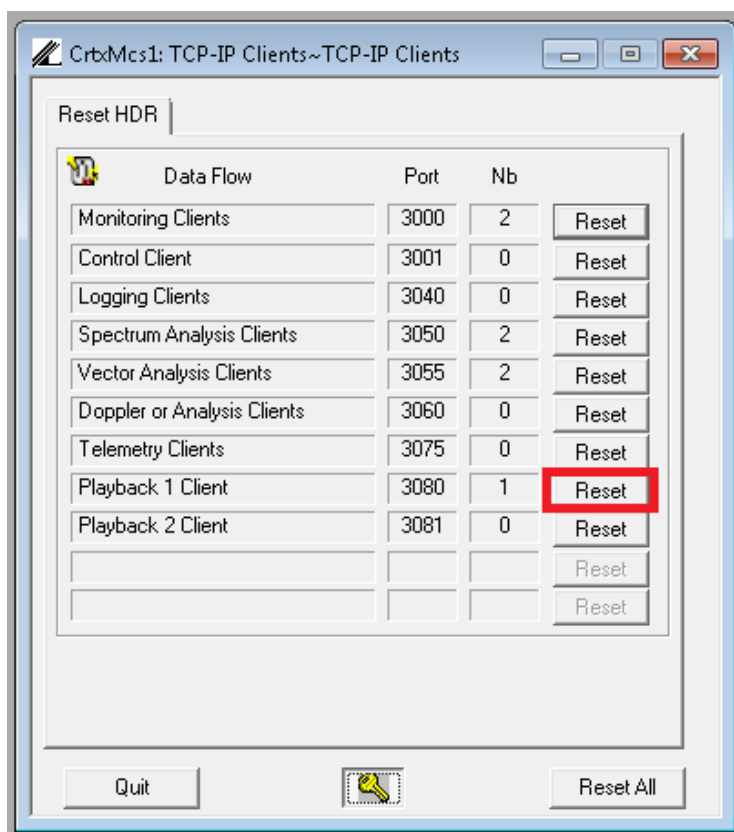


Figure 187: Playback LAN: "Reset playback client" button in "TCP-IP Clients" window

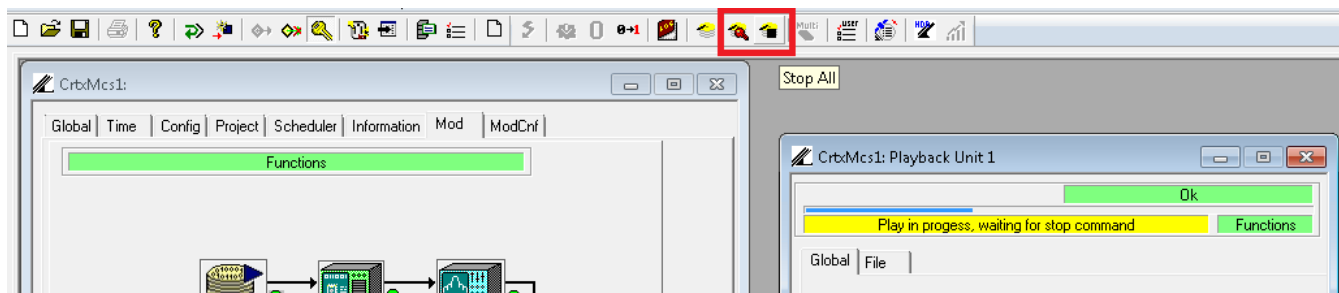


Figure 188: Playback LAN: "Acquisition" and "Stop All" buttons in the toolbar

4.5.9.2 "File" Window

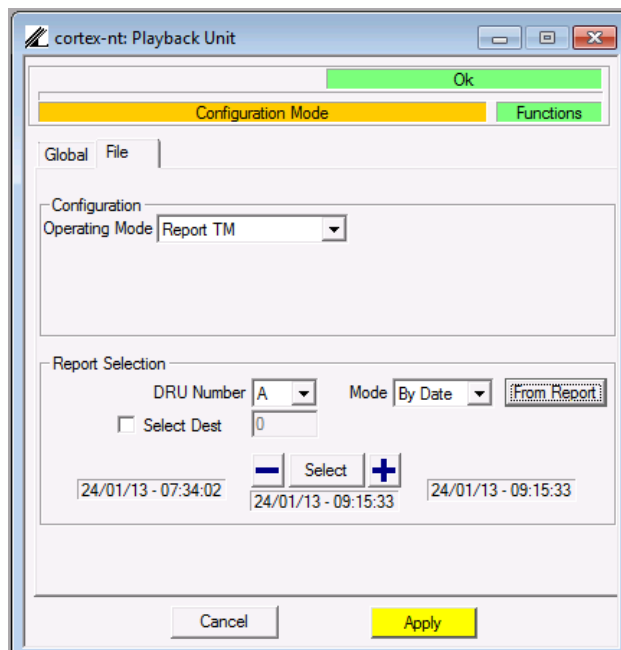


Figure 189: Playback: File Window in Report TM mode (By date)

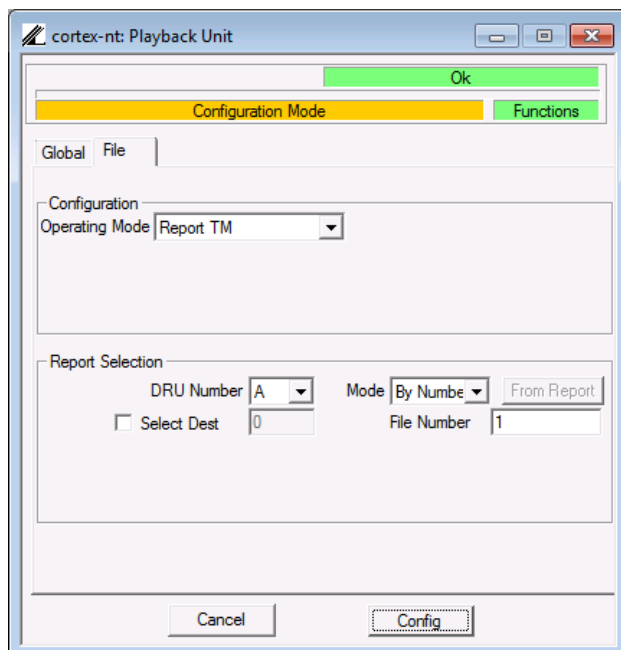


Figure 190: Playback: File Window in Report TM mode (By number)

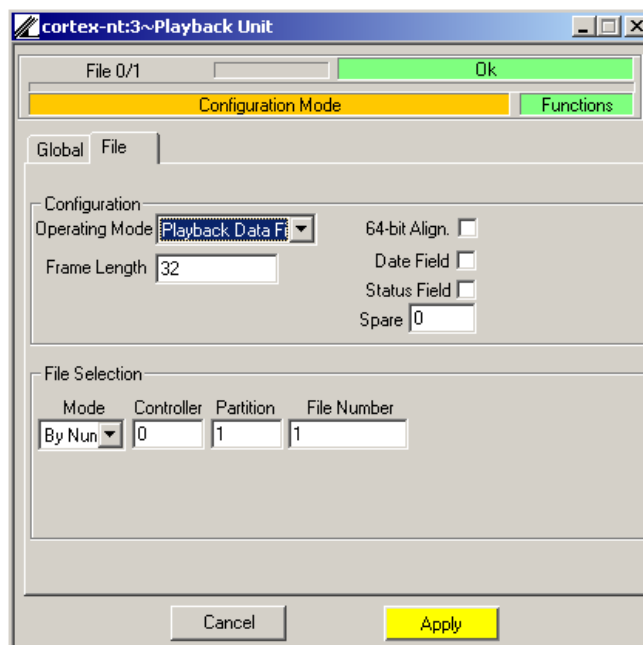


Figure 191: Playback: File Window in Playback Data File mode

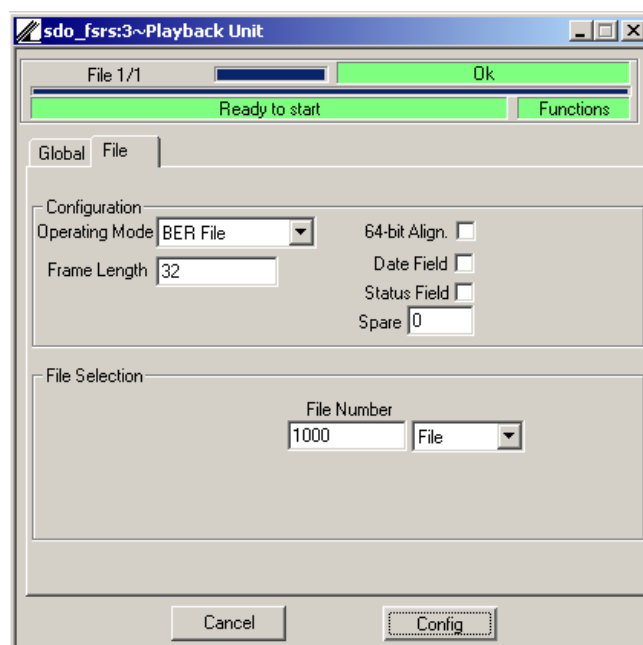


Figure 192: Playback: File Window in BER File mode

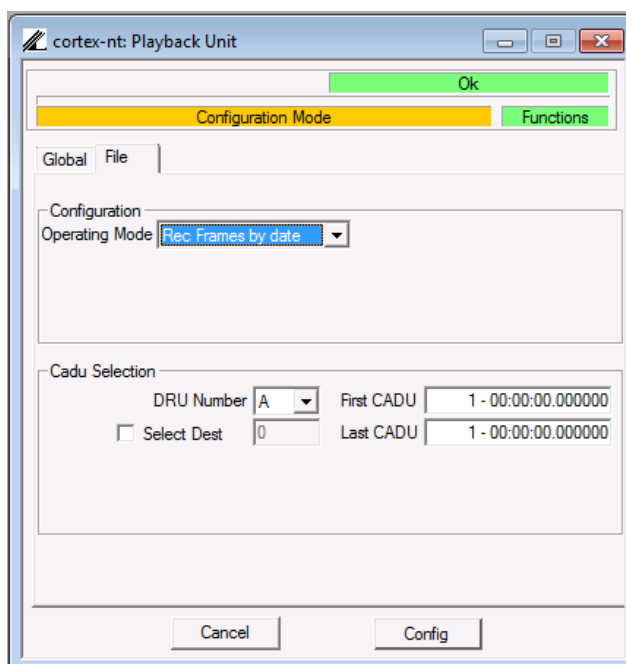


Figure 193: Playback: File Window in Recorded Frames mode

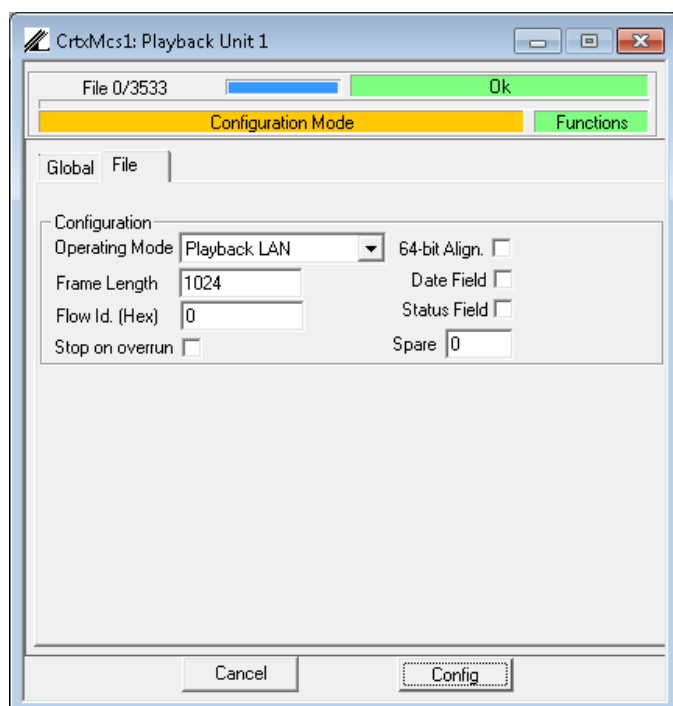


Figure 194: Playback: File Window in Playback LAN mode

4.5.9.2.1 Configuration

OPERATING MODE	Select the source of the data for the replay: ■ Report TM: Selects recorded data from a previous satellite pass (from the recorded report file of the pass – refer to “From report” command) ■ Playback Data from File: Selects data recorded inside a User-defined file ■ Playback Data from Directory: Selects data recorded inside a set of User-defined files recorded inside one directory ■ Rec Frames by date: Selects data CADUs by dates ■ BER file : Selects a PRN code or a data simulation file ■ Playback LAN
FRAME LENGTH	Selects the length of the data frames for the playback (synchronization word included).
64-BIT ALIGN	Enables to indicate that TM blocks of the file(s) to be replayed are 64-BIT aligned.
DATE FIELD	Enables to indicate that a date field (8-bytes word) is inserted in the TM block of the file(s) to be replayed.
STATUS FIELD	Enables to indicate that a status field (8-bytes word) is inserted in the TM block of the file(s) to be replayed.
SPARE	Enables to indicate if other spare 8-bytes words are inserted in the TM block of the file(s) to be replayed. Number of words from 0 to 8.
LOOP ON FILE	Enable or disable the loop replay on the selected file.
DRU NUMBER	A to H selection of the DRU from where the replayed data are extracted.
FIRST CADU	Date of the first CADU to replay (only in “Rec Frames by date” mode)
LAST CADU	Date of the last CADU to replay (only in “Rec Frames by date” mode)
FLOW ID. (HEX)	Identification of the Playback data flow. This value is reported in all TCP-IP messages transmitted over the Playback ports (PBK-1 or PBK-2). Enter any hexadecimal value from 0 to FFFFFFFF (only in “Playback LAN” mode)
STOP ON OVERRUN	Stop session if overrun is detected (only in “Playback LAN” mode)

4.5.9.2.2 File Selection

With "Report TM" mode:

CHANNEL

Selection of the Recording channel (DRU-A, B, C, ...H) where the pass report file has to be selected.

■ DRU-A to DRU-H

REPORT NUMBER

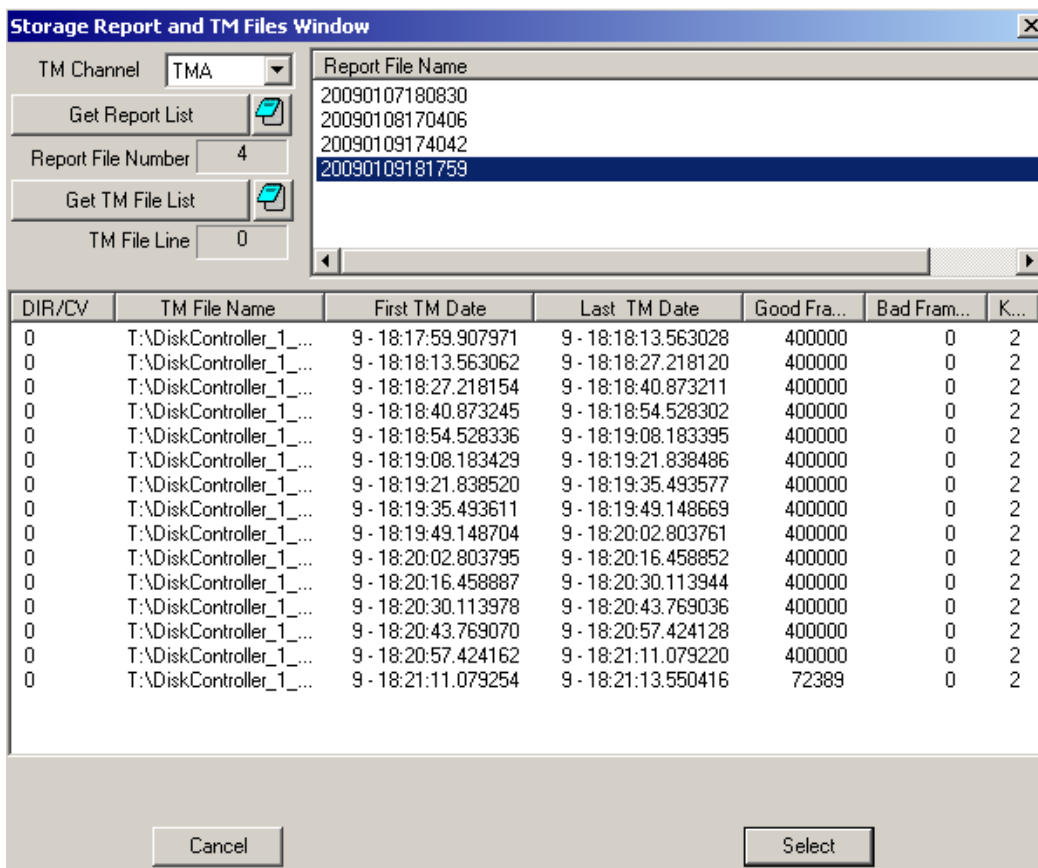
Selection of the report number: **1 to N**. Can be selected directly by clicking in the Storage report and TM files Window

LINE

Selection of a specific line (a specific file) inside the selected pass report: **0 to N**. Can be selected directly by clicking in the Storage report and TM files Window

From Report

Allows to open the Storage report and TM files Window:



The screenshot shows the 'Storage Report and TM Files Window'. It has a 'TM Channel' dropdown set to 'TMA'. Below it are buttons for 'Get Report List' and 'Get TM File List'. There are input fields for 'Report File Number' (set to 4) and 'TM File Line' (set to 0). A list box shows report file names, with '20090109181759' selected. Below the list box is a table with columns: DIR/CV, TM File Name, First TM Date, Last TM Date, Good Fra..., Bad Fram..., and K... The table contains 16 rows of data. At the bottom are 'Cancel' and 'Select' buttons.

DIR/CV	TM File Name	First TM Date	Last TM Date	Good Fra...	Bad Fram...	K...
0	T:\DiskController_1_...	9 - 18:17:59.907971	9 - 18:18:13.563028	400000	0	2
0	T:\DiskController_1_...	9 - 18:18:13.563062	9 - 18:18:27.218120	400000	0	2
0	T:\DiskController_1_...	9 - 18:18:27.218154	9 - 18:18:40.873211	400000	0	2
0	T:\DiskController_1_...	9 - 18:18:40.873245	9 - 18:18:54.528302	400000	0	2
0	T:\DiskController_1_...	9 - 18:18:54.528336	9 - 18:19:08.183395	400000	0	2
0	T:\DiskController_1_...	9 - 18:19:08.183429	9 - 18:19:21.838486	400000	0	2
0	T:\DiskController_1_...	9 - 18:19:21.838520	9 - 18:19:35.493577	400000	0	2
0	T:\DiskController_1_...	9 - 18:19:35.493611	9 - 18:19:49.148669	400000	0	2
0	T:\DiskController_1_...	9 - 18:19:49.148704	9 - 18:20:02.803761	400000	0	2
0	T:\DiskController_1_...	9 - 18:20:02.803795	9 - 18:20:16.458852	400000	0	2
0	T:\DiskController_1_...	9 - 18:20:16.458887	9 - 18:20:30.113944	400000	0	2
0	T:\DiskController_1_...	9 - 18:20:30.113978	9 - 18:20:43.769036	400000	0	2
0	T:\DiskController_1_...	9 - 18:20:43.769070	9 - 18:20:57.424128	400000	0	2
0	T:\DiskController_1_...	9 - 18:20:57.424162	9 - 18:21:11.079220	400000	0	2
0	T:\DiskController_1_...	9 - 18:21:11.079254	9 - 18:21:13.550416	72389	0	2

With "Report TM" mode:

CONTROLLER	Selection of the Controller for the replay.
PARTITION	Selection of the Partition for the replay.
FILE NBR.	Selection of the file number for the replay: 1 to N . The file must be recorded in the "PLAYBACK" directory of the selected partition.

With "BER File" mode:

MODE	BER measurement mode (simulated data for BER analysis): ■ File Simulated data from a file on the disk ■ PseudoDxx Pseudo random data, using from PN generator xx (where xx = 10, 11, 15 or 23. See Section 3.6.1.1.1)
FILE NUMBER	Selection of the simulation number: 1 to N .

4.5.9.2.3 Specific operating mode: Playback LAN

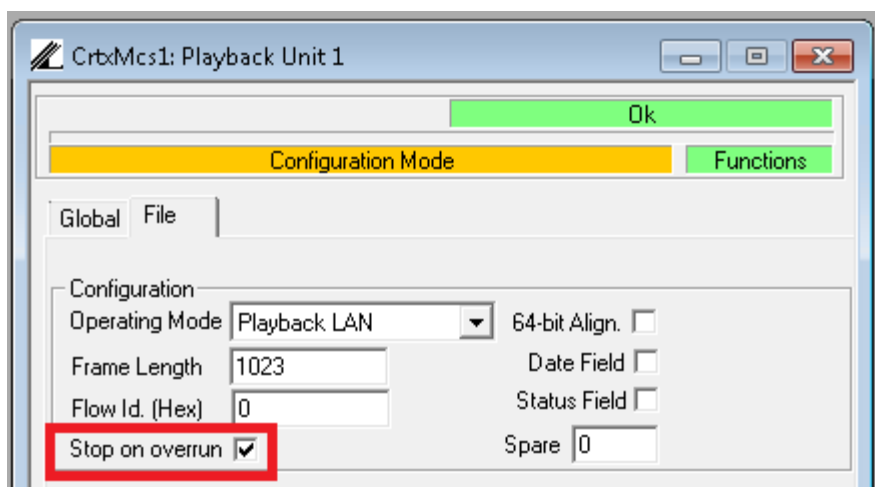


Figure 195: Playback LAN: Overrun option

In "Playback LAN" mode, user can choose the behavior if overrun is detected:

- Default: overrun counter is incremented and session continues.
- Stop on overrun: user is disconnected from Playback port (PBK-1 or PBK-2), session is immediately stopped and all remaining data are flushed.

4.5.10 Logging Windows

4.5.10.1 "Data Logging Control" Window

For details on the data logging function, refer to section 3.8 .

To control the data logging function, in the upper toolbar click on  . This operation opens the data logging control window:

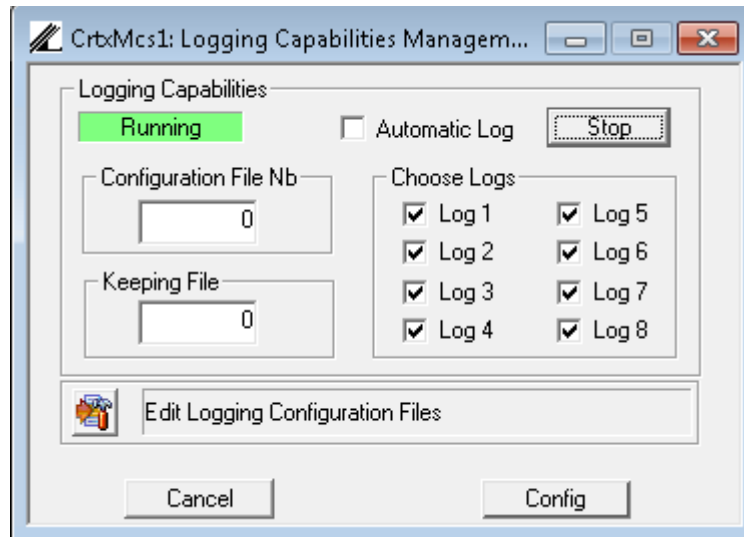


Figure 196: Logging status Window

4.5.10.1.1 Control

START / STOP Start or stop the data logging in manual mode.

AUTOMATIC LOG If selected, the recording starts automatically with the Acquisition mode.

CONFIGURATION FILE Select the number of the configuration file used for the current data logging operation

CHOOSE CHANNELS Select the log(s) Log 1, Log 2, Log 3, Log 4, Log 5, Log 6, Log 7 and/or Log 8 for the current logging operation.



Open the editor window to configure a new configuration file (see next section).

4.5.10.1.2 Status

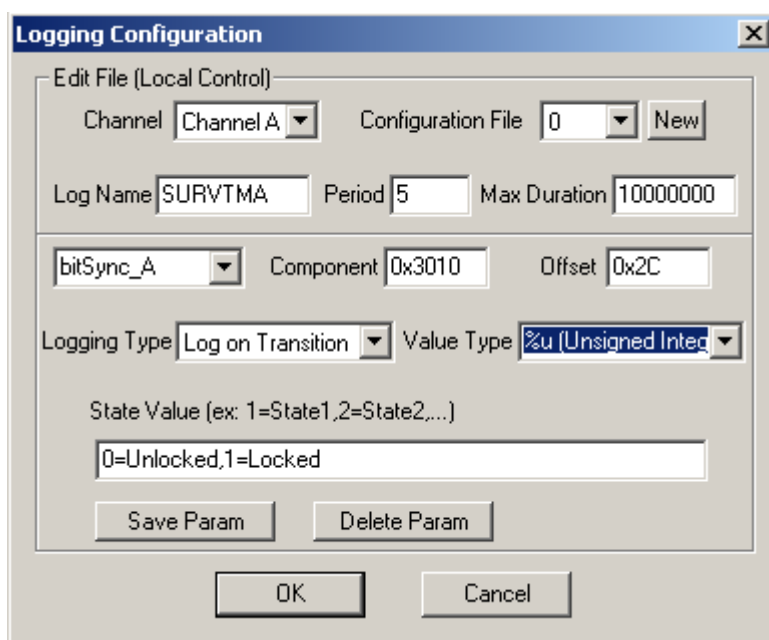


The status field can take the following values: Inactive, Running or Automatic. In case of a non-valid configuration file number, Error is reported in this status field.

4.5.10.2 "Data Logging Editor" Window

Note that the data logging edition can be performed only in local mode.
For details on the data logging editor function, refer to section 3.8.

To edit an already existing or a new logging configuration file, click on  in the previous data logging control window to open the following data logging editor window:



The dialog box is titled "Logging Configuration". It contains the following fields and controls:

- Edit File (Local Control):**
 - Channel:** A dropdown menu currently showing "Channel A".
 - Configuration File:** A dropdown menu currently showing "0", with a "New" button next to it.
- Log Name:** A text field containing "SURVTMA".
- Period:** A text field containing "5".
- Max Duration:** A text field containing "10000000".
- bitSync_A:** A dropdown menu currently showing "bitSync_A".
- Component:** A text field containing "0x3010".
- Offset:** A text field containing "0x2C".
- Logging Type:** A dropdown menu currently showing "Log on Transition".
- Value Type:** A dropdown menu currently showing "%u (Unsigned Integ)".
- State Value (ex: 1=State1,2=State2,...):** A text field containing "0=Unlocked,1=Locked".
- Buttons:** "Save Param", "Delete Param", "OK", and "Cancel".

Figure 197: Logging editor Window

CHANNEL	Select the channel: ChannelA , ChannelB , ChannelC or ChannelD on which the logging configuration file applies.
CONFIGURATION FILE	Select the number of the configuration file to edit: 0 to 100
LOG NAME	Enter a general label associated with the logging file (this label will appear inside the data logging text file).
PERIOD	Select a period (in seconds) for periodical recording.
MAX DURATION	If 0, the recording will stop only by clicking on the stop button in the control window. If a value is set (in seconds), the recording will stop at the end of this delay even is the stop button has not been clicked.
PARAMETER LABEL	Enter a label for the parameter to log (this label will appear inside the data logging text file).
COMPONENT AND OFFSET	Select the effective HDR control parameter or status by entering its component number (hexadecimal format) and offset (hexadecimal or decimal format) as defined in the STI document.

LOGGING TYPE

Select the logging type associated with the selected parameter

- Periodic Logging
- Log on Transition
- Log on Variation

VALUE TYPE

Select the logging format for the value of the selected parameter:

- Unsigned Integer
- Signed Integer
- Hexadecimal
- Float
- Double float

STATE VALUE

In “Log on Transition” mode, defined the status labels associated with the actual status values (these label will appear inside the data logging text file).



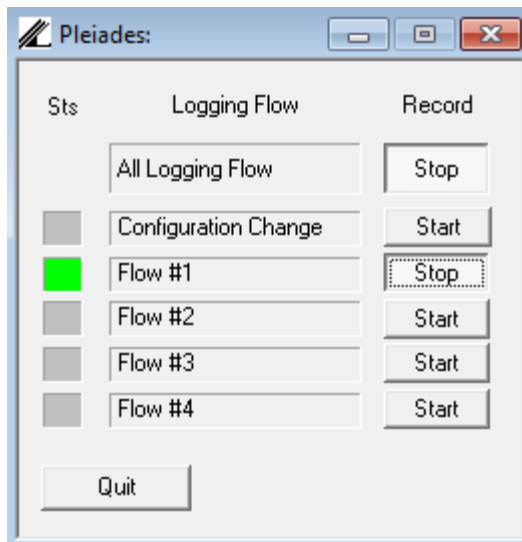
Save the current parameter definition in the current configuration file.



Delete the selected parameter from the current configuration file.

4.5.11 Extended Logging Windows

To open the control window of the Extended Logging function, click on  in the upper tab of the GUI



From this window, each logging flow, previously defined, can be individually controlled (Start/Stop).

5 MAINTENANCE PROCEDURES

5.1 General

5.1.1 *Mechanical Inspection and Cleaning*

It is recommended to take the following precautions to get the best performance from the system:

- ◆ Do not carry out any dust-producing work in the room housing the CORTEX HDR. Clean the air filters regularly to avoid electronic components overheating.
- ◆ Use only alcohol or a water-based cleaner on the front and rear panels of equipment. Other solvents may damage the keys, LCD and markings.
- ◆ Perform a regular operating check on the fan units (check for any abnormal temperature rise, check the air flow at the air exhaust). The system is not designed for permanent operation without ventilation.
- ◆ **Preventive maintenance:** it is recommended to change the Pentium cooling fan every 3 years.

5.1.2 *Handling Precautions*

WARNING:

**SOME OF THE HARDWARE COMPONENTS USE CMOS TECHNOLOGY AND CAN BE DAMAGED
BY ELECTROSTATIC DISCHARGE FROM YOUR CLOTHES OR BODY.**

**ALWAYS USE AN ANTI-STATIC BRACELET WHEN WORKING ON THE CORTEX
TO REPLACE A BOARD OR COMPONENT.**

- ◆ Never remove a board from the CORTEX HDR without switching it off first and removing the power cord!
- ◆ When a module has been removed from the chassis, it is essential to handle and store it with care. It is particularly important to ensure that it is not exposed to electrostatic shocks.
- ◆ Certain precautions must also be taken when reinstalling a board in the chassis: when plugging the board into the PCI bus connectors, never force it; screw the board strip attaching screws back into the frame; reinstall the connector locking screws. Only then may the CORTEX HDR be switched back on.

5.1.3 *CPU Load Considerations*

**SDSY DOES NOT WARRANTY CORTEX HDR PERFORMANCE
IF ADDITIONAL SOFTWARE IS LOADED / EXECUTED ON THE MACHINE.**

5.1.4 *Software Loading*

Avoid using floppy disks from unidentified sources. The presence of a virus may severely affect the performance of the equipment or crash the software.

5.2 Functional Investigation

The following check-list can be followed for checking that the CORTEX HDR functions correctly:

- Demodulator check: signal acquisition and demodulation.
- Viterbi decoder check (if the Viterbi license is supplied).
- Demodulator output ports.

5.2.1 Demodulator Check

Apply a modulated RF signal (input level between -10 and -50 dBm) to the Demodulator board and verify the acquisition and demodulation process:

1. If the IF level status is lower than 50 dBm, check the RF connecting cable. Check also that the IF input port is correctly programmed (Nominal or Alternate IF).
2. Check the correctness of the Demodulator status: Carrier Offset, BR Offset, PLL locked, Bit Synchronizer Locked.
3. Check that the displayed constellation is the expected one and remains stable (see Figure 126):
 - Four points on a circle for QPSK, OQPSK (SQPSK) and UQPSK (AQPSK, AUQPSK, AUSQPSK) modulations.
 - Two points on the abscissa axis for BPSK modulation.
 - Eight points on a circle for 8PSK modulations
4. Check that the Eb/No status is correct.

If the Demodulator fails to lock, verify the setting of the unit.

5.2.2 Viterbi Check

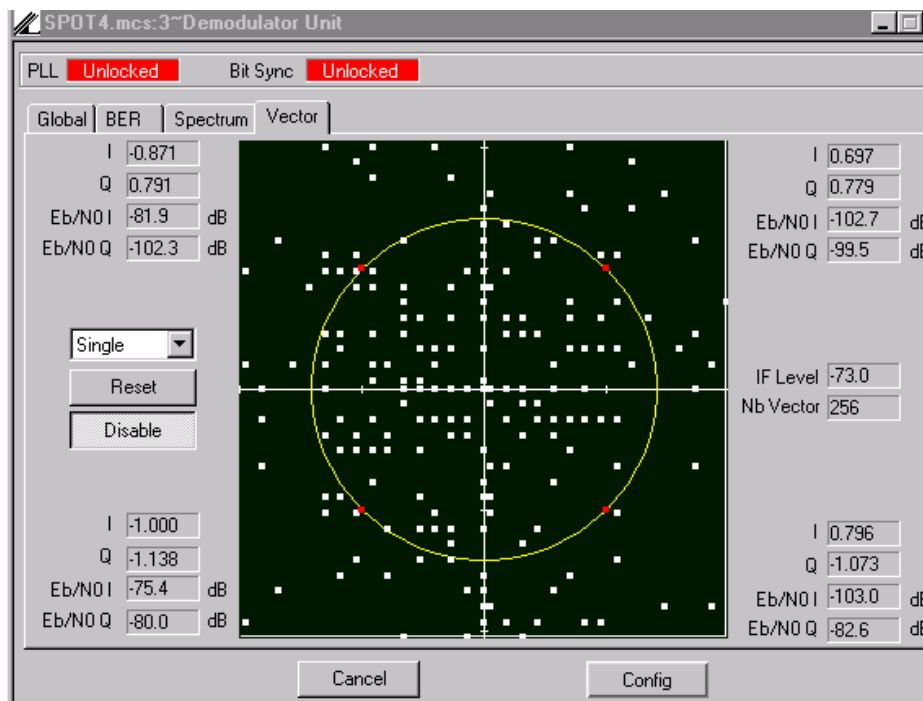
1. Apply an RF signal with Viterbi encoded data and verify the acquisition and demodulation process (see previous section).
2. Check the Viterbi Lock / Unlock status.
3. Check that the Viterbi BER status is in accordance with the estimated Eb/No status.
4. If the Viterbi fails to decode, check the setting of the Viterbi decoder: single or dual Viterbi mode, G2 inverted, etc...

5.2.3 Demodulator Output Ports

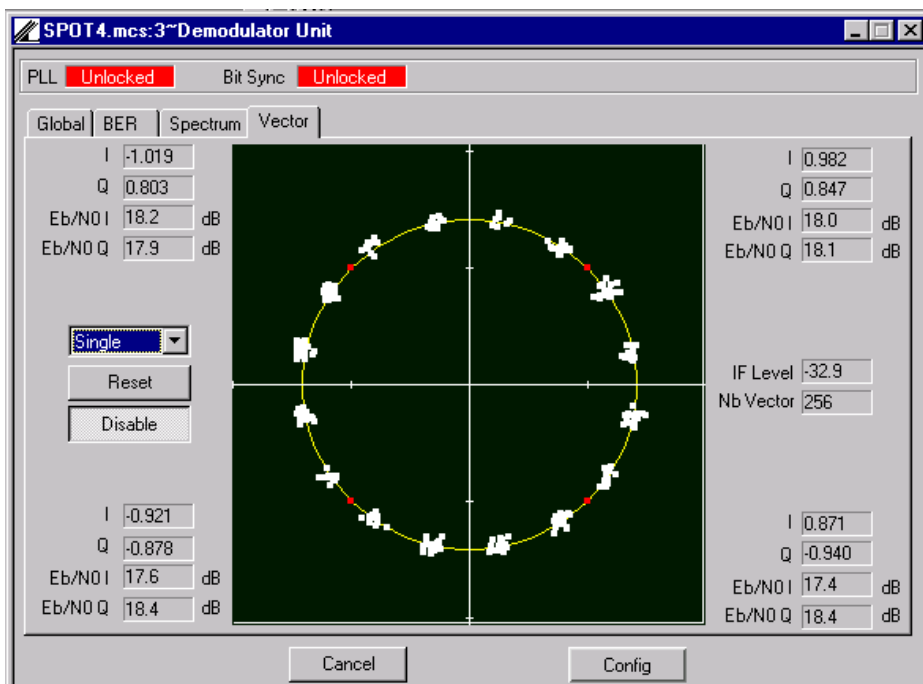
Once the Demodulator and the Viterbi decoder (if any) are locked, check that the Demodulator output signals are formatted as expected by the Ingestion System. In case of problem (wrong BER measured by the Data Ingestion System), check that:

1. The output ports setting is correct: Output Clock, I/Q polarity, Normal or I&Q merged, Output mode. If the problem remains, try different setting.
2. The PCM code is correct (NRZ-M, DNRZ, ...).
3. The hardware connections to the Data Ingestion System are correct.

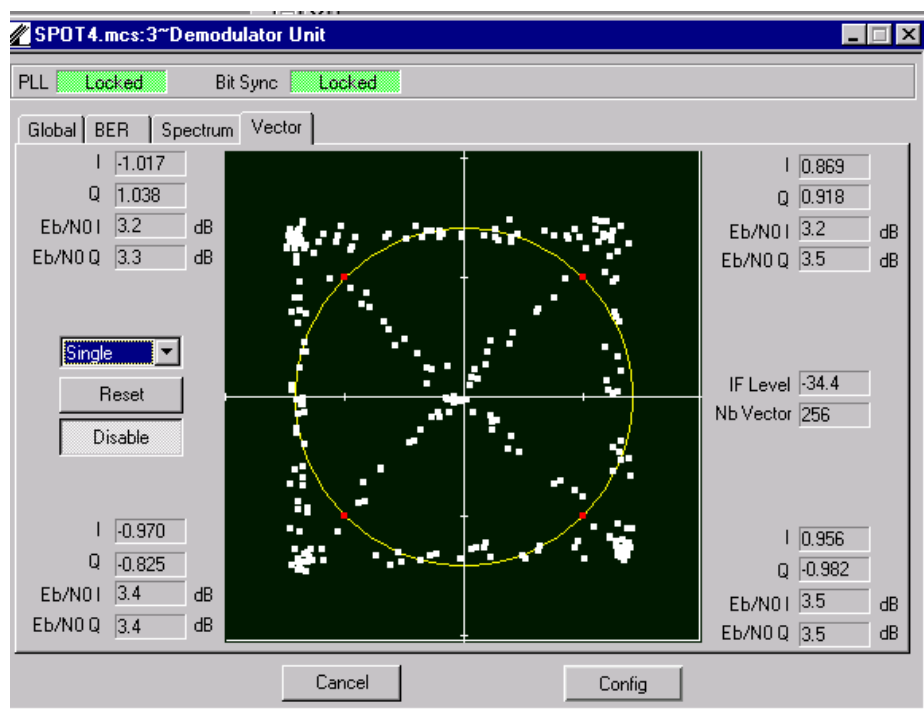
5.2.4 Interpretation of the Vector Display



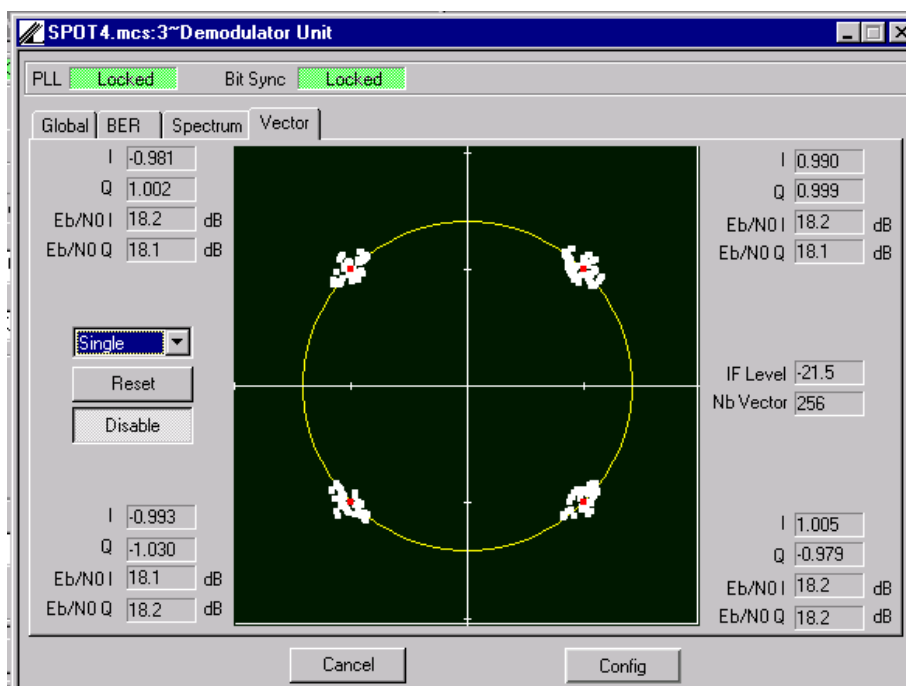
- Low IF input level (-73 dBm). No signal at Demodulator input
- I/Q display: random noise



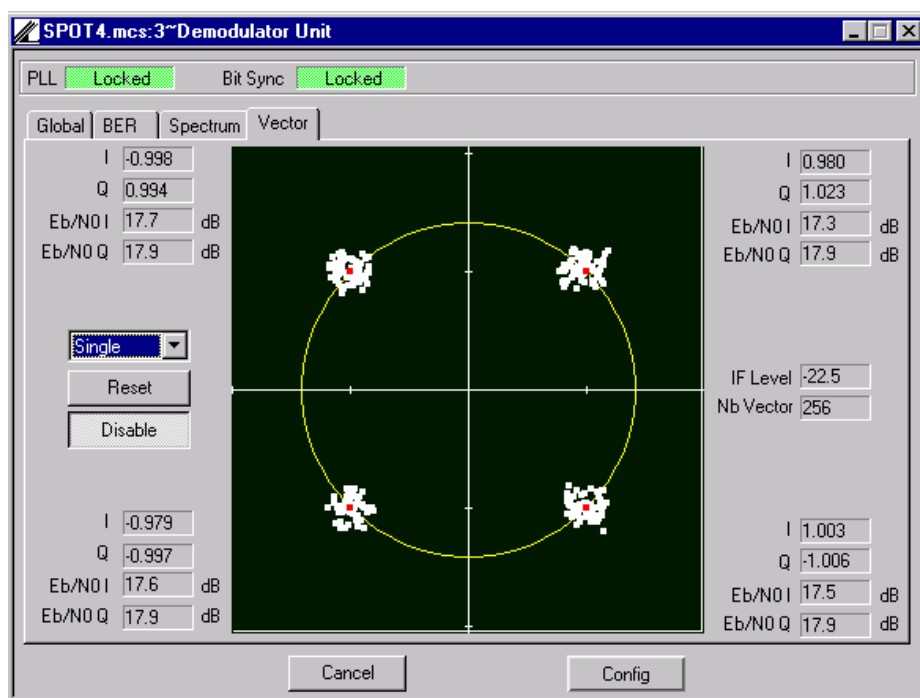
- Incorrect IF setting (signal = QPSK 720 MHz. Demodulator setting = 721 MHz)



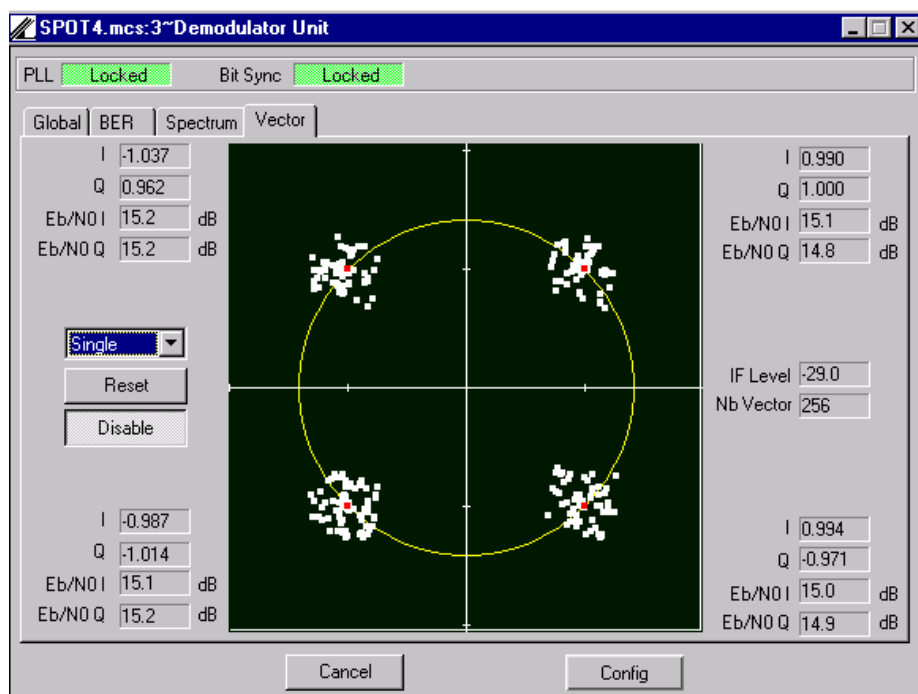
- Incorrect bit rate setting (signal = QPSK 50 Mbps. Demodulator setting: 47 Mbps). Matched filter inefficient.



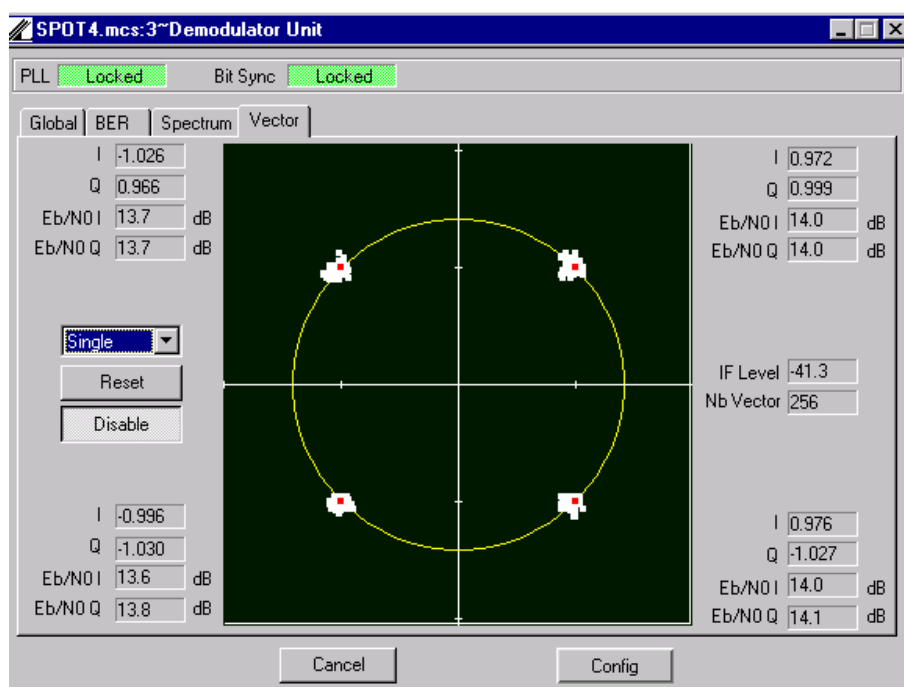
- QPSK input signal with unwanted MSK modulation appearing as phase noise on the I/Q display



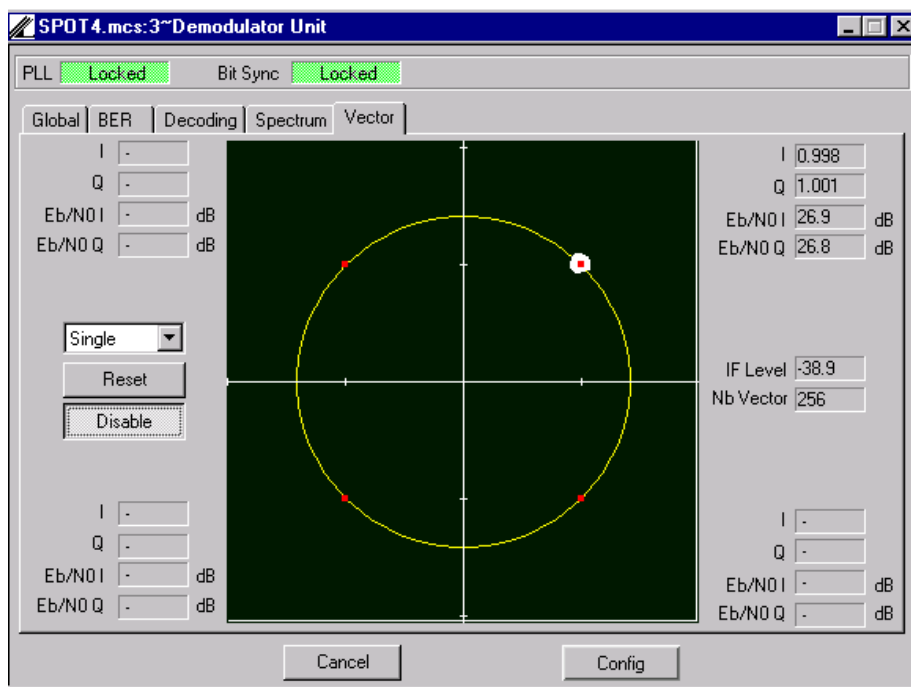
- Incorrect matched filter setting (input signal = RRC, Roll-off 0.3. Demodulator setting: I & D) provoking noise-looking inter-symbol interferences.



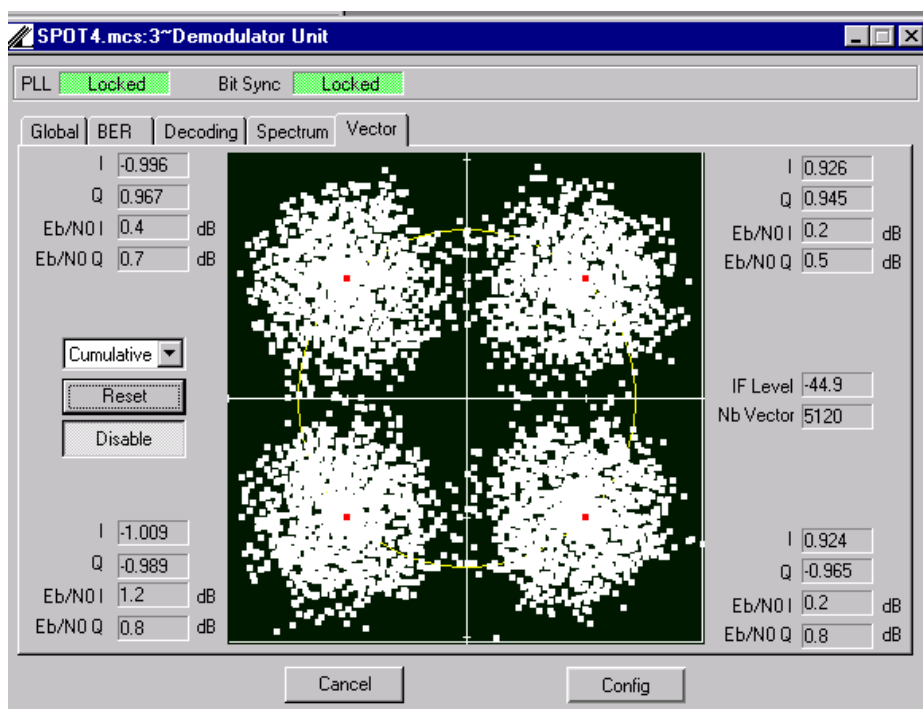
- Dual 50 Mbps QPSK signals at 720 and 770 MHz (adjacent channel interferences).



- Dual 50 Mbps QPSK signals at 720 and 770 MHz. Limited interferences thanks to an RRC filter (Roll-off 0.1) on the Demodulator.



- Reception of an un-modulated carrier.



- Demodulator locked to a heavily affected IF signal (E_b/N_0 close to 0).

5.3 Hardware Investigation & Repair

5.3.1 LED Indicators

5.3.1.1 Test Modulator Board

One LED indicator is located on the front panel of the board:

HW	Turns to green and flashes at 1-PPS to show that the hardware is correct.
	Unlit (or high rate flashing) in the case of Test Modulator board or CPU board failure.

5.3.1.2 Demodulator Board

One LED indicator is located on the front panel of the board:

HW	Turns to green and flashes at 1 PPS to show that the hardware is correct.
	Unlit (or high rate flashing) in the case of Demodulator board or CPU board failure.

5.3.1.3 CPU Board

Refer to Annex 4.

5.3.2 IPMIMonitor Tool. Temperature Alarm

The CPU board has several sensors for monitoring the Pentium processors and board temperature.

These sensors are permanently monitored by a dedicated software tool **IPMIMonitor**. To run this software, click on START → PROGRAM → IPMIMonitor → IPMIMonitor.

Check the three following status in the list:

STATUS NAME:	TYPICAL VALUE @ 22°C AMBIENT: ALARM THRESHOLD:	
Baseboard Temp.	34 °C	52 °C
Processor 1 Temp.	43 °C	72 °C
Processor 2 Temp.	43 °C	72 °C

These alarms are reported to the remote monitoring clients via the Global CORTEX Monitoring table (Annex 1, STI 100013_<product name>, Table 6, offset 41) and at the CORTEX GUI (bottom status bar. See section 4.3.5.5).

In case of temperature alarm:

- **Immediately shutdown and switch off the chassis to avoid any damage to the hardware.**
- Verify that the air exhaust system of the housing rack works properly,
- Verify that the air intake and exhaust grids are clear. Important: a free rack space must be provided above and below the chassis,
- Check the cooling fans on both sides of the chassis, at the rear panel of the chassis (power supply module) and on both CPU processors.
- Return the chassis to SDSY or its maintenance representative if the problem cannot be fixed on site.

5.3.3 *PC Workstation Maintenance*

Refer to Annex 4 for detailed instructions on how to change:

- ☐ The power supply module
- ☐ The fan units
- ☐ The disk & DVD
- ☐ The display assembly
- ☐ The keyboard/trackball.

5.4 Software Upgrade & Installation

The CORTEX HDR software is supplied in the form of DVD:

- ☐ **Windows OS** for re-installing the operating system in the case of disk failure.
- ☐ **HDR software.** The CD ROM contains the Signal Processing Software (SPS), Monitoring & Control Software (MCS), Demodulator board software (disk-resident part only) and the user documentation.

5.4.1 Loading or Upgrading the IN-SNEC Software

Strictly follow the procedure described on the back of the reference CD ROM.

5.4.2 Complete Software Re-installation

In the case of disk failure, complete software re-installation will be required:

1. Load Windows Operating System (procedure described at the back of the CD ROM).
2. Load the HDR software (procedure described on the back of the CD ROM).

6 SATELLITES CONFIGURATION EXAMPLES

Next table is a non-exhaustive list of in-orbit satellites with their main downlink configuration parameters: modulation, bit rate, PCM code, Viterbi mode, output mode.

These parameters can be entered manually on the CORTEX HDR or, more easily, by loading the MCS file indicated in the left column. These MCS files can be found in:

C:\Program Files\IN-SNEC\CrtxMcs\Default\Config

Reminder: procedure for configuring the CORTEX HDR from an MCS file:

1. Click on FILE, CLOSE. If necessary, save the current configuration to the active MCS document.
2. Click on FILE, OPEN. Select a new MCS document in the list and click on OK to load it. The configuration status turns to  (unless the new MCS document is strictly identical to the previous one).
3. Enable the configuration mode if necessary (click on  in the lower toolbar and enter the Configuration password).
4. Reconfigure the unit by clicking on . The configuration status turns to .

WARNING:

**SATELLITE INFORMATION LISTED IN THIS SECTION IS NOT CONTRACTUAL.
THE OPERATOR SHOULD KNOW THE DETAILED FEATURES OF THE TRACKED SATELLITE
AND PROGRAM THE CORTEX HDR ACCORDINGLY.**

Satellite + MCS Document to be loaded	Bit rate	Modulation	Viterbi Mode	Output mode	Basic Menu
Adeos II <i>Adeos2.mcs</i>	I: 30 Mbps Q: 30 Mbps	UQPSK: 6dB (or ¼ AUSQPSK) NRZ-M	OFF	I channel A Q channel B	5
AKARI <i>Akari.mcs</i>	8.388608 Mbps	QPSK NRZ-M	Dual	Normal	3
Alos <i>Alos.mcs</i>	138.76 Mbps	QPSK DNRZ	OFF	Merge	3
Alsat <i>Alsat2.mcs</i>	65 Mbps	8PSK TCM 2.00/3	OFF	Merge	1
Aqua DB <i>AquaDB.mcs</i>	15 Mbps	OQPSK NRZ-M	OFF	Merge	3
Aqua DP <i>AquaDP.mcs</i>	150 Mbps	OQPSK NRZ-M	OFF	Merge	3
Aqua DP & DB <i>AquaDP_DB.mcs</i>	15 Mbps 150 Mbps	OQPSK NRZ-M	OFF	Merge	3
ATV <i>ATV.mcs</i>	6 Mbps	OQPSK NRZ-M	OFF	Merge	3
Aura DP & DB <i>AuraDP_DB.mcs</i>	15 Mbps 150 Mbps	OQPSK NRZ-M	OFF	Merge	3
CBERS <i>CBERS.mcs</i>	53 Mbps	QPSK DNRZ	OFF	Normal	3
Coriolis 1 <i>Coriolis1.mcs</i>	51.2 Mbps	OQPSK NRZ-M	OFF	Merge	3

Satellite + MCS Document to be loaded	Bit rate	Modulation	Viterbi Mode	Output mode	Basic Menu
Coriolis 2 <i>Coriolis2.mcs</i> <i>Coriolis2_FS.mcs</i>	102.4 Mbps	OQPSK NRZ-M	Dual	Merge	3
Cosmo <i>Cosmo.mcs</i>	155 Mbps	QPSK DNRZ	OFF	Normal	3
Cryosat <i>Cryosat.mcs</i>	100 Mbps	QPSK DNRZ	OFF	Normal	3
Demeter <i>Demeter.mcs</i>	16 Mbps	8PSK TCM 2.00/3	OFF	Merge	1
Envisat <i>Envisat_Asar.mcs</i>	100 Mbps	QPSK DNRZ	OFF	Merge	3
Envisat <i>Envisat_Meris.mcs</i>	100 Mbps	QPSK DNRZ	OFF	Normal	3
Envisat <i>Envisat_Meris_Asar.mcs</i>	100 Mbps	QPSK DNRZ	OFF	Merge	3
EO1 <i>EO1merge.mcs</i> <i>EO1nonmerge.mcs</i> <i>EO1merge_nomerge.mcs</i>	105 Mbps	QPSK DNRZ	OFF	Merge or Normal or Both <i>depend on mcs file</i>	3
EROS <i>EROS.mcs</i>	87.5 Mbps	QPSK DNRZ	OFF	Merge	3
ERS-1 <i>ERS1merge.mcs</i> & <i>ERS1nonmerge.mcs</i>	105 Mbps	QPSK DNRZ	OFF	Merge or normal <i>depend on mode</i>	3

Satellite + MCS Document to be loaded	Bit rate	Modulation	Viterbi Mode	Output mode	Basic Menu
Feng Yun 3 <i>FengYun3_DPT_HRTP.mcs</i> <i>FengYun3_DPT_HRTP_FS.mcs</i>	124 Mbps 37.4 Mbps	QPSK DNRZ	Dual Rate 3/4 Rate 1/2	Normal	3
Feng Yun 3 <i>FengYun3_HRPT_MPT_FS.mcs</i>	5.6 Mbps 37.4 Mbps	QPSK DNRZ	Dual Rate 3/4 Rate 1/2	Normal	3
Formosat 2 <i>Formosat-2.mcs</i>	120 Mbps	QPSK DNRZ	OFF	Normal	3
GOES-R <i>GOES-R.mcs</i>	150 Mbps	OQPSK NRZ-L LDPC	OFF	Normal	2
GOSAT <i>Gosat.mcs</i>	138.76 Mbps	OQPSK NRZ-M	OFF	Merge	3
HY1 <i>HY1.mcs</i>	5.3232 Mbps	QPSK DNRZ	OFF	Merge	3
HY3 <i>HY3.mcs</i>	105 Mbps	QPSK DNRZ	Dual Rate 3/4	Normal	3
IceSat <i>Icesat.mcs</i> <i>Icesat_fs.mcs</i>	40 Mbps	OQPSK NRZM	OFF	Merge	3
Ikonos <i>Ikonos.mcs</i>	320 Mbps	OQPSK NRZ-M	OFF	Merge	3
IRS-1C <i>IRS-1C_liss3.mcs</i>	42.4515 Mbps	QPSK DNRZ	OFF	Merge	3

Satellite + MCS Document to be loaded	Bit rate	Modulation	Viterbi Mode	Output mode	Basic Menu
IRS-1C <i>IRS-1C_pan.mcs</i>	84.903 Mbps	QPSK DNRZ	OFF	Merge	3
IRS-1D <i>IRS-1D_liss3.mcs</i>	42.4515 Mbps	QPSK DNRZ	OFF	Normal	3
IRS-1D <i>IRS-1D_pan.mcs</i>	84.903 Mbps	QPSK DNRZ	OFF	Normal	3
IRSP4 <i>IRSP4.mcs</i>	20.8 Mbps	QPSK DNRZ	OFF	Normal	3
IRSP5 <i>IRSP5.mcs</i>	84.903 Mbps	QPSK DNRZ	OFF	Normal	3
IRSP6 <i>IRSP6.mcs</i>	105 Mbps	QPSK DNRZ	OFF	Normal	3
JEM <i>JEM.mcs</i>	100 Mbps	QPSK NRZ-M	Single	Merge	3
JEM FORWARD <i>JEMFordward.mcs</i>	6 Mbps	BPSK NRZ-M	ON	Normal	3
Kompsat-2 <i>Kompsat2.mcs</i>	320 Mbps	OQPSK NRZ-M	OFF	Merge	3
Landsat-5 <i>Landsat5.mcs</i>	I: 84.09 Mbps Q: 15.06 Mbps	UQPSK: 6dB NRZ-M	OFF	I channel A Q channel B	5
Landsat-7 <i>Landsat7.mcs</i> <i>Landsat7_FS.mcs</i>	I: 74.914 Mbps Q: 74.914 Mbps	UQPSK: 0dB (AQPSK) NRZ-L	OFF	I channel A Q channel B	5

Satellite + MCS Document to be loaded	Bit rate	Modulation	Viterbi Mode	Output mode	Basic Menu
LRO HR MR <i>LRO_High_Medium.mcs</i>	75 Mbps 50 Mbps	OQPSK NRZ-M	Dual	Merge	3
LRO MR LR <i>LRO_Medium_Low.mcs</i>	50 Mbps 25 Mbps	OQPSK NRZ-M	Dual	Merge	3
Metop <i>Metop.mcs</i> <i>Metop_FS.mcs</i>	70 Mbps	QPSK NRZ-L	OFF	Merge	3
Metop HRPT <i>Metop_HRPT_FS.mcs</i>	4.6666 Mbps	QPSK NRZ-L	Single	Normal	3
NPOES <i>NPOES.mcs</i> <i>NPOES_FS.mcs</i>	300 Mbps	OQPSK NRZ-M	OFF	Merge	3
NPP <i>NPP.mcs</i> <i>NPP_FS.mcs</i>	300 Mbps	OQPSK NRZ-M	OFF	Merge	3
NRL-1 <i>NRL1.mcs</i>	51.2 Mbps	OQPSK NRZ-M	OFF	Merge	3
NRL-2 <i>NRL2.mcs</i>	102.4 Mbps	OQPSK NRZ-M	Dual	Merge	3
OFEC <i>OFEC.mcs</i>	103 Mbps	QPSK DNRZ	OFF	Merge	3
Oryx <i>Oryx.mcs</i>	125.414 Mbps	QPSK NRZ-M	Dual	Normal	3

Satellite + MCS Document to be loaded	Bit rate	Modulation	Viterbi Mode	Output mode	Basic Menu
PLEIADES <i>Pleiades.mcs</i>	199.954 Mbps	8PSK TCM 2.50/3	OFF	Normal	1
PROBA <i>PROBA.mcs</i> <i>PROBA_FS.mcs</i>	2 Mbps	BPSK NRZ-L	ON	Normal	3
QuickBird <i>QuickBird.mcs</i>	320 Mbps	OQPSK NRZ-M	OFF	Normal	3
RadarSat <i>Radarsat.mcs</i>	105 Mbps	QPSK DNRZ	OFF	Merge	3
RazakSat <i>RazakSat.mcs</i>	30 Mbps	QPSK DNRZ	OFF	Normal	3
RESURS-DK <i>RESURS_DK.mcs</i>	153.6 Mbps	QPSK NRZ-L	OFF	Normal	3
SAC-C <i>SAC-C.mcs</i> <i>SAC-C_FS.mcs</i>	6.606 Mbps	BPSK NRZ-M	ON	Merge	3
SDO <i>sdo.mcs</i> <i>sdo_fsrs.mcs</i>	300 Mbps	OQPSK NRZ-M	Dual	Merge	3
SICH 2 <i>SICH-2.mcs</i>	61.44 Mbps	OQPSK NRZ-M	Dual	Normal	3
SICH 3 <i>SICH-3.mcs</i>	320 Mbps	OQPSK NRZ-M	OFF	Normal	3
SICH BPSK HR <i>SICH-BPSK-HR.mcs</i>	61.44 Mbps	BPSK NRZ-M	OFF	Normal	3

Satellite + MCS Document to be loaded	Bit rate	Modulation	Viterbi Mode	Output mode	Basic Menu
SICH BPSK LR <i>SICH-BPSK-LR.mcs</i>	15.36 Mbps	BPSK NRZ-M	OFF	Normal	3
SPOT-2 <i>SPOT2.mcs</i>	49.372 Mbps	QPSK DNRZ	OFF	Merge	3
SPOT-4 <i>SPOT4.mcs</i>	49.372 Mbps	QPSK DNRZ	OFF	Merge	3
SPOT-5 <i>spot5.mcs</i> <i>spot5_FS.mcs</i>	49.372 Mbps	QPSK DNRZ	OFF	Merge	3
SPOT VGT L band <i>SPOT_VGT_LBand.mcs</i>	1.0204 Mbps	BPSK NRZ-M	ON	Normal	3
SPOT VGT X band <i>SPOT_VGT_Band.mcs</i>	3.4021 Mbps	BPSK NRZ-M	OFF	Normal	3
SSTL BPSK <i>SSTL-Bpsk.mcs</i>	40 Mbps	BPSK NRZ-M	OFF	Merge	3
SSTL QPSK <i>SSTL-Qpsk.mcs</i>	80 Mbps	QPSK NRZ-M	Single	Normal	3
Terra <i>Terra.mcs</i> <i>Terra_FS.mcs</i>	I: 26.250 Mbps Q: 13.125 Mbps	UQPSK: 6dB (or 1/4 AUSQPSK) NRZ-M	I: ON Q: OFF	I channel A Q channel B MODIS is on channel A: data1 + CLK	5

Satellite <i>+ MCS Document to be loaded</i>	Bit rate	Modulation	Viterbi Mode	Output mode	Basic Menu
Terra High Rate <i>Terra_highdatarate.mcs</i>	150 Mbps	OQPSK NRZ-M	I: 8 // decod. Q: 8 // decod.	Merge	7
TerraSARX <i>TerraSARX.mcs</i>	300 Mbps	QPSK DNRZ	OFF	Merge	3
TopSat BPSK <i>TopSat-BPSK.mcs</i>	11.052 Mbps	BPSK NRZ-M	ON	Normal	3
TopSat QPSK <i>TopSat-BPSK.mcs</i>	22.104 Mbps	QPSK NRZ-M	Dual	Normal	3
Worldview 2 <i>Worldview2.mcs</i>	400 Mbps	OQPSK NRZ-M DEAF	OFF	Merge	3

7 ANNEX 1: CORTEX HDR ETHERNET INTERFACE

8 ANNEX 2: DEMODULATOR BOARD

9 ANNEX 3: TEST MODULATOR BOARD

10 ANNEX 4: PC-COMPATIBLE WORKSTATION

11 ANNEX 5: PROJECT-SPECIFIC DATA

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